ALGEBRAIC GEOMETRY I

LECTURE COURSE IN BONN, WS 2025

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1. Lecture 1: Presheaves and Sheaves

1.1. **Quick motivation.** Recall from commutative algebra that every ring A can be considered as a ring of functions on its spectrum $\operatorname{Spec} A$ equipped with the Zariski topology. Our main objects of interest in this lecture will be *schemes*. Roughly speaking, schemes are objects that "look locally" like pairs $(\operatorname{Spec} A, A)$, where A is a ring. Making precise what "looks locally" means is the topic of the first couple of weeks of the lecture. Examples of schemes include:

(1) $(\operatorname{Spec} A, A)$ itself: For example, open subsets of $\operatorname{Spec} A$ are not necessarily spectra, but they fall into the formalism of scheme theory. This way, commutative algebra becomes a part of algebraic geometry.
(2) Affine and projective varieties: Zero sets of polynomials in affine or projective space over an algebraically closed field k are the closed points of certain schemes over k. This way, classical algebraic geometry can be studied using scheme theory.

1.2. **Presheaves.** Presheaves are a way of keeping track of algebraic data on a topological space. More precisely:

Definition 1.1. Let X be a topological space and $\mathcal{C}$ a category. A *presheaf $\mathcal{F}$ of $\mathcal{C}$ on X* consists of

(1) For all $U \subseteq X$ open, an object $\mathcal{F}(U)$ in $\mathcal{C}$.
(2) For all $V \subseteq U \subseteq X$ open, a morphism $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$ in $\mathcal{C}$

such that

(i) For all $U \subseteq X$ open, $\rho_{UU} : \mathcal{F}(U) \to \mathcal{F}(U)$ is the identity.
(ii) For all $W \subseteq V \subseteq U \subseteq X$ open, we have $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$, that is, the following diagram commutes

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\rho_{UV}"] \arrow[rr, bend left, "\rho_{UW}"] & \mathcal{F}(V) \arrow[r, "\rho_{VW}"] & \mathcal{F}(W) \end{tikzcd}$$

Remark 1.2 (A note on the category $\mathcal{C}$)**.** In the cases relevant for the lecture, the category $\mathcal{C}$ will always be a category of sets with additional structure, for example the category (Set) of sets, the category (Ab) of abelian groups, the category (Ring) of rings, etc. In categorical language, this means we assume that $\mathcal{C}$ comes equipped with a faithful functor to (Set) (this just means that we can associate a set to every object in $\mathcal{C}$ and equality of morphisms in $\mathcal{C}$ can be checked on the corresponding morphisms of sets). In particular, whenever convenient, we will work with elements of objects of $\mathcal{C}$.

Remark/Notation 1.3.
- Elements $s \in \mathcal{F}(U)$ are called *sections* of $\mathcal{F}$ over U. The object $\mathcal{F}(U)$ is called *space of sections of $\mathcal{F}$ over U*.
- Elements $s \in \mathcal{F}(X)$ are called *global sections* of $\mathcal{F}$. The object $\mathcal{F}(X)$ is called *space of global sections of $\mathcal{F}$*.
- Alternative notations for $\mathcal{F}(U)$ are $\Gamma(U, \mathcal{F})$ and $H^0(U, \mathcal{F})$.
- The morphisms ρ_{UV} are called *restriction maps*. For $V \subseteq U$ open and $s \in \mathcal{F}(U)$, we will sometimes write $s|_V$ for $\rho_{UV}(s)$ and call $s|_V$ the *restriction* of s (from U to V).

Example 1.4. Let X be a topological space.

(1) For any object A in a category $\mathcal{C}$, the *constant presheaf with value A* is the presheaf $\underline{A}'$ with $\underline{A}'(U) = A$ and $\rho_{UV} = \mathrm{id}_A$ for all $V \subseteq U \subseteq X$ open.

(2) The *presheaf of continuous functions on* X is defined by setting $\mathcal{C}^0(U) := C^0(U) := \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$ and $\rho_{UV} : \mathcal{C}^0(U) \to \mathcal{C}^0(V), f \mapsto f|_V$. This can be considered as a presheaf of sets, abelian groups, or even rings.
(3) More generally, if $\pi : Y \to X$ is a continuous map of topological spaces, we can construct on X the presheaf (of sets) of continuous sections of π whose space of sections on every open $\iota_U : U \hookrightarrow X$ is $\mathcal{F}_\pi(U) := \{s : U \to Y \mid s \text{ continuous}, \pi \circ s = \iota_U\}$ with ρ_{UV} the obvious restriction maps. [1]

Remark 1.5. Let X be a topological space and let Ouv_X be the category of open subsets of X, that is, the category with:

- Objects: $U \subseteq X$ open
- Morphisms:

$$\mathrm{Hom}(U, V) = \begin{cases} \emptyset \text{ if } U \not\subseteq V \\ \iota \text{ if } U \subseteq V \text{ where } \iota \text{ denotes the inclusion} \end{cases}$$

Then, a presheaf on X with values in a category $\mathcal{C}$ is the same as a contravariant functor from Ouv_X to $\mathcal{C}$.

Definition 1.6. Let X be a topological space. A morphism $\varphi : \mathcal{F}_1 \to \mathcal{F}_2$ of presheaves of $\mathcal{C}$ on X consists of a collection of morphisms $\varphi(U) : \mathcal{F}_1(U) \to \mathcal{F}_2(U)$ in $\mathcal{C}$ for every $U \subseteq X$ open compatible with restrictions, i.e. such that for all $V \subseteq U \subseteq X$ open, the following diagram commutes:

$$\begin{tikzcd} \mathcal{F}_1(U) \arrow[r, "\varphi(U)"] \arrow[d, "\rho_{1,UV}"] & \mathcal{F}_2(U) \arrow[d, "\rho_{2,UV}"] \\ \mathcal{F}_1(V) \arrow[r, "\varphi(V)"] & \mathcal{F}_2(V) \end{tikzcd}$$

Example 1.7. Let X be a topological space.

(1) Every morphism $f : A \to B$ of objects in $\mathcal{C}$ yields a morphism of presheaves $\varphi : \underline{A}' \to \underline{B}'$ by setting $\varphi(U) := f$.
(2) If $X = \mathbb{R}^n$, the association $\mathcal{C}^\infty(U) := C^\infty(U)$ with the usual restriction maps defines a presheaf and the inclusions $C^\infty(U) \subseteq C^0(U)$ define a morphism of presheaves $\mathcal{C}^\infty \to \mathcal{C}^0$.
(3) If $Y_2 \xrightarrow{\pi_2} Y_1 \xrightarrow{\pi_1} X$ are continuous maps of topological spaces, then there is an associated morphism $\mathcal{F}_{\pi_1 \circ \pi_2} \to \mathcal{F}_{\pi_1}$ of the presheaves of continuous sections.

Remark 1.8. When viewing the $\mathcal{F}_i$ as functors from Ouv_X to $\mathcal{C}$, a morphism $\mathcal{F}_1 \to \mathcal{F}_2$ is just a natural transformation. In particular, the category of presheaves on X with values in $\mathcal{C}$ is the same as the category $\mathrm{Fun}(\mathrm{Ouv}_X^{op}, \mathcal{C})$ of contravariant functors from Ouv_X to $\mathcal{C}$.

1.3. **Sheaves.** The following sheaf axioms formalize the idea that certain algebraic data on a topological space is local in the sense that it can be recovered uniquely from compatible local data on an open covering.

[1] Every sheaf (to be defined below) of sets is of this form for a suitable continuous map π (see Exercise 5). This justifies calling the elements of $\mathcal{F}(U)$ sections.

Definition 1.9. Let X be a topological space. Let $\mathcal{C}$ be the category of sets, abelian groups, or rings.[2] A sheaf $\mathcal{F}$ of $\mathcal{C}$ on X is a presheaf of $\mathcal{C}$ on X satisfying the following two conditions for every $U \subseteq X$ open and every open covering $U = \cup_{i\in I} U_i$ of U:

(i) For all $s, t \in \mathcal{F}(U)$: If $s|_{U_i} = t|_{U_i}$ for all $i \in I$, then $s = t$.
(ii) For all $\{s_i\}_{i\in I}$ with $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$, there exists an $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$.

Notation 1.10. Condition (i) in Definition 1.9 is sometimes called *locality* and Condition (ii) is called the *glueing* condition.

Remark 1.11. (1) The section s in Condition (ii) of Definition 1.9 is unique by Condition (i).

(2) If $\mathcal{C}$ is a category with products, but not necessarily a category of sets with additional structure, one says that a presheaf $\mathcal{F}$ on X is a sheaf if for every open $U \subseteq X$ and every open cover $U = \cup_{i\in I} U_i$, the diagram

$$\mathcal{F}(U) \xrightarrow{(\rho_{UU_i})_{i\in I}} \prod_{i\in I} \mathcal{F}(U_i) \underset{\prod_{j\in I}(\rho_{U_jU_i\cap U_j})_{i\in I}}{\overset{\prod_{i\in I}(\rho_{U_iU_i\cap U_j})_{j\in I}}{\rightrightarrows}} \prod_{i,j\in I} \mathcal{F}(U_i \cap U_j)$$

is an equalizer diagram. This agrees with Definition 1.9 when they both make sense.

(3) If $\mathcal{C}$ is an Abelian category with products, then the equalizer condition (2) is equivalent to saying that ρ_{UU_i} realizes $\mathcal{F}(U)$ as the kernel of the difference of the two maps on the right.

Example 1.12. Let X be a topological space.

(1) If $\mathcal{F}$ is a sheaf of sets (or abelian groups, rings, etc.), then, by Condition (i) of Definition 1.9 applied to $I = \emptyset$, the underlying set of $\mathcal{F}(\emptyset)$ has at most one element, and by Condition (ii), it has at least one element.
(2) The presheaves $\mathcal{C}^0$ (and $\mathcal{C}^\infty$ if $X = \mathbb{R}^n$) on X are sheaves, since:
  (i) Two continuous functions f, g that coincide on an open cover of $U \subseteq X$ coincide at every point, hence they are equal.
  (ii) Given $U \subseteq X$ open, a cover $U = \cup_{i\in I} U_i$, and continuous functions $f_i : U_i \to \mathbb{R}$ that agree on $U_i \cap U_j$, the function

$$\begin{aligned} f : \quad U &\to \mathbb{R} \\ x &\mapsto f_i(x) \text{ if } x \in U_i \end{aligned}$$

  is well-defined. By construction, $f|_{U_i} = f_i$ and f is continuous (resp. differentiable), since continuity (resp. differentiability) at any point can be checked on an open neighborhood.

Definition 1.13. A morphism $\varphi : \mathcal{F}_1 \to \mathcal{F}_2$ of sheaves $\mathcal{F}_i$ is a morphism $\mathcal{F}_1 \to \mathcal{F}_2$ between the $\mathcal{F}_i$ considered as presheaves.

Notation 1.14. Let X be a topological space and let $\mathcal{C}$ be the category of sets (or abelian groups, rings, etc.). The symbols $\mathrm{PSh}_{\mathcal{C}}(X)$ and $\mathrm{Sh}_{\mathcal{C}}(X)$ denote the category of presheaves and sheaves of $\mathcal{C}$ on X, respectively. If $\mathcal{C}$ is the category (Ab) of abelian groups, we drop the index.

Remark 1.15. Note that there is a forgetful functor $\iota : \mathrm{Sh}_{\mathcal{C}}(X) \to \mathrm{PSh}_{\mathcal{C}}(X)$ that simply forgets the sheaf axioms. By our definition of morphism of sheaves, this functor is fully faithful.

[2] More generally, we can let $\mathcal{C}$ be any category with a faithful functor to (Set) that preserves limits.

1.4. **The structure sheaf on** $\operatorname{Spec} A$**.** Let A be a ring (always commutative, with 1). Recall that the Zariski topology on $\operatorname{Spec} A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ prime ideal}\}$ is obtained by defining the closed sets to be $V(I) := \{\mathfrak{p} \in \operatorname{Spec} A \mid I \subseteq \mathfrak{p}\}$ for any ideal $I \subset A$. A basis of the Zariski topology on $\operatorname{Spec} A$ is given by the subsets of the form $D(a) := \{\mathfrak{p} \in \operatorname{Spec} A \mid a \notin \mathfrak{p}\}$, where $a \in A$.

Definition 1.16. Let A be a ring and let $X = \operatorname{Spec} A$. The *structure sheaf* $\mathcal{O}_X$ on X is defined by setting

$$\mathcal{O}_X(U) := \{s : U \to \coprod_{\mathfrak{p} \in U} A_{\mathfrak{p}} \mid s \text{ satisfies (i) and (ii)}\},$$

where

(i) For all $\mathfrak{p} \in U$, we have $s(\mathfrak{p}) \in A_{\mathfrak{p}}$
(ii) For all $\mathfrak{p} \in U$, there exist $a, b \in A$ and $V \subseteq U$ open with $\mathfrak{p} \in V \subseteq D(b)$ such that $s(\mathfrak{q}) = \frac{a}{b} \in A_{\mathfrak{q}}$ for all $\mathfrak{q} \in V$,

and $\rho_{UV}(s) := s|_V$.

Remark 1.17. The object $\mathcal{O}_X$ is a sheaf of rings on X: The ring structure on $\mathcal{O}_X(U)$ is given by pointwise addition and multiplication. Clearly, ρ_{UV} is a ring homomorphism, so $\mathcal{O}_X$ is a presheaf of rings. The proof that the sheaf conditions are satisfied is essentially the same as in Example 1.12 (2).

Remark 1.18. Note that there is a natural ring homomorphism

$$\begin{array}{rcl} A & \to & \mathcal{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) \\ a & \mapsto & (\mathfrak{p} \mapsto \dfrac{a}{1} \in A_{\mathfrak{p}}). \end{array}$$

At the end of the next lecture, we will see that this map is in fact an isomorphism. Roughly speaking, this means that every function from $\operatorname{Spec} A$ that satisfies (i) above and is locally given by a fraction of ring elements is in fact induced by an element of the ring.

2. Lecture 2: Direct limits, stalks, and sheafification

The goal of this lecture is to understand (pre)sheaves on a topological space X near points $p \in X$. This leads to the idea of stalks of presheaves and germs of functions. To define these objects, we need to study direct limits. Using stalks of presheaves, we will find a natural way of associating a sheaf to a presheaf, the so-called *sheafification*. In categorical language, sheafification will turn out to be a left-adjoint functor to the forgetful functor of Remark 1.15.

2.1. Direct limits and stalks.

We begin by recalling some elementary notions on ordered sets.

Definition 2.1. Let $(I, \leq)$ be a pair where I is a set and $\leq$ is a binary relation on I.

(1) $(I, \leq)$ is called *preordered set* if $\leq$ is reflexive and transitive.
(2) $(I, \leq)$ is called *partially ordered set* (poset, for short), if it is a preordered set and $\leq$ is antisymmetric.
(3) $(I, \leq)$ is called *directed set* if it is preordered and for every $a, b \in I$, there exists $c \in I$ with $a, b \leq c$.

The two key examples for directed posets that will appear in the context of stalks are the following:

Example 2.2. Let X be a topological space.

(1) Let $I := \{U \subseteq X \mid U \text{ open}\}$ be the set of open subsets of X and define $U \leq V$ if and only if $V \subseteq U$. Then, $(I, \leq)$ is a directed poset.
(2) If $x \in X$ is a point and $I_x := \{U \subseteq X \mid U \text{ open}, x \in U\}$ is the set of open neighborhoods of x, let $\leq$ be as in (1). Then, $(I_x, \leq)$ is again a directed poset, because if $U, V \in I_x$, then $x \in U$ and $x \in V$, hence $x \in U \cap V$, and therefore $U \cap V \in I$.

Definition 2.3. Let $(I, \leq)$ be a directed set and let $\mathcal{C}$ be a category. A *direct system in $\mathcal{C}$ indexed by I* is a pair $(\{A_i\}_{i \in I}, \{\rho_{ij}\}_{i,j \in I, i \leq j})$, where the A_i are objects in $\mathcal{C}$ and $\rho_{ij} : A_i \to A_j$ are morphisms in $\mathcal{C}$ such that

(i) $\rho_{ii} = \mathrm{id}_{A_i}$ for all $i \in I$.
(ii) $\rho_{ik} = \rho_{jk} \circ \rho_{ij}$ for all $i, j, k, \in I$, $i \leq j \leq k$.

Example 2.4. Let X be a topological space and let $\mathcal{F}$ be a presheaf of $\mathcal{C}$ on X. For $(I, \leq)$ as in Example 2.2, the pair $(\{\mathcal{F}(U)\}_{U \in I}, \{\rho_{UV}\}_{U,V \in I, V \subseteq U})$ is a directed system in $\mathcal{C}$ indexed by I.

Similarly, for every point $x \in X$, the pair $(\{\mathcal{F}(U)\}_{U \in I_x}, \{\rho_{UV}\}_{U,V \in I_x, V \subseteq U})$ is a directed system in $\mathcal{C}$ indexed by I_x.

Definition 2.5. Let $(I, \leq)$ be a directed set and let $\mathcal{C}$ be a category. Let $(\{A_i\}_{i \in I}, \{\rho_{ij}\}_{i,j \in I})$ be a direct system in $\mathcal{C}$ indexed by I. The *direct limit* of $(\{A_i\}_{i \in I}, \{\rho_{ij}\}_{i,j \in I})$ is a pair $(\varinjlim_{i \in I} A_i, \{\rho_i\}_{i \in I})$ where $\varinjlim_{i \in I} A_i$ is an object in $\mathcal{C}$ and $\rho_i : A_i \to \varinjlim_{i \in I} A_i$ are morphisms in $\mathcal{C}$ such that:

(i) $\rho_j \circ \rho_{ij} = \rho_i$ for all $i, j \in I$, $i \leq j$..
(ii) For every object A in $\mathcal{C}$ and every collection of morphisms $f_i : A_i \to A$ such that $f_i = f_j \circ \rho_{ij}$ for all $i, j \in I$ with $i \leq j$, there exists a unique morphism $f : \varinjlim_{i \in I} A_i \to A$ such

that $f \circ \rho_i = f_i$ for all $i \in I$. In pictures:

$$\begin{tikzcd}
A_i \arrow[rr, "\rho_{ij}"] \arrow[dr, "\rho_i"] \arrow[ddr, bend right, "f_i"'] & & A_j \arrow[dl, "\rho_j"'] \arrow[ddl, bend left, "f_j"] \\
& \varinjlim_{i\in I} A_i \arrow[d, dashed, "\exists! f"] & \\
& A &
\end{tikzcd}$$

Remark 2.6. (1) In particular, a direct limit is unique up to unique isomorphism if it exists.

(2) Despite its name, a direct limit is, in general, not a limit in $\mathcal{C}$. Instead, it is a special kind of colimit in $\mathcal{C}$.

Example 2.7. Direct limits exist in the category (Set) and (Ab) of sets and abelian groups. More precisely:

(1) Let $(\{A_i\}_{i\in I}, \{\rho_{ij}\}_{i,j\in I})$ be a direct system in (Set). Then, $\varinjlim_{i\in I} A_i = \coprod_{i\in I} A_i/\sim$, where $a_i \sim a_j$ if and only if there exists $k \in I$ such that $i, j \leq k$ and $\rho_{ik}(a_i) = \rho_{jk}(a_j)$, together with the maps $\rho_i : A_i \to \coprod_{i\in I} A_i/\sim$ induced by the inclusions is a direct limit of $(\{A_i\}_{i\in I}, \{\rho_{ij}\}_{i,j\in I})$.

(2) Let $(\{A_i\}_{i\in I}, \{\rho_{ij}\}_{i,j\in I})$ be a direct system in (Ab). Then, $\varinjlim_{i\in I} A_i = \bigoplus_{i\in I} A_i/N$, where N is the subgroup generated by elements of the form $a_i - \rho_{ij}(a_i)$, together with the maps $\rho_i : A_i \to \bigoplus_{i\in I} A_i/N$ induced by the inclusions is a direct limit of $(\{A_i\}_{i\in I}, \{\rho_{ij}\}_{i,j\in I})$.

(3) Let $(\{A_i\}_{i\in I}, \{\rho_{ij}\}_{i,j\in I})$ be a direct system in (Ab). Forgetting the group structure, it becomes a direct system in (Set). The induced map $\coprod_{i\in I} A_i/\sim \to \bigoplus_{i\in I} A_i/N$ is a bijection that identifies the f_i in (1) and (2).

Similarly, if $(\{A_i\}_{i\in I}, \{\rho_{ij}\}_{i,j\in I})$ is a direct system of modules or rings, then the direct limit exists and the underlying set is the direct limit of the underlying sets.

Check/Fact 2.8. Taking direct limits in (Ab) is exact in the following sense:

Given directed systems M_i, N_i, P_i indexed by the same directed set I, and for all $i, j \in I, i \leq j$ commutative diagrams with exact rows of the form

$$\begin{tikzcd}
0 \arrow[r] & M_i \arrow[r] \arrow[d] & N_i \arrow[r] \arrow[d] & P_i \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & M_j \arrow[r] & N_j \arrow[r] & P_j \arrow[r] & 0,
\end{tikzcd}$$

the induced sequence $0 \to \varinjlim_{i\in I} M_i \to \varinjlim_{i\in I} N_i \to \varinjlim_{i\in I} P_i \to 0$ is exact.

Definition 2.9. Let X be a topological space, $\mathcal{C}$ a category with direct limits, $x \in X$ a point, and $\mathcal{F}$ a presheaf of $\mathcal{C}$ on X. The *stalk of $\mathcal{F}$ at x* is the direct limit $\mathcal{F}_x := \varinjlim_{x\in U} \mathcal{F}(U) := \varinjlim_{U\in I_x} \mathcal{F}(U)$, where $I_x = \{U \subseteq X \text{ open } \mid x \in U\}$

Notation 2.10. For every $x \in U$, the image of a section $s \in \mathcal{F}(U)$ under the canonical map $\mathcal{F}(U) \to \mathcal{F}_x$ is denoted by s_x and called *germ of s at x*. Note that s_x is independent of the choice of U (see Remark 2.12 below.).

Remark 2.11. Let X be a topological space and let $\varphi\colon \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves of $\mathcal{C}$. Then, for every $x \in X$, there is an induced morphism $\varphi_x\colon \mathcal{F}_x \to \mathcal{G}_x$ obtained by applying the universal property of $\mathcal{F}_x$ to the compositions $\mathcal{F}(U) \xrightarrow{\varphi(U)} \mathcal{G}(U) \to \mathcal{G}_x$.

The following important remark gives a concrete way of thinking about germs as equivalence classes of local sections:

Remark 2.12. Let $x \in X$ and let $\mathcal{F}$ be a sheaf of sets (or Abelian groups, rings, etc.) on X. The following follows immediately from the description of direct limits in Example 2.7

(1) For $U \subseteq X$ open and $s, t \in \mathcal{F}(U)$, we have $s_x = t_x$ if and only if there exists $x \in V \subseteq U$ open with $s|_V = t|_V$.
(2) For every $s \in \mathcal{F}_x$, there exists an open $x \in U$ and $t \in \mathcal{F}(U)$ such that $t_x = s$.

2.2. **Sheafification.**

Definition 2.13. Let X be a topological space and let $\mathcal{F}$ be a presheaf of sets (or Abelian groups, rings, etc.) on X. The *sheafification of* $\mathcal{F}$ is the sheaf $\mathcal{F}^+$ defined by setting

$$\mathcal{F}^+(U) := \{s : U \to \coprod_{x \in U} \mathcal{F}_x \mid s \text{ satisfies (i) and (ii)}\}$$

where

(i) For all $x \in U$, $s(x) \in \mathcal{F}_x$.
(ii) For all $x \in U$, there exists $V \subseteq U$ open with $x \in V$ and $t \in \mathcal{F}(V)$ such that $t_y = s(y)$ for all $y \in V$.

Roughly speaking, sections of $\mathcal{F}^+$ over U are collections of germs of $\mathcal{F}$ whose local variation is controlled by sections of $\mathcal{F}$.

Remark 2.14. (1) It is easy to check that $\mathcal{F}^+$ is indeed a sheaf (compare with Remark 1.17).
(2) Sheafification is functorial, because taking direct limits is. More precisely, given $\varphi : \mathcal{F} \to \mathcal{G}$, we define $\varphi^+ : \mathcal{F}^+ \to \mathcal{G}^+$ by setting $\varphi^+(U)(s) = \coprod_{x \in U} \varphi_x \circ s$.
(3) There exists a natural map $\iota_{\mathcal{F}} : \mathcal{F} \to \mathcal{F}^+$ defined by mapping $s \in \mathcal{F}(U)$ to $x \mapsto s_x$.
(4) For every $s \in \mathcal{F}^+(U)$, there exists an open cover $U = \cup_{i \in I} U_i$ and sections $s_i \in \mathcal{F}(U_i)$ such that $\iota_{\mathcal{F}}(U_i)(s_i) = s|_{U_i}$.
(5) For every $x \in X$, the map $\iota_{\mathcal{F},x} : \mathcal{F}_x \to \mathcal{F}_x^+$ is an isomorphism.

Proposition 2.15. *Let X be a topological space and let $\mathcal{F}$ be a presheaf of $\mathcal{C}$ on X, where $\mathcal{C}$ is a category with direct limits and products. The sheaf $\mathcal{F}^+$ satisfies the following universal property:*

For every morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a sheaf, there is a unique morphism $\varphi^+ : \mathcal{F}^+ \to \mathcal{G}$ such that $\varphi = \varphi^+ \circ \iota_{\mathcal{F}}$.

Proof. Let $U \subseteq X$ open and $s \in \mathcal{F}^+(U)$. Then, there exists an open cover $U = \cup_{i \in I} U_i$ and $s_i \in \mathcal{F}(U_i)$ such that $s|_{U_i} = \iota_{\mathcal{F}}(U_i)(s_i)$. Since we want $\varphi = \varphi^+ \circ \iota_{\mathcal{F}}$, we have to set $\varphi^+(U_i)(s|_{U_i}) = \varphi(U_i)(s_i)$.

Observe that $\varphi(U_i)(s_i)|_{U_i \cap U_j} = \varphi(U_j)(s_j)|_{U_i \cap U_j}$. Indeed, we know $\iota_{\mathcal{F}}(U_i \cap U_j)(s_i|_{U_i \cap U_j}) = \iota_{\mathcal{F}}(U_i \cap U_j)(s_j|_{U_i \cap U_j})$, so, by Remark 2.14 (5), $(s_i)_x = (s_j)_x$ for all $x \in U_i \cap U_j$. Hence, $(\varphi(U_i)(s_i))_x = (\varphi(U_j)(s_j))_x$ for all $x \in U_i \cap U_j$. Since $\mathcal{G}$ is a sheaf, this implies $\varphi(U_i)(s_i)|_{U_i \cap U_j} = \varphi(U_j)(s_j)|_{U_i \cap U_j}$. Applying one more time that $\mathcal{G}$ is a sheaf, we deduce from this that there exists a unique $t \in \mathcal{G}(U)$ such that $t|_{U_i} = \varphi(s_i)$. For φ^+ to be a morphism of sheaves, we have to set

$\varphi^+(U)(s) = t$. It remains to check that the $\varphi^+(U)$ defined this way give a morphism of sheaves, which we leave to the reader. $\square$

Remark 2.16. Let $F : \mathrm{Sh}_{\mathcal{C}}(X) \to \mathrm{PSh}_{\mathcal{C}}(X)$ be the forgetful functor. By Remark 2.14, sheafification is functor $(-)^+ : \mathrm{PSh}_{\mathcal{C}}(X) \to \mathrm{Sh}_{\mathcal{C}}(X)$. Then, Proposition 2.15 shows $(-)^+$ is left-adjoint to F, that is, for all presheaves $\mathcal{F}$ and sheaves $\mathcal{G}$, there is a natural bijection $\mathrm{Hom}_{\mathrm{Sh}_{\mathcal{C}}}(\mathcal{F}^+, \mathcal{G}) \to \mathrm{Hom}_{\mathrm{PSh}_{\mathcal{C}}}(\mathcal{F}, F(\mathcal{G}))$.

2.3. **The structure sheaf on** $\mathrm{Spec}\, A$ **revisited.** It turns out that the stalks of the structure sheaf on the spectrum of a ring are precisely its localizations at prime ideals. This is a consequence of the following result that we announced at the end of the previous lecture:

Proposition 2.17. *Let $X = \mathrm{Spec}\, A$ where A is a ring. Then, for all $a \in A$, there exist isomorphisms $\varphi_a : A_a \to \mathcal{O}_X(D(a))$ such that for all $D(b) \subseteq D(a)$, the following diagram commutes*

$$\begin{tikzcd} A_a \arrow[r, "\varphi_a"] \arrow[d] & \mathcal{O}_X(D(a)) \arrow[d] \\ A_b \arrow[r, "\varphi_b"] & \mathcal{O}_X(D(b)), \end{tikzcd} \tag{1}$$

where the vertical morphisms are the natural map of localizations on the left-hand side and the restriction morphism of $\mathcal{O}_X$ on the right-hand side. In particular, we have $\mathcal{O}_X(X) \cong A$.

Proof. We define

$$\begin{aligned} \varphi_a : \quad A_a &\to \mathcal{O}_X(D(a)) \\ \tfrac{s}{a^n} &\mapsto (\mathfrak{p} \mapsto \frac{s}{a^n} \in A_{\mathfrak{p}}). \end{aligned}$$

It is clear that these φ_a make diagram (1) commute. It remains to show that they are isomorphisms.

Injectivity: Assume $\varphi_a(\frac{s}{a^n}) = 0$. Let $I := \{r \in A \mid rs = 0\}$ be the annihilator of s. Since $\frac{s}{a^n} = 0$ for all $\mathfrak{p} \in D(a)$, we have $I \not\subseteq \mathfrak{p}$ for any $\mathfrak{p} \in D(a)$. In other words, $V(I) \subseteq V(a)$. Hence, $a \in \sqrt{I}$, so there exists $m \geq 1$ with $a^m s = 0$. But this means that $\frac{s}{a^n} = 0$ in A_a.

Surjectivity: Let $s \in \mathcal{O}_X(D(a))$. By definition of $\mathcal{O}_X$ and since the principal open sets form a basis of the topology, we find elements $a_i, g_i, h_i \in A$ such that $D(a) = \cup_{i \in I} D(h_i)$, $D(h_i) \subseteq D(g_i)$, and $s(\mathfrak{q}) = \frac{a_i}{g_i}$ for all $\mathfrak{q} \in D(h_i)$.

Claim 1: We may assume $g_i = h_i$ for all $i \in I$

Proof of Claim 1: Since $D(h_i) \subseteq D(g_i)$, we have $V(g_i) \subseteq V(h_i)$, hence $h_i \in \sqrt{(g_i)}$, that is, $h_i^n = c_i g_i$ for some $c_i \in A$ and $n \geq 1$. Replacing h_i by h_i^n [3] and a_i by $c_i a_i$, we may assume that $s(\mathfrak{q}) = \frac{a_i}{h_i}$ on $D(h_i)$. ■

Claim 2: We may assume that I is finite

Proof of Claim 2: We have $D(a) \subseteq \cup_{i \in I} D(h_i)$ if and only if $V(\sum_{i \in I}(h_i)) = \cap_{i \in I} V(h_i) \subseteq V(a)$ if and only if $a \in \sqrt{\sum_{i \in I}(h_i)}$ if and only if $a^n \in \sum_{i \in I}(h_i)$ for some $n \geq 0$. But $a^n \in$

[3] Recall that $D(h_i) = D(h_i^n)$.

$\sum_{i\in I}(h_i)$ holds if and only if there exist *finitely many* h_i and b_i such that $a^n = \sum b_i h_i$, so finitely many $D(h_i)$ suffice to cover $D(a)$. [4] ■

Claim 3: We may assume that $a_i h_j = a_j h_i$ for all $i, j \in I$

Proof of Claim 3: Now, on $D(h_i) \cap D(h_j) = D(h_i h_j)$, both $\frac{a_i}{h_i}$ and $\frac{a_j}{h_j}$ represent s, so, by the injectivity of $\varphi_{h_i h_j}$, we have $\frac{a_i}{h_i} = \frac{a_j}{h_j}$ in $A_{h_i h_j}$. Hence, there exists $N \geq 1$ such that

$$(h_i h_j)^N (h_j a_i - h_i a_j) = h_i^N h_j^{N+1} a_i - h_i^{N+1} h_j^N a_j = 0 \tag{2}$$

in A, and since there are only finitely many h_i, we can choose an N that works for all i, j. We can replace h_i by h_i^{N+1} and a_i by $h_i^N a_i$ without changing $\frac{a_i}{h_i}$, but Equation (2) simplifies to the equation $h_j a_i = h_i a_j$. ■

As noted in Claim 2, we can write $a^n = \sum b_i h_i$ for some $b_i \in A$. Write $c = \sum_i b_i a_i$. Then, we have

$$h_j c = \sum_i b_i a_i h_j = \sum_i b_i h_i a_j = a^n a_j.$$

This implies that $\frac{c}{a^n} = \frac{a_j}{h_j}$ on $D(h_j)$. So, $\frac{c}{a^n}$ represents s on all of $D(a)$, hence φ_a is surjective. □

Corollary 2.18. *Let $X = \operatorname{Spec} A$ where A is a ring. Then, for all $\mathfrak{p} \in A$, there exists an isomorphism $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \to \mathcal{O}_{X,\mathfrak{p}}$ such that for all $a \in A \setminus \mathfrak{p}$, the following diagram commutes*

$$\begin{tikzcd} A_a \arrow[r, "\varphi_a"] \arrow[d] & \mathcal{O}_X(D(a)) \arrow[d] \\ A_{\mathfrak{p}} \arrow[r, "\varphi_{\mathfrak{p}}"] & \mathcal{O}_{X,\mathfrak{p}}. \end{tikzcd}$$

Proof. Similarly to Example 2.4, the principal open neighborhoods $\mathfrak{p} \in D(a) \subseteq \operatorname{Spec} A$ form a direct system. We pass to direct limits in Diagram (1) and, by Check/Fact 2.8, we obtain an isomorphism

$$\varinjlim_{a\in A\setminus\mathfrak{p}} A_a \to \varinjlim_{a\in A\setminus\mathfrak{p}} \mathcal{O}_X(D(a)).$$

We have morphisms $A_a \to A_{\mathfrak{p}}$ and $\mathcal{O}_X(D(a)) \to \mathcal{O}_{X,\mathfrak{p}}$ which are compatible with further localization resp. restriction maps. This yields morphisms $\varinjlim_{a\in A\setminus\mathfrak{p}} A_a \to A_{\mathfrak{p}}$ and $\varinjlim_{a\in A\setminus\mathfrak{p}} \mathcal{O}_X(D(a)) \to \mathcal{O}_{X,\mathfrak{p}}$. The first morphism is an isomorphism because the universal property of $A_{\mathfrak{p}}$ is exactly the universal property of the direct limit over the A_a. The second morphism is easily seen to be an isomorphism using the fact that the $D(a)$ form a basis of the topology (hence, in particular, for every $x \in U$ open, there exists $a \in A$ such that $x \in D(a) \subseteq U$). □

[4] In fact, this shows that $D(a)$ is quasi-compact, a property we will come back to later.

3. Lecture 3: Kernels and cokernels, exactness

As usual, X denotes a topological space. The goal of this lecture is to show that the category $\mathrm{Sh}(X)$ of sheaves of abelian groups on X is an *abelian category*. This involves constructing kernels, cokernels, and images of morphisms of sheaves and will allow us to speak of exact sequences of sheaves. First of all, note that the category $\mathrm{Sh}(X)$ is *preadditive*, that is, for any two sheaves $\mathcal{F}, \mathcal{G}$, the set $\mathrm{Hom}_{\mathrm{Sh}(X)}(\mathcal{F}, \mathcal{G})$ has a natural structure of an abelian group given by $(\varphi + \psi)(U) := \varphi(U) + \psi(U)$ for all $U \subseteq X$ open. Note also that there is a 0-*sheaf* in $\mathrm{Sh}(X)$ that associates the 0-group to every open.

Definition 3.1. Let $\mathcal{F}$ be a sheaf of abelian groups. A *sub(pre)sheaf* $\mathcal{G} \subseteq \mathcal{F}$ is a (pre)sheaf $\mathcal{G}$ such that $\mathcal{G}(U) \subseteq \mathcal{F}(U)$ is a subgroup for all open $U \subseteq X$ and such that the restriction maps of $\mathcal{G}$ are induced by those of $\mathcal{F}$.

Definition 3.2. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves of abelian groups.

(1) The *presheaf kernel* of φ is the presheaf $\ker^{\mathrm{pre}}(\varphi)$ defined by $\ker^{\mathrm{pre}}(\varphi)(U) := \ker(\varphi(U))$ together with the natural map $\ker^{\mathrm{pre}}(\varphi) \to \mathcal{F}$.
(2) The *presheaf image* of φ is the presheaf $\mathrm{im}^{\mathrm{pre}}(\varphi)$ defined by $\mathrm{im}^{\mathrm{pre}}(\varphi)(U) := \mathrm{im}(\varphi(U))$ together with the natural map $\mathrm{im}^{\mathrm{pre}}(\varphi) \to \mathcal{G}$.
(3) The *presheaf cokernel* of φ is the presheaf $\mathrm{coker}^{\,\mathrm{pre}}(\varphi)$ defined by $\mathrm{coker}^{\,\mathrm{pre}}(\varphi)(U) := \mathrm{coker}\,(\varphi(U))$ together with the natural map $\mathcal{G} \to \mathrm{coker}^{\,\mathrm{pre}}(\varphi)$.

In each case, the restriction maps are the ones induced by the restriction maps on $\mathcal{F}$ or $\mathcal{G}$.

Lemma 3.3. *If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves, then $\ker^{\mathrm{pre}}(\varphi)$ is a sheaf.*

Proof. We consider $\ker^{\mathrm{pre}}(\varphi)$ as a subpresheaf of $\mathcal{F}$ via the natural map. Let $U \subseteq X$ be open and $U = \cup_{i \in I} U_i$ an open cover. Let $s_i \in \ker^{\mathrm{pre}}(\varphi)(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. By definition, we have $\ker^{\mathrm{pre}}(\varphi)(U_i) = \ker(\varphi(U_i))$, hence $\varphi(U_i)(s_i) = 0$. Since $\mathcal{F}$ is a sheaf, there is a unique section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$. We have $\varphi(U)(s)|_{U_i} = \varphi(U_i)(s_i) = 0$ for all $i \in I$. Since $\mathcal{G}$ is a sheaf, this implies $\varphi(U)(s) = 0$, hence $s \in \ker^{\mathrm{pre}}(\varphi)(U)$. □

Example 3.4. The corresponding statement for $\mathrm{im}(\varphi)$ and $\mathrm{coker}\,(\varphi)$ is false in general, see Exercise 4.

Definition 3.5. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups.

(1) The *kernel* of φ is the presheaf kernel $\ker(\varphi) := \ker^{\mathrm{pre}}(\varphi)$ together with the natural map $\ker(\varphi) \to \mathcal{F}$.
(2) The *image* of φ is the sheafification of the presheaf image $\mathrm{im}(\varphi) := (\mathrm{im}^{\mathrm{pre}}(\varphi))^+$ together with the natural map $\mathrm{im}(\varphi) \to \mathcal{G}$ induced by the universal property of sheafification.
(3) The *cokernel* of φ is the sheafification of the presheaf cokernel $\mathrm{coker}\,(\varphi) := (\mathrm{coker}^{\,\mathrm{pre}}(\varphi))^+$ together with the natural map $\mathcal{G} \to \mathrm{coker}\,(\varphi)$.

Lemma 3.6. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. The following statements hold in $\mathrm{Sh}(X)$:*

(1) The sheaf $\ker(\varphi)$ together with the morphism $\ker(\varphi) \to \mathcal{F}$ is the categorical kernel of φ.
(2) The sheaf $\mathrm{coker}\,(\varphi)$ together with the morphism $\mathcal{G} \to \mathrm{coker}\,(\varphi)$ is the categorical cokernel of φ.

Proof. Statements (1) and (2) mean that for each of the following commutative diagrams of solid arrows, where $\mathcal{F}'$ is a sheaf, the dotted arrow exists and is unique:

(1)

$$\begin{tikzcd}
 & \mathcal{F}' \arrow[dl, dotted] \arrow[d] \arrow[dr, "0"] & \\
\ker(\varphi) \arrow[r] & \mathcal{F} \arrow[r] & \mathcal{G}
\end{tikzcd}$$

(2)

$$\begin{tikzcd}
\mathcal{F} \arrow[r] \arrow[dr, "0"'] & \mathcal{G} \arrow[r] \arrow[d] & \operatorname{coker}(\varphi) \arrow[dl, dotted] \\
 & \mathcal{G}' &
\end{tikzcd}$$

Claim (1) is true, because we have the corresponding statement on every $U \subseteq X$ open, because $\ker(\varphi)(U) = \ker(\varphi(U))$. For the same reason, the dotted arrow in (2) exists if we replace $\operatorname{coker}(\varphi)$ by $\operatorname{coker}(\varphi)^{\mathrm{pre}}$. But then we get the desired dotted arrow from the universal property of sheafification. $\square$

Proposition 3.7. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. The following are equivalent:*

(1) φ is a monomorphism in $\mathrm{Sh}(X)$.
(2) $\ker(\varphi) = 0$.
(3) $\ker(\varphi(U)) = 0$ for all $U \subseteq X$ open.
(4) $\ker(\varphi_x) = 0$ for all $x \in X$.

Proof. Recall that φ is called *monomorphism* if for every $\psi : \mathcal{F}' \to \mathcal{F}$, the equality $\varphi \circ \psi = 0$ implies $\psi = 0$.

(1) $\Rightarrow$ (2): Apply the above property of monomorphisms to $\psi : \ker(\varphi) \to \mathcal{F}$.

(2) $\Rightarrow$ (1): If $\varphi \circ \psi = 0$, then ψ factors through $\ker(\varphi) \to \mathcal{F}$ by Lemma 3.6, so $\psi = 0$.

(2) $\Leftrightarrow$ (3): Follows immediately from $\ker(\varphi)(U) = \ker(\varphi(U))$.

(3) $\Rightarrow$ (4): Follows from exactness of direct limits Check/Fact 2.8.

(4) $\Rightarrow$ (3): Let $s \in \mathcal{F}(U)$ with $\varphi(U)(s) = 0$. Then, $\varphi(U)(s)_x = \varphi_x(s_x) = 0$ for all $x \in U$. Since φ_x is injective, this implies $s_x = 0$ for all $x \in U$. By Remark 2.12, this means that every point in x has an open neighborhood V_x such that $s|_{V_x} = 0$. Since $\mathcal{F}$ is a sheaf, this implies $s = 0$. $\square$

Proposition 3.8. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. The following are equivalent:*

(1) φ is an epimorphism in $\mathrm{Sh}(X)$.
(2) $\operatorname{coker}(\varphi) = 0$.
(3) $\operatorname{coker}(\varphi_x) = 0$ for all $x \in X$.

Proof. Recall that φ is called *epimorphism* if for every $\psi : \mathcal{G} \to \mathcal{G}'$, the equality $\psi \circ \varphi = 0$ implies $\psi = 0$.

(1) $\Rightarrow$ (2): Apply the above property of epimorphisms to $\psi : \mathcal{G} \to \operatorname{coker}(\varphi)$.

(2) $\Rightarrow$ (3): Clearly, (2) implies $\operatorname{coker}(\varphi)_x = 0$. We have $\operatorname{coker}^{\mathrm{pre}}(\varphi)_x \cong \operatorname{coker}(\varphi)_x$ by Remark 2.14 and $\operatorname{coker}^{\mathrm{pre}}(\varphi)_x \cong \operatorname{coker}(\varphi_x)$ by exactness of direct limits (Check/Fact 2.8).

(3) $\Rightarrow$ (1): Let $\psi : \mathcal{G} \to \mathcal{G}'$ be a morphism such that $\psi \circ \varphi = 0$. Then, $(\psi \circ \varphi)_x = \psi_x \circ \varphi_x = 0$ for all $x \in X$. Hence, $\psi_x = 0$ for all $x \in X$. Since $\mathcal{H}om(\mathcal{G}, \mathcal{G}')$ is a sheaf (Exercise 2) and $\psi \in \mathcal{H}om(\mathcal{G}, \mathcal{G}')(X)$ is a section all of whose germs are 0, Remark 2.12 implies that $\psi = 0$. $\square$

Remark 3.9. By definition, we have $\operatorname{coker}(\varphi(U)) = 0$ for all $U \subseteq X$ open if and only if $\operatorname{coker}^{\text{pre}}(\varphi) = 0$. Since sheafification is functorial, this implies $\operatorname{coker}(\varphi) = 0$. However, by Exercise 4, the converse is *not* true in general.

Corollary 3.10. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. The following are equivalent:*

(1) φ is an isomorphism.
(2) $\varphi(U)$ is an isomorphism for all $U \subseteq X$.
(3) φ_x is an isomorphism for all $x \in U$.

Proof. Since taking sections over U is a functor, we have (1) $\Rightarrow$ (2). Conversely, if (2) holds, then the $(\varphi(U))^{-1}$ define a morphism of sheaves which is the inverse to φ, hence (2) $\Rightarrow$ (1).

Since taking stalks at x is also a functor, we have (1) $\Rightarrow$ (3). It remains to show (3) $\Rightarrow$ (2). In fact, by Proposition 3.7, it suffices to prove that $\varphi(U)$ is surjective.

So, let $t \in \mathcal{G}(U)$. Since all φ_x are isomorphisms, we find a unique $s_x \in \mathcal{F}_x$ with $\varphi_x(s_x) = t_x$. By Remark 2.12 (2), there is a section s_{V_x} of $\mathcal{F}$ on some open neighborhood $x \in V_x \subseteq U$ whose germ is s_x. By Remark 2.12 (1) and since $\varphi(V_x)(s_{V_x})_x = \varphi_x(s_x) = t_x$, we can choose V_x sufficiently small so that $\varphi(V_x)(s_{V_x}) = t|_{V_x}$. By injectivity of $\varphi(V_x \cap V_y)$, the sections s_{V_x} and s_{V_y} coincide on $V_x \cap V_y$, hence we find $s \in \mathcal{F}(U)$ such that $\varphi(U)(s)|_{V_x} = t|_{V_x}$. Since $\mathcal{G}$ is a sheaf, this implies $\varphi(U)(s) = t$. $\square$

Combining the universal properties of kernel and sheafification, we see that if we have a sequence of sheaves

$$\mathcal{F}_1 \xrightarrow{\varphi_1} \mathcal{F}_2 \xrightarrow{\varphi_2} \mathcal{F}_3$$

with $\varphi_2 \circ \varphi_1 = 0$, then the natural map $\operatorname{im}(\varphi_1) \to \mathcal{F}_2$ factors uniquely through $\ker(\varphi_2) \to \mathcal{F}_2$.

Definition 3.11. A sequence of sheaves $\mathcal{F}_1 \xrightarrow{\varphi_1} \mathcal{F}_2 \xrightarrow{\varphi_2} \mathcal{F}_3$ is called *exact* (at $\mathcal{F}_2$) if $\varphi_2 \circ \varphi_1 = 0$ and the natural map $\operatorname{im}(\varphi_1) \to \ker(\varphi_2)$ is an isomorphism. A *short exact sequence* is an everywhere exact sequence of the form

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0.$$

Corollary 3.12. *A sequence of sheaves*

$$\mathcal{F}_1 \xrightarrow{\varphi_1} \mathcal{F}_2 \xrightarrow{\varphi_2} \mathcal{F}_3$$

is exact at $\mathcal{F}_2$ if and only if for all $x \in X$, the induced sequence on stalks

$$\mathcal{F}_{1,x} \xrightarrow{\varphi_{1,x}} \mathcal{F}_{2,x} \xrightarrow{\varphi_{2,x}} \mathcal{F}_{3,x}$$

is exact at $\mathcal{F}_{2,x}$.

Proof. Since $\ker(\varphi_{2,x}) = \ker(\varphi_2)_x$ and $\operatorname{im}(\varphi_{1,x}) = \operatorname{im}(\varphi_1)_x$, exactness on the level of stalks follows from exactness on the level of sheaves. Conversely, assume that $\varphi_{2,x} \circ \varphi_{1,x} = 0$ for all $x \in X$ and that the induced map $\operatorname{im}(\varphi_{1,x}) \to \ker(\varphi_{2,x})$ is an isomorphism. Then, $(\varphi_2 \circ \varphi_1)_x = \varphi_{2,x} \circ \varphi_{1,x} = 0$ for all $x \in X$. Since $\mathcal{H}om(\mathcal{F}_1, \mathcal{F}_3)$ is a sheaf (Exercise 2), we obtain $\varphi_2 \circ \varphi_1 = 0$. Thus, there is a natural morphism of sheaves $\operatorname{im}(\varphi_1) \to \ker(\varphi_2)$ which, on the level of stalks,

coincides with the natural isomorphism $\mathrm{im}(\varphi_{1,x}) \to \ker(\varphi_{2,x})$. By Corollary 3.10, this implies that $\mathrm{im}(\varphi_1) \to \ker(\varphi_2)$ is an isomorphism. □

Corollary 3.13. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. Then,* $\mathrm{im}(\varphi) = \ker(\mathcal{G} \to \mathrm{coker}\,(\varphi))$.

Proof. The sequence $\mathcal{F} \xrightarrow{\varphi} \mathcal{G} \to \mathrm{coker}\,(\varphi)$ induces sequences $\mathcal{F}_x \xrightarrow{\varphi_x} \mathcal{G}_x \to \mathrm{coker}\,(\varphi_x) = \mathrm{coker}\,(\varphi)_x$ which are exact at $\mathcal{G}_x$ for all $x \in X$. Hence, $\mathcal{F} \xrightarrow{\varphi} \mathcal{G} \to \mathrm{coker}\,(\varphi)$ is exact at $\mathcal{G}$, which is precisely the claim. □

Corollary 3.14. *For every topological space X, the category* $\mathrm{Sh}(X)$ *is abelian.*

Proof. This means the following:

- $\mathrm{Sh}(X)$ is *preadditive*. This means that the Hom-sets have the structure of abelian groups in a way that makes the composition bilinear. This can be checked directly from our definition of the group law on $\mathrm{Hom}_{\mathrm{Sh}(X)}(\mathcal{F}, \mathcal{G})$ as in the beginning of this lecture.
- $\mathrm{Sh}(X)$ has a *zero object*. This means that there is a sheaf $\mathcal{F}$ such that for every sheaf $\mathcal{G}$, there are unique morphisms $\mathcal{F} \to \mathcal{G}$ and $\mathcal{G} \to \mathcal{F}$. Clearly, the 0-sheaf does the job.
- Every morphism admits a kernel and a cokernel. This holds by Lemma 3.6.
- $\mathrm{Sh}(X)$ admits all *binary biproducts*. For this, check that the sheaf $\mathcal{F} \oplus \mathcal{G}$ of Exercise 1 is a product and a coproduct in $\mathrm{Sh}(X)$.
- Every monomorphism is a kernel and every epimorphism is a cokernel. For this, use Proposition 3.7 to show that if φ is a monomorphism, then it is the kernel of its cokernel, and use Proposition 3.8 to show that if φ is an epimorphism, it is the cokernel of its kernel.

□

4. Lecture 4: Direct and inverse image, ringed spaces

In this lecture, we will learn how to compare sheaves on two different topological spaces by taking *direct and inverse images* of sheaves along continuous morphisms $X \to Y$. This will allow us to define the category of *ringed spaces* and *schemes*.

4.1. **Direct and inverse image.**

Definition 4.1. Let $f : X \to Y$ be a continuous map between topological spaces. Let $\mathcal{F}$ be a sheaf on X. The *direct image of $\mathcal{F}$ along f* is the sheaf $f_*\mathcal{F}$ defined by $f_*\mathcal{F}(V) \coloneqq \mathcal{F}(f^{-1}(V))$.

Check/Fact 4.2. The presheaf $f_*\mathcal{F}$ really is a sheaf, with restriction maps $\rho_{f^{-1}(V)f^{-1}(W)}$. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves on X, then the $(f_*\varphi)(V) \coloneqq \varphi(f^{-1}(V)) : \mathcal{F}(f^{-1}(V)) \to \mathcal{G}(f^{-1}(V))$ define a morphism of sheaves $f_*\varphi : f_*\mathcal{F} \to f_*\mathcal{G}$. This turns f_* into a functor $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$.

Definition 4.3. Let $f : X \to Y$ be a continuous map between topological spaces. Let $\mathcal{G}$ be a sheaf of abelian groups on Y. The *inverse image of $\mathcal{G}$ along f* is the sheafification $f^{-1}\mathcal{G}$ of the presheaf $f^{-1,\mathrm{pre}}\mathcal{G}$ defined by $f^{-1,\mathrm{pre}}\mathcal{G}(U) = \varinjlim_{f(U)\subseteq V} \mathcal{G}(V)$.

Check/Fact 4.4. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves on Y, then there is an associated morphism $f^{-1}\varphi : f^{-1}\mathcal{F} \to f^{-1}\mathcal{G}$ of sheaves on X, constructed on U from the $\varphi(V)$ with $f(U) \subseteq V$ by first passing to direct limits and then to the sheafification. This turns f^{-1} into a functor $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$.

Lemma 4.5. *Let $f : X \to Y$ be a continuous map between topological spaces, let $\mathcal{F}$ be a sheaf on X and $\mathcal{G}$ a sheaf on Y.*

1. *For all $y \in Y$, there is a natural isomorphism $(f_*\mathcal{F})_y \cong \varinjlim_{y\in V\subseteq Y} \mathcal{F}(f^{-1}(V))$. In particular, for all $x \in X$, there is a natural morphism $(f_*\mathcal{F})_{f(x)} \to \mathcal{F}_x$.*
2. *For all $x \in X$, there is a natural isomorphism $(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$.*

Proof. The isomorphisms are immediate from the definitions (for (2), convince yourself that the nested direct limit that occurs satisfies the same universal property as the usual direct limit in the definition of $\mathcal{G}_{f(x)}$). The morphism $(f_*\mathcal{F})_{f(x)} \to \mathcal{F}_x$ is given by the natural map

$$(f_*\mathcal{F})_{f(x)} = \varinjlim_{f(x)\in V} \mathcal{F}(f^{-1}(V)) = \varinjlim_{x\in f^{-1}(V)} \mathcal{F}(f^{-1}(V)) \to \varinjlim_{x\in U} \mathcal{F}(U) = \mathcal{F}_x.$$

□

Proposition 4.6. *If $f : X \to Y$ is a continuous map between topological spaces, then $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ is right-adjoint to $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$.*

Proof. This is Exercise 11. □

Corollary 4.7. *Let $f : X \to Y$ be a continuous map between topological spaces. The functor $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ is exact.*

Proof. Let

$$0 \to \mathcal{G}_1 \to \mathcal{G}_2 \to \mathcal{G}_3 \to 0$$

be a short exact sequence of sheaves on Y. By Corollary 3.12, the associated sequence of stalks

$$0 \to (\mathcal{G}_1)_y \to (\mathcal{G}_2)_y \to (\mathcal{G}_3)_y \to 0$$

is exact for all $y \in Y$. In particular, it is exact for all y of the form $f(x)$ with $x \in X$. Hence, by Lemma 4.5 and Corollary 3.12, the associated sequence

$$0 \to f^{-1}\mathcal{G}_1 \to f^{-1}\mathcal{G}_2 \to f^{-1}\mathcal{G}_3 \to 0$$

of sheaves on X is exact. □

Corollary 4.8. *Let $f : X \to Y$ be a continuous map between topological spaces. The functor $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ is left-exact.*

Proof. Let

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

be a short exact sequence of sheaves on X. Taking sections over $U \subseteq X$, we get an exact sequence

$$0 \to \mathcal{F}_1(U) \to \mathcal{F}_2(U) \to \mathcal{F}_3(U).$$

Since taking direct limits is exact and by Lemma 4.5 (1), we get an exact sequence

$$0 \to (f_*\mathcal{F}_1)_y \to (f_*\mathcal{F}_2)_y \to (f_*\mathcal{F}_3)_y$$

for all $y \in Y$. By Corollary 3.12, this shows that the sequence

$$0 \to f_*\mathcal{F}_1 \to f_*\mathcal{F}_2 \to f_*\mathcal{F}_3$$

is exact. □

Example 4.9. Let $f : X \to \{*\}$ be a morphism from any topological space X to the 1-point space $\{*\}$. Let $\mathcal{F}$ be a sheaf on X. Then, $f_*\mathcal{F}(\emptyset) = 0$ and $f_*\mathcal{F}(\{*\}) = \mathcal{F}(X)$. In particular, Exercise 4 shows that f_* is in general not exact on the right.

4.2. **Ringed spaces and schemes.** Now that we can compare sheaves on different topological spaces, we can introduce the concept of ringed spaces.

Definition 4.10. A *ringed space* is a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings on X. A *morphism of ringed spaces* $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a pair $(f, f^\sharp)$, where $f : X \to Y$ is a continuous map and $f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is a morphism of sheaves of rings on Y.

We think of a ringed space as a pair consisting of a topological space and a collection of functions on it. A morphism of ringed spaces consists of a morphism of topological spaces together with a specified way of pulling back functions from Y to X.

Remark 4.11. Ringed spaces form a category. For $(f, f^\sharp) : X \to Y$ and $(g, g^\sharp) : Y \to Z$, The composition $(g, g^\sharp) \circ (f, f^\sharp)$ is defined on the topological level by $g \circ f$. On the level of sheaves of rings, we need a map $\mathcal{O}_Z \to (g \circ f)_*\mathcal{O}_X = g_*(f_*\mathcal{O}_X)$. For this, we take $g_*(f^\sharp) \circ g^\sharp$.

Example 4.12.
(1) For every ring A, $(\mathrm{Spec}\ A, \mathcal{O}_{\mathrm{Spec}\ A})$ is a ringed space.
(2) For any field k, there is a functor from the category of topological spaces to the category of ringed spaces (over $(*, k)$) defined by mapping a topological space X to $(X, \mathrm{Fun}_{X,k})$, where $\mathrm{Fun}_{X,k}(U) := \{s : U \to k\}$ and $f : X \to Y$ is mapped to $(f, f^\sharp)$ with $f^\sharp(U)(s) = s \circ f$.

(3) The pair $(X, \mathcal{C}^0)$, where X is a topological space and $\mathcal{C}^0$ is the sheaf of continuous functions from X to $\mathbb{R}$ is a ringed space. As before, every continuous map $f : X \to Y$ yields a morphism of ringed spaces $(f, f^\sharp)$ by setting $f^\sharp(U) : \mathcal{C}^0(U) \to \mathcal{C}^0(f^{-1}(U)), s \mapsto s \circ f$.
(4) A similar example works for C^∞-manifolds and smooth maps between them, or complex manifolds and holomorphic maps.

By Proposition 2.17 and Exercise 9, many of the above examples have the property that the stalks of $\mathcal{O}_X$ are in fact local rings. In analogy with Exercise 9, we think of a function s whose germ at a point $x \in X$ lies in the maximal ideal of $\mathcal{O}_{X,x}$ as a function that vanishes at x (not to be confused with $s_x = 0$). Note that there is no condition in Definition 4.10 that guarantees that if $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces such that $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,y}$ are local rings for all $x \in X$ and $y \in Y$, the pullback of a function s which vanishes at a point of the form $f(x)$ also vanishes at x. To remedy this we recall the notion of local homomorphism.

Definition 4.13. A morphism of local rings $\varphi : A \to B$ with maximal ideals $\mathfrak{m}_A$ and $\mathfrak{m}_B$ is called *local* if $\mathfrak{m}_A = \varphi^{-1}\mathfrak{m}_B$, or, equivalently, $\varphi(\mathfrak{m}_A) \subseteq \mathfrak{m}_B$.

Example 4.14. (1) For any ring homomorphism $\varphi : A \to B$ and prime ideal $\mathfrak{q} \subseteq B$, the induced homomorphism on localizations $A_{\varphi^{-1}(\mathfrak{q})} \to B_\mathfrak{q}$ is local.
(2) A homomorphism $\varphi : A \to K$ from a local ring A to a field K is local if and only if $\mathfrak{m}_A = \ker(\varphi)$. In particular, the inclusion from a local integral domain into its field of fractions is usually not local.

Definition 4.15. A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that for every point $x \in X$, the stalk $\mathcal{O}_{X,x}$ is a local ring. A *morphism of locally ringed spaces* is a morphism of ringed spaces $(f, f^\sharp) : X \to Y$ such that for all $x \in X$, the homomorphism $f^\sharp_x$ [5] is a local homomorphism, where $f^\sharp_x$ is the composition $\mathcal{O}_{Y,f(x)} \to (f_*\mathcal{O}_X)_{f(x)} \to \mathcal{O}_{X,x}$.

Remark 4.16. Locally ringed spaces form a subcategory of the category of ringed spaces. Indeed, if $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ and $(g, g^\sharp) : (Y, \mathcal{O}_Y) \to (Z, \mathcal{O}_Z)$ are morphisms of locally ringed spaces, then

$$(g_*(f^\sharp) \circ g^\sharp)_x = f^\sharp_x \circ g^\sharp_{f(x)}$$

since taking direct limits and pushforward are functors, and the homomorphism $(g_*\mathcal{F})_{g(x)} \to \mathcal{F}_x$ is natural in the sheaf $\mathcal{F}$. Now, it suffices to observe that the composition of two local homomorphisms is again local.

Now, we are finally ready to define schemes.

Definition 4.17. An *affine scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that there exists a ring A and an isomorphism $(X, \mathcal{O}_X) \xrightarrow{\sim} (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that there exists an open cover $X = \cup_{i \in I} U_i$ such that each $(U_i, \mathcal{O}_X|_{U_i})$ is isomorphic to an affine scheme. A *morphism of schemes* is a morphism of the corresponding locally ringed spaces.

Example 4.18. (1) Let $(X, \mathcal{O}_X)$ be a scheme and let $U \subseteq X$ be an open subset. Then, $(U, \mathcal{O}_X|_U)$ is a scheme [6], since $(D(a), \mathcal{O}_{\operatorname{Spec} A}|_{D(a)}) \cong (\operatorname{Spec} A_a, \mathcal{O}_{\operatorname{Spec} A_a})$ for all rings A and $a \in A$.

[5] There is no ambiguity of notation, since x is a point on X and $f^\sharp$ is a morphism of sheaves on Y. In particular, note that $f^\sharp_x \neq f^\sharp_{f(x)}$.

[6] But not necessarily an affine scheme, even if $(X, \mathcal{O}_X)$ is. We will see an example for this in an exercise.

(2) If $(X, \mathcal{O}_X)$ is a scheme and $X = \{*\}$, then $(X, \mathcal{O}_X)$ is affine. Moreover, note that $\operatorname{Spec} A \cong \{*\}$ for a ring A if and only if every element of A is either a unit or nilpotent. Even though the topological space underlying this scheme is just a point, we can recover the ring from $\mathcal{O}_X$, because $\mathcal{O}_X(X) \cong A$ by Proposition 2.17. This shows that a locally ringed space can carry immensely more information than its underlying topological space.

5. Lecture 5: The category of affine schemes and affinization

In the beginning of this lecture, we will see that morphisms from a locally ringed space $(X, \mathcal{O}_X)$ to the spectrum of a ring A are in natural 1-to-1 correspondence with morphisms of rings $A \to \mathcal{O}_X(X)$. This implies that every locally ringed space X admits a natural map, called the *affinization*, to the affine scheme $\operatorname{Spec} \mathcal{O}_X(X)$, that this morphism is an isomorphism if and only if X is affine (so we can think of X as being "equal" to $\operatorname{Spec} \mathcal{O}_X(X)$), and that the category of affine schemes is dual to the category of rings. Finally, we will study a couple of examples of schemes, see how to study affine algebraic sets using schemes, and learn how to glue two schemes together.

Notation 5.1. We will often use the common abuse of terminology and say that "X is a (locally) ringed space/an (affine) scheme", keeping in mind that there is a sheaf of rings $\mathcal{O}_X$ that is part of the datum.

We will need the following technical lemma.

Lemma 5.2. *Let X be a topological space and let $\{V_i\}_{i \in I}$ be a basis of the topology on X. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. For any collection of morphisms $\varphi_i : \mathcal{F}(V_i) \to \mathcal{G}(V_i)$ such that $\rho_{V_i V_j} \circ \varphi_i = \varphi_j \circ \rho_{V_i V_j}$ for all $V_j \subseteq V_i$, there exists a unique morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ such that $\varphi(V_i) = \varphi_i$.*

Proof. Let $U \subseteq X$ be any open subset. Define

$$\begin{aligned} \varphi(U) : \quad \mathcal{F}(U) &\to \mathcal{G}(U) \\ s &\mapsto t, \end{aligned}$$

where t is the unique section with $t|_{V_i} = \varphi_i(s|_{V_i})$ for all $V_i \subseteq U$.

Since $\mathcal{G}$ is a sheaf and U admits a cover by V_i, the section t is unique. By assumption, we have $\varphi_i(s|_{V_i})|_{V_k} = \varphi_j(s|_{V_j})|_{V_k}$ for all $V_k \subseteq V_i \cap V_j$. Since $V_i \cap V_j$ admits a cover by V_k, this implies $\varphi_i(s|_{V_i})|_{V_i \cap V_j} = \varphi_j(s|_{V_j})|_{V_i \cap V_j}$ for all $i, j \in I$, so t exists. □

Proposition 5.3. *Let $(X, \mathcal{O}_X)$ be a locally ringed space and let A be a ring. Then, the map*

$$\begin{aligned} \operatorname{Hom}((X, \mathcal{O}_X), (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})) &\to \operatorname{Hom}(A, \mathcal{O}_X(X)) \\ (f, f^\sharp) &\mapsto f^\sharp(\operatorname{Spec} A), \end{aligned}$$

where the left-hand side denotes morphisms of locally ringed spaces and the right-hand side denotes morphisms of rings, is a natural bijection.

Proof. Throughout, we use the natural identifications of Proposition 2.17. Since taking global sections is functorial, the map is natural in X and A. It remains to construct an inverse. Let $\varphi : A \to \mathcal{O}_X(X)$ be a morphism of rings. We need to define $(f, f^\sharp) : (X, \mathcal{O}_X) \to (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$.

We set

$$\begin{aligned} f : \quad X &\to \operatorname{Spec} A \\ x &\mapsto \ker(A \xrightarrow{\varphi} \mathcal{O}_X(X) \to \mathcal{O}_{X,x}/\mathfrak{m}_x), \end{aligned}$$

where $\mathfrak{m}_x$ is the maximal ideal in $\mathcal{O}_{X,x}$. To show that f is continuous, it suffices to check that

$$X_{\varphi(a)} := f^{-1}(D(a)) = \{x \in X \mid \varphi(a)_x \notin \mathfrak{m}_x\}$$

is open, where $a \in A$ is any element, because the $D(a)$ form a basis of the topology on $\operatorname{Spec} A$. To see that $X_{\varphi(a)}$ is open, let $x \in X_{\varphi(a)}$ be a point. Since $\varphi(a)_x \notin \mathfrak{m}_x$, we have $\varphi(a)_x \in \mathcal{O}_{X,x}^\times$. So, by

Remark 2.12, there exists an open neighborhood $x \in V \subseteq X$ and a section $b \in \mathcal{O}_X(V)$ such that $\varphi(a)|_V \cdot b = 1 \in \mathcal{O}_X(V)$. But then $\varphi(a)_y \notin \mathfrak{m}_y$ for all $y \in V$, so $V \subseteq X_{\varphi(a)}$, hence $X_{\varphi(a)}$ is open.

Next, we define $f^\sharp(D(a))$ for $a \in A$. For this, observe that the element $\varphi(a)|_{X_{\varphi(a)}} \in \mathcal{O}_X(X_{\varphi(a)})$ is a unit in every stalk, hence a unit in $\mathcal{O}_X(X_{\varphi(a)})$. Therefore, the universal property of the localization $A \to A_a$ shows that there is a unique dotted arrow making the following diagram commute:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & \mathcal{O}_X(X) \arrow[d] \\ A_a \arrow[r, dotted] & \mathcal{O}_X(X_{\varphi(a)}). \end{tikzcd} \tag{3}$$

We let $f^\sharp(D(a))$ be this dotted arrow. By Lemma 5.2, the uniqueness of the dotted arrow in the above diagram implies that the $f^\sharp(D(a))$ glue to a morphism of sheaves $f^\sharp : \mathcal{O}_{\operatorname{Spec} A} \to f_*\mathcal{O}_X$. [7]

We still have to show that $(f, f^\sharp)$ is a morphism of locally ringed spaces. For this, note that for every $x \in X$ the image $f(x)$ is a prime ideal in A, and the diagram (3) induces a commutative diagram

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & \mathcal{O}_X(X) \arrow[d] \\ A_{f(x)} \arrow[r, "f_x^\sharp"] & \mathcal{O}_{X,x}. \end{tikzcd}$$

The ideal $f(x)$ is the preimage of the maximal ideal under $A \to A_{f(x)}$. On the other hand, by definition of f, it is the preimage of $\mathfrak{m}_x$ under the composition $A \to \mathcal{O}_X(X) \to \mathcal{O}_{X,x}$. Hence, $(f_x^\sharp)^{-1}(\mathfrak{m}_x) = f(x)A_{f(x)}$, so $f_x^\sharp$ is local, as desired.

It remains to show that our construction is inverse to $(f, f^\sharp) \mapsto f^\sharp(\operatorname{Spec} A)$. Note that the map on global sections of the morphism of ringed spaces associated to φ is exactly φ by diagram (3) (with $a = 1$). So, let $(f, f^\sharp) : X \to \operatorname{Spec} A$ be a morphism of locally ringed spaces and let $(f', f'^\sharp) : X \to \operatorname{Spec} A$ be the morphism of locally ringed spaces associated to $f^\sharp(\operatorname{Spec} A)$. We want to show that $(f, f^\sharp) = (f', f'^\sharp)$. We have a commutative diagram

$$\begin{tikzcd} & A \arrow[r, "f^\sharp(\operatorname{Spec} A)"] \arrow[dl] \arrow[d] & \mathcal{O}_X(X) \arrow[d] \\ A_{f'(x)} \arrow[rr, bend right, "f_x'^\sharp"'] & A_{f(x)} \arrow[r, "f_x^\sharp"] & \mathcal{O}_{X,x}. \end{tikzcd}$$

Both $f_x^\sharp$ and $f_x'^\sharp$ are local, hence $f(x)$ and $f'(x)$ both coincide with the preimage of $\mathfrak{m}_x$ under $A \to \mathcal{O}_X(X) \to \mathcal{O}_{X,x}$, so $f(x) = f'(x) \subseteq A$.

Finally, both $f^\sharp(D(a))$ and $f'^\sharp(D(a))$ fit as the dotted arrow into diagram (3) (with $\varphi = f^\sharp(\operatorname{Spec} A)$), hence they are equal, so $f^\sharp = f'^\sharp$ by Lemma 5.2. $\square$

[7] To see that the $f^\sharp$ satisfy the compatibility condition, use that the localization maps $A \to A_a$ are epimorphisms in the category of rings.

Proposition 5.3 shows that $\mathrm{Spec}(-)$ is right-adjoint to the functor that takes a locally ringed space to its ring of global sections. In particular, for every locally ringed space $(X, \mathcal{O}_X)$, there is a natural *affinization* morphism $\mathrm{aff}_X : X \to \mathrm{Spec}\ \mathcal{O}_X(X)$ induced by the identity on $\mathcal{O}_X(X)$.

Corollary 5.4. *Every morphisms of locally ringed spaces $X \to \mathrm{Spec}\ A$ factors uniquely through aff_X.*

If $X = \mathrm{Spec}\ A$, then the affinization aff_X is just the identity. From this, we get the following criterion for a scheme to be affine.

Corollary 5.5. *A locally ringed space $(X, \mathcal{O}_X)$ is an affine scheme if and only if aff_X is an isomorphism.*

For the following, we let (AffSch) be the category of affine schemes, where the morphisms are morphisms of locally ringed spaces, and (Ring) is the category of rings.

Corollary 5.6. *The functor*

$$\begin{array}{rcl} (\mathrm{AffSch}) & \to & (\mathrm{Ring})^{op} \\ (X, \mathcal{O}_X) & \mapsto & \mathcal{O}_X(X) \\ (f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y) & \mapsto & f^\sharp(Y) : \mathcal{O}_Y(Y) \to f_*\mathcal{O}_X(Y) = \mathcal{O}_X(X) \end{array}$$

is an equivalence of categories.

Proof. The functor is fully faithful (i.e. induces bijections on Hom-sets) by Proposition 5.3. By Proposition 2.17, the functor is essentially surjective (i.e. every ring is isomorphic to $\mathcal{O}_X(X)$ for some affine scheme X). □

Example 5.7.
(1) Every locally ringed space X admits a unique morphism to $\mathrm{Spec}\ \mathbb{Z}$. This follows from Proposition 5.3, since there is a unique morphism of rings $\mathbb{Z} \to \mathcal{O}_X(X)$.
(2) Every locally ringed space X such that $\mathcal{O}_X(U)$ has characteristic $p > 0$ for all $U \subseteq X$ open admits a unique morphism to $\mathrm{Spec}\ \mathbb{F}_p$. Conversely, if X admits a morphism to $\mathrm{Spec}\ \mathbb{F}_p$, then every $\mathcal{O}_X(U)$ has characteristic $p > 0$.

Definition 5.8. Let S be a scheme. The category of schemes over S, denoted by (Sch/S), is the category with

- Objects: Morphisms $\pi_X : X \to S$.
- Morphisms: Commutative diagrams

$$\begin{tikzcd} X \arrow[rr] \arrow[dr, "\pi_X"'] & & Y \arrow[dl, "\pi_y"] \\ & S & \end{tikzcd}$$

In particular, Example 5.7 shows that $(\mathrm{Sch}/\mathrm{Spec}\ \mathbb{Z}) = (\mathrm{Sch})$ and that $(\mathrm{Sch}/\mathrm{Spec}\ \mathbb{F}_p)$ consists of schemes X such that every $\mathcal{O}_X(U)$ has characteristic p. The following example gives a first taste on how scheme theory is related to the classical study of varieties in affine space.

Example 5.9. Let k be a field.

(1) The *affine n-space* over k is $\mathbb{A}^n_k := \operatorname{Spec} k[x_1, \dots, x_n]$. If k is algebraically closed, we have natural identifications

$$\begin{array}{rcl} k^n \to & \operatorname{Spec}_{\max} k[x_1, \dots, x_n] & \cong \operatorname{Hom}_{k-alg}(k[x_1, \dots, x_n], k) \cong \mathbb{A}^n_k(k) \\ (a_1, \dots, a_n) \mapsto & \mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n) & \mapsto (f \mapsto f(a_1, \dots, a_n)), \end{array}$$

where $\operatorname{Spec}_{\max}$ denotes the set of maximal ideals.

(2) If $\varphi : A \to B$ is a surjection of rings, then, from commutative algebra, we know that the induced map $\operatorname{Spec} B \to \operatorname{Spec} A$ is a homeomorphism onto $V(I) \subseteq \operatorname{Spec} A$, where $I = \ker(\varphi)$.

(3) In particular, if $I \subseteq k[x_1, \dots, x_n]$ is an ideal and k is algebraically closed, then

$$V(I) = \{(a_1, \dots, a_n) \in k^n \mid f(a_1, \dots, a_n) = 0 \text{ for all } f \in I\}$$

is the image of $\operatorname{Spec}_{\max} k[x_1, \dots, x_n]/I \to \mathbb{A}^n_k(k) \cong k^n$. In particular, zero sets of polynomials in k^n can be studied using scheme theory.

We finish this lecture by looking at two simple examples of non-affine schemes obtained by glueing two affine schemes.

Example 5.10 (Glueing two schemes)**.** If X_1, X_2 are schemes, $U_i \subseteq X_i$ are open subsets, and $(f, f^\sharp) : (U_1, \mathcal{O}_X|_{U_1}) \to (U_2, \mathcal{O}_X|_{U_2})$ is an isomorphism, we define the *scheme obtained by glueing X_1 to X_2 along φ* by setting

$$X := (X_1 \coprod X_2)/\sim,$$

where $x_1 \sim x_2$ if and only if $x_2 = \varphi(x_1)$. This X becomes a topological space by defining $U \subseteq X$ to be open if and only if $\pi_i^{-1}(U) \subseteq U$ is open for $i = 1, 2$. With this topology, the natural maps $\pi_i : X_i \to X$ are homeomorphisms onto open subsets.

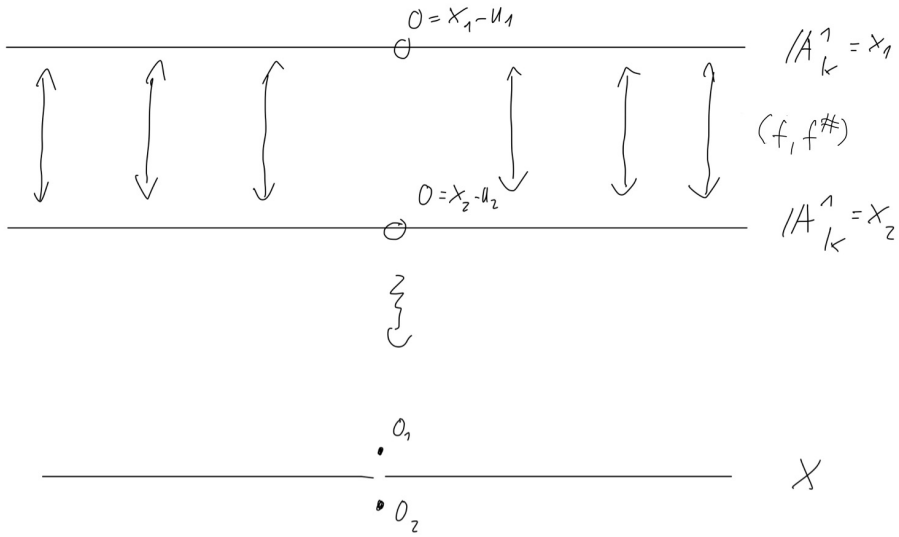
We define a the structure sheaf on X by setting

$$\begin{array}{rcl} \mathcal{O}_X(U) := \ker(\mathcal{O}_{X_1}(\pi_1^{-1}(U)) \oplus \mathcal{O}_{X_2}(\pi_2^{-1}(U)) & \to & \mathcal{O}_{X_1}(\pi_1^{-1}(U) \cap U_1)) \\ (a, b) & \mapsto & a|_{\pi_1^{-1}(U) \cap U_1} - f^\sharp(\pi_2^{-1}(U) \cap U_2)(b|_{\pi_2^{-1}(U) \cap U_2}). \end{array}$$

In particular, note that $X = \pi_1(X_1) \cup \pi_2(X_2)$ and $\mathcal{O}_X|_{\pi_i(X_i)} \cong \mathcal{O}_{X_i}$, so X is a scheme.

Let us make this explicit in two examples. Let k be an algebraically closed field, $X_1 = X_2 = \mathbb{A}^1_k$, and $U_1 = U_2 = \mathbb{A}^1_k \setminus \{0\} \cong \operatorname{Spec} k[x, x^{-1}]$.

(1) If we take $(f, f^\sharp) : U_1 \to U_2$ induced by the identity on $k[x, x^{-1}]$, we get the *affine line with two origins*. Here is a sketch of the process:

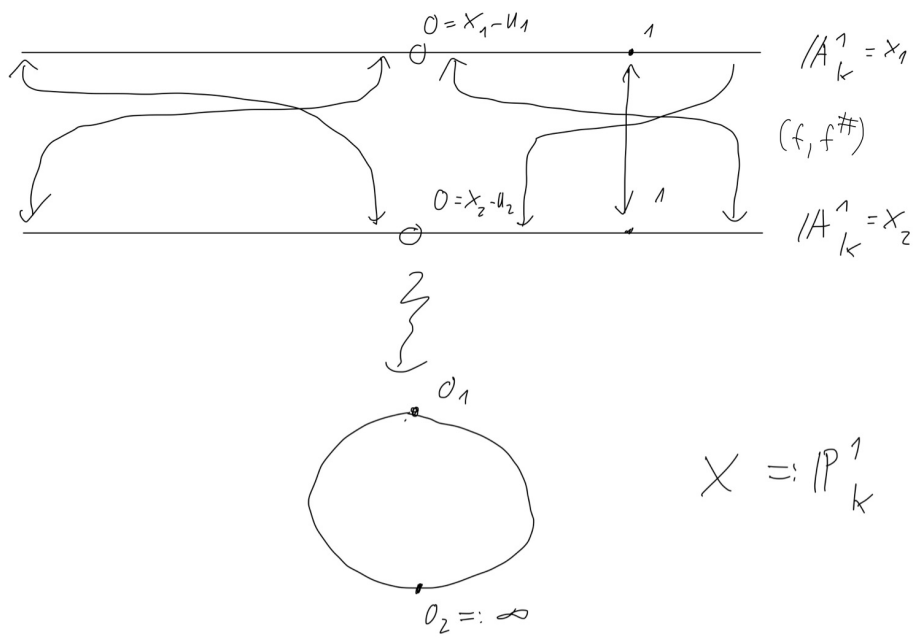


We can also compute the affinization in this case:

$$\mathcal{O}_X(X) \quad = \quad \ker(k[x] \oplus k[x] \to k[x, x^{-1}], (f, g) \mapsto (f - g)) \cong k[x].$$

So $\operatorname{Spec} \mathcal{O}_X(X) = \mathbb{A}^1_k$ and the restriction of aff_X to both charts X_1, X_2 is just the identity.

(2) If we take $(f, f^\sharp) : U_1 \to U_2$ induced by $x \mapsto x^{-1}$, we get the *projective line* $\mathbb{P}^1_k$. Here is a sketch of the process:



Again, we compute

$$\mathcal{O}_X(X) \quad = \quad \ker(k[x] \oplus k[x] \to k[x, x^{-1}], (f, g) \mapsto (f(x) - g(x^{-1})).$$

In particular, if (f, g) is in the above kernel, then the polynomial f can be written as a polynomial in x^{-1}. This is only possible if f is already constant, hence $\mathcal{O}_X(X) = k$. The affinization map in this case is thus $\mathrm{aff}_{\mathbb{P}^1_k} : \mathbb{P}^1_k \to \mathrm{Spec}\, k$.

6. Lecture 6: Properties of schemes

In this lecture, we start studying properties of schemes. Since a scheme is a pair consisting of a topological space and a sheaf of rings, we can define topological as well as algebraic properties, and study the interplay between them.

6.1. Topological properties.

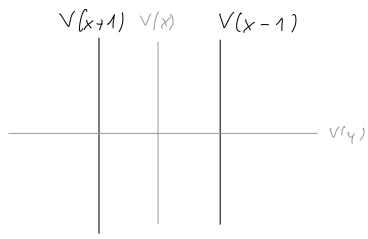
Definition 6.1. A scheme $(X, \mathcal{O}_X)$ is called

(1) *connected* if X is connected, i.e., there exist no non-empty open subsets $U_1, U_2 \subseteq X$ such that $U_1 \cup U_2 = X$ and $U_1 \cap U_2 = \emptyset$.
(2) *irreducible* if X is irreducible, i.e., for all non-empty open subsets $U_1, U_2 \subseteq X$ we have $U_1 \cap U_2 \neq \emptyset$.
(3) *quasi-compact* if X is quasi-compact, i.e., for all open covers $X = \cup_{i \in I} U_i$, there exists a finite subset $J \subseteq I$ such that $X = \cup_{j \in J} U_j$.
(4) *quasi-separated* if X is quasi-separated, i.e., for all $U_1, U_2 \subseteq X$ open and quasi-compact, the intersection $U_1 \cap U_2$ is quasi-compact.

Remark 6.2. Note that every irreducible scheme is connected.

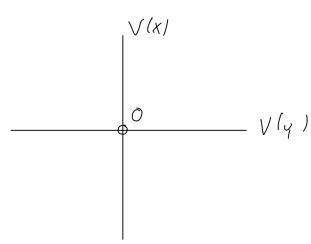
Example 6.3. Let k be an algebraically closed field.

(1) If $\operatorname{char} k \neq 2$, then $X = \operatorname{Spec} k[x, y]/(x^2 - 1)$ is not connected. Indeed, we have $(x^2 - 1) = (x + 1)(x - 1)$, so $X = V(x - 1) \cup V(x + 1) \subseteq \mathbb{A}^2_k$ and $V(x - 1) \cap V(x + 1) = \emptyset$.



As an easier example, take $X = \operatorname{Spec}(k \times k)$. Here, the underlying space is a discrete set with two points.

(2) $X = \operatorname{Spec} k[x, y]/(xy)$ is not irreducible. Indeed, $(V(x) \setminus \{0\}), (V(y) \setminus \{0\}) \subseteq X \subseteq \mathbb{A}^2_k$ are non-empty opens with empty intersection.



Lemma 6.4. *Affine schemes are quasi-compact and quasi-separated.*

Proof. Let $X = \operatorname{Spec} A$ be an affine scheme. By the proof of Claim 2 in Proposition 2.17, the principal open subset $D(a) \subseteq X$ is quasi-compact for every $a \in A$. [8] If $U \subseteq X$ is open and quasi-compact, then $U = \cup_{i \in I_U} D(a_i)$ for some finite set I_U and elements $a_i \in I_U$. In particular, if $U_1, U_2 \subseteq X$ are open and quasi-compact, we can write

$$U_1 \cap U_2 = \cup_{i \in I_{U_1}, j \in I_{U_2}} D(a_i) \cap D(a_j) = \cup_{i \in I_{U_1}, j \in I_{U_2}} D(a_i a_j).$$

Hence, $U_1 \cap U_2$ is a finite union of quasi-compact open subsets of X. Any such union is again quasi-compact, hence X is quasi-separated. □

Example 6.5. Let k be a field.

(1) The scheme $\mathbb{A}_k^\infty := \operatorname{Spec} k[x_1, \ldots, x_n, \ldots]$ is quasi-compact and quasi-separated, since it is affine.
(2) The scheme $X = \mathbb{A}_k^\infty \setminus \{0\}$ is not quasi-compact. Indeed, we have $X = \cup_{i=1}^\infty D(x_i)$, since the unique prime ideal that contains all x_i is exactly the origin in $\mathbb{A}_k^\infty$. The ideal $\mathfrak{p}_n := (x_1, \ldots, \hat{x}_n, \ldots)$ generated by all x_i except x_n is prime and $\mathfrak{p}_n \in D(x_i)$ if and only if $i = n$. So, $\cup_{i=1, i \neq n}^\infty D(x_i) \neq X$, hence the original open cover does not admit a finite subcover.
(3) Let X be the scheme obtained by glueing two copies of $\mathbb{A}_k^\infty$ via the identity along $\mathbb{A}_k^\infty \setminus \{0\}$. Then, X is not quasi-separated. Indeed, we can write $X = X_1 \cup X_2$ with $X_i \cong \mathbb{A}_k^\infty$, which are quasi-compact, but $X_1 \cap X_2 \cong \mathbb{A}_k^\infty \setminus \{0\}$, which is not quasi-compact.

The constructions in Example 6.5 use very large rings. This is no coincidence: In Exercise 18, we will see that every subset of a scheme which is covered by spectra of Noetherian rings is quasi-compact and quasi-separated.

Remark 6.6. Let X be a topological space. Then, the following hold:

(1) If $V \subseteq X$ is any subset and $\overline{V}$ is its closure, then an open $U \subseteq X$ intersects V if and only if it intersects $\overline{V}$. In particular, V is irreducible (with respect to the subset topology) if and only if $\overline{V}$ is irreducible.
(2) If X is irreducible, every non-empty open $U \subseteq X$ is dense, i.e. $\overline{U} = X$.

6.2. Scheme theoretic properties.

Definition 6.7. An *open subscheme* of a scheme $(X, \mathcal{O}_X)$ is a pair $(U, \mathcal{O}_U)$, where $U \subseteq X$ is open and $\mathcal{O}_U := \mathcal{O}_X|_U$.

Let $\mathcal{P}$ be a property of rings or of open affine subschemes of a scheme $(X, \mathcal{O}_X)$. When defining what it means that a scheme $(X, \mathcal{O}_X)$ satisfies $\mathcal{P}$, we have (at least) the following five possibilities:

(1) For all $U \subseteq X$ open, $\mathcal{O}_X(U)$ satisfies $\mathcal{P}$.
(2) For all $U \subseteq X$ affine open, $\mathcal{O}_X(U)$ satisfies $\mathcal{P}$.
(3) There exists an open affine cover $X = \cup U_i$ such that each $\mathcal{O}_X(U_i)$ satisfies $\mathcal{P}$.
(4) Every point $x \in X$ has an open affine neighborhood U such that $\mathcal{O}_X(U)$ satisfies $\mathcal{P}$.
(5) For every point $x \in X$, the stalk $\mathcal{O}_{X,x}$ satisfies $\mathcal{P}$.

We have the following implications: $(1) \Rightarrow (2) \Rightarrow (3) \Leftrightarrow (4)$.

Lemma 6.8. *For $\mathcal{P}$ = "reduced", all five conditions are equivalent.*

[8] Observe that, to prove quasi-compactness of a topological space X, it suffices to prove that every open cover by opens in a basis of the topology admits a finite subcover.

Proof. Since a ring is reduced if and only if all its localizations are, conditions $(2), \dots, (5)$ are equivalent (recall Proposition 2.17).

It remains to show $(2) \Rightarrow (1)$. Let $U \subseteq X$ be open and $s \in \mathcal{O}_X(U)$ such that $s^n = 0$. Then, $s^n|_V = 0$ for all $V \subseteq U \subseteq X$ open affine. Since $\mathcal{O}_X(V)$ is reduced, we have $s|_V = 0$. Since $\mathcal{O}_X$ is a sheaf, this implies $s = 0$. □

So, it is clear how to define reducedness for schemes.

Definition 6.9. A scheme $(X, \mathcal{O}_X)$ is called *reduced*, if $\mathcal{O}_X(U)$ is reduced for all $U \subseteq X$ open.

In general, the situation is more complicated. Still, for many properties of rings, Conditions (2) and (3) are equivalent. In the following, we make this precise:

Definition 6.10. Let $\mathcal{P}$ be a property of rings or of open affine subschemes $\operatorname{Spec} A \hookrightarrow X$ of a scheme X.

(1) $\mathcal{P}$ is called *affine-local* if for all $a_1, \dots, a_n \in A$ such that $(a_1, \dots, a_n) = A$, the ring A (or the open subscheme $\operatorname{Spec} A \hookrightarrow X$) satisfies $\mathcal{P}$ if and only if A_{a_i} (or $\operatorname{Spec} A_{a_i} \hookrightarrow X$) satisfies $\mathcal{P}$ for all $i = 1, \dots n$.
(2) $\mathcal{P}$ is called *stalk-local* if A satisfies $\mathcal{P}$ if and only if $A_{\mathfrak{p}}$ satisfies $\mathcal{P}$ for all $\mathfrak{p} \in \operatorname{Spec} A$.

Remark 6.11. Since the localization maps $A \to A_a$ induce isomorphisms $A_{\mathfrak{p}} \cong (A_a)_{\mathfrak{p}A_a}$ for all $\mathfrak{p} \in D(a)$, stalk-local properties are affine-local.

Example 6.12. The following properties are stalk-local:

(1) Reduced
(2) Normal [9]
(3) etc.

Example 6.13.
(1) "Integral" is not affine-local: $A = k \times k$ is not an integral domain, even though $A_{((1,0))} \cong A_{((0,1))} \cong k$.
(2) "Factorial" is not affine-local: $A = \mathbb{Z}[\sqrt{-5}]$ is not factorial, since $3^2 = 9 = (2+\sqrt{-5})(2-\sqrt{-5})$. But A_2 and A_3 are factorial [10] and cover A, since any ideal that contains 2 and 3 also contains $3 - 2 = 1$.
(3) "Noetherian" is not stalk-local: $A = \prod_{i=1}^{\infty} \mathbb{F}_2$ is not Noetherian, but every $A_{\mathfrak{p}}$ is a local ring consisting of idempotents, hence isomorphic to $\mathbb{F}_2$. This last implication is a surprisingly easy exercise in commutative algebra.

Lemma 6.14. *"Noetherian" is an affine-local property of rings.*

Proof. We know from commutative algebra that localizations of Noetherian rings are again Noetherian. So, let A be a ring, let $a_1, \dots, a_n \in A$ be elements generating the unit ideal, and assume that A_{a_i} is Noetherian for $i = 1, \dots, n$. We have to show that then A is also Noetherian. Let $\varphi_i : A \to A_{a_i}$ be the localization homomorphism.

<u>Claim:</u> For all ideals $I \subseteq A$, we have $I = \cap_{i=1}^{n} \varphi_i^{-1}(\varphi_i(I)A_{a_i})$.

[9] Recall: A ring A is called *normal* if for all $\mathfrak{p} \in \operatorname{Spec} A$, the localization $A_{\mathfrak{p}}$ is an integral domain which is integrally closed in its field of fractions, so normality is stalk-local by definition. In the special case where A is an integral domain, A is normal if and only if it is integrally closed in its field of fractions.

[10] This is a non-trivial exercise in number theory (e.g. use that the map $\mathbb{Z}/2\mathbb{Z} \cong \operatorname{Cl}(A) \to \operatorname{Cl}(A_n)$ with $n \in \{2, 3\}$ is surjective and find a non-trivial element in the kernel).

The inclusion " $\subseteq$" is clear. So, let $b \in \cap_{i=1}^n \varphi_i^{-1}(\varphi_i(I)A_{a_i})$. Then, $b = \frac{b_i}{a_i^N} \in A_{a_i}$ for some $b_i \in I$ and $N \geq 0$. In particular, note that we can choose N independent of i, since there are only finitely many i. Hence, there exists $M > 0$ (again independent of i), such that $a_i^M(a_i^N b - b_i) = 0 \in A$. Set $K = M + N$ and note that $1 = (a_1^K, \ldots, a_n^K)$. Write $1 = \sum_{i=1}^n c_i a_i^K$ for suitable $c_i \in A$. Then, $b = \sum_{i=1}^n c_i a_i^K b = \sum_{i=1}^n c_i a_i^M b_i \in I$, as claimed.

Now, if $I_1 \subseteq \ldots \subseteq I_k \subseteq \ldots$ is any ascending chain of ideals in A, then $\varphi_i(I_1) \subseteq \ldots \subseteq \varphi_i(I_k) \subseteq \ldots$ is an ascending chaing in A_{a_i} for all $i = 1, \ldots, n$. Since each A_{a_i} is Noetherian and since there are only finitely many i, we find $N \geq 0$ such that $\varphi_i(I_k) = \varphi_i(I_N)$ for all $k \geq N$. By the claim, this implies $I_k = I_N$ for all $k \geq N$, so A is Noetherian. $\square$

As announced before, the reason why we care about affine-local properties is that for such properties, conditions (2) and (3) of the beginning of this subsection are equivalent, that is, we have the following:

Lemma 6.15. *Let $\mathcal{P}$ be an affine-local property of rings (or open affine subschemes $\operatorname{Spec} A \hookrightarrow X$ of a scheme X). Let $(X, \mathcal{O}_X)$ be a scheme. Then, the following are equivalent:*

(1) Every open affine $\operatorname{Spec} A \hookrightarrow X$ satisfies $\mathcal{P}$.
(2) There exists an open affine cover $X = \cup \operatorname{Spec} A_i$ such that each $\operatorname{Spec} A_i \hookrightarrow X$ satisfies $\mathcal{P}$.

Proof. $(1) \Rightarrow (2)$ is clear. For the converse, let $\operatorname{Spec} A \subseteq X$ be any open affine. Write $\operatorname{Spec} A = \cup_{i=1}^n \operatorname{Spec} A_{a_i}$ with $a_i \in A$ such that there exist elements $b_i \in A_i$ and isomorphisms $(A_i)_{b_i} \cong A_{a_i}$ making the following diagrams commute:

$$\begin{tikzcd}
\operatorname{Spec}(A_i)_{b_i} \arrow[r] \arrow[d] & \operatorname{Spec} A_{a_i} \arrow[d] \\
\operatorname{Spec} A_i \arrow[dr] & \operatorname{Spec} A \arrow[dl] \\
& X
\end{tikzcd}.$$

This is possible by Lemma 6.16. Since $\mathcal{P}$ is affine-local and all the $\operatorname{Spec} A_i \hookrightarrow X$ satisfy $\mathcal{P}$, all the $\operatorname{Spec} A_{a_i} \cong \operatorname{Spec}(A_i)_{b_i} \hookrightarrow X$ satisfy $\mathcal{P}$. Since the $\operatorname{Spec} A_{a_i}$ cover $\operatorname{Spec} A$, the $a_1, \ldots, a_n$ generate the unit ideal, hence the fact that $\mathcal{P}$ is affine-local shows that also $\operatorname{Spec} A \hookrightarrow X$ satisfies $\mathcal{P}$. $\square$

Lemma 6.16. *Let $\operatorname{Spec} A, \operatorname{Spec} B \subseteq X$ be open affine subschemes. Then, for every point $x \in \operatorname{Spec} A \cap \operatorname{Spec} B$, there exist $a \in A$, $b \in B$, and an isomorphism $A_a \cong B_b$ such that $x \in D(b) \subseteq \operatorname{Spec} B$ and $x \in D(a) \subseteq \operatorname{Spec} A$ and the diagram*

$$\begin{tikzcd}
\operatorname{Spec} A_a \arrow[r] \arrow[d] & \operatorname{Spec} B_b \arrow[d] \\
\operatorname{Spec} A \arrow[dr] & \operatorname{Spec} B \arrow[dl] \\
& X
\end{tikzcd}$$

commutes.

Proof. Since $\operatorname{Spec} A \cap \operatorname{Spec} B \subseteq \operatorname{Spec} A$ is open, there is a principal open neighborhood of x in $\operatorname{Spec} A$ which is contained in $\operatorname{Spec} B$, so, after replacing $\operatorname{Spec} A$ by this principal open neighborhood, we may assume without loss of generality that $\operatorname{Spec} A \to X$ factors through $\operatorname{Spec} B$. Since $\operatorname{Spec} A \subseteq \operatorname{Spec} B$ open, say induced by $\varphi : B \to A$ via the correspondence of Corollary 5.6, there exists a $b \in B$ such that $\operatorname{Spec} B_b \to \operatorname{Spec} B$ factors through $\operatorname{Spec} A$. By Corollary 5.6, this means that the localization map $B \to B_b$ factors through φ. But then the map $A \to B_b$ satisfies the universal property of the localization of A at $\varphi(b)$, so there is a commutative square

$$\begin{tikzcd} B \arrow[r, "\varphi"] \arrow[d] & A \arrow[d] \\ B_b \arrow[r] & A_{\varphi(b)} \end{tikzcd}$$

where the bottom arrow is an isomorphism. This yields the desired commutative square on the level of spectra by Corollary 5.6. □

Definition 6.17. Let $\mathcal{P}$ be an affine-local property of rings. A scheme $(X, \mathcal{O}_X)$ is called *locally* $\mathcal{P}$ if every affine open $\operatorname{Spec} A \hookrightarrow X$ satisfies $\mathcal{P}$.

By Lemma 6.15, we can check whether $(X, \mathcal{O}_X)$ is locally $\mathcal{P}$ by checking that every $\operatorname{Spec} A_i \hookrightarrow X$ in a single open affine cover $X = \cup \operatorname{Spec} A_i$ satisfies $\mathcal{P}$. Sometimes a scheme $(X, \mathcal{O}_X)$ is said to satisfy $\mathcal{P}$ if it is quasi-compact and locally $\mathcal{P}$. An example is the following.

Definition 6.18. A scheme $(X, \mathcal{O}_X)$ is called *Noetherian* if it is locally Noetherian and quasi-compact.

It remains to deal with the notion of integrality, which, as we have noted before, is not affine-local.

Definition 6.19. A scheme $(X, \mathcal{O}_X)$ is called *integral* if $\mathcal{O}_X(U)$ is an integral domain for all $U \subseteq X$ open and non-empty. [11]

Since integrality is not affine-local, we cannot check it on an arbitrary open affine cover. Instead, we have the following criterion, which allows us to check integrality by checking reducedness (which is stalk-local) and irreducibility (which is purely topological).

Lemma 6.20. *For a scheme $(X, \mathcal{O}_X)$, the following are equivalent:*

(1) X is integral.
(2) X is irreducible and reduced.
(3) For all $U \subseteq X$ open, affine, and non-empty, the ring $\mathcal{O}_X(U)$ is integral.

Proof. $(1) \Rightarrow (3)$ is clear.

For $(3) \Rightarrow (2)$, reducedness is clear, since it is stalk-local. To see that X is irreducible, let $U_1, U_2 \subseteq X$ be open subsets with $U_1 \cap U_2 = \emptyset$. After shrinking the U_i, we may assume that they are affine. Note that $U_1 \cup U_2 \cong \operatorname{Spec} \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$ is also affine, since $U_1 \cap U_2 = \emptyset$. In particular, $\mathcal{O}_X(U_1 \cup U_2) \cong \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$ is either 0 or an integral domain, so $\mathcal{O}_X(U_1) = 0$ or $\mathcal{O}_X(U_2) = 0$. By Proposition 2.17, this implies that $U_1 = \emptyset$ or $U_2 = \emptyset$, so X is irreducible.

[11] Note that the 0-ring is not an integral domain, so we have to exclude the empty set.

Finally, to show that $(2) \Rightarrow (1)$, let $U \subseteq X$ be open and assume that there exist $a, b \in \mathcal{O}_X(U)$ with $ab = 0$. Let $U_a := \{x \in U \mid a_x \notin \mathfrak{m}_x\}$ and define U_b analogously. As in the proof of Proposition 5.3, U_a and U_b are open. Since $ab = 0$, also $a_x b_x = 0$ for all $x \in U$, so a_x and b_x are not both units. In other words, $U_a \cap U_b = \emptyset$. Since U is irreducible by Remark 6.6, we deduce that $U_a = \emptyset$ or $U_b = \emptyset$. Assume that $U_a = \emptyset$. Then, for all $\operatorname{Spec} A \subseteq U_a$ open and affine, and for all $\mathfrak{p} \in \operatorname{Spec} A$, we have $(a|_{\operatorname{Spec} A})_{\mathfrak{p}} \in \mathfrak{p}A_{\mathfrak{p}}$ (here we use Proposition 2.17 for the isomorphism $A_{\mathfrak{p}} \cong \mathcal{O}_{U,\mathfrak{p}}$). But this means that $a|_{\operatorname{Spec} A} \in \mathcal{O}_U(A)$ is contained in every prime ideal of A, so it is nilpotent. Since A is reduced, we deduce that $a|_{\operatorname{Spec} A} = 0$. As U is covered by open subsets of the form $\operatorname{Spec} A$, we obtain that $a = 0$. Similarly, if $U_b = \emptyset$, then $b = 0$. Therefore, $\mathcal{O}_X(U)$ is integral. $\square$

7. Lecture 7: Open and closed subschemes and immersions

Recall the definition of open subscheme that we gave last lecture:

Definition 7.1. An *open subscheme* of a scheme $(X, \mathcal{O}_X)$ is a pair $(U, \mathcal{O}_U)$, where $U \subseteq X$ is open and $\mathcal{O}_U := \mathcal{O}_X|_U$.

As we mentioned before, open subschemes of schemes are indeed schemes, essentially because principal open subschemes of affine schemes are. Note that if $\iota : U \hookrightarrow X$ is an open subset, then $(\iota, \iota^\sharp) : (U, \mathcal{O}_U) \to (X, \mathcal{O}_X)$, where we define

$$\iota^\sharp(V) : \mathcal{O}_X(V) \to \mathcal{O}_X(V \cap U) = \iota_* \mathcal{O}_U(V)$$

using the restriction map $\rho_{VV\cap U}$ on $\mathcal{O}_X$, is a morphism of schemes.

Caution 7.2. The sheaf $\mathcal{O}_X|_U$ is, in general, not the only sheaf of rings $\mathcal{F}$ such that $(U, \mathcal{F})$ is a scheme and such that there exists a $\iota^\sharp : \mathcal{O}_X \to \iota_* \mathcal{F}$ that makes $(\iota, \iota^\sharp) : (U, \mathcal{F}) \to (X, \mathcal{O}_X)$ into a morphism of schemes. For example, consider $X = \operatorname{Spec} k$ and equip $U = X$ with the structure sheaf of $\operatorname{Spec} k[x]/(x^2)$.

Definition 7.3. An *open immersion* (or open embedding) is a morphism of schemes $(f, f^\sharp) : X \to Y$ such that f is a homeomorphism onto an open subset U of Y and such that $f^\sharp$ induces an isomorphism $\mathcal{O}_Y|_U \cong (f_* \mathcal{O}_X)|_U$.

Equivalently, an open immersion is a morphism of schemes that factors through an isomorphism with an open subscheme of the target. The definition of closed subschemes and closed immersions is more subtle, because there is no obvious choice of structure sheaf:

Example 7.4. Let k be a field and let $\iota : \operatorname{Spec} k \to X := \mathbb{A}^n_k$ be the (closed) rational point corresponding to 0. Then, we have

$$(\mathcal{O}_X)|_{\operatorname{Spec} k}(\operatorname{Spec} k) := (\iota^{-1}\mathcal{O}_X)(\operatorname{Spec} k) = \varinjlim_{0 \in U \subseteq \mathbb{A}^n_k} \mathcal{O}_X(U) \cong k[x_1, \ldots, x_n]_{(x_0, \ldots, x_n)}.$$

Note that $\operatorname{Spec} k[x_1, \ldots, x_n]_{(x_0,\ldots,x_n)}$ has more than one point if $n \geq 1$. In particular, $(\operatorname{Spec} k, \iota^{-1}\mathcal{O}_X)$ cannot be a scheme, since its topological space is a singleton, so it would have to be an affine scheme and hence coincide with $\operatorname{Spec} k[x_1, \ldots, x_n]_{(x_0,\ldots,x_n)}$.

Instead, note that if $Z \subseteq \operatorname{Spec} A$ is a closed subset of an affine scheme, then $Z = V(I)$ for some ideal $I \subseteq A$ and the map induced by $A \to A/I$ induces a morphism $\operatorname{Spec} A/I \to \operatorname{Spec} A$ which is a homeomorphism onto $V(I)$. Thus, the choice of ideal I with $V(I) = Z$ gives us a scheme structure on the closed subset. Here, it is important to note that if we choose $I \neq J$ with $V(I) = V(J)$, then, as schemes, $\operatorname{Spec} A/I \ncong \operatorname{Spec} A/J$, so the scheme structure on $V(I)$ depends on our choice of I.

Definition 7.5. Let $(X, \mathcal{O}_X)$ be a scheme.

(1) An *ideal sheaf* on $(X, \mathcal{O}_X)$ is a subsheaf $\mathcal{I} \subseteq \mathcal{O}_X$ such that for each $U \subseteq X$ open, $\mathcal{I}(U) \subseteq \mathcal{O}_X(U)$ is an ideal.
(2) For an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$, the *quotient sheaf* $\mathcal{O}_X/\mathcal{I}$ is defined as the sheafification of the presheaf $U \mapsto \mathcal{O}_X(U)/\mathcal{I}(U)$.

Definition 7.6. Let $(X, \mathcal{O}_X)$ be a scheme. A *closed subscheme* of $(X, \mathcal{O}_X)$ consists of a subset $Z \subseteq X$ and an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ such that

(1) $Z = \{x \in X \mid (\mathcal{O}_X/\mathcal{I})_x \neq 0\}$.
(2) $(Z, (\mathcal{O}_X/\mathcal{I})|_Z)$ is a scheme. [12]

Remark 7.7. Condition (1) implies that Z is closed. Indeed, for $1 \in (\mathcal{O}_X/\mathcal{I})(X)$, we have $\{x \in X \mid (\mathcal{O}_X/\mathcal{I})_x \neq 0\} = \mathrm{supp}(1) \subseteq X$, which is closed by Exercise 8.

Remark 7.8. By condition (1) in Definition 7.6, the closed subset $Z \subseteq X$ can be recovered from the ideal sheaf $\mathcal{I}$. Moreover, the unit of the adjunction between ι^{-1} and ι_* (Exercise 11) is a morphism $\mathcal{O}_X/\mathcal{I} \to \iota_*((\mathcal{O}_X/\mathcal{I})|_Z)$, which is an isomorphism on stalks by Exercise 12, hence an isomorphism. In particular, the quotient map $\iota^\sharp : \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}$ defines a morphism of schemes $(\iota, \iota^\sharp) : (Z, (\mathcal{O}_X/\mathcal{I})|_Z) \to (X, \mathcal{O}_X)$.

Example 7.9. Condition (2) in Definition 7.6 is not automatic, even if X is affine:

Let $X = \mathrm{Spec}\ A$ where A is a discrete valuation ring (e.g. $k[[t]]$). Then, $\mathrm{Spec}\ A = \{(0), \mathfrak{m}_A\}$, the point $\{(0)\}$ is open, and the point $\mathfrak{m}_A$ is closed. Define an ideal sheaf $\mathcal{I}$ by setting $\mathcal{I}(\{(0)\}) = \mathrm{Frac}(A)$ and equal to 0 for the other opens $\emptyset$ and X. Then, $(\mathcal{O}_X/\mathcal{I})_{\mathfrak{m}_A} = A$ and $(\mathcal{O}_X/\mathcal{I})_{(0)} = 0$, so the closed subset of X determined by $\mathcal{I}$ is $\{\mathfrak{m}_A\}$, but the pair $(\{\mathfrak{m}_A\}, (\mathcal{O}_X/\mathcal{I})|_{\{\mathfrak{m}_A\}})$ is not a scheme, because if it were, then it would have to be affine (since its topological space is a single point), hence isomorphic to the spectrum of $(\mathcal{O}_X/\mathcal{I})|_{\{\mathfrak{m}_A\}}(\{\mathfrak{m}_A\}) \cong A$, which is absurd, since $\mathrm{Spec}\ A$ has two points.

We will see later that the key property that makes this sheaf of ideals not define a closed subscheme is that it is not induced by an actual ideal of the ring A.

As for open subschemes, we can now define the notion of closed immersion.

Definition 7.10. A *closed immersion* (or closed embedding) is a morphism of schemes $(f, f^\sharp) : X \to Y$ such that f is a homeomorphism onto a closed subset Z of Y and such that $f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective. [13]

Remark 7.11. Note that the morphism $(\iota, \iota^\sharp)$ in Remark 7.8 is a closed immersion. Conversely, one can check that the kernel of the map $f^\sharp$ in Definition 7.10 defines an ideal sheaf $\mathcal{I}$ on Y which determines a closed subscheme Z of Y such that $(f, f^\sharp)$ factors through an isomorphism with Z (compare this with Exercise 23). To see this, use that the natural map $\mathcal{O}_Y/\ker(f^\sharp) \to f_*\mathcal{O}_X$ is an isomorphism on stalks, hence an isomorphism and that the support of $f_*\mathcal{O}_X$ equals the image of f, because the image of f is closed.

Example 7.12. If A is a ring and $I \subseteq A$ is an ideal, then the morphism $(f, f^\sharp) : \mathrm{Spec}\ A/I \to \mathrm{Spec}\ A$ induced by the quotient map $A \to A/I$ is a closed immersion. Indeed, we know from commutative algebra that f is a homeomorphism onto $V(I)$. Moreover, we note that for all $\mathfrak{p} \in V(I)$, the map $f^\sharp_{f(\mathfrak{p})}$ is obtained by localizing $A \to A/I$, hence $f^\sharp$ is surjective.

Notation 7.13. From now on, we denote by $V(I)$ the closed subscheme of $\mathrm{Spec}\ A$ determined by the ideal I.

The following proposition says that the ideal sheaves on affine schemes that define closed subschemes are exactly the ideal sheaves that arise from ideals of the ring:

[12] Here, the "restriction" $\mathcal{F}|_Z$ of a sheaf on X to Z is defined as the inverse image $\iota^{-1}(\mathcal{F})$ along the inclusion $\iota : Z \to X$.

[13] We call a morphism of sheaves *surjective* if it is surjective on stalks. Recall that this does not imply that all the morphisms $f^\sharp(U)$ are surjective.

Proposition 7.14. *If $X = \operatorname{Spec} A$ is affine, then $I \mapsto V(I)$ is a bijection between ideals in A and closed subschemes of X.*

Proof. Let $Z \subseteq X$ be a closed subscheme determined by an ideal sheaf $\mathcal{I}$. Let $I_Z = \ker(\mathcal{O}_X(X) \xrightarrow{\varphi} (\iota_* \mathcal{O}_Z)(X) = \mathcal{O}_Z(Z))$. Note that if $Z = V(I)$ for some ideal I, then $I_Z = I$, so it will be enough to show that $Z = V(I_Z)$.

Now, by the universal property of affinization, the morphism $\iota : Z \to X$ factors through $\operatorname{aff}_Z : Z \to \operatorname{Spec} \mathcal{O}_Z(Z)$ and the morphism $\operatorname{Spec} \mathcal{O}_Z(Z) \to X$ which is an isomorphism onto the closed subscheme $V(I_Z)$. It follows that aff_Z is also a closed immersion. [14] Thus, to prove the proposition, we have to prove the following claim:

<u>Claim:</u> If Z is a scheme with $A := \mathcal{O}_Z(Z)$ such that $f := \operatorname{aff}_Z : Z \to \operatorname{Spec} A =: X$ is a closed immersion, then f is an isomorphism.

First, we show that f is in fact surjective. Since f is a homeomorphism onto a closed subset, it suffices to show that, as sets, $V(a) = X$ for all $a \in A$ with $f(Z) \subseteq V(a)$. Recall from the proof of Proposition 5.3 that

$$Z_a := f^{-1}(D(a)) = \{z \in Z \mid a_z \notin \mathfrak{m}_z\},$$

hence

$$Z \subseteq f^{-1}(V(a)) = \{z \in Z \mid a_z \in \mathfrak{m}_z\}.$$

This implies that a is in fact nilpotent: Indeed, since X is affine, it is quasi-compact by Lemma 6.4 and closed subspaces of quasi-compact spaces are quasi-compact, hence Z is quasi-compact. Thus, we can choose a finite cover $Z = \bigcup_{i=1}^n U_i$ by open affines U_i and then $(a|_{U_i})_z \in \mathfrak{m}_z$ for all $z \in U_i$ implies that $a|_{U_i}$ is nilpotent, hence a is nilpotent, and hence $V(a) = X$.

Thus, we may identify Z with $\operatorname{Spec} A$ as topological spaces and assume that f is the identity on the level of topological spaces. We still have to show that f is an isomorphism of schemes, that is, that $f^\sharp : \mathcal{O}_{\operatorname{Spec} A} \to \mathcal{O}_Z$ is an isomorphism. Since f is a closed immersion, we already know that $f^\sharp$ is surjective and it remains to check injectivity. By Corollary 3.10, we can do this on stalks, so let $\mathfrak{p} \in \operatorname{Spec} A$ be a point, let $\varphi = f^\sharp(\operatorname{Spec} A)$[15] and $\varphi_\mathfrak{p} = f^\sharp_\mathfrak{p}$.

It suffices to show that for every $a \in A$ with $\varphi_\mathfrak{p}(\frac{a}{1}) = 0$, we have $\frac{a}{1} = 0$. Since $\varphi_\mathfrak{p}(\frac{a}{1}) = 0$, we know that there exists an open $U \subseteq Z$ such that $\mathfrak{p} \in U$ and $\varphi(a)|_U = 0$.

Now, as $Z = \operatorname{Spec} A$ and principal opens form a basis of the topology, we find $s \in A$ such that $Z_{\varphi(s)} \subseteq U$. For $V \subseteq Z$ open and affine, we have $V \cap Z_{\varphi(s)} = D(\varphi(s)|_V)$ and so $\varphi(a)|_{V \cap Z_{\varphi(s)}} = 0$ means that there exists an $n \geq 0$ such that $(\varphi(s)|_V)^n \cdot \varphi(a)|_V = 0 \in \mathcal{O}_V(V)$. Since Z is quasi-compact, we can cover Z by finitely many open affines and find an $N \geq 0$ such that

$$\varphi(s^N a) = \varphi(s)^N \cdot \varphi(a) = 0 \in \mathcal{O}_Z(Z).$$

Now, since φ is injective, we deduce that $s^N a = 0 \in A$, hence $\frac{a}{1} = 0 \in A_s$. As $f^{-1}(D(s)) = Z_s$, we have $\mathfrak{p} \in D(s)$, hence $\frac{a}{1} = 0 \in A_\mathfrak{p}$, as desired. □

Corollary 7.15. *Closed subschemes of affine scheme are affine. Moreover, if $\varphi : A \to B$ is a morphism of rings, then $\operatorname{Spec} B \to \operatorname{Spec} A$ is a closed immersion if and only if φ is surjective.*

[14] This is a general topological fact on the topological level and a small exercise on the sheaf-theoretic level (use that you can compute the stalk of direct images along homeomorphisms onto closed subsets)

[15] In fact, this φ is just the identity, but it is helpful in this proof to be very clear about whether we are working on the scheme $\operatorname{Spec} A$ or the scheme Z, even though the global sections of their structure sheaves are equal.

Remark 7.16. Every closed subset $Z \subseteq X$ of a scheme X is the underlying topological space of a closed subscheme. By Proposition 7.14, this holds if X is affine. If X is not affine, consider an open affine cover $X = \cup U_i$ and show that the unique radical ideals $I_i \subseteq \mathcal{O}_X(U_i)$ such that $Z \cap U_i = V(I_i)$ determine a sheaf of ideals $\mathcal{I}$ on X such that $Z = \{x \in X \mid (\mathcal{O}_X/\mathcal{I})_x \neq 0\}$ and $(Z, (\mathcal{O}_X/\mathcal{I})|_Z)$ is a scheme. This scheme structure on Z is also called *reduced induced scheme structure*.

8. Lecture 8: Fiber products

The goal of this lecture is to show that fiber products exist in the category of schemes. We begin by reviewing the notion of fiber product in a category and motivate why we care about the existence of fiber products in (Sch). At the end of this lecture, we will compare our new notion of (scheme-theoretic) fiber with the corresponding set-theoretic fiber.

Definition 8.1. Let $\mathcal{C}$ be a category, let S be an object in $\mathcal{C}$, and let $\pi_X : X \to S$ and $\pi_Y : Y \to S$ two objects over S in $\mathcal{C}$. A *fiber product* of X and Y over S is an object $X \times_S Y$ together with morphisms $p_X : X \times_S Y \to X$ and $p_Y : X \times_S Y \to Y$ such that $\pi_Y \circ p_Y = \pi_X \circ p_X$ satisfying the following universal property:

For every object Z in $\mathcal{C}$ and every pair of morphisms $f_Y : Z \to Y$ and $f_X : Z \to X$ such that $\pi_Y \circ f_Y = \pi_X \circ f_X$, there exists a unique morphism $f : Z \to X \times_S Y$ such that $p_Y \circ f = f_Y$ and $p_X \circ f = f_X$.

In other words, for every commutative diagram with solid arrows in $\mathcal{C}$ of the following form, the dotted arrow exists and is unique:

$$\begin{tikzcd}
Z \arrow[dr, dotted, "\exists! f"] \arrow[drr, bend left, "f_Y"] \arrow[ddr, bend right, "f_X"'] & & \\
& X \times_S Y \arrow[r, "p_Y"] \arrow[d, "p_X"'] & Y \arrow[d, "\pi_Y"] \\
& X \arrow[r, "\pi_X"'] & S
\end{tikzcd}$$

Remark 8.2. (1) A fiber product is also called fibre product, fibered product, or pullback [16].

(2) A fiber product is unique up to unique isomorphism if it exists.

(3) If every object in $\mathcal{C}$ admits a unique morphism to S (i.e. if S is *terminal*), then $X \times_S Y = X \times Y$.

(4) If $X = S$ and $\pi_X : S \to S$ is the identity, then Y together with id_Y and π_Y is a fiber product of Y over S with S, i.e., $Y \times_S S \cong Y$.

Notation 8.3. If, in a commutative diagram of the form

$$\begin{tikzcd}
Z \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & Y \arrow[d] \\
X \arrow[r] & S
\end{tikzcd},$$

the object Z, together with $Z \to Y$ and $Z \to X$, is a fiber product of X and Y over S, we call the diagram *Cartesian* and denote this by a square in the center of the diagram (or by a small "right–angle" sign in the top-left corner).

[16] In the algebraic geometry literature, the pullback of $X \to S$ along $Y \to S$ usually denotes the object $X \times_S Y$ together with the morphism $X \times_S Y \to Y$

Lemma 8.4. *Let $\mathcal{C}$ be a category with fiber products. Assume we are given a diagram*

$$\begin{tikzcd}
X_1 \arrow[r] \arrow[d] & X_0 \arrow[d] & X_2 \arrow[l] \arrow[d] \\
S_1 \arrow[r] & S_0 & S_2 \arrow[l] \\
Y_1 \arrow[r] \arrow[u] & Y_0 \arrow[u] & Y_2 \arrow[l] \arrow[u]
\end{tikzcd}$$

Then,

$$(X_1 \times_{S_1} Y_1) \times_{X_0 \times_{S_0} Y_0} (X_2 \times_{S_2} Y_2) \cong (X_1 \times_{X_0} X_2) \times_{S_1 \times_{S_0} S_2} (Y_1 \times_{Y_0} Y_2).$$

Proof. It suffices to observe that one side of the isomorphism satisfies the universal property of the other side. This follows from a diagram chase. □

As special cases, we obtain:

Corollary 8.5. *If $\mathcal{C}$ admits fiber products, then the following identities hold:*

(1) For all diagrams

$$\begin{tikzcd}
 & Y \arrow[d] \\
X \arrow[r] & S,
\end{tikzcd}$$

we have $X \times_S Y = Y \times_S X$.

(2) For all diagrams

$$\begin{tikzcd}
 & & Y \arrow[d] \\
X \arrow[r] & S \arrow[r] & T,
\end{tikzcd}$$

we have $(Y \times_T S) \times_S X = X \times_T Y$.

(3) For all diagrams

$$\begin{tikzcd}
X \arrow[dr] & & Y \arrow[dl] \arrow[dr] & & Z \arrow[dl] \\
 & S & & T &
\end{tikzcd},$$

we have $(X \times_S Y) \times_T Z = X \times_S (Y \times_T Z)$.

(4) For all diagrams

$$\begin{tikzcd}
 & Y \arrow[d] & \\
X \arrow[r] & S \arrow[r] & T,
\end{tikzcd}$$

the square

$$\begin{tikzcd} X \times_S Y \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & X \times_T Y \arrow[d] \\ S \arrow[r] & S \times_T S \end{tikzcd}$$

is Cartesian. This is sometimes called magic square.

Proof. Claim (1) is clear by the universal property. For Claim (2), apply Lemma 8.4 with $X_1 = X, X_0 = X_2 = S$, $S_1 = S_0 = S_2 = Y_1 = Y_0 = T$ and $Y_2 = Y$. For Claim (3), apply (2) to see that both sides of the identity are equal to $(X \times_S Y) \times_Y (Y \times_T Z)$. For Claim (4), apply Lemma 8.4 with $X_1 = X_0 = S_1 = Y_1 = Y_0 = S$, $X_2 = X$, $S_0 = S_2 = T$, and $Y_2 = Y$. $\square$

Example 8.6. (1) If $\pi_X : X \to S$ and $\pi_Y : Y \to S$ are in (Set), then $X \times_S Y := \{(x, y) \mid \pi_X(x) = \pi_Y(y)\} \subseteq X \times Y$ together with the projections to X and Y is the fiber product.

(2) If X, Y, and S are groups (or rings), and π_X and π_Y are morphisms of groups, then the set $X \times_S Y$ as defined in (1) admits a group structure compatible with the one on $X \times Y$, and the resulting group is the fiber product of X and Y over S.

The goal of this lecture is to prove the following:

Theorem 8.7. *Fiber products in* (Sch) *exist. In particular, for every scheme S, fiber products exist in* (Sch/S).

Why do we care about this result? Once we have fiber products in a category, we can talk about fibers, pullbacks, graphs, diagonals, etc. For the following definition, recall from Exercise 15 that a point $y \in Y$ of a scheme Y comes with a scheme structure given by the residue field $k(y) := \mathcal{O}_{Y,y}/\mathfrak{m}_y$, such that the inclusion $\{y\} \to Y$ is the continuous map underlying a morphism of schemes $\operatorname{Spec} k(y) \to Y$. Moreover, a *generic point* η of a topological space Y is a point such that $\overline{\eta} = Y$. In Exercise 24, we will see that every integral scheme admits a unique generic point.

Definition 8.8. Let $f : X \to Y$ be a morphism of schemes over S.

(1) For any point $y \in Y$, let $k(y) := \mathcal{O}_{Y,y}/\mathfrak{m}_y$.
  (a) The *fiber* of f over $y \in Y$ is $f^{-1}(y) := X_y := X \times_Y \operatorname{Spec} k(y) \to \operatorname{Spec} k(y)$.
  (b) The *geometric fiber* of f over $y \in Y$ is $X_{\bar{y}} := X \times_Y \operatorname{Spec} \overline{k(y)} \to \operatorname{Spec} \overline{k(y)}$, where $k(y) \hookrightarrow \overline{k(y)}$ is an algebraic closure of $k(y)$.
  (c) A *closed fiber* of f is a fiber over a closed point $y \in Y$.
  (d) If Y is integral, the *generic fiber* of f is the fiber over the generic point of Y.

(2) The morphism $\Gamma_f := (\mathrm{id}, f) : X \to X \times_S Y$ is called *graph* of f.

(3) The morphism $\Delta_{X/Y} := \Gamma_{\mathrm{id}} : X \to X \times_Y X$ is called *diagonal* of f.

To prove Theorem 8.7, we first construct the fiber product in the affine case.

Proposition 8.9. *In the situation of Definition 8.1, let $X = \operatorname{Spec} A, Y = \operatorname{Spec} B$, and $S = \operatorname{Spec} C$. Then, the fiber product $X \times_S Y$ exists in* (Sch) *and is given by* $\operatorname{Spec}(A \otimes_C B)$ *together with the morphisms induced by the maps $A \to A \otimes_C B, a \mapsto a \otimes 1$ and $B \to A \otimes_C B, b \mapsto 1 \otimes b$.*

Proof. Recall that the tensor product $A \otimes_C B$ together with the maps $A \to A \otimes_C B$ and $B \to A \otimes_C B$ satisfies the universal property that for every C-algebra D and every pair of morphisms of C-algebras $A \to D$ and $B \to D$, there exists a unique compatible morphism of C-algebras

$A \otimes_C B \to D$. By Proposition 5.3, this implies that $\operatorname{Spec}(A \otimes_C B)$ is not only the fiber product of X and Y over S in (AffSch), but also in (Sch). In diagrams, the proof can be summarized as follows:

The following diagram describes the universal property of the tensor product $A \otimes_C B$:

$$\begin{tikzcd}
D & & \\
& A \otimes_C B \arrow[ul, dotted, "\exists!"'] & B \arrow[l] \arrow[ull, bend right] \\
& A \arrow[u] \arrow[uul, bend left] & C \arrow[l] \arrow[u]
\end{tikzcd}$$

Applying Spec and using Corollary 5.6, we obtain the following diagram in (AffSch):

$$\begin{tikzcd}
\operatorname{Spec} D \arrow[dr, dotted, "\exists!"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& \operatorname{Spec}(A \otimes_C B) \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
& \operatorname{Spec} A \arrow[r] & \operatorname{Spec} C
\end{tikzcd}$$

Finally, for an arbitrary scheme Z, and any diagram of solid arrows as follows, the dotted arrows exist and are unique because of Proposition 5.3 and because of the universal property of $\operatorname{Spec}(A \otimes_C B)$ in (AffSch):

$$\begin{tikzcd}
Z \arrow[dr] \arrow[drrr, bend left] \arrow[ddrr, bend right] & & & \\
& \operatorname{Spec} \mathcal{O}_Z(Z) \arrow[dr, dotted, "\exists!"] \arrow[drrr, dotted, "\exists!"] \arrow[ddrr, dotted, "\exists!"] & & \\
& & \operatorname{Spec}(A \otimes_C B) \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
& & \operatorname{Spec} A \arrow[r] & \operatorname{Spec} C
\end{tikzcd}$$

□

Example 8.10. (1) If $X = Y = \mathbb{A}^1_k$ and $S = \operatorname{Spec} k$ with the maps $X, Y \to S$ induced by the standard inclusion $k \hookrightarrow k[x]$, then $X \times_S Y = \operatorname{Spec}(k[x] \otimes_k k[y]) \cong \operatorname{Spec} k[x,y] = \mathbb{A}^2_k$. However, the topology on $\mathbb{A}^2_k$ is not the product topology (try to see why)!

(2) If $X = Y = \operatorname{Spec} \mathbb{C}$ and $S = \operatorname{Spec} \mathbb{R}$ with the maps induced by the inclusion $\mathbb{R} \to \mathbb{C}$. Then, $X \times_S Y \cong \operatorname{Spec}(\mathbb{R}[x]/(x^2+1) \otimes_{\mathbb{R}} \mathbb{C}) \cong \operatorname{Spec} \mathbb{C}[x]/(x+i)(x-i) \cong \operatorname{Spec}(\mathbb{C} \times \mathbb{C})$. In particular, note that the underlying topological spaces of X,Y, and S have only one point, but $X \times_S Y$ has two.

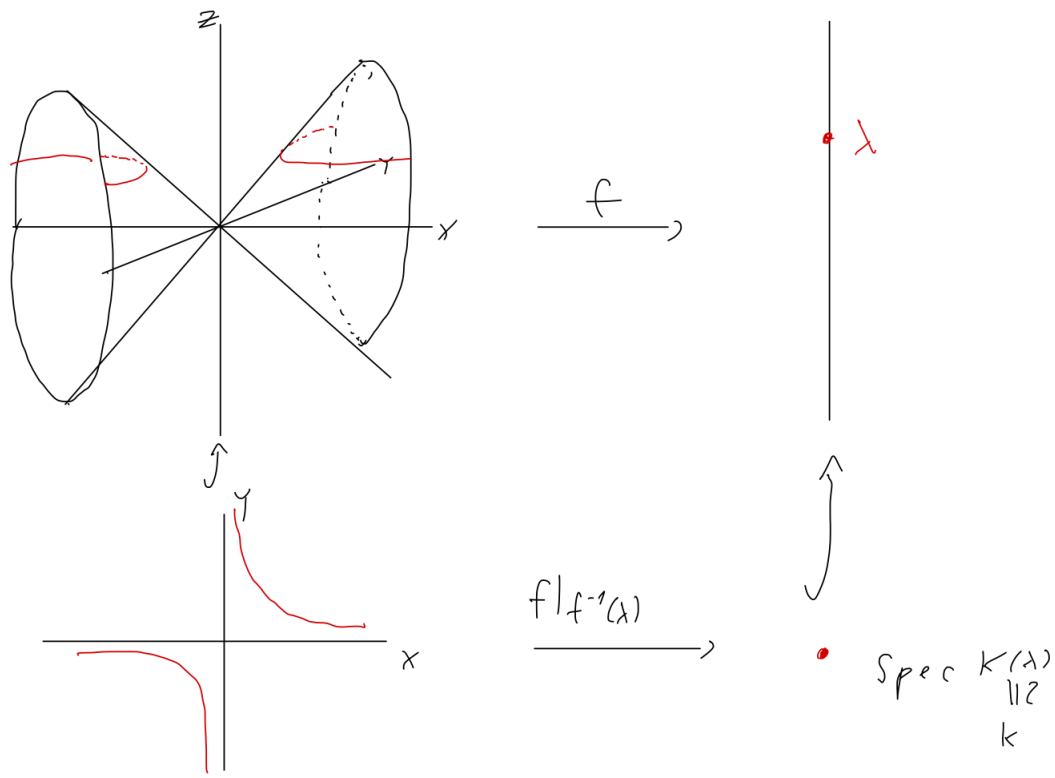
(3) If k is a field, $X = \operatorname{Spec} k[x,y,z]/(z^2 - xy)$, $Y = \operatorname{Spec} k[z]$ and $f : X \to Y$ is induced by $z \mapsto z$, then the fiber of f over the closed point $\lambda := (z - \lambda) \in Y$ with $\lambda \in k$ is

$$\begin{aligned} f^{-1}(\lambda) &= X \times_Y \operatorname{Spec} k[z]/(z-\lambda) \cong \operatorname{Spec} k[x,y,z]/(z^2 - xy) \otimes_{k[z]} k[z]/(z-\lambda) \\ &\cong \operatorname{Spec} k[x,y]/(\lambda^2 - xy). \end{aligned}$$

The generic fiber of f is

$$X \times_Y k(z) \cong \operatorname{Spec} k(z)[x,y]/(z^2 - xy)$$

Geometrically, X is a cone in $\mathbb{A}^3_k$, the map $X \to Y$ is the projection onto the z-coordinate, and its fiber over λ is the intersection of X with the plane $V(z-\lambda) \subseteq \mathbb{A}^3_k$. Note that, for $\lambda \neq 0$, we have $f^{-1}(\lambda) \cong \mathbb{A}^1_k - \{0\}$, because $y = \frac{\lambda^2}{x}$. Similarly, the generic fiber is $\mathbb{A}^1_{k(z)} - \{0\}$. However, the closed fiber over 0 is $f^{-1}(0) \cong \operatorname{Spec} k[x,y]/(xy)$, which is not irreducible. Here is a picture of the induced map on real-valued points if $k = \mathbb{R}$:



(4) In Example (3), assume $k = \mathbb{R}$ and consider the closed point $(z^2+1) \in Y$. The fiber over (z^2+1) is

$$f^{-1}((z^2+1)) = \operatorname{Spec} \mathbb{R}[x,y,z]/(z^2-xy) \otimes_{\mathbb{R}[z]} \mathbb{R}[z]/(z^2+1) \cong \operatorname{Spec} \mathbb{R}[x,y,z]/(-1-xy, z^2+1) \cong \mathbb{A}^1_{\mathbb{C}} - \{0\}.$$

Now that we convinced ourselves of the importance of fiber products, we are ready to prove Theorem 8.7.

Proof of Theorem 8.7.

Claim 1: If $X \times_S Y$ exists and $U \subseteq X$ is an open subscheme, then the open subscheme $p_X^{-1}(U) \subseteq X \times_S Y$ is a fiber product of U and Y over S.

Consider the following commutative diagram of solid arrows, where Z is any scheme:

$$\begin{tikzcd}
Z \arrow[dr, dotted, "(2)"'] \arrow[drr, dotted, "(1)"] \arrow[ddr, bend right] \arrow[drrr, bend left] & & & \\
& p_X^{-1}(U) \arrow[r] \arrow[d] & X \times_S Y \arrow[r] \arrow[d] & Y \arrow[d] \\
& U \arrow[r] & X \arrow[r] & S
\end{tikzcd}$$

The dotted arrow (1) exists and is unique, because $X \times_S Y$ is a fiber product. By definition of the preimage $p_X^{-1}(U)$, the dotted arrow (2) exists and is unique on the topological level, so it exists and is unique as a morphism of schemes by Exercise 23.

Claim 2: If $X = \cup_{i \in I} U_i$ is an open cover such that $U_i \times_S Y$ exists for all $i \in I$, then $X \times_S Y$ exists.

By Claim 1, the schemes $p_{U_i}^{-1}(U_i \cap U_j)$ and $p_{U_j}^{-1}(U_i \cap U_j)$ are isomorphic via a unique isomorphism compatible with the two projections. In particular, these isomorphisms satisfy the cocycle conditions of Exercise 20. Hence, there is a scheme T with morphisms $p_X : T \to X$ and $p_Y : T \to Y$ such that $p_X^{-1}(U_i) \cong U_i \times_S Y$, compatibly with the two projections. We claim that T, together with p_X and p_Y, is the fiber product of X and Y over S: If Z is a scheme and $f_X : Z \to X$ and $f_Y : Z \to Y$ are two morphism such that $\pi_X \circ f_X = \pi_Y \circ f_Y$, then we can cover Z by the $V_i := f_X^{-1}(U_i)$. Since $U_i \times_S Y$ is a fiber product, we get unique morphisms $f_i : V_i \to U_i \times_S Y \to T$ such that $p_X \circ f_i = f_X|_{V_i}$ and $p_Y \circ f_i = f_Y|V_i$. By Claim 1, f_i and f_j agree on $V_i \cap V_j$, hence by Exercise 14, we obtain a unique morphism $f : Z \to T$ such that $f_X = p_X \circ f$ and $f_Y = p_Y \circ f$, as desired.

Now, if S is affine, consider open affine covers $X = \cup U_i$ and $Y = \cup V_i$. Then, the $U_i \times_S V_i$ exist by Proposition 8.9, hence the $U_i \times_S Y$ exist by Claim 2, so $X \times_S Y$ exists, again by Claim 2.

If S is not affine, let $S = \cup W_i$ be an open affine cover and let $U_i = \pi_X^{-1}(W_i)$ and $V_i = \pi_Y^{-1}(W_i)$. By the previous paragraph, the fiber products $U_i \times_{W_i} V_i$ exist. Claim 1, applied to the case $X = S$ shows that $V_i = W_i \times_S Y$, so by Corollary 8.5, we have $U_u \times_{W_i} V_i = U_i \times_S Y$, hence $U_i \times_S Y$ exists. Applying Claim 2 one final time, we see that $X \times_S Y$ exists. □

From the proof of Theorem 8.7, we get the following statement about open coverings of fiber products:

Corollary 8.11. *Let $\pi_X : X \to S$ and $\pi_Y : Y \to S$ be S-schemes. Let $S = \cup_{i \in I} W_i$ be an open cover, let $U_i = \pi_X^{-1}(W_i)$ and $V_i = \pi_Y^{-1}(W_i)$, and let $U_i = \cup_{j \in J_{U_i}} U_{ij}$ and $V_i = \cup_{j \in J_{V_i}} V_{ij}$ be open covers. Then,*

$$X \times_S Y = \bigcup_{i \in I} \left(\bigcup_{j \in J_{U_i}, k \in J_{V_i}} U_{ij} \times_{W_i} V_{ik} \right)$$

is an open cover of the fiber product.

Finally, as a reality check, we note that our definition of fiber for a morphism of schemes is close to the fiber of the map of underlying topological spaces.

Proposition 8.12. *Let $f : X \to Y$ be a morphism of schemes. Then, for every $y \in Y$, the map $g : f^{-1}(y) \to X$ is a homeomorphism onto the set-theoretic fiber $f^{-1}_{set}(y)$ (with the topology induced by X).*

Proof. Replacing Y by an open affine neighborhood of y, we may assume that $Y = \operatorname{Spec} A$ is affine. Moreover, if $X = \cup U_i$ is an open cover and $g|_{g^{-1}(U_i)} : g^{-1}(U_i) \to U_i$ is a homeomorphism onto $f^{-1}_{set}(y) \cap U_i$, then g is a homeomorphism onto $f^{-1}_{set}(U_i)$.

So, it suffices to prove the case where $X = \operatorname{Spec} B$. Let $\mathfrak{p} \in \operatorname{Spec} A$ be the prime ideal corresponding to y.

<u>Claim 1:</u> $B \otimes_A k(y) \cong S^{-1}B/\mathfrak{p}S^{-1}B$, compatibly with the maps from B and $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$, where $S = \operatorname{im}(A \setminus \mathfrak{p} \to B)$.

To see this, we check that $S^{-1}B/\mathfrak{p}S^{-1}B$ together with the natural maps from B and $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ satisfies the universal property of the tensor product. So, consider the following commutative diagram in (Ring):

$$\begin{tikzcd}
C & & \\
& S^{-1}B/\mathfrak{p}S^{-1}B & B \arrow[l, "p_B"'] \arrow[ull, bend right, "f_B"'] \\
& A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} \arrow[u, "p_A"] \arrow[uul, bend left, "f_A"] & A \arrow[l, "\pi_A"] \arrow[u, "\pi_B"']
\end{tikzcd}$$

Since $f_B \circ \pi_B = f_A \circ \pi_A$ and π_A inverts all elements in $A \setminus \mathfrak{p}$ and sends all elements in $\mathfrak{p}$ to 0, we have $f_B(S) \in C^\times$ and $f_B(\mathfrak{p}S^{-1}B) = 0$. Thus, by the universal property of p_B, there exists a unique morphism $f : S^{-1}B/\mathfrak{p}S^{-1}B \to C$ such that $f \circ p_B = f_B$. Now, $f \circ p_A \circ \pi_A = f \circ p_B \circ \pi_B = f_B \circ \pi_B = f_A \circ \pi_A$ and π_A is an epimorphism (being a composition of a localization and a quotient map), so $f \circ p_A = f_A$, as desired.

<u>Claim 2:</u> $\operatorname{Spec} S^{-1}B/\mathfrak{p}S^{-1}B \to \operatorname{Spec} B$ is a homeomorphism onto $f^{-1}_{set}(y)$.

The map is the composition $\operatorname{Spec} S^{-1}B/\mathfrak{p}S^{-1}B \to \operatorname{Spec} S^{-1}B \to \operatorname{Spec} B$ and we know from commutative algebra that both morphisms are homeomorphisms onto their image. Moreover, the image can be described as $\{\mathfrak{q} \in \operatorname{Spec} B \mid S \cap \mathfrak{q} = \emptyset, \mathfrak{p}S^{-1}B \subseteq \mathfrak{q}S^{-1}B\}$. Since $S = \operatorname{im}(A \setminus \mathfrak{p} \to B)$, the condition $S \cap \mathfrak{q} = \emptyset$ is equivalent to $\pi_B^{-1}(\mathfrak{q}) \subseteq \mathfrak{p}$. On the other hand, the condition $\mathfrak{p}S^{-1}B \subseteq \mathfrak{q}S^{-1}B$ is equivalent to $\pi_B^{-1}(\mathfrak{q}) \supseteq \mathfrak{p}$. So, the image of $\operatorname{Spec} S^{-1}B/\mathfrak{p}S^{-1}B \to \operatorname{Spec} B$ is $\{\mathfrak{q} \in \operatorname{Spec} B \mid S \cap \mathfrak{q} = \emptyset, \mathfrak{p}S^{-1}B \subseteq \mathfrak{q}S^{-1}B\} = \{\mathfrak{q} \in \operatorname{Spec} B \mid \pi_B^{-1}(\mathfrak{q}) = \mathfrak{p}\} = f^{-1}_{set}(\mathfrak{p}) = f^{-1}_{set}(y)$. $\square$

9. Lectures 9 and 10: Properties of morphisms

In these lectures, we study properties of morphisms of schemes. First, we take a step back and look at what properties properties of morphisms can have. Then, we study topological properties and scheme theoretic properties of morphisms. The advantage of this approach is that it allows us to distinguish between properties of properties of morphisms which are purely formal (for example, all the good properties of separatedness follow immediately from the corresponding properties of closed immersions via Lemma 9.7) and those which are really non-trivial (such as the fact that the property of being quasi-finite is stable under base change).

9.1. **Properties of properties of morphisms.**

Notation 9.1. If $f : X \to Y$ and $g : Y' \to Y$ are morphisms of schemes, we let $X_{Y'} := X \times_Y Y'$ and $f_{Y'} : X_{Y'} \to Y'$. We call $f_{Y'}$ the *base change of f along g*.

Remark 9.2. In the following, whenever we say that $\mathcal{P}$ is a property of morphisms of schemes, we assume that $\mathcal{P}$ is satisfied by isomorphisms.

Definition 9.3. Let $\mathcal{P}$ be a property of morphisms of schemes. Then, we say that $\mathcal{P}$ satisfies

(COMP) if it is *stable under composition*, that is, if for all morphisms $f : X \to Y$ and $g : Y \to Z$ that satisfy $\mathcal{P}$, also $g \circ f$ satisfies $\mathcal{P}$.

(CANC) if it is *cancellable*, that is, if for all morphisms $f : X \to Y$ and $g : Y \to Z$ such that $g \circ f$ satisfies $\mathcal{P}$, also f satisfies $\mathcal{P}$.

(BC) if it is *stable under base change*, that is, if for all morphisms $f : X \to Y$ and $g : Y' \to Y$ such that f satisfies $\mathcal{P}$, also $f_{Y'}$ satisfies $\mathcal{P}$.

(LOCT) if it is (Zariski-)*local on the target*, that is, if for all morphisms $f : X \to Y$ and all open covers $Y = \cup_{i \in I} V_i$, the morphism f satisfies $\mathcal{P}$ if and only if f_{V_i} satisfies $\mathcal{P}$ for all $i \in I$.

(LOCS) if it is (Zariski-)*local on the source*, that is, if for all morphisms $f : X \to Y$ and all open covers $X = \cup_{i \in I} U_i$, the morphism f satisfies $\mathcal{P}$ if and only if $f|_{U_i}$ satisfies $\mathcal{P}$ for all $i \in I$.

Any property $\mathcal{P}$ can be made to satisfy (BC) by asking for it to hold "universally" in the following sense:

Definition 9.4. Let $\mathcal{P}$ be a property of morphisms of schemes. A morphism $f : X \to Y$ is called *universally* $\mathcal{P}$ if $f_{Y'}$ satisfies $\mathcal{P}$ for all morphisms of schemes $g : Y' \to Y$.

Lemma 9.5. *Let $f : X \to Y$ be a morphism of schemes over S. Then, the following square is Cartesian:*

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"] \arrow[dr, phantom, "\square"] & X \times_S Y \arrow[d, "f \times \mathrm{id}_Y"] \\ Y \arrow[r, "\Delta_{Y/S}"] & Y \times_S Y \end{tikzcd}$$

Proof. This is a special case of the magic square from Corollary 8.5 (4). In the following commutative diagram, the top left arrow is an isomorphism ($\Gamma_{f/Y}$ denotes the graph of f considered as a morphism of Y-schemes), the square is the magic square, and the composition of the two arrows in

the first row is Γ_f:

$$\begin{tikzcd}
X \arrow[r, "\Gamma_{f/Y}"] \arrow[dr, "f"'] & X \times_Y Y \arrow[r, "\mathrm{id}_X \times_S \mathrm{id}_Y"] \arrow[d, "p_Y"] \arrow[dr, phantom, "\square"] & X \times_S Y \arrow[d, "f \times \mathrm{id}_Y"] \\
& Y \arrow[r, "\Delta_{Y/S}"] & Y \times_S Y
\end{tikzcd}$$

□

Definition 9.6. Let $\mathcal{P}$ be a property of morphisms of schemes. We say that a morphism $f : X \to Y$ satisfies $\Delta_{\mathcal{P}}$ if $\Delta_{X/Y}$ satisfies $\mathcal{P}$.

Lemma 9.7. *The following hold:*

(1) If $\mathcal{P}$ satisfies (BC), then $\Delta_{\mathcal{P}}$ satisfies (BC).
(2) If $\mathcal{P}$ satisfies (BC) and (COMP), then $\Delta_{\mathcal{P}}$ satisfies (COMP).
(3) If $\mathcal{P}$ satisfies (LOCT), then $\Delta_{\mathcal{P}}$ satisfies (LOCT).
(4) If $\mathcal{P}$ satisfies (BC) and (COMP) and if $f : X \to Y, g : X \to Z$ are morphisms of schemes over S that satisfy $\mathcal{P}$ and $X \to S$ satisfies $\Delta_{\mathcal{P}}$, then $(f, g) : X \to Y \times_S Z$ satisfies $\mathcal{P}$.

Proof. This is Exercise 29. □

Lemma 9.8. *Let $\mathcal{P}$ be a property of morphisms of schemes which satisfies (COMP) and (BC). Let $f : X \to Y$ and $g : X' \to Y'$ be morphisms of schemes satisfying $\mathcal{P}$. Then, the product $f \times_S g : X \times_S X' \to Y \times_S Y'$ satisfies $\mathcal{P}$.*

Proof. Consider the following commutative diagram:

$$\begin{tikzcd}
& X \times_S X' \arrow[r, "\mathrm{id}_X \times_S g"] \arrow[dl, "p_{X'}"'] & X \times_S Y' \arrow[r, "f \times_S \mathrm{id}_{Y'}"] \arrow[dl, "p_{Y'}"] \arrow[dr, "p_X"'] & Y \times_S Y' \arrow[dr, "p_Y"] & \\
X' \arrow[rr, "g"] & & Y' & X \arrow[r, "f"] & Y
\end{tikzcd}$$

The composition of the two arrows in the first row is $f \times_S g$ and both squares are Cartesian by Corollary 8.5 (2). Since $\mathcal{P}$ satisfies (COMP) and (BC), this implies that $f \times_S g$ satisfies $\mathcal{P}$. □

Theorem 9.9 (Cancellation Theorem)**.** *Let $\mathcal{P}$ be a property of morphisms of schemes which satisfies (COMP) and (BC). Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes such that $g \circ f$ and $\Delta_{Y/Z}$ satisfy $\mathcal{P}$. Then, f satisfies $\mathcal{P}$.*

Proof. Consider the following commutative diagram:

$$\begin{tikzcd}
& X \arrow[r, "\Gamma_f"] \arrow[dl, "f"'] & X \times_Z Y \arrow[r, "p_Y"] \arrow[dl, "f_Y"] \arrow[dr, "p_X"'] & Y \arrow[dr, "g"] & \\
Y \arrow[rr, "\Delta_{Y/Z}"] & & Y \times_Z Y & X \arrow[r, "g \circ f"] & Z
\end{tikzcd}$$

The composition of the two arrows in the first row is f by definition. The left square is Cartesian by Lemma 9.5 and the right square is Cartesian by definition. Hence f satisfies $\mathcal{P}$. □

9.2. **Topological properties.**

Definition 9.10. A morphism $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ of schemes

(1) is *injective (resp. surjective, resp. bijective)* if f is.
(2) is *open (resp.closed)* if f is.
(3) is *quasi-compact* if $f^{-1}(V)$ is quasi-compact for every quasi-compact open $V \subseteq Y$.
(4) is *quasi-separated* if $\Delta_{X/Y} : X \to X \times_Y X$ is quasi-compact.
(5) has *finite fibers* if $f^{-1}(y)$ is finite (as a set) for all $y \in Y$.

Lemma 9.11. *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent:*

(1) f is quasi-compact.
(2) For every open affine $V \subseteq Y$ the preimage $f^{-1}(V)$ is quasi-compact.
(3) There exists an open affine cover $Y = \cup_{i\in I} V_i$ such that each $f^{-1}(V_i)$ is quasi-compact.

Proof. (1) $\Rightarrow$ (2) is clear. For (2) $\Rightarrow$ (1), let $V \subseteq Y$ be any quasi-compact open. Then, $V = \cup_{i=1}^n V_i$ with V_i open affine, so $f^{-1}(V) = \cup_{i=1}^n f^{-1}(V_i)$, hence $f^{-1}(V)$ is quasi-compact.

To show that (2) $\Leftrightarrow$ (3), we want to apply Lemma 6.15 and for this we have to show that "$f^{-1}(V)$ is quasi-compact" is an affine-local property of open affine subschemes $V \subseteq Y$.

So, let $V = \operatorname{Spec} A$ and $a_i \in A$ such that $(a_1, \dots, a_n) = A$. If each $f^{-1}(D(a_i))$ is quasi-compact, then so is $f^{-1}(V)$, being a finite union of quasi-compact opens. Conversely, assume $f^{-1}(V)$ is quasi-compact and let $f^{-1}(V) = \cup_{i=1}^m U_i$ be a finite cover by open affines. Then, $(f|_{U_i})^{-1}(D(a_i))$ is a principal open in U_i, hence affine, so $f^{-1}(D(a_i))$ is a finite union of affine schemes, hence quasi-compact. □

Remark 9.12. If $U, V \subseteq X$ are open subschemes and X is a scheme over S, then $\Delta_{X/S}^{-1}(U \times_S V) = U \cap V$ (check that $U \cap V$ satisfies the universal property of the fiber product). So, if X is quasi-separated (in the sense of Definition 6.1), then $X \to S$ is quasi-separated (in the sense of Definition 9.10) for every scheme S, since we can cover $X \times_S X$ by fiber products $U \times_W V \cong U \times_S V$, where U, V, W are affine (Corollary 8.11), which have quasi-compact preimage under $\Delta_{X/S}$. Conversely, if S is affine and $X \to S$ is quasi-separated, then X is quasi-separated.

Proposition 9.13. *The properties of Definition 9.10 satisfy the following properties:*

Name	*(COMP)*	*(BC)*	*(CANC)*	*(LOCS)*	*(LOCT)*
injective	✓	×	✓	×	✓
surjective	✓	✓	×	×	✓
bijective	✓	×	×	×	✓
open	✓	×	×	✓	✓
closed	✓	×	×	×	✓
quasi-compact	✓	✓	×	×	✓
quasi-separated	✓	✓	✓	×	✓
finite fibers	✓	×	✓	×	✓

Proof. We only prove the ✓entries and leave the task of finding counterexamples to prove the × entries to the reader. The properties (COMP), (BC), and (LOCT) of quasi-separatedness follow from the properties of quasi-compactness by Lemma 9.7. The property (CANC) follows from Exercise 32 and Theorem 9.9. The only remaining properties which are not straightforward are the following:

- quasi-compactness: (LOCT) and (BC)

- surjectivity: (BC)

(LOCT) for quasi-compactness: This follows from the alternative characterizations in Lemma 9.11.

(BC) for quasi-compactness: Let $f : X \to Y$ be a quasi-compact morphism and $g : Y' \to Y$ any morphism. Since quasi-compactness is (LOCT), we may assume that Y is affine. In this case, X is quasi-compact by Lemma 9.11. Now, let $V' \subseteq Y'$ be open and affine. Its preimage in $X \times_Y Y'$ is isomorphic to $X \times_Y V'$ by Claim 1 in the proof of Theorem 8.7. By Corollary 8.11, a finite open affine cover $X = \cup_{i=1}^n U_i$ yields a finite open affine cover $X \times_Y V' = \bigcup_{i=1}^n U_i \times_Y V'$. The union is finite and each $U_i \times_Y V'$ is affine by Proposition 8.9, so $X \times_Y V'$ is quasi-compact.

(BC) for surjectivity is part of Exercise 30. □

Example 9.14. (1) Open (resp. closed) immersions are open (resp. closed). By Exercise 25, they are even universally open (resp. closed).

(2) Closed immersions are quasi-compact:
If $f : X \to Y$ is a closed immersion and $V \subseteq Y$ is open affine, then $f^{-1}(V)$ is affine (Corollary 7.15), hence quasi-compact (Lemma 6.4), so f is quasi-compact (Lemma 9.11).

(3) Open immersions are not necessarily quasi-compact:
The morphism $f : \mathbb{A}_k^\infty \setminus \{0\} \to \mathbb{A}_k^\infty$ is an open immersion, but the preimage of $\mathbb{A}_k^\infty$ is not quasi-compact (compare this with Example 6.5).

(4) Every morphism $f : X \to Y$ where X is Noetherian is quasi-compact and quasi-separated by Exercise 18.

9.3. **Scheme theoretic properties.**

Definition 9.15. A morphism $f : X \to Y$ of schemes is called

(1) *(locally closed) immersion* if $f = g \circ h$, where g is an open immersion and h is a closed immersion.
(2) *locally of finite type* if for all affine open $\operatorname{Spec} A = V \subseteq Y$, the preimage $f^{-1}(V)$ can be covered by open affines $U_i = \operatorname{Spec} B_i$ such that each B_i is a finitely generated A-algebra.
(3) *locally of finite presentation* if for all affine open $\operatorname{Spec} A = V \subseteq Y$, the preimage $f^{-1}(V)$ can be covered by open affines $U_i = \operatorname{Spec} B_i$ such that each B_i is an A-algebra of finite presentation.
(4) *of finite type* if it is locally of finite type and quasi-compact.
(5) *of finite presentation* if it is locally of finite presentation and quasi-compact.
(6) *quasi-finite* if it is of finite type with finite fibers.
(7) *affine* if $f^{-1}(V)$ is affine for all $V \subseteq Y$ open affine.
(8) *finite* if $f^{-1}(V) = \operatorname{Spec} B$ is affine for all $\operatorname{Spec} A = V \subseteq Y$ open affine and B is finitely generated as an A-module.
(9) *separated* if $\Delta_{X/Y} : X \to X \times_Y X$ is a closed immersion.
(10) *proper* if it is separated, of finite type, and universally closed.

Remark 9.16. If f is a morphism such that $f = h \circ g$ with h a closed immersion with image $Z \subseteq Y$ and g an open immersion with image $U \subseteq Z$, then f is an immersion in the sense of Definition 9.15. Indeed, let $V \subseteq Y$ be open with $V \cap Z = U$. Then, $V \cap Z \to V$ is a closed immersion (Exercise 25), $V \to Y$ is an open immersion, and hence $U = V \cap Z \to Y$ is an immersion.

The converse is not true in general (see Tag01QW on the StacksProject), but it holds whenever there is a good notion of scheme-theoretic image for f.

In the following lemma, we say that a property $\mathcal{P}$ of morphisms of schemes is affine-local if for every morphism of schemes $f : X \to Y$ the property "f_V satisfies $\mathcal{P}$" is an affine-local property of open affine subschemes $V \subseteq Y$.

Proposition 9.17. *The properties of Definition 9.15 satisfy the following properties:*

Name	*(COMP)*	*(BC)*	*(CANC)*	*(LOCS)*	*(LOCT)*
open immersion	✓	✓	×	×	✓
closed immersion	✓	✓	×	×	✓
immersion	✓	✓	✓	×	✓
locally of finite type	✓	✓	✓	✓	✓
locally of finite presentation	✓	✓	×	✓	✓
finite type	✓	✓	×	×	✓
finite presentation	✓	✓	×	×	✓
quasi-finite	✓	✓	×	×	✓
affine	✓	✓	×	×	✓
finite	✓	✓	×	×	✓
separated	✓	✓	✓	×	✓
proper	✓	✓	×	×	✓

Moreover, the properties "locally of finite type", "locally of finite presentation", "affine", and "finite" are affine-local.

Proof. Again, we only prove the ✓entries and leave the task of finding counterexamples for the × entries to the reader.

For the first three rows, (COMP) follows from Remark 9.16. (BC) and (LOCT) were treated in Exercise 25. Proving the properties of "locally of finite type" is the content of Exercise 31 and the properties of "locally of finite presentation" can be proved in a similar way. The properties of "finite type" (resp. "finite presentation") follow from the previous rows and the corresponding properties of "quasi-compact".

For "quasi-finite", (COMP) and (LOCT) follow from the properties of "finite type" and "quasi-compact". Let us show that "quasi-finite" satisfies (BC) [17]:

Assume $f : X \to Y$ is quasi-finite and let $g : Y' \to Y$ be any morphism. We have to show that $f_{Y'}$ is quasi-finite. Finite type is clear by (BC) for "finite type", so we have to check that $f_{Y'}$ has finite fibers. Hence, we may assume that $Y' = \operatorname{Spec} K$ for some field K. Then, we have the following diagram with Cartesian squares, where $y \in Y$ is the image of g:

$$\begin{tikzcd} X_K \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & X_y \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & X \arrow[d] \\ \operatorname{Spec} K \arrow[r] & \operatorname{Spec} k(y) \arrow[r] & Y \end{tikzcd}$$

The space underlying X_y is finite, so $X_y = \cup_{i=1}^n \operatorname{Spec} A_i$ with A_i a finitely generated $k(y)$-algebra with finite spectrum. From commutative algebra (see Proposition 24.2 in Appendix A), we know that such an A_i satisfies $\dim_{k(y)} A_i < \infty$. Hence, $\dim_K A_i \otimes_{k(y)} K < \infty$, so $\operatorname{Spec}(A_i \otimes_{k(y)} K)$

[17] The fact that "quasi-finite" satisfies (BC) is the reason for adding the finite type hypothesis to "finite fibers", because the latter does not satisfy (BC)

is finite. Since these open subschemes cover X_K, the topological space underlying X_K is finite, as claimed.

For the properties "affine" and "finite", (COMP) is clear and we omit the proof of (LOCT) and affine-local, which is similar to the corresponding proof for "locally of finite presentation". [18] Then, (BC) follows from the description of fiber products of affine schemes as spectra of tensor products (Proposition 8.9).

For "separated", (COMP), (BC), and (LOCT) follow from the properties of closed immersions and Lemma 9.7. (CANC) is part of Exercise 32.

Finally, the properties of "proper" follow from the corresponding properties of "separated", "finite presentation" and "closed". □

There are some clear implications between various properties, such as the fact that closed immersions are immersions. Others are not as clear, such as the following, which requires some facts on integral ring extensions:

Proposition 9.18. *Finite morphisms are quasi-finite and proper.*

Proof. Let $f : X \to Y$ be a finite morphism. Then, f is of finite type and affine, hence separated (by Exercise 32). It remains to show that f is closed with finite fibers. Indeed, being finite satisfies (BC), so this will show that f is universally closed.

Since "closed", "finite fibers", and "finite" satisfy (LOCT) and (COMP), and closed immersions satisfy all three properties, we can assume that Y (hence X) is affine, say $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$ and then replace A by the image of $A \to B$ to assume that $A \to B$ is injective. Now, we have a finite ring extension $A \subseteq B$ and we want to show that $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is closed with finite fibers.

To see that f has finite fibers, we use that "finite" satisfies (BC), so for every $\mathfrak{p} \in \operatorname{Spec} A$ we have $\dim_{k(\mathfrak{p})} B \otimes_A k(\mathfrak{p}) < \infty$. In particular, $\operatorname{Spec} B \otimes_A k(\mathfrak{p})$, which is nothing but $f^{-1}(\mathfrak{p})$ by Proposition 8.12, is finite.

To see that f is closed, note that if $V(I) \subseteq \operatorname{Spec} B$ is closed, then $V(I) \to \operatorname{Spec} A$ is finite, so it suffices to show that $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective if $A \subseteq B$ is a finite ring extension. This is the "lying over" theorem from commutative algebra. □

Remark 9.19. In the next term, as an application of Zariski's main theorem, we will prove that proper quasi-finite morphisms are finite.

Here is a summary of some relations between some properties of morphisms. All implications are either not hard to check, were already proven in the lecture, or follow from the exercises on Sheet 6.

[18] See Vakil's lecture notes, August 29, 2022 version, Proposition 8.3.4. for a proof of Claim (7).

$$
\begin{tikzcd}
 & & \text{quasi-finite} \arrow[r, Rightarrow] \arrow[dr, Rightarrow, bend left] & \text{finite fibers} \\
\text{closed immersion} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{finite} \arrow[r, Rightarrow] \arrow[d, Rightarrow] \arrow[ur, Rightarrow] & \text{affine} \arrow[r, Rightarrow] \arrow[d, Rightarrow, bend left] \arrow[dl, Rightarrow, bend right] & \text{quasi-compact} \\
\text{immersion} \arrow[dr, Rightarrow] \arrow[drr, Rightarrow, bend right] & \text{proper} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{finite type} \arrow[r, Rightarrow] \arrow[ur, Rightarrow] & \text{loc. finite type} \\
\text{open immersion} \arrow[u, Rightarrow] \arrow[drr, Rightarrow, bend right] & \text{separated} \arrow[d, Rightarrow] & \text{finite presentation} \arrow[r, Rightarrow] \arrow[u, Rightarrow] & \text{loc. finite presentation} \arrow[u, Rightarrow] \\
 & \text{quasi-separated} & &
\end{tikzcd}
$$

10. Lecture 11: Valuative criteria

The purpose of this lecture is to give geometric meaning to the notions of separatedness and properness. We start by introducing the topological concept of specialization and generalization of points in topological spaces and define what it means for specializations and generalizations to lift along a continuous map. Then, we introduce valuation rings and observe that specialization of points on a scheme is "witnessed" by spectra of valuation rings. Then, the *valuative criterion* formalizes the concept of checking whether specializations witnessed by valuation rings lift (uniquely) along a morphism of schemes. We will prove that for a morphism of schemes $f : X \to Y$, satisfying the existence part of the valuative criterion is equivalent to the property that specializations lift along any base change of f. As a corollary, we obtain valuative criteria for universal closedness, separatedness, and properness, thereby giving further geometric meaning to these notions.

Definition 10.1. Let X be a topological space.

(1) Let $x, x' \in X$. If $x' \in \overline{\{x\}}$, we say that x' is a *specialization* of x (or x *specializes* to x') and that x is a *generalization* of x' and we write $x \rightsquigarrow x'$.
(2) A subset $V \subseteq X$ is called *closed under specialization* (resp. *closed under generalization*) if it contains all specializations (resp. generalizations) of all its points.

Remark 10.2. Closed subsets are closed under specialization and open subsets are closed under generalization. The converse is not true in general (consider for example $\mathbb{Q} \subseteq \mathbb{R}$ and its complement).

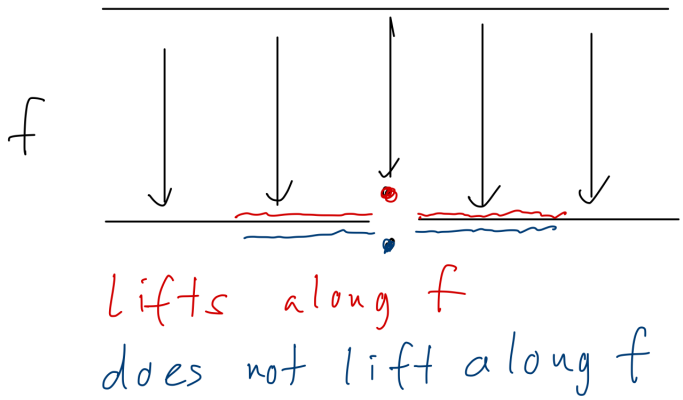
Definition 10.3. Let $f : X \to Y$ be a continuous map. We say that

(1) *specializations lift along* f if for every $x \in X$ and every specialization $f(x) \rightsquigarrow y'$ in Y, there exists a point $x' \in X$ with $x \rightsquigarrow x'$ and $f(x') = y'$.
(2) *generalizations lift along* f if for every $x' \in X$ and every generalization $y \rightsquigarrow f(x')$ in Y, there exists a point $x \in X$ with $x \rightsquigarrow x'$ and $f(x) = y$.

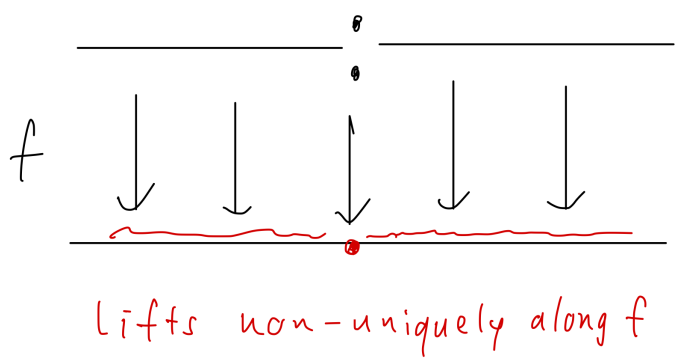
Remark 10.4. Both concepts are stable under composition and local on the target.

Example 10.5. Let k be an algebraically closed field. Let X be the affine line with two origins over k (see Example 5.10)

(1) Consider the inclusion $f : \mathbb{A}^1_k \to X$ of one of the affine charts. Let $x \in \mathbb{A}^1_k$ be the generic point. Since $\overline{f(x)} = X$, the image $f(x)$ is the generic point of X. We have $f(x) \rightsquigarrow y'$, where y' is any of the two origins in X. Only one of those specializations lifts along f. The other does not. In pictures:



(2) Consider the morphism $f : X \to \mathbb{A}^1_k$ defined by glueing the identities on both open affines we used to construct X. Let $x \in X$ be the generic point and note that $f(x) \rightsquigarrow y'$, where y' is the origin in $\mathbb{A}^1_k$. This specialization lifts along f, but not uniquely, since both origins of X are preimages of y'.



We will study liftability of generalizations later, when studying (formal) smoothness. For now, we want to relate the liftability of specializations to the closedness of a morphism of schemes. We will need the following fact:

Lemma 10.6. *Let X be a scheme. Then, X is quasi-compact if and only if there exists a ring A and a surjection* $\operatorname{Spec} A \to X$.

Proof. One implication is clear, since images of quasi-compact spaces under continuous maps are quasi-compact. Conversely, assume that X is quasi-compact. We can write $X = \bigcup_{i=1}^n \operatorname{Spec} A_i$. Set $A = \prod_{i=1}^n A_i$. Then, $\operatorname{Spec} A = \coprod_{i=1}^n \operatorname{Spec} A_i$ and we get the surjection

$$\operatorname{Spec} A = \coprod_{i=1}^n \operatorname{Spec} A_i \to \bigcup_{i=1}^n \operatorname{Spec} A_i = X.$$

□

Lemma 10.7. *Let $f : X \to Y$ be a quasi-compact morphism of schemes. Then, f is closed if and only if specializations lift along f.*

Proof. First, assume that f is closed, let $x \in X$, and let $f(x) \rightsquigarrow y'$ be a specialization in Y. Then, $y' \in \overline{\{f(x)\}} = \overline{f(\overline{\{x\}})} = f(\overline{\{x\}})$, so there exists $x' \in \overline{\{x\}}$ with $f(x') = y'$, i.e., specializations lift along f.

Conversely, assume that specializations lift along f and let $Z \subseteq X$ be a closed subset. Equip Z with the reduced induced scheme structure (Remark 7.16). Since closed immersions are quasi-compact and closed, we can assume without loss of generality that $Z = X$. Then, $f(X)$ is stable under specialization and we have to show that it is closed. Since being closed and quasi-compact are (LOCT) and liftability of specializations is also local on the target, we may assume that Y is affine. Then, quasi-compactness of f implies that X is quasi-compact, so there is a surjection $\operatorname{Spec} B \to X$. As the image $f(X)$ does not change if we precompose with this surjection, we may assume without loss of generality that X is affine. We have reduced the problem to the following algebraic claim:

<u>Claim:</u> Let $\varphi : A \to B$ be a morphism of rings. If the image T of $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is closed under specialization, then T is closed.

By replacing $\operatorname{Spec} A$ by the closure of the image of T, we may assume without loss of generality that φ is injective. Recall that a prime ideal $\mathfrak{p} \in \operatorname{Spec} A$ is called *minimal* if it is minimal with respect to inclusion. In particular, every prime ideal $\mathfrak{q} \in \operatorname{Spec} A$ is a specialization of some minimal prime, so we only have to show that T contains all minimal primes of A. So, let $\mathfrak{p} \in \operatorname{Spec} A$ be a minimal prime and consider the Cartesian square

$$\begin{tikzcd} \operatorname{Spec} B_{\mathfrak{p}} \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & \operatorname{Spec} B \arrow[d] \\ \operatorname{Spec} A_{\mathfrak{p}} \arrow[r] & \operatorname{Spec} A \end{tikzcd},$$

where $B_{\mathfrak{p}}$ is the localization of B considered as an A-module. Since localization is exact, $B_{\mathfrak{p}}$ is non-zero, so $\operatorname{Spec} B_{\mathfrak{p}}$ is non-empty. But the underlying topological space of $\operatorname{Spec} A_{\mathfrak{p}}$ is just a single point, since $\mathfrak{p}$ is minimal, so $\mathfrak{p} \in T$. □

If X is a scheme, then specialization of points is "witnessed" by spectra of valuation rings. First, we recall the definition of valuation rings.

Definition 10.8. Let A be a local integral domain with field of fractions K and maximal ideal $\mathfrak{m}_A$.

(1) If $B \subseteq K$ is a local integral domain with maximal ideal $\mathfrak{m}_B$, we say that B *dominates* A if $A \subseteq B$ and $\mathfrak{m}_B \cap A = \mathfrak{m}_A$
(2) The ring A is called *valuation ring* (in K) if it is maximal with respect to domination of local subrings of K.
(3) The ring A is called *discrete valuation ring* if it is a Noetherian valuation ring and not a field.

See Exercise 35 for an explanation of the name (discrete) valuation ring.

Lemma 10.9. *Let $A \subseteq K$ be a local subring of a field. Then A is dominated by a valuation ring with fraction field K.*

Proof. We want to apply Zorn's lemma to the set $M = \{A_i \mid A_i \subseteq K, A_i \text{ local ring}, A_i \text{ dominates } A\}$, ordered by the relation of domination.

So, let $\{A_i\}_{i\in I} \subseteq M$ be a totally ordered subset. Let $B = \cup_{i\in I} A_i$ and $\mathfrak{m}_B = \cup_{i\in I}\mathfrak{m}_{A_i}$. Then, the following properties hold:

- B is a ring.
- $\mathfrak{m}_B$ is an ideal.
- $b \in B$ is a unit $\iff \exists c \in B$ with $bc = 1$ in $B \iff \exists i \in I$ such that $b, c \in A_i$ and $bc = 1$ in $A_i \iff \exists i \in I$ such that $b \in A_i \setminus \mathfrak{m}_{A_i} \iff b \in B \setminus \mathfrak{m}_B$.

Hence, B is a local ring that dominates every A_i, or in other words $B \in M$. By Zorn's lemma, this implies that M admits a maximal element C.

It remains to show that the field of fractions K' of C is K. Let $x \in K$. If x is transcendental over K', then $C[x] \subseteq K$ and $C[x]_{(x,\mathfrak{m}_C)}$ strictly contains C and dominates it, contradicting the fact that C is maximal in M. If x is algebraic over K', then, after clearing denominators in a minimal polynomial for x, there exist $c_i \in C$ such that $\sum_{i=0}^{n} c_i x^i = 0$. Multiplying this equation by c_n^{n-1}, we see that the element $c_n x$ is integral over C. Let $C \subseteq C' \subseteq K$ be the integral ring extension of C obtained by adjoining $c_n x$. By the "Lying Over" theorem, there exists a prime $\mathfrak{p}$ of C' lying over $\mathfrak{m}_C$. But then $C'_{\mathfrak{p}} \subseteq K$ dominates C, so $C = C'_{\mathfrak{p}}$, hence $x \in K'$. Therefore, $K = K'$, so C is a valuation ring with field of fractions K and C dominates A by construction. [19] $\square$

Note that for every valuation ring A, the spectrum $\operatorname{Spec} A$ contains a unique generic point η_A (since A is an integral domain) and a unique closed point $\mathfrak{m}_A$ (since A is local). Here is a rough picture of what $\operatorname{Spec} A$ looks like:

If you want, in the following you can think of the spectrum of a valuation ring as the algebro-geometric analogue of a complex disk. As promised, we now show that specializations of points on schemes are witnessed by such spectra.

Lemma 10.10. *Let X be a scheme and let $x, x' \in X$ be points with $x \rightsquigarrow x'$. Then, there exists a valuation ring A and a morphism $f : \operatorname{Spec} A \to X$ such that $f(\eta_A) = x$ and $f(\mathfrak{m}_A) = x'$. More*

[19] This proof also shows that a valuation ring A is integrally closed in its field of fractions.

precisely, given a field extension $g : k(x) \hookrightarrow K$, we may choose a valuation ring A with field of fractions K such that f induces g.

Proof. Since $x' \in \overline{\{x\}}$, we have a morphism $\varphi : \mathcal{O}_{X,x'} \to k(x) \to K$ (recall that $k(x) := \mathcal{O}_{X,x}/\mathfrak{m}_x$ and that there is a natural morphism $\mathcal{O}_{X,x'} \to \mathcal{O}_{X,x}$, since both of these rings are localizations of a ring whose spectrum is an open affine neighborhood of $x' \in X$). By Lemma 10.9, there exists a valuation ring $A \subseteq K$ with fraction field K that dominates the image of φ. The resulting map $\mathcal{O}_{X,x'} \to A$ induces a morphism $f : \operatorname{Spec} A \to \operatorname{Spec} \mathcal{O}_{X,x'} \to X$. This f has the desired properties by construction. □

Definition 10.11. Let $f : X \to Y$ be a morphism of schemes. We say that f *satisfies the existence part of the valuative criterion* if for every diagram of solid arrows of the following form

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} A \arrow[r] \arrow[ur, dotted] & Y \end{tikzcd}$$

where A is a valuation ring and K is its field of fractions, there is a dotted arrow making the diagram commute. We say that f *satisfies the uniqueness part of the valuative criterion* if there is at most one dotted arrow making the diagram commutes.

Theorem 10.12 (Valuative criterion)**.** *Let $f : X \to Y$ be a morphism of schemes. The following are equivalent:*

(1) Specializations lift along any base change of f.
(2) f satisfies the existence part of the valuative criterion.

Proof. (1) $\Rightarrow$ (2): Without loss of generality, we may assume that $Y = \operatorname{Spec} A$. Then, we are looking for a dotted arrow in the diagram

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r, "g"] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} A \arrow[r, "="] \arrow[ur, dotted] & \operatorname{Spec} A \end{tikzcd} \tag{4}$$

Let $x \in X$ be the image of g. Since specializations lift along f, there exists a specialization $x \rightsquigarrow x'$ in X such that $f(x')$ is the closed point in $\operatorname{Spec} A$. We obtain a commutative diagram of solid arrows

$$\begin{tikzcd} K & \mathcal{O}_{X,x'} \arrow[l, "g^\sharp_{x'}"'] \arrow[dl, dotted] \\ A \arrow[u] & A \arrow[l, "="] \arrow[u, "f^\sharp_{x'}"'] \end{tikzcd} \tag{5}$$

Since $f^\sharp_{x'}$ is local, $g^\sharp_{x'}(\mathcal{O}_{X,x'})$ dominates A. [20] But A is a valuation ring, hence maximal with respect to domination, so $g^\sharp_{x'}(\mathcal{O}_{X,x'}) = A$. In other words, there is a dotted arrow making the diagram (5)

[20] Recall that images of local rings under ring homomorphisms are again local.

commute. The induced morphism of spectra composed with the natural map $\operatorname{Spec} \mathcal{O}_{X,x'} \to X$ makes the original diagram (4) commute.

(2) $\Rightarrow$ (1): If $Y' \to Y$ is any morphism, then $f_{Y'}$ satisfies the existence part of the valuative criterion by the universal property of the fiber product:

$$\begin{tikzcd}
\operatorname{Spec} K \arrow[r] \arrow[d] & X_{Y'} \arrow[r] \arrow[d] & X \arrow[d] \\
\operatorname{Spec} A \arrow[r] \arrow[ur, dotted] \arrow[urr, dotted] & Y' \arrow[r] & Y
\end{tikzcd}$$

So, it suffices to show that specializations lift along f. Let $x \in X$ and let $f(x) \rightsquigarrow y'$ be a specialization of $f(x)$. Let $K = k(x)$ and consider the field extension $g : k(f(x)) \to k(x)$ induced by f. Then, by Lemma 10.10, we get a commutative square with solid arrows

$$\begin{tikzcd}
\operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d] \\
\operatorname{Spec} A \arrow[r] \arrow[ur, dotted] & Y
\end{tikzcd}$$

where A is a valuation ring such that $\mathfrak{m}_A$ maps to y' under $\operatorname{Spec} A \to Y$. Since the dotted arrow exists, the image of $\mathfrak{m}_A$ under $\operatorname{Spec} A \to X$ is a preimage x' of y' and $x \rightsquigarrow x'$ by construction. □

Corollary 10.13 (Valuative criterion for universal closedness)**.** *Let $f : X \to Y$ be a quasi-compact morphism. Then, the following are equivalent:*

(1) f is universally closed.
(2) f satisfies the existence part of the valuative criterion.

Proof. Combine Theorem 10.12 and Lemma 10.7 and the fact that being quasi-compact is stable under base change. □

Corollary 10.14 (Valuative criterion for separatedness)**.** *Let $f : X \to Y$ be a quasi-separated morphism. Then, the following are equivalent:*

(1) f is separated.
(2) f satisfies the uniqueness part of the valuative criterion.

Proof. (1) $\Rightarrow$ (2): Assume there are two dotted arrows g and h making the following diagram of solid arrows commute:

$$\begin{tikzcd}
\operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d] \\
\operatorname{Spec} A \arrow[r] \arrow[ur, dotted] & Y
\end{tikzcd}$$

Since g and h coincide on the open dense subscheme $\operatorname{Spec} K \subseteq \operatorname{Spec} A$, the valuation ring A is integral, and X is separated, Exercise 33 (iii) shows that $g = h$.

(2) $\Rightarrow$ (1): By Exercise 32, we have to show that $\Delta_{X/Y}$ is closed. Since f is quasi-separated, $\Delta_{X/Y}$ is quasi-compact, so by Corollary 10.13, we have to show that $\Delta_{X/Y}$ satisfies the existence

part of the valuative criterion. So, consider the following diagram of solid arrows:

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} A \arrow[r, "(g,h)"] \arrow[ur, dotted] & X \times_Y X \end{tikzcd} \tag{6}$$

Both g and h fit as dotted arrows into the diagram

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} A \arrow[r] \arrow[ur, dotted] & Y \end{tikzcd}$$

hence they are equal by assumption, so we can take any of them as the dotted arrow in Diagram (6). □

Corollary 10.15 (Valuative criterion for properness)**.** *Let $f : X \to Y$ be a quasi-separated morphism of finite type. Then, the following are equivalent:*

(1) f is proper.
(2) f satisfies the existence and uniquess parts of the valuative criterion.

Proof. Combine Corollary 10.13 and Corollary 10.14. □

11. Lectures 12 and 13: (Quasi-)coherent sheaves

In Lecture 5, we studied the category of affine schemes and saw that it is dual to the category of rings (Corollary 5.6). What happens to modules over rings under this equivalence? The answer to this question leads to the notion of quasi-coherent sheaves, which we will introduce and study in these lectures.

Definition 11.1. Let $(X, \mathcal{O}_X)$ be a ringed space.

(1) A *(pre)sheaf of $\mathcal{O}_X$-modules on X* (or simply $\mathcal{O}_X$-module in the sheaf case) is a (pre)sheaf $\mathcal{F}$ on X such that
   (a) for all $U \subseteq X$ open, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module,
   (b) for all $V \subseteq U \subseteq X$ open, the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$ is an $\mathcal{O}_X(U)$-module homomorphism, where $\mathcal{F}(V)$ is considered as an $\mathcal{O}_X(U)$-module through the restriction map $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$.

(2) A *morphism of (pre)sheaves of $\mathcal{O}_X$-modules* is a morphism of (pre)sheaves $\varphi : \mathcal{F} \to \mathcal{G}$ such that for all $U \subseteq X$ open, the map $\varphi(U)$ is an $\mathcal{O}_X(U)$-module homomorphism.

(3) For two sheaves of $\mathcal{O}_X$-modules $\mathcal{F}$ and $\mathcal{G}$, the presheaf of $\mathcal{O}_X$-modules
$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) : U \quad \mapsto \quad \mathrm{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$$
is a sheaf of $\mathcal{O}_X$-modules called *Hom-sheaf*. We also write $\mathcal{H}om_X(\mathcal{F}, \mathcal{G}) := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$.

(4) The category of $\mathcal{O}_X$-modules on X is denoted by $\mathrm{Mod}(X, \mathcal{O}_X)$.

Example 11.2. (1) If $X = \mathrm{Spec}\, K$ where K is a field, then $\mathrm{Mod}(X, \mathcal{O}_X)$ is equivalent to the category of K-vector spaces.

(2) The sheaf $\mathcal{O}_X$ is an $\mathcal{O}_X$-module. So is $\mathcal{O}_X^{\oplus I}$ for any set I and any quotient of these sheaves.

(3) Sheaves of $\mathcal{O}_X$-modules that are subsheaves of $\mathcal{O}_X$ are nothing but ideal sheaves in the sense of Definition 7.5.

Remark 11.3. (1) There is a forgetful functor $\mathrm{Mod}(X, \mathcal{O}_X) \to \mathrm{Sh}(X)$.

(2) The sheafification of a presheaf of $\mathcal{O}_X$-modules is a sheaf of $\mathcal{O}_X$-modules in a natural way (i.e. such that the sheafification map is a morphism of presheaves of $\mathcal{O}_X$-modules).

(3) $\mathrm{Mod}(X, \mathcal{O}_X)$ is an Abelian category (check that all the constructions from Lecture 3 work in this category).

Now, we can try to perform constructions that we know from the theory of modules in the setting of sheaves of modules.

Definition 11.4. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of $\mathcal{O}_X$-modules. The *tensor product* $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ of $\mathcal{F}$ and $\mathcal{G}$ is the sheafification of the presheaf $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$.

Recall that if $A \to B$ is a morphism of rings and $\mathrm{Mod}(A)$ and $\mathrm{Mod}(B)$ are the categories of modules over A and B, then we have a restriction of scalars
$$\begin{aligned} {}_A- : \quad \mathrm{Mod}(B) &\to \mathrm{Mod}(A) \\ N &\mapsto {}_AN \end{aligned}$$
mapping a B-module N to N considered as an A-module through $A \to B$, and an extension of scalars
$$- \otimes_A B : \mathrm{Mod}(A) \to \mathrm{Mod}(B), M \mapsto M \otimes_A B$$

mapping an A-module M to $M \otimes_A B$ considered as a B-module through the second factor.

If $\mathcal{O}_X \to \mathcal{O}'_X$ is a morphism of sheaves of rings on a topological space X, we can perform these constructions for every open $U \subseteq X$ (and sheafify) to obtain a restriction of scalars

$$_{\mathcal{O}_X}- : \mathrm{Mod}(X, \mathcal{O}'_X) \to \mathrm{Mod}(X, \mathcal{O}_X)$$

and an extension of scalars

$$- \otimes_{\mathcal{O}_X} \mathcal{O}'_X : \mathrm{Mod}(X, \mathcal{O}_X) \to \mathrm{Mod}(X, \mathcal{O}'_X).$$

Next, observe that if $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces, $\mathcal{F}$ is a sheaf of $\mathcal{O}_X$-modules, and $\mathcal{G}$ is a sheaf of $\mathcal{O}_Y$-modules, then $f^{-1}\mathcal{G}$ is a sheaf of $f^{-1}\mathcal{O}_Y$-modules and $f_*\mathcal{F}$ is a sheaf of $f_*\mathcal{O}_X$-modules in a natural way. This leads us the following definition.

Definition 11.5. Let $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces.

(1) The *direct image* functor

$$f_* : \mathrm{Mod}(X, \mathcal{O}_X) \to \mathrm{Mod}(Y, \mathcal{O}_Y)$$

is the composition

$$\mathrm{Mod}(X, \mathcal{O}_X) \xrightarrow{f_*} \mathrm{Mod}(Y, f_*\mathcal{O}_X) \xrightarrow{{}_{\mathcal{O}_Y}-} \mathrm{Mod}(Y, \mathcal{O}_Y).$$

(2) The *inverse image* functor

$$f^* : \mathrm{Mod}(Y, \mathcal{O}_Y) \to \mathrm{Mod}(X, \mathcal{O}_X)$$

is the composition

$$\mathrm{Mod}(Y, \mathcal{O}_Y) \xrightarrow{f^{-1}} \mathrm{Mod}(X, f^{-1}\mathcal{O}_Y) \xrightarrow{-\otimes_{f^{-1}\mathcal{O}_Y}\mathcal{O}_X} \mathrm{Mod}(X, \mathcal{O}_X).$$

Definition 11.6. Let $(X, \mathcal{O}_X)$ be a ringed space. A sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ is called

(1) *free* if $\mathcal{F} \cong \mathcal{O}_X^{\oplus I}$ for some set I.
(2) *locally free* if there exists an open cover $X = \cup_{i \in I} U_i$ such that each $\mathcal{F}|_{U_i}$ is free.
(3) *locally free of rank r* if there exists an open cover $X = \cup U_i$ such that $\mathcal{F}|_{U_i} \cong \mathcal{O}_{U_i}^{\oplus r}$.
(4) *invertible* (or *line bundle*) if $\mathcal{F}$ is locally free of rank 1.

For completeness, let us state what we mean by direct sums and products of $\mathcal{O}_X$-modules:

Definition 11.7. Let $(X, \mathcal{O}_X)$ be a ringed space and $\{\mathcal{F}_i\}_{i \in I}$ a family of $\mathcal{O}_X$-modules. Then,

(1) the *direct sum* of the $\mathcal{F}_i$ is the sheafification $\bigoplus_{i \in I} \mathcal{F}_i$ of the presheaf

$$U \mapsto \bigoplus_{i \in I} \mathcal{F}_i(U).$$

(2) the *product* of the $\mathcal{F}_i$ is the sheaf

$$\prod_{i \in I} \mathcal{F}_i : U \mapsto \prod_{i \in I} \mathcal{F}_i(U).$$

Remark 11.8. Even though we need to sheafify in the definition of the direct sum, one can check that for every $x \in X$, there is a natural isomorphism $(\bigoplus_{i \in I} \mathcal{F}_i)_x \cong \bigoplus_{i \in I} (\mathcal{F}_i)_x$. On an abstract level, this is because the direct sum we defined is indeed the categorical direct sum in the category of $\mathcal{O}_X$-modules and "limits commutes with limits".

For a similar reason, we do not need to sheafify in the definition of products, since the sheaf condition involves only products and products commutes with products.

Definition 11.9. Let $(X, \mathcal{O}_X)$ be a ringed space. A sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ is called

(1) *quasi-coherent* if there exists an open cover $X = \cup U_i$ and sets I_i, J_i such that each $\mathcal{F}|_{U_i}$ appears in an exact sequence of the form
$$\mathcal{O}_{U_i}^{\oplus I_i} \to \mathcal{O}_{U_i}^{\oplus J_i} \to \mathcal{F}|_{U_i} \to 0$$
(2) *coherent* if
   (a) there exists an open cover $X = \cup U_i$, $n_i \in \mathbb{N}$, and a surjection $\mathcal{O}_{U_i}^{\oplus n_i} \to \mathcal{F}|_{U_i}$.
   (b) for all $U \subseteq X$ open, $n \in \mathbb{N}$, and all morphisms $\varphi : \mathcal{O}_U^{\oplus n} \to \mathcal{F}|_U$, the kernel sheaf $\ker(\varphi)$ satisfies (a).

We let $\mathrm{Coh}(X, \mathcal{O}_X)$ and $\mathrm{QCoh}(X, \mathcal{O}_X)$ be the categories of coherent and quasi-coherent sheaves of $\mathcal{O}_X$-modules on X. Note that every coherent sheaf is also quasi-coherent. In general, the category $\mathrm{QCoh}(X, \mathcal{O}_X)$ is not an Abelian category, but we will see later that it is if $(X, \mathcal{O}_X)$ is a scheme. Next, we study these notions for schemes.

Definition 11.10. Let $X = \mathrm{Spec}\ A$ be an affine scheme and let M be an A-module. The *sheaf of $\mathcal{O}_X$-modules associated to M* is defined as
$$\widetilde{M} : \quad U \mapsto \quad \{s : U \to \coprod_{\mathfrak{p} \in U} M_{\mathfrak{p}} \mid \text{ s satisfies (i) and (ii)}\},$$
where

(i) For all $\mathfrak{p} \in U$, we have $s(\mathfrak{p}) \in M_{\mathfrak{p}}$.
(ii) For all $\mathfrak{p} \in U$, there exist $m \in M$, $a \in A$, and $V \subseteq U$ open with $\mathfrak{p} \in V \subseteq D(a)$ such that $s(\mathfrak{q}) = \frac{m}{a} \in M_{\mathfrak{q}}$ for all $\mathfrak{q} \in V$,

and $\rho_{UV}(s) := s|_V$.

In particular, note that $\widetilde{A} = \mathcal{O}_X$. Mimicking Proposition 2.17 and Corollary 2.18, we obtain

Proposition 11.11. *Let $X = \mathrm{Spec}\ A$ where A is a ring and let M be an A-module. Then,*

(1) for all $a \in A$, there exist natural isomorphisms $\varphi_a : M_a \to \widetilde{M}(D(a))$ such that for all $D(b) \subseteq D(a)$, the following diagram commutes
$$\begin{tikzcd} M_a \arrow[r, "\varphi_a"] \arrow[d] & \widetilde{M}(D(a)) \arrow[d] \\ M_b \arrow[r, "\varphi_b"] & \widetilde{M}(D(b)), \end{tikzcd}$$
(2) for all $\mathfrak{p} \in \mathrm{Spec}\ A$, there exists a natural isomorphism $\varphi_{\mathfrak{p}} : M_{\mathfrak{p}} \to \widetilde{M}_{\mathfrak{p}}$ such that for all $a \in A \setminus \mathfrak{p}$, the following diagram commutes
$$\begin{tikzcd} M_a \arrow[r, "\varphi_a"] \arrow[d] & \widetilde{M}(D(a)) \arrow[d] \\ M_{\mathfrak{p}} \arrow[r, "\varphi_{\mathfrak{p}}"] & \widetilde{M}_{\mathfrak{p}}. \end{tikzcd}$$

Note that a morphism of A-modules $M \to N$ yields a morphism of $\mathcal{O}_X$-modules $\widetilde{M} \to \widetilde{N}$ by post-composition with the the maps $M_{\mathfrak{p}} \to N_{\mathfrak{p}}$. This turns $\widetilde{(\ )}$ into a functor $\mathrm{A-Mod} \to \mathrm{Mod}(X, \mathcal{O}_X)$.

Lemma 11.12. *Let $X = \operatorname{Spec} A$. Then, for all A-modules M and N, the maps*

$$\begin{array}{rcl} \operatorname{Hom}_A(M,N) & \leftrightarrow & \operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \widetilde{N}) \\ \varphi & \mapsto & \widetilde{\varphi} \\ f(X) & \leftarrow\!\shortmid & f \end{array}$$

where we use the identifications of Proposition 11.11 are mutually inverse bijections.

Proof. This follows from Exercise 38, so we omit the proof. □

Lemma 11.13. *Let $X = \operatorname{Spec} A$ be an affine scheme.*

(1) A sequence of A-modules

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

is exact if and only if the associated sequence of $\mathcal{O}_X$-modules

$$0 \to \widetilde{M_1} \to \widetilde{M_2} \to \widetilde{M_3} \to 0$$

is exact.

(2) If $\varphi : M \to N$ is a morphism of A-modules, then

$$\begin{array}{rcl} \widetilde{\ker(\varphi)} & = & \ker(\widetilde{\varphi}) \\ \widetilde{\operatorname{coker}(\varphi)} & = & \operatorname{coker}(\widetilde{\varphi}) \\ \widetilde{\operatorname{im}(\varphi)} & = & \operatorname{im}(\widetilde{\varphi}). \end{array}$$

(3) If $\{M_i\}_{i \in I}$ is a family of A-modules, then $(\widetilde{\bigoplus_{i \in I} M_i}) = \bigoplus_{i \in I} \widetilde{M_i}$.

(4) If $(\{M_i\}_{i \in I}, \{\varphi_{ij}\}_{i,j \in I, i \leq j})$ is a directed system of A-modules, then $(\widetilde{\varinjlim_{i \in I} M_i}) = \varinjlim_{i \in I} \widetilde{M_i}$.

Proof. To prove Claim (1), it suffices to observe that exactness of the first sequence can be checked on localizations at $\mathfrak{p} \in \operatorname{Spec} A$ and exactness of the second sequence can be checked on stalks at $\mathfrak{p} \in \operatorname{Spec} A$ by Corollary 3.12. Now, it suffices to recall that by Proposition 11.11, we have natural isomorphisms $(M_i)_{\mathfrak{p}} \cong (\widetilde{M_i})_{\mathfrak{p}}$.

Claim (2) follows immediately from Claim (1).

For Claim (3), we use that the maps $M_i \to \bigoplus_{i \in I} M_i$ yield, by functoriality of $\widetilde{(\ )}$, a morphism $\widetilde{M_i} \to (\widetilde{\bigoplus_{i \in I} M_i})$ and thus, by the universal property of direct sums, a morphism $\bigoplus_{i \in I} \widetilde{M_i} \to (\widetilde{\bigoplus_{i \in I} M_i})$. Since localization commutes with direct sums, this morphism is an isomorphism on stalks, hence an isomorphism by Corollary 3.12.

The proof of Claim (4) is similar to Claim (3). □

It turns out that quasi-coherent sheaves on a scheme are exactly the ones that arise locally via the above $\widetilde{(\ )}$-construction. This is what makes the category of quasi-coherent sheaves on schemes well-behaved.

Theorem 11.14. *Let X be a scheme and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Then, the following are equivalent:*

1. *For every open affine* $\operatorname{Spec} A = U \subseteq X$*, there exists an A-module M such that $\mathcal{F}|_U \cong \widetilde{M}$.*
2. *There exists an open affine cover $X = \cup_{i \in I} U_i$ with $U_i = \operatorname{Spec} A_i$ such that for all $i \in I$ there exists an A_i-module M_i with $\mathcal{F}|_{U_i} \cong \widetilde{M_i}$.*
3. *$\mathcal{F}$ is quasi-coherent.*
4. *For every open affine* $\operatorname{Spec} A = U \subseteq X$ *and every $a \in A$, the map $\mathcal{F}(U)_a \to \mathcal{F}(D(a))$ induced*[21] *by restriction is an isomorphism.*

Proof. $(1) \Rightarrow (2)$ is clear.

For $(2) \Rightarrow (3)$, choose a presentation

$$A_i^{\oplus I_i} \to A_i^{\oplus J_i} \to M_i \to 0.$$

Since $\widetilde{(\ \ )}$ is exact, we get a presentation

$$\mathcal{O}_{U_i}^{\oplus I_i} \to \mathcal{O}_{U_i}^{\oplus J_i} \to \mathcal{F}|_{U_i} \to 0$$

so $\mathcal{F}$ is quasi-coherent.

To see $(4) \Rightarrow (1)$, set $M := \mathcal{F}(U)$. Since the isomorphisms $M_a \to \mathcal{F}(D(a))$ are obtained from the restriction maps of $\mathcal{F}$, they are compatible with restriction to smaller principal opens in U, so by Lemma 5.2, they glue to an isomorphism $\widetilde{M} \to \mathcal{F}$, where we use Proposition 11.11 to identify $\widetilde{M}(D(a))$ with M_a.

It remains to prove $(3) \Rightarrow (4)$. Without loss of generality, we may assume that $X = \operatorname{Spec} A$. Since X is quasi-compact and $\mathcal{F}$ is quasi-coherent, there exists a finite cover $X = \cup_{i=1}^n D(b_i)$ by principal open subsets and exact sequences

$$\widetilde{A_{b_i}}^{\oplus I_i} \to \widetilde{A_{b_i}}^{\oplus J_i} \to \mathcal{F}|_{D(b_i)} \to 0.$$

By Lemma 11.12, there exists a morphism $\varphi : A_{b_i}^{\oplus I_i} \to A_{b_i}^{\oplus J_i}$ such that $\widetilde{\varphi}$ is the left-most morphism in the above presentation, so $\mathcal{F}|_{D(b_i)} = \operatorname{coker}(\widetilde{\varphi}) = \widetilde{\operatorname{coker}(\varphi)}$ by Lemma 11.13.

Now, let $a \in A$. Since $\mathcal{F}$ is a sheaf, we have the following diagram with exact rows:

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(X) \arrow[r] \arrow[d] & \prod_{i=1}^n \mathcal{F}(D(b_i)) \arrow[r] \arrow[d] & \prod_{i,j=1}^n \mathcal{F}(D(b_ib_j)) \arrow[d] \\
0 \arrow[r] & \mathcal{F}(X)_a \arrow[r] \arrow[d, "f_1"] & \prod_{i=1}^n \mathcal{F}(D(b_i))_a \arrow[r] \arrow[d, "f_2"] & \prod_{i,j=1}^n \mathcal{F}(D(b_ib_j))_a \arrow[d, "f_3"] \\
0 \arrow[r] & \mathcal{F}(D(a)) \arrow[r] & \prod_{i=1}^n \mathcal{F}(D(ab_i)) \arrow[r] & \prod_{i,j=1}^n \mathcal{F}(D(ab_ib_j))
\end{tikzcd}$$

Here, the arrows in the first and third row are induced by the restriction morphisms (note that $D(b_ib_j) = D(b_i) \cap D(b_j)$) and exactness follows from the sheaf condition (see Remark 1.11). The second row is obtained by localizing the first row. The vertical arrows are induced by restriction morphisms, so the whole diagram commutes.

[21] Note that $\mathcal{F}(D(a))$ is an A-module through the localization map $A \to A_a$, so multiplication by a is a bijection on $\mathcal{F}(D(a))$, which is exactly what we need for the restriction map $\mathcal{F}(U) \to \mathcal{F}(D(a))$ to factor through the localization.

We have shown that the $\mathcal{F}|_{D(b_i)}$ (and hence the $\mathcal{F}|_{D(b_ib_j)}$) are of the form $\widetilde{M_i}$ for some A_{b_i} (resp. $A_{b_ib_j}$)-modules M_i. Hence, by Proposition 11.11, f_2 and f_3 are isomorphisms. By the 5-lemma, this implies that f_1 is an isomorphism, as desired. □

In particular, if X itself is affine, then every quasi-coherent sheaf is in the essential image of $\widetilde{(\ )}$.

Corollary 11.15. *Let $X = \operatorname{Spec} A$ be an affine scheme. Then $\widetilde{(\ )}$ induces an equivalence of categories $A - \operatorname{Mod} \to \operatorname{QCoh}(X, \mathcal{O}_X)$.*

Proof. Combine Theorem 11.14 and Lemma 11.12. □

Remark 11.16. Let $X = \operatorname{Spec} A$ be an affine scheme.

(1) If $\widetilde{M} \in \operatorname{Coh}(X, \mathcal{O}_X)$, then M is a finitely generated A-module. Indeed, by condition (a) in Definition 11.9, there exist $a_1, \dots, a_n$ such that the $D(a_i)$ cover $\operatorname{Spec} A$ and such that M_{a_i} is finitely generated as A_{a_i}-module. This implies that M is finitely generated as A-module (similarly to how "finite" is an affine-local property of morphisms).
(2) If A is Noetherian and M is a finitely generated A-module, then $\widetilde{M} \in \operatorname{Coh}(X, \mathcal{O}_X)$.

In particular, if A is Noetherian, then the equivalence of Corollary 11.15 restricts to an equivalence between the category of finitely generated modules over A and the category of coherent sheaves on X.

Moreover, if X is a, not necessarily affine, locally Noetherian scheme, then an $\mathcal{O}_X$-module $\mathcal{F}$ is coherent if and only if, for every open affine $\operatorname{Spec} A = U \subseteq X$, it holds that $\mathcal{F}|_U \cong \widetilde{M}$ for some finitely generated A-module M. As in the quasi-coherent case, coherence can be checked on an open affine cover.

Corollary 11.17. *If $(X, \mathcal{O}_X)$ is a scheme, then $\operatorname{QCoh}(X, \mathcal{O}_X)$ is an Abelian category.*

Proof. Since $\operatorname{QCoh}(X, \mathcal{O}_X)$ is a full subcategory of the category of $\mathcal{O}_X$-modules and it contains the 0-module, it suffices to show that all kernels, cokernels, and binary biproducts of (quasi-)coherent sheaves in the category of $\mathcal{O}_X$-modules are again (quasi-)coherent. (Quasi-)coherence is local on X and taking kernels, cokernels, and binary biproducts is compatible with restriction, so we may assume that X is affine. In this setting, the claim follows from Corollary 11.15. □

Next, we want to understand how (quasi-)coherence behaves under the direct and inverse image functors. To do this, we first give a more precise description of direct and inverse images under morphisms of affine schemes.

Proposition 11.18. *Let $\varphi : A \to B$ be a morphism of rings and let $f : X = \operatorname{Spec} B \to \operatorname{Spec} A = Y$ be the induced morphism of affine schemes. Let $\mathcal{F} = \widetilde{M} \in \operatorname{QCoh}(X, \mathcal{O}_X)$ and $\mathcal{G} = \widetilde{N} \in \operatorname{QCoh}(Y, \mathcal{O}_Y)$. Then,*

(1) there is a natural isomorphism $f_*(\widetilde{M}) \cong \widetilde{({}_AM)}$,
(2) there is a natural isomorphism $f^*(\widetilde{N}) \cong \widetilde{N \otimes_A B}$.

In particular, $f_*\mathcal{F}$ is quasi-coherent.

Proof. Recall that $f^{-1}(D(a)) = D(\varphi(a))$. For Claim (1) calculate

$$(f_*\widetilde{M})(D(a)) = \widetilde{M}(D(\varphi(a))) \cong M_{\varphi(a)} = ({}_AM)_a \cong \widetilde{({}_AM)}(D(a)).$$

Here, the two isomorphisms are the natural isomorphisms of Proposition 11.11, so they are natural and compatible with restriction, hence they glue to a natural isomorphism $f_*(\widetilde{M}) \cong \widetilde{({}_AM)}$.

To prove Claim (2), we use the chain of natural isomorphisms

$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_X}(f^*(\widetilde{N}), \mathcal{F}) &\overset{Ex.37}{\cong} \mathrm{Hom}_{\mathcal{O}_Y}(\widetilde{N}, f_*\mathcal{F}) \overset{(1)}{\cong} \mathrm{Hom}_{\mathcal{O}_Y}(\widetilde{N}, \widetilde{({}_AM)}) \\ &\overset{Ex.38}{\cong} \mathrm{Hom}_A(N, {}_AM) \cong \mathrm{Hom}_B(N \otimes_A B, M) \overset{Ex.38}{\cong} \mathrm{Hom}_{\mathcal{O}_X}(\widetilde{N \otimes_A B}, \mathcal{F}).\end{aligned}$$

The isomorphism without a reference is the usual adjunction between extension and restriction of scalars for modules. By the Yoneda lemma in the category $\mathrm{QCoh}(X, \mathcal{O}_X)$, this yields the natural isomorphism of Claim (2). More explicitly: Choosing $\mathcal{F} = f^*(\widetilde{N})$ and the identity on $\mathcal{F}$, the above yields a morphism $\widetilde{N \otimes_A B} \to f^*(\widetilde{N})$. Choosing $\mathcal{F} = \widetilde{N \otimes_A B}$ and the identity on this $\mathcal{F}$, we get a morphism $f^*(\widetilde{N}) \to \widetilde{N \otimes_A B}$. By naturality of the chain of isomorphisms, these two morphisms are inverse to each other. □

Proposition 11.19. *Let $f : X \to Y$ be a morphism of schemes. Then, the following hold:*

(1) For all $\mathcal{G} \in \mathrm{QCoh}(Y, \mathcal{O}_Y)$, we have $f^*\mathcal{G} \in \mathrm{QCoh}(X, \mathcal{O}_X)$.
(2) If X is locally Noetherian and $\mathcal{G} \in \mathrm{Coh}(Y, \mathcal{O}_Y)$, we have $f^*\mathcal{G} \in \mathrm{Coh}(X, \mathcal{O}_X)$.
(3) If f is quasi-compact and quasi-separated and $\mathcal{F} \in \mathrm{QCoh}(X, \mathcal{O}_X)$, then $f_*\mathcal{F} \in \mathrm{QCoh}(Y, \mathcal{O}_Y)$.[22]

Proof. Note that, if $\iota : V \hookrightarrow Y$ is an affine open and $\iota' : f^{-1}(V) \hookrightarrow X$ is the induced open immersion of the preimage of V, then

$$(f_V)^*(\mathcal{G}|_V) = (f_V^* \circ \iota^*)(\mathcal{G}) = (\iota'^* \circ f^*)(\mathcal{G}) = (f^*\mathcal{G})|_{f^{-1}(V)}$$

and

$$((f_V)_*(\mathcal{F}|_{f^{-1}(V)}))(W) = \mathcal{F}|_{f^{-1}(V)}(f^{-1}(W)) = \mathcal{F}(f^{-1}(W)) = (f_*\mathcal{F})(W)$$

for all $W \subseteq V$ open, so $(f_V)_*(\mathcal{F}|_{f^{-1}(V)}) = (f_*\mathcal{F})|_V$. In particular, it suffices to prove the claims if Y is affine.

For Claims (1) and (2), we use that (quasi-)coherence is local on X, so we may assume that X affine. Then, the claim follows immediately from Proposition 11.18.

It remains to prove Claim (3). Since f is quasi-compact, we find a finite open affine cover $X = \cup_{i=1}^n U_i$ with $U_i = \mathrm{Spec}\, B_i$. The intersections $U_i \cap U_j$ are quasi-compact, since f is quasi-separated, so we find a finite open affine cover $U_i \cap U_j = \cup_{k=1}^{n_{ij}} U_{ijk}$ with $U_{ijk} = \mathrm{Spec}\, B_{ijk}$. Since $f_*\mathcal{F}$ is a sheaf, the sheaf condition gives an exact sequence

$$0 \to f_*\mathcal{F} \to \prod_{i=1}^n (f|_{U_i})_*(\mathcal{F}|_{U_i}) \to \prod_{i,j=1}^n \prod_{k=1}^{n_{ij}} (f|_{U_{ijk}})_*(\mathcal{F}|_{U_{ijk}}).$$

Now, the morphisms $f|_{U_i}$ and $f|_{U_{ijk}}$ are morphisms of affine schemes, so taking direct images along these morphisms preserves quasi-coherence by Proposition 11.18. Since the products in the above sequence are finite, they are direct sums, hence by Lemma 11.13, the resulting sheaves are again quasi-coherent. Thus, $f_*\mathcal{F}$, being a kernel of a morphism of quasi-coherent sheaves, is quasi-coherent. □

[22] The corresponding statement for coherent sheaves on locally Noetherian schemes requires strong assumptions on f (e.g. proper) and is harder to prove. We will come back to this later.

As a final remark in this lecture, we observe that the ideal sheaves cutting out closed subschemes of a scheme are exactly the quasi-coherent ideal sheaves. More precisely:

Proposition 11.20. *Let $(X, \mathcal{O}_X)$ be a scheme and let $\mathcal{I} \subseteq \mathcal{O}_X$ be a sheaf of ideals. Let*

$$\begin{aligned} Z &:= \operatorname{Supp}(\mathcal{O}_X/\mathcal{I}) := \{x \in X \mid (\mathcal{O}_X/\mathcal{I})_x \neq 0\} \\ \mathcal{O}_Z &:= (\mathcal{O}_X/\mathcal{I})|_Z. \end{aligned}$$

Then, $(Z, (\mathcal{O}_X/\mathcal{I})|_Z)$ is a scheme if and only if $\mathcal{I}$ is quasi-coherent.

Proof. The properties "Z is a scheme" and "$\mathcal{I}$ is quasi-coherent" are local on X, so we may assume without loss of generality that $X = \operatorname{Spec} A$ is affine.

If $\mathcal{I}$ is quasi-coherent, then, by Theorem 11.14, there exists an ideal $I \subseteq A$ such that $\mathcal{I} \cong \widetilde{I}$. [23] Then, $\mathcal{O}_X/\mathcal{I} \cong \widetilde{A/I}$ by Lemma 11.13 and

$$\operatorname{Supp}(\mathcal{O}_X/\mathcal{I}) = \{\mathfrak{p} \in \operatorname{Spec} A \mid (A/I)_{\mathfrak{p}} \neq 0\} = V(I),$$

so $Z \cong \operatorname{Spec} A/I$ is a scheme.

Conversely, if Z is a scheme, then, by Proposition 7.14, there exist an ideal $I \subseteq A$ such that $Z \cong \operatorname{Spec} A/I$. Then, by Lemma 11.13, we have

$$\mathcal{I} \cong \ker(\widetilde{A} \to \widetilde{A/I}) \cong \widetilde{I},$$

so $\mathcal{I}$ is quasi-coherent. □

Corollary 11.21. *Let $(X, \mathcal{O}_X)$ be a scheme. Then, there is a bijection between quasi-coherent ideal sheaves on X and closed subschemes of X.*

[23] Actually, Theorem 11.14 a priori only yields an A-module M with $\widetilde{M} = \widetilde{I}$, but since $M \cong \widetilde{M}(X) \cong \mathcal{I}(X) \hookrightarrow \mathcal{O}_X(X) = A$, M is a submodule of A, hence an ideal.

12. Lectures 14 and 15: The Proj construction

Up until this point of the lecture course, most of the explicit examples of schemes we have seen have been affine schemes, the notable exceptions being the affine line with double origin and the projective line $\mathbb{P}^1_k$. In this lecture, we will learn how to systematically produce (non-affine, in general) schemes from graded rings. As a special case, we will recover the projective line and, more generally, *projective n-space* $\mathbb{P}^n_k$. At the end, we will recover the classical definition of projective space as the set $(k^{n+1} \setminus \{0\})/k^\times$ of lines through the origin in k^{n+1} as the set of k-rational points $\mathbb{P}^n_k(k)$ of our scheme-theoretic version of projective space. In particular, as for zero sets of polynomials in affine space $\mathbb{A}^n_k$, this will allow us to use the language and tools we developed so far to study zero sets of homogeneous polynomials in projective space.

12.1. Algebraic preliminaries.

Before studying the projective spectrum of a graded ring and the Zariski topology on it, we need to recall some background on graded rings from commutative algebra.

Definition 12.1.
(1) A *($\mathbb{N}$-)graded ring* is a ring A together with a decomposition $A = \bigoplus_{d=0}^\infty A_d$ of its underlying additive group such that for all $d, d' \in \mathbb{N}$ we have $A_d \cdot A_{d'} \subseteq A_{d+d'}$.
(2) A *graded morphism of graded rings* is a ring morphism $\varphi : A \to B$ of graded rings such that $\varphi(A_d) \subseteq B_d$ for all $d \geq 0$.
(3) A *graded module* over a graded ring A is an A-module M together with a decomposition $M = \bigoplus_{d\in\mathbb{Z}} M_d$ such that $A_d \cdot M_{d'} \subseteq M_{d+d'}$ for all $d \geq 0, d' \in \mathbb{Z}$.
(4) A *graded morphism of graded A-modules* is an A-linear map $\varphi : M \to N$ of graded modules such that $\varphi(M_d) \subseteq N_d$ for all $d \in \mathbb{Z}$.
(5) The *irrelevant ideal* of $A = \bigoplus_{d=0}^\infty A_d$ is $A_+ := \bigoplus_{d=1}^\infty A_d$.
(6) An element $a \in A$ is called *homogeneous* if $a \in A_d$ for some d. If $a \neq 0$, then d is called *degree* of a and denoted by $\deg(a)$.
(7) An ideal $I \subseteq A$ is called *homogeneous* if $I = \bigoplus_{d=0}^\infty (A_d \cap I)$.
(8) The *homogenization* of an ideal $I \subseteq A$ is $I^h := \bigoplus_{d=0}^\infty (A_d \cap I) \subseteq I$.

Example 12.2. For any ring A_0, the polynomial ring $A = A_0[x_0, \ldots, x_n]$ is a graded ring if we define $A_d \subseteq A$ to be the subgroup of polynomials which are sums of monomials of total degree d.

Lemma 12.3. *Let $A = \bigoplus_{d=0}^\infty A_d$ be a graded ring. Then, the following hold:*
(1) $A_0 \subseteq A$ is a subring.
(2) If $\mathfrak{p} \subseteq A$ is a prime ideal, then so is $\mathfrak{p}^h \subseteq A$.
(3) An ideal $I \subseteq A$ is homogeneous if and only if it can be generated by homogeneous elements.
(4) A homogeneous ideal $I \subseteq A$ is prime if and only if for all homogeneous $a, b \in A$ with $a \cdot b \in I$, either $a \in I$ or $b \in I$.
(5) If $I \subseteq A$ is a homogeneous ideal, then so is its radical $\sqrt{I}$.

Proof. Claim (1) is clear. Claim (2) will follow from Claim (4).

For Claim (3), assume that $I = (\{a_i\}_{i\in J})$ with a_i homogeneous. We claim that $I = \bigoplus_{d=0}^\infty (I \cap A_d)$, where the inclusion "$\supseteq$" is clear. To show the other inclusion, let $a \in I$ and write $a = \sum_{i\in J} a_i b_i$, where only finitely many b_i are non-zero. Write b_i as the sum of its non-zero homogeneous parts $b_i = \sum_{j=1}^{n_i} b_{ij}$. Then, $a = \sum_{i\in J} \sum_{j=1}^{n_i} a_i b_{ij}$ and each $a_i b_{ij}$ is homogeneous and contained in I. Hence, $a \in \bigoplus_{d=0}^\infty (I \cap A_d)$.

Conversely, assume that $I = \bigoplus_{d=0}^{\infty}(I \cap A_d)$. Then, I is generated by $\bigcup_{d=0}^{\infty}(I \cap A_d)$, hence generated by homogeneous elements.

For Claim (4), it is clear that if I is prime and $a, b \in A$ are homogeneous, then $a \in I$ or $b \in I$. Conversely, assume this condition holds. Consider $a, b \in A$ not necessarily homogeneous and assume that $ab \in I$. Write $a = \sum_{i=0}^{d} a_i$ with a_i homogeneous of degree i (or zero) and $a_d \neq 0$, and $b = \sum_{i=0}^{e} b_i$ with b_i homogeneous of degree i (or zero) and $b_e \neq 0$. Since $ab \in I$ and I is homogeneous, we can write $ab = \sum_{i=0}^{f} c_i$ with c_i homogeneous of degree i (or zero) and $c_f \neq 0$. Then, either $a_d b_e = 0$ or $a_d b_e = c_f$. In any case, either $a_d \in I$ or $b_e \in I$. In the first case, replace a by $a - a_d$, and in the second case, replace b by $b - b_e$. Then, ab is still in I, but the new $d + e$ is smaller. Repeat this argument until $d = e = 0$. Then, a and b are homogeneous and $ab \in I$, hence $a \in I$ or $b \in I$.

For Claim (5), let $a \in \sqrt{I}$. We have to show that if we decompose $a = \sum a_i$ into its non-zero homogeneous parts a_i of degree i, then $a_i \in \sqrt{I}$. Let d be maximal such that $a_d \neq 0$ and choose $n > 0$ such that $a^n \in I$. Then, either $a_d^n = 0$ or a_d^n is the homogeneous part of highest degree in a^n. Since I is homogeneous, we obtain that $a_d^n \in I$. In particular, $a_d \in \sqrt{I}$. Now, repeat the argument for $(a - a_d) \in \sqrt{I}$. By induction, this shows that $a_i \in \sqrt{I}$ for all i, as claimed. □

Definition 12.4. Let $A = \bigoplus_{d=0}^{\infty} A_d$ be a graded ring and $M = \bigoplus_{d \in \mathbb{Z}} M_d$ a graded A-module.

(1) Let $S \subseteq A$ be a multiplicatively closed subset consisting of homogeneous elements. Then, the *homogeneous localization of M at S* is the submodule $(S^{-1}M)_0 \subseteq S^{-1}M$ defined by
$$(S^{-1}M)_0 := \{\frac{a}{b} \in S^{-1}M \mid \deg(a) = \deg(b)\}$$
(2) If $a \in A$ is a homogeneous element, then $S := \{a^i\}_{i \in \mathbb{N}}$ consists of homogeneous elements and we set $M_{(a)} := (S^{-1}M)_0$ and call it the *homogeneous localization of M at a*.
(3) If $\mathfrak{p} \subseteq A$ is a prime ideal, we let S be the set of homogeneous elements in $A \setminus \mathfrak{p}$ and we set $M_{(\mathfrak{p})} := (S^{-1}M)_0$ and call it the *homogeneous localization of M at $\mathfrak{p}$*.

If $A = M$, then all these localizations are (ungraded) rings.

Example 12.5. Let $A = k[x, y]$ be the graded polynomial ring as in Example 12.2. Then, $A_{(x)} \cong k[\frac{y}{x}]$ and $A_{(y)} \cong k[\frac{x}{y}]$. Note that the spectra of both localizations contain a principal open U with corresponding ring $k[\frac{x}{y}, \frac{y}{x}]$. Now, our notation already suggests that we want to glue $\operatorname{Spec} A_{(x)}$ to $\operatorname{Spec} A_{(y)}$ along the map $\frac{y}{x} \mapsto \frac{x}{y}$ over the open U. This is exactly what we will do for general graded rings below. You should compare this with Example 5.10 (2).

12.2. **The projective spectrum.** In this subsection, we define the projective spectrum of a graded ring and equip it with a topology, called *Zariski topology*, in analogy with the affine case.

Definition 12.6. Let $A = \bigoplus_{d=0}^{\infty} A_d$ be a graded ring.

(1) The *projective spectrum* of A is $\operatorname{Proj} A := \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \text{ homogeneous}, A_+ \not\subseteq \mathfrak{p}\}$.
(2) The *Zariski topology* on $\operatorname{Proj} A$ is defined by defining the closed subsets to be
$$V_+(I) := \{\mathfrak{p} \in \operatorname{Proj} A \mid I \subseteq \mathfrak{p}\}$$
for subsets $I \subseteq A$.
(3) The *principal open subset* of $\operatorname{Proj} A$ associated to a homogeneous $a \in A_+$ is
$$D_+(a) := \operatorname{Proj} A \setminus V_+(\{a\}) = \{\mathfrak{p} \in \operatorname{Proj} A \mid a \notin \mathfrak{p}\}.$$

Lemma 12.7. *The following identities hold:*

(1) For any subset $I \subseteq A$, we have $V_+(I) = V_+((I)) = V_+(I') = V_+(I' \cap A_+)$, where I' is the smallest homogeneous ideal containing I.
(2) For any two subsets $I, J \subseteq A$, we have $V_+(I) \cup V_+(J) = V_+(I \cap J)$.
(3) For arbitrarily many subsets $\{J_i\}_{i \in I}$, $J_i \subseteq A$, we have $\cap_{i \in I} V_+(J_i) = V_+(\sum_{i \in I}(J_i))$.
(4) $V_+(A_+) = V_+(A) = \emptyset$.
(5) $V_+((0)) = \mathrm{Proj}\ A$.
(6) If $a = \sum a_i \in A$ is an element and the a_i are homogeneous of degree i, then

$$D(a) \cap \mathrm{Proj}\ (A) = (D(a_0) \cap \mathrm{Proj}\ (A)) \cup (\bigcup_{i \geq 1} D_+(a_i)).$$

(7) If $a = a_0 \in A$ is homogeneous of degree 0, then

$$D(a_0) \cap \mathrm{Proj}\ (A) = \bigcup_{b \in A_d, d \geq 1} D_+(a_0 b).$$

In particular, the Zariski topology on $\mathrm{Proj}\ A$ is well-defined and the $D_+(a)$ with $a \in A_+$ homogeneous form a basis of this topology.

Proof. We omit the proofs. Each of these facts either follows immediately from the definitions or from the corresponding statements for affine schemes. □

As for affine spectra, we know exactly when $V_+(I) \subseteq V_+(J)$ holds:

Lemma 12.8. *Let $I, J \subseteq A$ be homogeneous ideals in a graded ring A. Then, $V_+(I) \subseteq V_+(J)$ if and only if $J \cap A_+ \subseteq \sqrt{I}$. In particular, if $a, b \in A_+$ are homogeneous, then $D_+(b) \subseteq D_+(a)$ if and only if $D(b) \subseteq D(a)$.*

Proof. Assume $J \cap A_+ \subseteq \sqrt{I}$ and let $\mathfrak{p} \in V_+(I)$. Then, $I \subseteq \mathfrak{p}$. Since $\mathfrak{p}$ is prime, this implies $\sqrt{I} \subseteq \mathfrak{p}$, so $J \cdot A_+ \subseteq J \cap A_+ \subseteq \mathfrak{p}$. Since $A_+ \not\subseteq \mathfrak{p}$, this implies $J \subseteq \mathfrak{p}$, so $\mathfrak{p} \in V_+(J)$.

Conversely, assume $V_+(I) \subseteq V_+(J)$. We need to show that $J \cap A_+ \subseteq \mathfrak{p}$ for every $\mathfrak{p} \in \mathrm{Spec}\ A$ with $I \subseteq \mathfrak{p}$. Since I is homogeneous, we have $I \subseteq \mathfrak{p}^h$. If $A_+ \not\subseteq \mathfrak{p}^h$, then $\mathfrak{p}^h \in V_+(I) \subseteq V_+(J)$, so $J \cap A_+ \subseteq J \subseteq \mathfrak{p}^h \subseteq \mathfrak{p}$. If $A_+ \subseteq \mathfrak{p}^h$, then $J \cap A_+ \subseteq \mathfrak{p}^h \subseteq \mathfrak{p}$ as well.

Finally, we note that $D_+(b) \subseteq D_+(a)$ if and only if $V_+(a) \subseteq V_+(b)$, which, by the above, happens if and only if $(b) \cap A_+ \subseteq \sqrt{(a)}$. Since $b \in A_+$, this is equivalent to $(b) \subseteq \sqrt{(a)}$, which we know is equivalent to $D(b) \subseteq D(a)$. □

Corollary 12.9. *Let A be a graded ring. Then, $\mathrm{Proj}\ (A) = \emptyset$ if and only if $A_+ \subseteq \sqrt{0}$.*

Proof. By Lemma 12.7, we have $\mathrm{Proj}\ (A) = V_+((0))$ and $V_+(A_+) = \emptyset$. By Lemma 12.8, we thus have $\mathrm{Proj}\ (A) = \emptyset$ if and only if $A_+ \subseteq \sqrt{0}$. □

12.3. **Proj as a scheme.** Next, we need a sheaf of rings on $\mathrm{Proj}\ A$. In order to avoid having to work twice to define sheaves of modules on $\mathrm{Proj}\, A$ later, we immediately extend the $\widetilde{(\ \ )}$ functor to graded modules over graded rings and obtain the structure sheaf on $\mathrm{Proj}\, A$ as a special case. For the following statement, recall from Lemma 12.8 that $D_+(b) \subseteq D_+(a)$ is equivalent to $D(b) \subseteq D(a)$.

Lemma 12.10. *Let $A = \bigoplus_{d=0}^{\infty} A_d$ be a graded ring, $M = \bigoplus_{d \in \mathbb{Z}} M_d$ a graded module, and $a, b \in A_+$ homogeneous with $D_+(b) \subseteq D_+(a)$. Then, the composition $M_{(a)} \to M_a \to M_b$*

factors through an $A_{(a)}$-linear map $M_{(a)} \to M_{(b)}$. This map satisfies the universal property of the localization of $M_{(a)}$ at $\frac{b^{\deg(a)}}{a^{\deg(b)}}$, hence it induces a natural isomorphism $(M_{(a)})_{\frac{b^{\deg(a)}}{a^{\deg(b)}}}$.

Proof. Since $D_+(b) \subseteq D_+(a)$, we have $D(b) \subseteq D(a)$, so there exists $n > 0$ and $c \in A$ such that $b^n = ac$. Replacing c with its homogeneous part of the appropriate degree, we may assume that c is homogeneous.

Now, let $\frac{e}{a^m} \in M_{(a)}$, so $\deg e = m \deg a$. Then, its image in M_b is $\frac{e}{a^m} = \frac{c^m e}{(ac)^m} = \frac{c^m e}{b^{nm}}$. Since $m \deg(c) + \deg(e) = m(n \deg(b) - \deg(a)) + m \deg(a)) = mn \deg(b)$, this element is in $M_{(b)}$, as claimed. Checking the universal property of the localization is an explicit calculation, which we omit. □

In particular, Lemma 12.10 implies that for every $D_+(b) \subseteq D_+(a)$ we have a commutative diagram of the form

$$\begin{tikzcd} D_+(a) \arrow[r] \arrow[rrr, bend left, "\varphi_a"] & D(a) \arrow[r] & \operatorname{Spec} A_a \arrow[r] & \operatorname{Spec} A_{(a)} \\ D_+(b) \arrow[r] \arrow[u] & D(b) \arrow[r] \arrow[u] & \operatorname{Spec} A_b \arrow[r] \arrow[u] & \operatorname{Spec} A_{(b)} \arrow[u] \end{tikzcd} \tag{7}$$

where the middle horizontal arrows are homeomorphisms. The key result that bridges the gap between $\operatorname{Proj} A$ and schemes is the following observation.

Proposition 12.11. *Let $A = \bigoplus_{d=0}^{\infty} A_d$ be a graded ring and let $a \in A_d$ for some $d > 0$.* [24] *Then, the morphism φ_a in Diagram (7) is a homeomorphism.*

Proof. Since the Diagram (7) commutes, since $\operatorname{Spec} A_{(b)} \to \operatorname{Spec} A_{(a)}$ is a homeomorphism onto a principal open subset by Lemma 12.10, and since the $D_+(b)$ with $D_+(b) \subseteq D_+(a)$ form a basis of the Zariski topology on $D_+(a)$, it suffices to show that φ_a is a bijection.

We define a map

$$\begin{aligned} \psi_a : \quad \operatorname{Spec} A_{(a)} &\to D_+(a) \\ \mathfrak{q} &\to \bigoplus_{d=0}^{\infty} \{b \in A_d \mid \frac{b^{\deg(a)}}{a^d} \in \mathfrak{q}\} \end{aligned}$$

and claim that ψ_a is an inverse for φ_a.

Note that $\psi_a(\mathfrak{q})$ is well-defined: If $b, b' \in \psi_a(\mathfrak{q})_d$, then the fact that $\mathfrak{q}$ is prime implies that all summands of

$$\frac{(b+b')^{\deg(a)}}{a^d} = \sum_{i=0}^{\deg(a)} \binom{\deg(a)}{i} \frac{b^i (b')^{\deg(a)-i}}{a^d}$$

are again in $\mathfrak{q}$, hence $b + b' \in \psi_a(\mathfrak{q})_d$. Since $\mathfrak{q}$ is an ideal, $\psi_a(\mathfrak{q})$ is also absorbing for the multiplication on A, hence $\psi_a(\mathfrak{q})$ is an ideal. By definition, it is homogeneous. The prime property of $\mathfrak{q}$ immediately implies the prime property for $\psi_a(\mathfrak{q})$ on homogeneous elements, hence $\psi_a(\mathfrak{q})$ is a

[24] Note: $d > 0$ is necessary. For example, $A_{(1)} = A_0$ and $D_+(1) = \operatorname{Proj}(A)$, so if $A = k[x, y]$ with the standard grading, then $\operatorname{Spec} A_0 = \operatorname{Spec} k$ but $\operatorname{Proj}(A)$ has infinitely many points.

homogeneous prime ideal. Finally, note that $a \notin \psi_a(\mathfrak{q})$, since $\frac{1}{1} \notin \mathfrak{q}$. Since a has positive degree, this implies that $A_+ \not\subseteq \psi_a(\mathfrak{q})$.

We calculate

$$((\psi_a \circ \varphi_a)(\mathfrak{p}))_d \quad = \quad (\psi_a(\mathfrak{p}A_a \cap A_{(a)}))_d = \{b \in A_d \mid \frac{b^{\deg(a)}}{a^d} \in \mathfrak{p}A_a \cap A_{(a)}\}.$$

Now, $\frac{b^{\deg(a)}}{a^d} \in \mathfrak{p}A_a \cap A_{(a)}$ implies that there exists $c \in \mathfrak{p}$ and $m, m' \geq 0$ such that $a^m(a^d c - a^{m'} b^{\deg(a)}) = 0$ in A. If this holds, then $b^{\deg(a)} a^{m+m'} \in \mathfrak{p}$ for some $m \geq 0$, hence $b \in \mathfrak{p}_d$. Conversely, if $b \in \mathfrak{p}_d$, then clearly $\frac{b^{\deg(a)}}{a^d} \in \mathfrak{p}A_a \cap A_{(a)}$. So, $((\psi_a \circ \varphi_a)(\mathfrak{p}))_d = \mathfrak{p}_d$. Since $\mathfrak{p}$ is homogeneous, this implies $(\psi_a \circ \varphi_a)(\mathfrak{p}) = \mathfrak{p}$.

Conversely, we have

$$(\varphi_a \circ \psi_a)(\mathfrak{q}) = \varphi_a(\bigoplus_{d=0}^{\infty} \{b \in A_d \mid \frac{b^{\deg(a)}}{a^d} \in \mathfrak{q}\}) = (\bigoplus_{d=0}^{\infty} \{b \in A_d \mid \frac{b^{\deg(a)}}{a^d} \in \mathfrak{q}\})A_a \cap A_{(a)}.$$

One checks that this is in fact equal to $\mathfrak{q}$, using explicit calculations similar to the above.

□

Since the homeomorphism φ_a of Proposition 12.11 is natural and compatible with localization, we will use it to identify $D_+(a) \subseteq \operatorname{Proj} A$ with $\operatorname{Spec} A_{(a)}$ from now on. If A is a graded ring and M is a graded A-module, we can write $\operatorname{Proj} A = \bigcup_{a \in A_d, d \geq 1} D_+(a)$. For each $a \in A_d, d \geq 1$, we can define the quasi-coherent sheaf $\widetilde{M_{(a)}}$ on $D_+(a)$. If $D_+(b) \subseteq D_+(a)$, then, by Lemma 12.10 and Proposition 11.18, the restriction $\widetilde{M_{(a)}}|_{D_+(b)}$ is naturally isomorphic to $\widetilde{M_{(b)}}$. In particular, the collection of all such isomorphisms satisfies the cocycle condition of Exercise 3, so the sheaves $\widetilde{M_{(a)}}$ glue to a sheaf on $\operatorname{Proj} A$.

Definition 12.12. Let A be a graded ring and let M be a graded A-module. Let $X = \operatorname{Proj} A$ and $n \in \mathbb{Z}$.

(1) The *structure sheaf $\mathcal{O}_X$ on X* is the unique sheaf of rings on X that agrees with $\widetilde{A_{(a)}}$ on $D_+(a)$.
(2) The *sheaf of $\mathcal{O}_X$-modules associated to M* is the unique sheaf $\widetilde{M}$ of $\mathcal{O}_X$-modules on X that agrees with $\widetilde{M_{(a)}}$ on $D_+(a)$.
(3) The *n-th twisting sheaf $\mathcal{O}_X(n)$ on X* is the $\mathcal{O}_X$-module associated to the graded A-module $A(n) = \bigoplus_{d=0}^{\infty} A(n)_d$ where $A(n)_d := A_{n+d}$.
(4) The *n-th twist $\widetilde{M(n)}$ of $\widetilde{M}$* is the $\mathcal{O}_X$-module associated to the graded A-module $M(n) = \bigoplus_{d \in \mathbb{Z}} M(n)_d$ where $M(n)_d := M_{n+d}$.

From now on, we write $\operatorname{Proj} A$ for the pair $(\operatorname{Proj} A, \mathcal{O}_{\operatorname{Proj} A})$.

By construction, the ringed space $(\operatorname{Proj} A, \mathcal{O}_{\operatorname{Proj} A})$ is a scheme. Similarly, every sheaf of $\mathcal{O}_X$-modules of the form $\widetilde{M}$, where M is a graded A-module, is a quasi-coherent sheaf.

Corollary 12.13. *Let A be a graded ring and let M be a graded A-module. Let $X = \operatorname{Proj} A$. For every $\mathfrak{p} \in D_+(b) \subseteq D_+(a) \subseteq \operatorname{Proj} A$, we have a natural commutative diagram*

$$\begin{tikzcd}
M_0 \arrow[r] \arrow[d] & \widetilde{M}(X) \arrow[d] \\
M_{(a)} \arrow[r, "\cong"] \arrow[d] & \widetilde{M}(D_+(a)) \arrow[d] \\
M_{(b)} \arrow[r, "\cong"] \arrow[d] & \widetilde{M}(D_+(b)) \arrow[d] \\
M_{(\mathfrak{p})} \arrow[r, "\cong"] & \widetilde{M}_{\mathfrak{p}},
\end{tikzcd}$$

where the left vertical arrows are induced by localization and the right vertical arrows are induced by restriction maps. The bottom three horizontal arrows are isomorphisms.

Proof. By definition of $\widetilde{M}$, we have $\widetilde{M}|_{D_+(a)} = \widetilde{M_{(a)}}$, so the commutativity of the middle square as well as the fact that the horizontal morphisms are isomorphisms follows from Proposition 11.11.

The bottom horizontal arrow is obtained by passing to directed limits and observing that $M_{(\mathfrak{p})} = \varinjlim_{\mathfrak{p} \in D_+(a)} M_{(a)}$.

Finally, an element $m \in M_0$ gives elements $\frac{m}{1} \in M_{(a)}$ for every $D_+(a)$, hence elements in every $\widetilde{M}(D_+(a))$ that agree on intersections, hence an element in $\widetilde{M}(X)$. This defines the top horizontal arrow. $\square$

Remark 12.14. Using that $\mathcal{O}_{\operatorname{Proj} A}$ and $\widetilde{M}$ are sheaves, we obtain a description of $\mathcal{O}_{\operatorname{Proj} A}(U)$ and $\widetilde{M}(U)$ as spaces of functions from Definition 2.13.

Definition 12.15. Let A_0 be a ring. Let $n \geq 1$ and let $A = A_0[x_0, \ldots, x_n]$. Equip A with the grading $A = \bigoplus_{d=0}^{\infty} A_0[x_0, \ldots, x_n]_d$, where $A_0[x_0, \ldots, x_n]_d$ is the group of polynomials of degree d. Then, the scheme $\operatorname{Proj} A$ is called *projective n-space over A_0* and denoted by $\mathbb{P}^n_{A_0}$.

Remark 12.16. One can check that $\mathbb{P}^n_{A_0} \cong \mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} A_0$. So, as a generalization, we can define projective n-space over an arbitrary scheme X as $\mathbb{P}^n_X := X \times_{\operatorname{Spec} \mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$.

Remark 12.17. Let A_0 be a ring and consider $\mathbb{P}^n_{A_0} = \operatorname{Proj} A_0[x_0, \ldots, x_n]$.

(1) We have isomorphisms $D_+(x_i) \cong \operatorname{Spec} A_{(x_i)} \cong \operatorname{Spec} A_0[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}] \cong \mathbb{A}^n_{A_0}$.
(2) The graded morphism $A_0[x_0, \ldots, x_n] \to A_0[x_0, \ldots, \ldots, x_n]/(x_i)$ induces a closed immersion $\mathbb{P}^{n-1}_{A_0} \to \mathbb{P}^n_{A_0}$ whose image is exactly $V_+(x_i)$.
(3) In particular, projective space admits the *standard cover* $\mathbb{P}^n_{A_0} = \bigcup_{i=0}^n D_+(x_i)$ where each $D_+(x_i)$ is affine n-space. The complement of each $D_+(x_i)$ is $V_+(x_i)$, a projective space of one dimension less. So, alternatively, we can write $\mathbb{P}^n_{A_0}$ as a disjoint union of locally closed subschemes as follows $\mathbb{P}^n_{A_0} = \bigcup_{i=0}^n (V(\sum_{j=0}^{i-1}(x_j)) \setminus V(\sum_{j=0}^{i}(x_j)))$

Next, we want to clarify the relation between $\operatorname{Spec} A$ and $\operatorname{Proj} A$ if A is a graded ring.

Definition 12.18. Let A be a graded ring. Then, $\operatorname{Spec} A$ is called *affine cone over* $\operatorname{Proj} A$. [25]

Recall that, for every $a, b \in A_+$ homogeneous, we have a commutative diagram

$$\begin{tikzcd} A_{(a)} \arrow[r] & A_a \\ A_{(b)} \arrow[u] \arrow[r] & A_b \arrow[u] \end{tikzcd}$$

so the morphisms of schemes $\operatorname{Spec} A_a \to \operatorname{Spec} A_{(a)}$ induced by $A_{(a)} \to A_a$ glue to a morphism of schemes

$$\operatorname{Spec} A \supseteq \bigcup_{a \in A_d, d \geq 1} D(a) = \bigcup_{a \in A_d, d \geq 1} \operatorname{Spec} A_a \to \bigcup_{a \in A_d, d \geq 1} \operatorname{Spec} A_{(a)} = \bigcup_{a \in A_d, d \geq 1} D_+(a) = \operatorname{Proj} A.$$

Note that $\mathfrak{p} \in \operatorname{Spec} A$ is contained in some $D(a)$ with $a \in A_+$ homogeneous if and only if $A_+ \subsetneq \mathfrak{p}$. So, we can write $\bigcup_{a \in A_d, d \geq 1} D(a) = \operatorname{Spec} A \setminus V(A_+)$ and we get a morphism

$$\operatorname{Spec} A \setminus V(A_+) \to \operatorname{Proj} A.$$

Example 12.19. Let us explore the morphism above in the case where $k = A_0$ is a field and $A = k[x_0, \ldots, x_n]$ is the standard graded polynomial ring. Then,

- $\operatorname{Spec} A = \mathbb{A}_k^{n+1}$,
- $\operatorname{Proj} A = \mathbb{P}_k^n$, and
- $V(A_+) = \{(x_0, \ldots, x_n)\}$ is the rational point corresponding to $0 \in k^{n+1} \cong \mathbb{A}_k^{n+1}(k)$.

Considering $\mathbb{A}_k^{n+1}$ as the affine cone over $\operatorname{Proj} A$, we obtain a morphism of schemes

$$\pi : \mathbb{A}_k^{n+1} \setminus \{(x_0, \ldots, x_n)\} \to \mathbb{P}_k^n.$$

We want to understand what π looks like on rational points. So, let $P = (x_0 - a_0, \ldots, x_n - a_n)$ be a rational point of $\mathbb{A}_k^{n+1}$. Since $P \notin V_+(A_+)$, at least one of the a_i is non-zero, say $a_0 \neq 0$. Then, $\pi(P) = \varphi^{-1}(x_0 - a_0, \ldots, x_n - a_n)$, where $\varphi : A_{(x_0)} \to A_{x_0}$ is the inclusion. We claim that

$$\varphi^{-1}(x_0 - a_0, \ldots, x_n - a_n) = (\frac{x_1}{x_0} - \frac{a_1}{a_0}, \ldots, \frac{x_n}{x_0} - \frac{a_n}{a_0}).$$

Indeed, the right hand side is a maximal ideal, so it suffices to show that it is contained in the left-hand side, which is straightforward. Under the natural identification

$$\operatorname{Spec} A_{(x_0)} \to D_+(x_0) \subseteq \operatorname{Proj} A,$$

the point $\pi(P)$ corresponds to the homogeneous prime ideal $(x_1 - \frac{a_1}{a_0} x_0, \ldots, x_n - \frac{a_n}{a_0} x_0)$. To restore the symmetry we destroyed by singling out a_0 among the a_i, we rewrite this ideal as

$$\pi(P) = (\{a_i x_j - a_j x_i\}_{i,j=0}^n).$$

Now, we observe that π is surjective, since every k-rational point of $\operatorname{Spec} A_{(x_i)}$ is of the form $(\frac{x_1}{x_0} - \frac{a_1}{a_0}, \ldots, \frac{x_n}{x_0} - \frac{a_n}{a_0})$ for suitable a_i. Moreover, our explicit description of $\pi(P)$ shows that $\pi((x_0 - a_0, \ldots, x_n - a_n)) = \pi((x_0 - b_0, \ldots, x_n - b_n))$ if and only if there exists $\lambda \in k^\times$ such that $\lambda a_i = b_i$ for all $i = 0, \ldots, n$.

[25] Despite its name, the affine cone actually depends on A and not just on $\operatorname{Proj} A$.

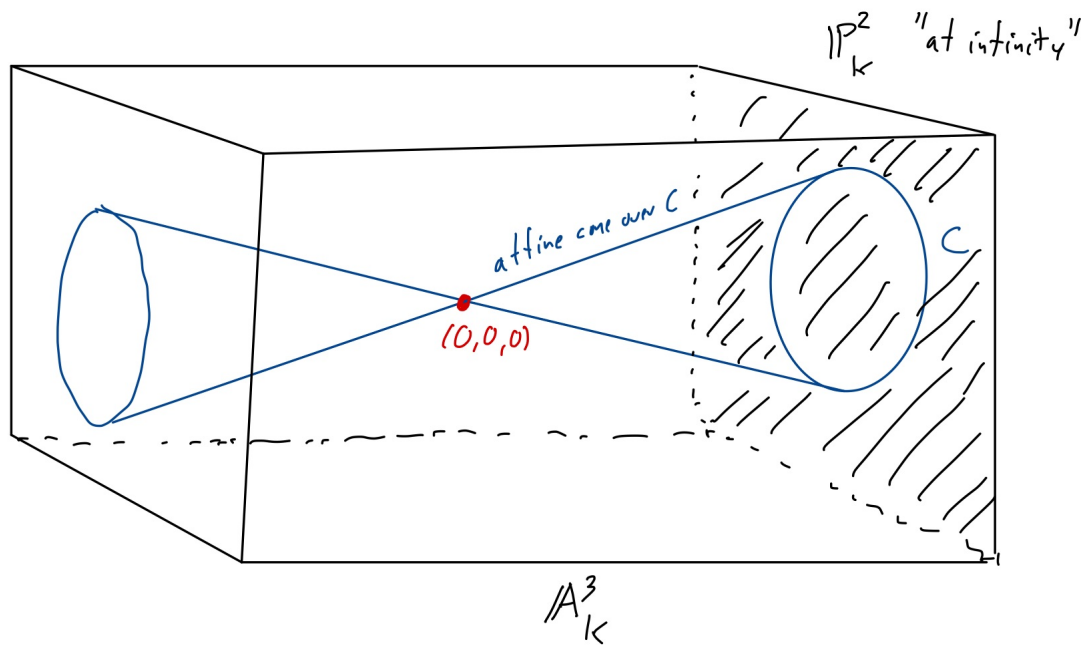
Summarizing, we have shown that the map π induces a bijection

$$(k^{n+1} \setminus \{0\})/k^\times \cong \mathbb{P}^n(k),$$

so we recover the classical description of projective space as the space of lines through the origin in k^{n+1}.

The point in $\mathbb{P}^n_k(k)$ corresponding to $(a_0, \ldots, a_n)$ under this bijection is denoted by $[a_0 : \ldots : a_n]$ and we keep in mind that for every $\lambda \in k^\times$, we have $[a_0 : \ldots : a_n] = [\lambda a_0 : \ldots : \lambda a_n]$.

Example 12.20. One can prove that taking affine cones is compatible with morphisms of projective spectra induced by graded ring homomorphisms. Using this, the reason for the name also becomes more apparent. For example, consider the graded ring homomorphism $k[x_0, x_1, x_2] \to k[x_0, x_1, x_2]/(x_2^2 - x_0x_1)$ and note that $(x_2^2 - x_0x_1)$ is the equation of what is classically called a *conic* in $\mathbb{P}^2_k$. We thus have the curve $C = \operatorname{Proj} k[x_0, x_1, x_2]/(x_2^2 - x_0x_1) \subseteq \mathbb{P}^2_k$ and the equation of the affine cone over C in $\mathbb{A}^3$ is $x_2^2 - x_0x_1$. In pictures:



If A is a graded ring, then we have seen in Corollary 12.13 that there is a natural map $A_0 \to \mathcal{O}_{\operatorname{Proj} A}(\operatorname{Proj} A)$. By Proposition 5.3, this induces a natural map $\pi_A : \operatorname{Proj} A \to \operatorname{Spec} A_0$. In the following, we want to study this morphism. First, we single out a particularly nice class of graded rings.

Definition 12.21. A graded ring A is called *finitely generated in degree* 1 if it is generated as an A_0-algebra by finitely many elements of A_1.

Lemma 12.22. *Let A be a graded ring which is finitely generated in degree* 1*. Then, there exists $n \geq 0$ and a closed immersion* $\operatorname{Proj} A \to \mathbb{P}^n_{A_0}$ *over* $\operatorname{Spec} A_0$.

Proof. Since A is finitely generated in degree 1, there exists a surjection $A_0[x_0, \ldots, x_n] \to A$, which, by Exercise 44, induces a morphism $\operatorname{Proj} A \to \mathbb{P}^n_{A_0}$. On principal opens, this morphism is induced by a surjective morphism of rings, hence it is a closed immersion by Corollary 7.15. $\square$

Remark 12.23. If A is a graded ring which is finitely generated over A_0, then $A^{(n)} := \bigoplus_{d\geq 0} A_d^{(n)}$ with $A_d^{(n)} := A_{dn}$ is finitely generated in degree 1 for a suitable $n > 0$. One can show (see Exercise 48), that there exists an isomorphism $v_n : \operatorname{Proj} A \cong \operatorname{Proj} A^{(n)}$, so Lemma 12.22 holds as soon as A is finitely generated over A_0.

We finish this lecture with one of the key applications of the valuative criterion.

Theorem 12.24 (Main theorem of elimination theory)**.** *Let A be a graded ring which is finitely generated in degree 1 over A_0. Then, $\pi_A : \operatorname{Proj} A \to \operatorname{Spec} A_0$ is proper.*

Proof. By Lemma 12.22 and Remark 12.16, we have a commutative diagram with Cartesian square of the following form

$$\begin{tikzcd} \operatorname{Proj} A \arrow[r] \arrow[dr] & \mathbb{P}^n_{A_0} \arrow[r] \arrow[d] \arrow[dr, phantom, "\square"] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d] \\ & \operatorname{Spec} A_0 \arrow[r] & \operatorname{Spec} \mathbb{Z} \end{tikzcd}$$

Since "proper" satisfies (BC) and (COMP) and closed immersions are finite, hence proper (Proposition 9.18), it suffices to show that $\pi : \mathbb{P}^n_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$ is proper.

<u>Claim 1: π is separated.</u>

It suffices to show that $D_+(x_ix_j) \xrightarrow{\Delta_\pi} D_+(x_i) \times_{\operatorname{Spec} \mathbb{Z}} D_+(x_j)$ is a closed immersion. By Corollary 7.15, we have to show that the multiplication map

$$\mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}] \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{x_0}{x_j}, \dots, \frac{x_n}{x_j}] \to \mathbb{Z}[\frac{x_0^2}{x_ix_j}, \frac{x_0x_1}{x_ix_j}, \dots, \frac{x_n^2}{x_ix_j}]$$

is surjective. This is obvious.

<u>Claim 2: π is of finite type.</u>

Recall that $\mathbb{P}^n_{\mathbb{Z}} = \cup_{i=0}^n D_+(x_i)$. Each $D_+(x_i)$ is affine, hence quasi-compact, hence so is $\mathbb{P}^n_{\mathbb{Z}}$. Since being locally of finite type is (LOCS) and $D_+(x_i) \cong \mathbb{A}^n_{\mathbb{Z}}$, we also see that π is of finite type.

<u>Claim 3: π satisfies the existence part of the valuative criterion.</u>

Let A be a valuation ring with field of fractions K. We have a commutative diagram of solid arrows of the form

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d] \\ \operatorname{Spec} A \arrow[r] \arrow[ur, dotted] & \operatorname{Spec} \mathbb{Z} \end{tikzcd} \tag{8}$$

and we have to construct the dotted arrow. Let ζ be the image of $\operatorname{Spec} K \to \mathbb{P}^n_{\mathbb{Z}}$. By induction, we may assume that $\zeta \notin V_+(x_i) \cong \mathbb{P}^{n-1}_{\mathbb{Z}}$ for any i. The morphism $\operatorname{Spec} K \to \mathbb{P}^n_{\mathbb{Z}}$ corresponds to a field extension $\varphi : k(\zeta) \to K$. Since $\zeta \notin V_+(x_i)$, we have $\zeta \in \bigcap_{i=0}^n D_+(x_i)$, so the functions $\frac{x_i}{x_j} \in \mathbb{Z}[x_0, \dots, x_n]_{(x_j)}$ restrict to elements $(\frac{x_i}{x_j})_\zeta \notin \mathfrak{m}_\zeta$. We let $f_{ij} = \varphi((\frac{x_i}{x_j})_\zeta)$ and note that these are non-zero with $f_{ij}f_{jk} = f_{ik}$.

Let $v : K^\times \to \Gamma$ be the valuation associated to A (see Exercise 35). Let $g_i = v(f_{i0})$ and choose k such that g_k is minimal. Then, $v(f_{ik}) = v(f_{i0}f_{k0}^{-1}) = v(f_{i0}) - v(f_{k0}) = g_i - g_k \geq 0$. Hence,

$f_{ik} \in A$ for all $i = 0, \ldots, n$. We define $\psi : \mathbb{Z}[\frac{x_0}{x_k}, \ldots, \frac{x_n}{x_k}] \to A$ by mapping $\frac{x_i}{x_k}$ to f_{ik}. Then, the following diagram commutes

$$\begin{tikzcd}
 & & \left(\frac{x_i}{x_k}\right)_\zeta \arrow[dl, mapsto] & \\
 & k(\zeta) \arrow[dl, "\varphi"'] & & \\
f_{ik} & K & \mathbb{Z}[\frac{x_0}{x_k}, \ldots, \frac{x_n}{x_k}] \arrow[l] \arrow[ul] \arrow[dl, "\psi"'] & \frac{x_i}{x_k} \arrow[uul, mapsto] \\
f_{ik} \arrow[u, mapsto] & A \arrow[u] & \mathbb{Z} \arrow[l] \arrow[u] \arrow[uul, bend right] &
\end{tikzcd}$$

Then, $\operatorname{Spec}(\psi)$ composed with $D_+(x_k) \to \mathbb{P}^n_{\mathbb{Z}}$ fits as the dotted arrow in Diagram (8). $\square$

Remark 12.25. Theorem 12.24 implies that the morphism $\pi : \mathbb{P}^n_{A_0} \to \operatorname{Spec} A_0$ is closed for all rings A_0. In particular, for every set of homogeneous polynomials $f_1, \ldots, f_m \in A_0[x_0, \ldots, x_n]$ there is an ideal $I \subseteq A_0$ such that for every morphism $\varphi : A_0 \to K$ to a field K with induced morphism $\varphi' : A_0[x_0, \ldots, x_n] \to K[x_0, \ldots, x_n]$ obtained by applying φ to the coefficients, the following holds:

The $\varphi(f_1), \ldots, \varphi(f_n)$ have a common solution in $\overline{K}$ (i.e. the fiber of $\pi|_{V_+(f_1,\ldots,f_m)}$ over the corresponding K-rational point of $\operatorname{Spec} A_0$ is non-empty) if and only if $\varphi(I) = 0$ (i.e. if the K-rational point lies in $V(I)$). In particular, we can see Theorem 12.24 as a generalization of the existence of the resultant of two polynomials.

13. Lectures 16 and 17: Quasi-coherent sheaves on Proj A

In the last lectures, we learned how to associate to a graded module $M = \bigoplus_{d\in\mathbb{Z}} M_d$ over a graded ring $A = \bigoplus_{d=0}^{\infty} A_d$ a quasi-coherent sheaf $\widetilde{M}$ on $\operatorname{Proj} A$. If $M \to N$ is a graded morphism of graded A-modules, then for every $a \in A_+$ homogeneous, we have compatible morphisms of $A_{(a)}$-modules $M_{(a)} \to N_{(a)}$, so, similarly to the affine case, the $\widetilde{(\ )}$-construction gives a functor

$$\widetilde{(\ )} : (\text{graded } A-\text{modules}) \to \operatorname{QCoh}(\operatorname{Proj} A, \mathcal{O}_{\operatorname{Proj} A}).$$

The goal of this lecture is to get a better understanding of this functor and to describe its essential image in the case where A is finitely generated in degree 1. We begin with the following observation:

Example 13.1. Let A be a graded ring and let $M = \bigoplus_{d\in\mathbb{Z}} M_d$ and $N = \bigoplus_{d\in\mathbb{Z}} N_d$ be graded A-modules. Assume that there exist $d_0 > 0$ such that $M_d = N_d$ for all $d \geq d_0$. Then, for all $a \in A_+$ homogeneous, the $A_{(a)}$-modules $M_{(a)}$ and $N_{(a)}$ are isomorphic via

$$\begin{array}{rcl} M_{(a)} & \to & N_{(a)} \\ \dfrac{m}{a^k} = \dfrac{ma^{d_0}}{a^{d_0+k}} & \mapsto & \dfrac{ma^{d_0}}{a^{d_0+k}} \end{array}$$

and these isomorphisms are compatible with restriction to $A_{(b)}$ with $D_+(b) \subseteq D_+(a)$, so they glue to an isomorphism $\widetilde{M} \cong \widetilde{N}$.

In particular, $\widetilde{(\ )}$ is usually not fully faithful. On the other hand, for $A = A_0[x_0, \dots, x_n]$ with the standard grading, we can recover A from $\operatorname{Proj} A$ (and the $\mathcal{O}_{\operatorname{Proj} A}(d)$) as follows.

Proposition 13.2. *Let $A = A_0[x_0, \dots, x_n]$ be the standard graded polynomial ring. Then, there is a natural isomorphism $A_d \cong \Gamma(\operatorname{Proj} A, \mathcal{O}_{\operatorname{Proj} A}(d))$.*

Proof. By Corollary 12.13, we have a natural map $\varphi : A_d \to \Gamma(\operatorname{Proj} A, \mathcal{O}_{\operatorname{Proj} A}(d))$ which, composed with restriction to $D_+(x_i)$, is

$$\begin{array}{rcl} \varphi_i : A_d & \to & A(d)_{(x_i)} = \widetilde{A(d)}|_{D_+(x_i)}(D_+(x_i)) \\ a & \mapsto & \dfrac{a}{1} \end{array}$$

We have a commutative diagram

$$\begin{tikzcd} A_d \arrow[r, "\varphi_i"] \arrow[d] & A(d)_{(x_i)} \arrow[d] \\ A \arrow[r] & A_{x_i}, \end{tikzcd}$$

where the bottom horizontal arrow is an injective localization map, so φ_i, and hence φ, is injective.

To see that φ is surjective, let $s \in \Gamma(\operatorname{Proj} A, \mathcal{O}_{\operatorname{Proj} A}(d))$ and $s_i = s|_{D_+(x_i)}$. Then, we can write $s_i = \frac{t_i}{x_i^{m_i}}$ for some $t_i \in A_0[x_0, \dots, x_n]$ homogeneous and $m_i \geq 0$ with $\deg(t_i) = m_i + d$. Without loss of generality, we may assume $x_i \nmid t_i$. On the other hand, we have $s_i|_{D_+(x_i)\cap D_+(x_j)} = s_j|_{D_+(x_i)\cap D_+(x_j)}$, so

$$\frac{t_i}{x_i^{m_i}} = \frac{t_j}{x_j^{m_j}} \in A(d)_{(x_i x_j)},$$

hence

$$t_i x_j^{m_i} = t_j x_i^{m_j} \in A.$$

Since $x_i \nmid t_i$ for all i, this implies that $m_i = 0$ for all i, hence $t_i = t_j$ for all i, j. Let $t = t_i$ and note that $t \in A_d$. By construction, $\varphi(t) = s$, so φ is surjective. □

This result suggests that we may be able to recover a module whose tilde is a quasi-coherent sheaf $\mathcal{F}$ on $\operatorname{Proj} A$ under suitable assumptions on A from the global section of certain "twists" of $\mathcal{F}$.

Definition 13.3. Let A be a graded ring and $X = \operatorname{Proj} A$. Let $\mathcal{F} \in \operatorname{QCoh}(X, \mathcal{O}_X)$.

(1) The *n-th twist of $\mathcal{F}$* is $\mathcal{F}(n) \coloneqq \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$.
(2) The *graded module associated to $\mathcal{F}$* is $\Gamma_*(\mathcal{F}) \coloneqq \bigoplus_{d \in \mathbb{Z}} \Gamma(X, \mathcal{F}(d))$.

Even though we called $\Gamma_*(\mathcal{F})$ the graded module associated to $\mathcal{F}$, we did not yet explain its A-module structure. To do this in a clean way, we need to introduce tensor products of graded modules.

Definition 13.4. Let M and N be graded modules over a graded ring A. Then, the degree d part of the graded tensor product $M \otimes_A N$ is defined as follows:

$$(M \otimes_A N)_d \coloneqq \Big(\bigoplus_{d_1 + d_2 = d} M_{d_1} \otimes_{\mathbb{Z}} M_{d_2} \Big) / (ma \otimes n - m \otimes an \mid u + v + w = d, m \in M_u, n \in N_v, a \in A_w).$$

Remark 13.5. The graded tensor product $M \otimes_A N$ satisfies the universal property of the tensor product in the category of graded A-modules.

Example 13.6. If M is a graded A-module, then $M(n) \cong M \otimes_A A(n)$.

Lemma 13.7. *Let A be a graded ring and M, N graded A-modules. Let $X = \operatorname{Proj} A$. Then, there is a natural morphism $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} \to \widetilde{M \otimes_A N}$.*

Proof. For $a \in A_+$ homogeneous, we can define

$$\begin{array}{rcl} (\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N})(D_+(a)) \cong M_{(a)} \otimes_{A_{(a)}} N_{(a)} & \to & (M \otimes N)_{(a)} \cong (\widetilde{M \otimes_A N})(D_+(a)) \\ \dfrac{m}{a^k} \otimes \dfrac{n}{a^l} & \mapsto & \dfrac{m \otimes n}{a^{k+l}} \end{array}$$

where we use Exercise 43 for the first isomorphism on the left. These maps are compatible with restriction and natural in M and N, so they glue to a natural morphism $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} \to \widetilde{M \otimes_A N}$ □

Hence, for A a graded ring, M a graded A-module, and $\mathcal{F}$ a quasi-coherent sheaf on $X = \operatorname{Proj} A$, we obtain the following natural maps:

(a) Applying Lemma 13.7 to $M = A(m), N = A(n)$:

$$\mathcal{O}_X(m) \otimes_{\mathcal{O}_X} \mathcal{O}_X(n) \to \mathcal{O}_X(m+n)$$

(b) By tensoring the map in (a) with $\mathcal{F}$:

$$\mathcal{O}_X(m) \otimes_{\mathcal{O}_X} \mathcal{F}(n) \to \mathcal{F}(m+n)$$

(c) Applying Lemma 13.7 to $N = A(n)$:

$$\widetilde{M}(n) \to \widetilde{M(n)}$$

Composing the natural maps $A_d \to \Gamma(X, \mathcal{O}_X(d))$ of Corollary 12.13 with the map on global sections from the presheaf tensor product of $\mathcal{O}_X(m)$ with $\mathcal{F}(n)$ to the sheafified tensor product $\mathcal{O}_X(m) \otimes_{\mathcal{O}_X} \mathcal{F}(n)$ and with the map on global sections induced by the natural map in (b) above, we obtain a natural graded A-module structure on $\Gamma_*(\mathcal{F})$.

In good cases, these natural morphisms are isomorphisms.

Lemma 13.8. *For every $d > 0, d \mid n$, and $a \in A_d$, we have $\mathcal{O}_X(n)|_{D_+(a)} \cong \mathcal{O}_X|_{D_+(a)}$ and the morphisms in (a),(b), and (c) above restrict to isomorphisms over $D_+(a)$.*

Proof. Multiplication by $a^{\frac{n}{d}}$ defines graded morphisms of graded A-modules $M \to M(n)$. If we invert a, this morphism becomes an isomorphism, hence $M_{(a)} \cong M(n)_{(a)}$. Now, it suffices to observe that these isomorphisms fit into a commutative diagram of the form

$$\begin{tikzcd} \widetilde{M}|_{D_+(a)} \arrow[r, "\cdot a^{\frac{n}{d}}"] \arrow[rrr, bend right, "\cdot a^{\frac{m+n}{d}}"'] & \widetilde{M(n)}|_{D_+(a)} \arrow[rr, "\mathrm{id}\otimes(\cdot a^{\frac{m}{d}})"] & & (\widetilde{M(n)} \otimes_{\mathcal{O}_X} \widetilde{A(m)})|_{D_+(a)} \arrow[r] & \widetilde{M(n+m)} \end{tikzcd}$$

where $d \mid m$ and the arrow on the right is the natural morphism in (a), (b), or (c) above for suitable choices of $M, n,$ and m. □

Corollary 13.9. *Let A be a graded ring which is generated in degree 1. Let $X = \operatorname{Proj} A$. Then,*

(1) $\mathcal{O}_X(n)$ is an invertible sheaf for all $n \in \mathbb{Z}$.
(2) There is a natural isomorphism $\mathcal{O}_X(n_1) \otimes_{\mathcal{O}_X} \mathcal{O}_X(n_2) \cong \mathcal{O}_X(n_1 + n_2)$ for all $n_1, n_2 \in \mathbb{Z}$.
(3) There is a natural isomorphism $\widetilde{M(n)} \cong \widetilde{M}(n)$ for all $n \in \mathbb{Z}$.

Recall from Corollary 12.13 that there are natural morphisms $M_d \to \Gamma(\operatorname{Proj} A, \widetilde{M(d)})$. Using Corollary 13.9, we obtain a natural map $\alpha : M \to \Gamma_*(\widetilde{M})$.

Definition 13.10. Let A be a graded ring which is generated in degree 1. An A-module M is called *saturated* if the natural map $\alpha : M \to \Gamma_*(\widetilde{M})$ is an isomorphism.

Example 13.11. In Proposition 13.2, we have proved that $A = A_0[x_0, \dots, x_n]$ with the standard grading is saturated as a module over itself.

We let $(\mathrm{gr.\ sat.}\ A - \mathrm{Mod})$ be the category of graded saturated A-modules.

Theorem 13.12. *Let A be a graded ring which is finitely generated in degree 1 and let $X = \operatorname{Proj} A$. Then, the functors*

$$\begin{array}{rcl} (\mathrm{gr.\ sat.}\ A - \mathrm{Mod}) & \leftrightarrow & \mathrm{QCoh}(X, \mathcal{O}_X) \\ M & \mapsto & \widetilde{M} \\ \Gamma_*(\mathcal{F}) & \leftarrow\!\shortmid & \mathcal{F} \end{array}$$

are essential inverses to each other. In particular, they induce an equivalence of categories.

Proof. Since A is finitely generated in degree 1, X admits a closed immersion to some $\mathbb{P}^n_{A_0}$ by Lemma 12.22, so X is quasi-compact and quasi-separated.

For any $\mathcal{F} \in \mathrm{QCoh}(X, \mathcal{O}_X)$, $a \in A_d$, and $d \geq 1$, we define

$$\begin{aligned} \Gamma_*(\mathcal{F})_{(a)} = (\bigoplus_{n\in\mathbb{Z}} \Gamma(X, \mathcal{F}(n)))_{(a)} &\to \Gamma(D_+(a), \mathcal{F}) \\ \frac{m}{a^n} &\mapsto a^{-n} \cdot (m|_{D_+(a)})^{26} \end{aligned}$$

These morphisms are compatible with restriction to other principal opens and they are natural, hence, by Lemma 5.2, they glue to a natural morphism of sheaves

$$\beta : \widetilde{\Gamma_*(\mathcal{F})} \to \mathcal{F}.$$

Claim: β is an isomorphism.

To prove this, it suffices to check that $\beta(D_+(a))$ is bijective where $a \in A_1$. We set $s = \alpha(a)$ and note

$$\begin{aligned} X_s &:= \{x \in X \mid s_x \notin \mathfrak{m}_x \mathcal{O}_X(1)\} = \bigcup_{b\in A_1} \{x \in D_+(b) \mid \left(\frac{a}{b}\right)_x \notin \mathfrak{m}_x\} \\ &= \bigcup_{b\in A_1} (D_+(a) \cap D_+(b)) = D_+(a), \end{aligned}$$

so we can use the contents of Exercise 47.

For the proof of surjectivity, pick $m' \in \Gamma(D_+(a), \mathcal{F})$. Then, by Exercise 47 (iii), there exists $n > 0$ and $m \in \Gamma(X, \mathcal{F}(n))$ such that $m|_{D_+(a)} = m' \otimes s|_{D_+(a)}^{\otimes n} = a^n \cdot m'$. In other words, $\beta(D_+(a))(\frac{m}{a^n}) = m'$.

For the proof of injectivity, let $\frac{m}{a^n} \in \Gamma_*(X, \mathcal{F})_{(a)}$ with $a^{-n} \cdot (m|_{D_+(a)}) = 0 \in \Gamma(D_+(a), \mathcal{F})$. Then, $m|_{D_+(a)} = 0 \in \Gamma(D_+(a), \mathcal{F}(n))$. By Exercise 47 (ii), there exists $n' > 0$ such that $a^{n'} \cdot m = m \otimes s^{\otimes n'} = 0 \in \Gamma(X, \mathcal{F}(n + n'))$. Hence, $\frac{m}{a^n} = 0 \in \Gamma_*(\mathcal{F})_{(a)}$. This finishes the proof of the claim.

Now, note that, by definition of α and β, the composition

$$\Gamma_*(\mathcal{F}) \xrightarrow{\alpha} \Gamma_*(\widetilde{\Gamma_*(\mathcal{F})}) \xrightarrow{\Gamma_*(\beta)} \Gamma_*(\mathcal{F})$$

is the identity. In particular, α is an isomorphism, that is, for every quasi-coherent sheaf on X, the associated graded module is saturated. This finishes the proof, since now both Γ_* and $\widetilde{(\quad)}$ are well-defined and we have natural isomorphisms $\alpha : M \to \Gamma_*(\widetilde{M})$ and $\beta : \widetilde{\Gamma_*(\mathcal{F})} \to \mathcal{F}$ for all saturated graded A-modules M and all quasi-coherent sheaves $\mathcal{F}$ on X. □

Corollary 13.13. *Let A be a graded ring which is finitely generated in degree* 1.

(1) *For every closed subscheme $Z \subseteq \mathrm{Proj}\, A$, given by an ideal sheaf $\mathcal{I}_Z$, there exists a homogeneous ideal $I_Z \subseteq A$ such that $Z \cong \mathrm{Proj}\, A/I_Z$ and such that $Z \to \mathrm{Proj}\, A$ is induced by $A \to A/I_Z$.*

[26] Recall that multiplication by a^n defines a morphism $\Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}(dn))$ which becomes invertible when inverting a, so it makes sense to take "$a^{-n} \cdot (m|_{D_+(a)})$" to mean taking the image of $m|_{D_+(a)} \in \Gamma(D_+(a), \mathcal{F}(dn))$ under the inverse of the isomorphism $\cdot a^n : \Gamma(D_+(a), \mathcal{F}) \to \Gamma(D_+(a), \mathcal{F}(dn))$

(2) *If A is saturated as an A-module, then we can choose $I_Z \cong \Gamma_*(\mathcal{I}_Z)$ and then I_Z is the unique saturated homogeneous ideal $I_Z \subseteq A$ such that $Z = \operatorname{Proj} A/I_Z$.*

Proof. If $\alpha : A \to \Gamma_*(\mathcal{O}_X)$ is the natural map on $X = \operatorname{Proj} A$, we can let $I_Z := \alpha^{-1}(\Gamma_*(\mathcal{I}_Z))$ in (1). In (2), α is an isomorphism, hence $I_Z \cong \Gamma_*(\mathcal{I}_Z)$ and the uniqueness follows from Theorem 13.12. □

Example 13.14. By Proposition 13.2, Corollary 13.13 (2) applies in particular to projective space $\mathbb{P}^n_{A_0} = \operatorname{Proj} A = \operatorname{Proj} A_0[x_0, \dots, x_n]$. Hence, we know that the closed subschemes of $\mathbb{P}^n_{A_0}$ correspond bijectively to saturated homogeneous ideals.

If $I \subseteq A$ is any homogeneous ideal, then we call $I^{\mathrm{sat}} := \Gamma_*(\widetilde{I})$ the saturation of I. Since $\widetilde{I}|_{D_+(x_i)} = I_{(x_i)}$ and $\widetilde{I}(d)(\operatorname{Proj} A) \subseteq A_d$, we can write

$$I^{\mathrm{sat}} = \{f \in A_0[x_0, \dots, x_n] \mid \exists N > 0 \text{ such that for all } i = 0, \dots, n : x_i^N f \in I\}.$$

The natural morphism $\operatorname{Proj} A/I^{\mathrm{sat}} \to \operatorname{Proj} A/I$ induced by the quotient map is an isomorphism, as it is one on each $D_+(x_i)$. In particular, choosing any ideal J with saturation I^{sat} will equip the closed subset $V_+(I)$ with the same scheme structure.

Example 13.15. If $f \in A_0[x_0, \dots, x_n]$ is a homogeneous polynomial, then the ideals (f) and $(fx_0, fx_1, \dots, fx_n)$ have the same saturation, hence they define the same closed subscheme of $\mathbb{P}^n_{A_0}$.

14. Lecture 18: Morphisms to projective space via invertible sheaves

In Lecture 5, we studied morphisms from an arbitrary scheme X to affine schemes. In particular, we saw in Proposition 5.3 that giving a morphism of schemes $X \to \operatorname{Spec} A$ is the same as giving a morphism of rings $A \to \mathcal{O}_X(X)$. In particular, as a special case, and using the universal property of the fiber product, we deduce that if X is a scheme over $\operatorname{Spec} A$, then giving a morphism $X \to \mathbb{A}^n_A$ over $\operatorname{Spec} A$ is the same as giving a morphism to $X \to \mathbb{A}^n_{\mathbb{Z}}$, which, in turn, is the same as giving n global sections of $\mathcal{O}_X(X)$. It turns out that morphisms to $\mathbb{P}^n_A$ can be described similarly, using global sections of invertible sheaves.

If $f : X \to Y$ is a morphism of schemes and $\mathcal{F}$ is a quasi-coherent sheaf on Y, then the unit of the adjunction between f_* and f^* (Exercise 37) gives a natural map $\mathcal{F} \to f_* f^* \mathcal{F}$. Taking global sections of this morphism, we obtain the *pullback of sections*

$$f^* : \Gamma(Y, \mathcal{F}) \to \Gamma(X, f^*\mathcal{F}).$$

More explicitly, this is defined as

$$\mathcal{F}(Y) \to (f^{-1}\mathcal{F})(X) \otimes_{(f^{-1}\mathcal{O}_Y)(X)} \mathcal{O}_X(X) \to f^*\mathcal{F}(X),$$

where the first map sends s to $s' \otimes 1$, where s' is the equivalence class of s in the direct limit $(f^{-1}\mathcal{F})(X) = \varinjlim_{f(X) \subseteq V} \mathcal{F}(V)$, and the second map exists by definition of $f^*\mathcal{F}$ as the tensor product of $f^{-1}\mathcal{F}$ and $\mathcal{O}_X$.

Proposition 14.1. *Let A be a ring and let $f : X \to \mathbb{P}^n_A$ be a morphism of schemes over $\operatorname{Spec} A$. Then, the following hold:*

(1) The sheaf $\mathcal{L} := f^*\mathcal{O}_{\mathbb{P}^n_A}(1)$ is invertible.
(2) For $s_i := f^* x_i \in \Gamma(X, \mathcal{L})$, the germs $(s_i)_x$ generate $\mathcal{L}_x$ as an $\mathcal{O}_{X,x}$-module for all $x \in X$.

Proof. Claim (1) follows from the general fact that the inverse image of an invertible sheaf under any morphism of schemes is again invertible. Indeed, it suffices to check this for morphisms between affine schemes $\operatorname{Spec} B \to \operatorname{Spec} A$, where it follows from the fact that $A \otimes_A B \cong B$.

By Lemma 4.5 and Exercise 43, we have a natural isomorphism $\mathcal{L}_x \cong (\mathcal{O}_{\mathbb{P}^n}(1))_{f(x)} \otimes_{\mathcal{O}_{\mathbb{P}^n, f(x)}} \mathcal{O}_{X,x}$. Using our explicit description of $f^* x_i$ above, one checks that, under this isomorphism, $(s_i)_x = (x_i)_{f(x)} \otimes 1$. So, it suffices to check that the germs of the x_i on $\mathbb{P}^n$ generate the stalk $\mathcal{O}_{\mathbb{P}^n}(1)$ at every point. This is clear. $\square$

The goal of this lecture is to show that, conversely, the datum of an invertible sheaf and generating global sections is enough to define a morphism to projective space and to study the properties of this induced morphism in terms of $\mathcal{L}$ and the s_i. Note that if $s \in \Gamma(X, \mathcal{F})$ is a global section of a quasi-coherent sheaf $\mathcal{F}$ on X, then there is an associated morphism $\mathcal{O}_X \to \mathcal{F}$ which, on every open $U \subseteq X$, is given by multiplication with s. Conversely, given a morphism $\varphi : \mathcal{O}_X \to \mathcal{F}$ of $\mathcal{O}_X$-modules, we obtain an associated global section $s = \varphi(X)(1)$. These two constructions are inverse to each other. In other words, there is a natural isomorphism of $\mathcal{O}_X(X)$-modules

$$\Gamma(X, \mathcal{F}) \cong \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}).$$

Definition 14.2. Let $\mathcal{F}$ be a quasi-coherent sheaf on a scheme X.

(1) A collection of global sections $\{s_i\}_{i \in I}$ of $\mathcal{F}$ is said to *generate* $\mathcal{F}$ if the associated morphism $\bigoplus_{i \in I} \mathcal{O}_X \to \mathcal{F}$ is surjective.

(2) The sheaf $\mathcal{F}$ is called *(finitely) globally generated* if it can be generated by (finitely many) global sections.

Recall that we call a morphism of sheaves surjective if it is surjective on stalks, so Condition (2) in Proposition 14.1 can be rephrased as saying that the $s_i = f^*(x_i)$ generate $\mathcal{L}$. For the following theorem, we call two $(n+2)$-tuples $(\mathcal{L}, s_0, \dots, s_n)$ and $(\mathcal{M}, t_0, \dots, t_n)$ isomorphic if there exists an isomorphism $\varphi : \mathcal{L} \to \mathcal{M}$ such that $\varphi(X)(s_i) = t_i$ for all i.

Theorem 14.3. *Let X be a scheme over* $\operatorname{Spec} A$. *Then, the association*

$$\Theta : (f : X \to \mathbb{P}^n_A) \mapsto (f^*\mathcal{O}_{\mathbb{P}^n_A}(1), f^*x_0, \dots, f^*x_n)$$

defines a natural bijection between $\operatorname{Hom}_{(\mathrm{Sch}/A)}(X, \mathbb{P}^n_A)$ *and the set of isomorphism classes of $(n+2)$-tuples $(\mathcal{L}, s_0, \dots, s_n)$, where $\mathcal{L}$ is an invertible sheaf on X and $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$ are global sections that generate $\mathcal{L}$.*

Proof. Let $f : X \to \mathbb{P}^n_A$ be any morphism of schemes. Then,

$$X_i := \{x \in X \mid (f^*(x_i))_x \notin \mathfrak{m}_x\mathcal{L}_x\} = \{x \in X \mid (x_i)_{f(x)} \notin \mathfrak{m}_{f(x)}\mathcal{O}_{\mathbb{P}^n_A}(1)_{f(x)}\} = f^{-1}(D_+(x_i)).$$

The induced map $f|_{X_i} : X_i \to D_+(x_i)$ is determined by a morphism of rings

$$\begin{array}{rcl} A[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}] & \to & \Gamma(X_i, \mathcal{O}_{X_i}) \\ \frac{x_j}{x_i} & \mapsto & s_{ji}. \end{array}$$

By naturality of the pullback of sections, this morphism is part of the following commutative diagram

$$\begin{tikzcd} A[x_0, \dots, x_n]_{(x_i)} \arrow[r, "\cong"] & \Gamma(D_+(x_i), \mathcal{O}_{\mathbb{P}^n}) \arrow[r] \arrow[d, "\cdot x_i"] & \Gamma(X_i, \mathcal{O}_X) \arrow[d, "\cdot f^*x_i"] \\ A[x_0, \dots, x_n](1)_{(x_i)} \arrow[r, "\cong"] & \Gamma(D_+(x_i), \mathcal{O}_{\mathbb{P}^n}(1)) \arrow[r, "f^*"] & \Gamma(X_i, f^*\mathcal{O}_{\mathbb{P}^n}(1)). \end{tikzcd}$$

The vertical arrows are isomorphisms, hence $s_{ji} = (f^*x_j)|_{X_i} \cdot (f^*x_i)|_{X_i}^{-1}$.

In particular, note that the s_{ji} are uniquely determined by the isomorphism class of the $(n+2)$-tuple $(f^*\mathcal{O}_{\mathbb{P}^n_A}(1), f^*x_0, \dots, f^*x_n)$, hence the morphism f is uniquely determined by this isomorphism class, so Θ is injective.

To see that Θ is surjective, let $\mathcal{L}$ and $s_0, \dots, s_n$ are as in the statement of the theorem. We set $X_{s_i} := \{x \in X \mid (s_i)_x \notin \mathfrak{m}_x\mathcal{L}_x\}$ and define $f_i : X_{s_i} \to \mathbb{P}^n_A$ to be the morphism induced by the A-linear map defined via

$$\begin{array}{rcl} A[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}] & \to & \Gamma(X_{s_i}, \mathcal{O}_{X_{s_i}}) \\ \frac{x_j}{x_i} & \mapsto & s_j|_{X_{s_i}} \cdot (s_i|_{X_{s_i}})^{-1}. \end{array}$$

The f_i coincide on $X_{s_i} \cap X_{s_j}$, hence they glue to a morphism of schemes $f : X \to \mathbb{P}^n_A$. Using this explicit description (and Exercise 3), one can show that there is an isomorphism $f^*\mathcal{O}_{\mathbb{P}^n_A}(1) \cong \mathcal{L}$ that sends f^*x_i to s_i, hence this f is indeed a preimage of the $(n+2)$-tuple $(\mathcal{L}, s_0, \dots, s_n)$ under Θ. We omit the proof of naturality of Θ. $\square$

Remark 14.4. By the Yoneda Lemma, we can take Theorem 14.3 as the definition of projective space.

Remark 14.5. If we let $X = \operatorname{Spec} k$ with k a field in Theorem 14.3, we obtain a bijection between k-valued points of $\mathbb{P}^n_A$ and 1-dimensional quotients of k^{n+1}. Taking duals, we see once again that there is a bijection between lines through the origin in k^{n+1} and k-rational points of projective n-space (compare this with Example 12.19).

Remark 14.6. Going through the proof of Theorem 14.3, one can check that the morphism $f : X \to \mathbb{P}^n_A$ associated to $(\mathcal{L}, s_0, \ldots, s_n)$ factors through $V_+(I) \subseteq \mathbb{P}^n_A$ with I a homogeneous saturated ideal if and only if $g(s_0, \ldots, s_n) = 0$ for all $g \in I$. In particular, $f(X)$ is contained in a hyperplane, i.e. in $V_+(g)$ for some $g \in A[x_0, \ldots, x_n]_1$, if and only if the s_i are A-linearly dependent.

Remark 14.7. Even if the s_i do not generate $\mathcal{L}$, the proof of Theorem 14.3 gives a morphism $X \supseteq \bigcup X_{s_i} \to \mathbb{P}^n_A$.

Remark 14.8. If $A = k$ is a field and we replace the s_i by t_i with $\langle s_0, \ldots, s_n\rangle_k = \langle t_0, \ldots, t_n\rangle_k$, then the two induced morphisms $X \to \mathbb{P}^n_k$ differ by an automorphism of $\mathbb{P}^n_k$.

Remark 14.9. If $A = k$ is a field, then the morphism $X \to \mathbb{P}^n_k$ induced by $s_0, \ldots, s_n$ is given on k-rational points by $x \mapsto [s_0(x) : \ldots : s_n(x)]$, where $s_i(x)$ is the image of s_i in $\mathcal{L}_x/\mathfrak{m}_x\mathcal{L}_x \cong k$. Here, the choice of isomorphism does not matter, since any two isomorphisms differ by multiplication with a scalar and $[s_0(x) : \ldots : s_n(x)] = [\lambda s_0(x) : \ldots : \lambda s_n(x)]$ for all $\lambda \in k^\times$.

Example 14.10. If $A = k$ is a field, $X = \mathbb{P}^{n+1}_k$, $\mathcal{L} = \mathcal{O}_X(1)$, and $(s_0, \ldots, s_n) = (x_0, \ldots, x_n)$, then the s_i generate $\mathcal{L}$ outside the k-rational point $[0 : \ldots : 0 : 1]$ and the morphism $f : X \setminus \{[0 : \ldots : 0 : 1]\} \to \mathbb{P}^n_k$ corresponds to the projection from a point of Exercise 50.

Next, it is interesting to ask when the morphism of Theorem 14.3 is a closed immersion. The existence of such a closed immersion would allow us to describe X as the projective spectrum of a graded ring and then all the results of the previous lectures could be applied to X.

Proposition 14.11. *Let X be a scheme over A and let $f : X \to \mathbb{P}^n_A$ be the morphism induced by an invertible sheaf $\mathcal{L}$ and generating global sections $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$. Then, f is a closed immersion if and only if the following two conditions hold:*

(1) Each X_{s_i} is affine.
(2) For each i, the A-linear map defined by

$$\begin{array}{rcl} A[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}] & \to & \Gamma(X_{s_i}, \mathcal{O}_{X_{s_i}}) \\ \frac{x_j}{x_i} & \mapsto & s_j|_{X_{s_i}} \cdot (s_i|_{X_{s_i}})^{-1}. \end{array}$$

is surjective.

Proof. First, recall from the proof of Theorem 14.3 that $X_{s_i} = f^{-1}(D_+(x_i))$. Then, since being a closed immersion is (LOCT), f is a closed immersion if and only if $f_{D_+(x_i)}$ is a closed immersion. By Corollary 7.15, the latter holds if and only if both conditions in the proposition are satisfied. □

Even though the above Proposition 14.11 gives a criterion for a morphism to projective space to be a closed immersion, this criterion is hard to check in practice. Under stronger assumptions, we can give a local criterion. For this, we need three preliminary lemmas.

Lemma 14.12. *Let X be a scheme locally of finite type over a field k. Let $x \in X$ be a point. Then, the following are equivalent:*

(1) $x \in X$ is closed.
(2) $k \hookrightarrow k(x)$ is a finite field extension.
(3) $k \hookrightarrow k(x)$ is an algebraic field extension.

In particular, if k is algebraically closed, then the set of closed points of X is exactly the set of k-rational points $X(k)$.

Proof. $(1) \Rightarrow (2)$: If $x \in X$ is closed, we find $\operatorname{Spec} A \subseteq X$ open affine with $x \in \operatorname{Spec} A$. Then, x corresponds to a maximal ideal $\mathfrak{m}$ in $\operatorname{Spec} A$. By Hilbert's Nullstellensatz, $k(x) = A/\mathfrak{m}$ is a finite extension of k.

$(2) \Rightarrow (3)$: This is clear.

$(3) \Rightarrow (1)$: For $\operatorname{Spec} A \subseteq X$ open affine with $x \in \operatorname{Spec} A$, the point x corresponds to a prime ideal $\mathfrak{p} \in A$. Consider the composition

$$k \hookrightarrow A \to A/\mathfrak{p} \hookrightarrow \operatorname{Frac}(A/\mathfrak{p}) = k(x).$$

Since $k(x)$ is algebraic over k, the ring $A/\mathfrak{p}$ is integral over k. But an integral domain which is integral over a field is a field [27], so $A/\mathfrak{p}$ is a field, hence x is closed in $\operatorname{Spec} A$. Since this holds for all affine open neighborhoods of x in X, x is also closed in X. □

In particular, note that every scheme X locally of finite type over a field k contains a closed point: If $X = \operatorname{Spec} A$ is affine, this amounts to the existence of a maximal ideal. For general X, Lemma 14.12 shows that a point x which is closed in some open affine in X is closed in all open affines since $k(x)$ does not depend on the chosen neighborhood, so x is closed in X. This implies that certain properties of morphisms can be checked on closed points in this setting:

Lemma 14.13. *Let $f : X \to Y$ be a closed morphism of schemes locally of finite type over a field k. Then, f is injective if and only if it is injective on closed points.*

Proof. Clearly, if f is injective, then it is injective on closed points. Conversely, assume that f is injective on closed points and let $x, y \in X$ be arbitrary points with $f(x) = f(y)$. Then,

$$f(\overline{\{x\}}) = \overline{\{f(x)\}} = \overline{\{f(y)\}} = f(\overline{\{y\}}).$$

We claim that the sets of closed points $\overline{\{x\}}_{\mathrm{cl}} \subseteq \overline{\{x\}}$ and $\overline{\{y\}}_{\mathrm{cl}} \subseteq \overline{\{y\}}$ coincide.

So, let $z \in \overline{\{x\}}_{\mathrm{cl}}$. Then, $f(z) \in f(\overline{\{x\}}) = f(\overline{\{y\}})$. Hence, there exists $v \in \overline{\{y\}}$ such that $f(v) = f(z)$. Since $f(z)$ is closed and f is a closed map, we have $\overline{\{v\}} \subseteq f^{-1}(f(z))$. Therefore, there exists $w \in \overline{\{v\}}_{\mathrm{cl}} \subseteq \overline{\{y\}}_{\mathrm{cl}}$ such that $f(w) = f(v) = f(z)$. Now, since w and z are both closed points, we deduce that $w = z$. Hence, $z \in \overline{\{y\}}_{\mathrm{cl}}$. Repeating this for all points in $\overline{\{x\}}_{\mathrm{cl}}$ and $\overline{\{y\}}_{\mathrm{cl}}$, we obtain $\overline{\{x\}}_{\mathrm{cl}} = \overline{\{y\}}_{\mathrm{cl}}$.

Now, take any open affine $\operatorname{Spec} A \subseteq X$, where, by assumption, A is a finitely generated algebra over k. Then, A is *Jacobson*, which implies that every closed subset of $\operatorname{Spec} A$ is the closure of its set of closed points. Thus,

$$\overline{\{x\}} \cap \operatorname{Spec} A = \overline{\overline{\{x\}}_{cl} \cap \operatorname{Spec} A} = \overline{\overline{\{y\}}_{cl} \cap \operatorname{Spec} A} = \overline{\{y\}} \cap \operatorname{Spec} A.$$

[27] If A is an integral domain which is integral over a field k and $a \in A$, then multiplication by a is injective on $k(a) \subseteq A$. Since a is integral over k, $k(a)$ is finite-dimensional over k, so every injective endomorphism of $k(a)$ is an isomorphism, hence multiplication by a is invertible in A, so there exists an inverse to a.

As $\operatorname{Spec} A$ was arbitrary, we get that $\overline{\{x\}} = \overline{\{y\}}$. Since x and y are the unique generic points of their closure, we deduce that $x = y$. $\square$

Lemma 14.14. *Let $\varphi : A \to B$ be a local homomorphism of Noetherian local rings. Assume that the following conditions hold:*

(1) The induced map $A/\mathfrak{m}_A \to B/\mathfrak{m}_B$ is an isomorphism.
(2) The induced map $\mathfrak{m}_A \to \mathfrak{m}_B/\mathfrak{m}_B^2$ is surjective.
(3) B is finitely generated as an A-module.

Then, φ is surjective.

Proof. Let $I = \mathfrak{m}_A B \subseteq B$. Since φ is local, we have $I \subseteq \mathfrak{m}_B$, and by Assumption (2), the induced map $I \to \mathfrak{m}_B/\mathfrak{m}_B^2$ is surjective, hence $\mathfrak{m}_B = I + \mathfrak{m}_B^2$. Since B is Noetherian, $\mathfrak{m}_B$ is finitely generated, so Nakayama's lemma shows $I = \mathfrak{m}_B$.

Now, by Assumption (1), the element $1 \in B$ generates $B/\mathfrak{m}_A B = B/\mathfrak{m}_B B = A/\mathfrak{m}_A$ as an A-module. By Assumption (3), B is a finitely generated A-module, hence Nakayama's lemma shows that $1 \in B$ also generates B as an A-module. Hence, φ is surjective. $\square$

Now, we are ready to prove the local criterion.

Proposition 14.15. *Let X be a proper scheme over an algebraically closed field k. Let $f : X \to \mathbb{P}^n_k$ be the morphism induced by an invertible sheaf $\mathcal{L}$ and generating global sections $s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$. Assume that $f_*\mathcal{O}_X$ is coherent* [28] *and let $V = \langle s_0, \ldots, s_n \rangle \subseteq \Gamma(X, \mathcal{L})$. Then, f is a closed immersion if and only if the following two conditions hold:*

(1) Elements of V separate points*, i.e., for any two distinct closed points $x, y \in X$, there exists $s \in V$ with $s_x \in \mathfrak{m}_x\mathcal{L}_x$ but $s_y \notin \mathfrak{m}_y\mathcal{L}_y$ or vice versa.*
(2) Elements of V separate tangent vectors*, i.e., for each closed point $x \in X$, the set $\{s \in V \mid s_x \in \mathfrak{m}_x\mathcal{L}_x\}$ spans the k-vector space $\mathfrak{m}_x\mathcal{L}_x/\mathfrak{m}_x^2\mathcal{L}_x$.*

Proof. By Lemma 14.12, the closed points of X and $\mathbb{P}^n_k$ are exactly the k-rational points.

First, assume that f is a closed immersion.

Let $x, y \in X$ be closed points with $f(x) = [a_0 : \ldots : a_n]$ and $f(y) = [b_0 : \ldots : b_n]$. Since f is injective, we have $f(x) \neq f(y)$. Choose a linear form $\ell = \sum_{i=0}^n \alpha_i x_i$ such that $\ell(f(x)) = 0$ and $\ell(f(y)) \neq 0$. Then, $(f^*\ell)_x \in \mathfrak{m}_x\mathcal{L}_x$ and $(f^*\ell)_y \notin \mathfrak{m}_y\mathcal{L}_y$. Hence, elements of V separate points.

To see that elements of V separate tangent vectors, choose $x \in X$. After postcomposing with an automorphism of $\mathbb{P}^n_k$, we may assume that $f(x) = [1 : 0 : \ldots : 0]$. Then, $x \in X_{s_0}$. By our construction of, the map $f|_{X_{s_0}} : X_{s_0} \to D_+(x_0)$ induces

$$\begin{aligned} \langle \frac{\overline{x_1}}{x_0}, \ldots, \frac{\overline{x_n}}{x_0} \rangle = \mathfrak{m}_{\mathbb{P}^n, f(x)}/\mathfrak{m}^2_{\mathbb{P}^n, f(x)} &\to \mathfrak{m}_x/\mathfrak{m}_x^2 \xrightarrow{\cdot s_0} \mathfrak{m}_x\mathcal{L}_x/\mathfrak{m}_x^2\mathcal{L}_x \\ \frac{\overline{x_i}}{x_0} &\mapsto \frac{\overline{s_i}}{s_0} \mapsto \overline{s_i} \end{aligned}$$

The first map is surjective because f is a closed immersion and the second map is an isomorphism because multiplication by s_0 is.

Conversely, assume that the collection $\mathcal{L}, s_0, \ldots, s_n$ satisfies (1) and (2).

Claim 1: f is a homeomorphism onto a closed subset.

[28] As mentioned in Proposition 11.19, this follows from the assumption that X is proper over $\operatorname{Spec} k$.

Since X and $\mathbb{P}^n_k$ are proper over $\operatorname{Spec} k$, so is f. Hence, f is closed so by Lemma 14.13 it suffices to check that f is injective on rational points. So, let $x, y \in X$ with $[s_0(x) : \ldots : s_n(x)] = [s_0(y) : \ldots : s_n(y)]$. Then, for all $s \in V$, we have $s(x) = 0$ if and only if $s(y) = 0$, i.e., $s_x \in \mathfrak{m}_x \mathcal{L}_x$ if and only if $s_y \in \mathfrak{m}_y \mathcal{L}_y$. Since V separates points, this implies $x = y$.

Claim 2: $f^\sharp : \mathcal{O}_{\mathbb{P}^n} \to f_* \mathcal{O}_X$ is surjective.

Since $f_*\mathcal{O}_X$ is coherent, so is $\operatorname{Coker}(f^\sharp)$. By Exercise 41, the support of $\operatorname{Coker}(f^\sharp)$ is closed, so $\operatorname{Coker}(f^\sharp) = 0$ if and only if $\operatorname{Coker}(f^\sharp)_x = \operatorname{Coker}(f^\sharp_x) = 0$ for all closed points $x \in X$. Hence, it suffices to show that $f^\sharp_x : \mathcal{O}_{\mathbb{P}^n, f(x)} \to \mathcal{O}_{X,x}$ is surjective for all closed points $x \in X$. Now, we note that

(i) $f^\sharp_x$ induces an isomorphism of residue fields, since both x and $f(x)$ are closed points, hence k-rational (Lemma 14.12).
(ii) The induced map $\mathfrak{m}_{\mathbb{P}^n, f(x)} \to \mathfrak{m}_x/\mathfrak{m}_x^2 \cong \mathfrak{m}_x \mathcal{L}_x / \mathfrak{m}_x^2 \mathcal{L}_x$ is surjective, since elements of V separate tangent vectors.
(iii) $\mathcal{O}_{X,x}$ is finitely generated as an $\mathcal{O}_{\mathbb{P}^n, f(x)}$-module, since $f_*\mathcal{O}_X$ is coherent.

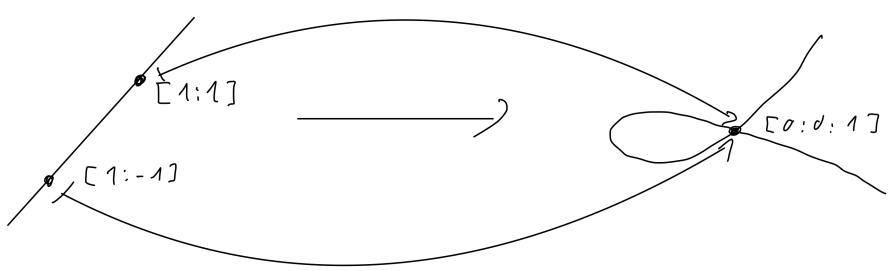
Therefore, we can apply Lemma 14.14 to deduce that $f^\sharp_x$ is surjective, as desired. □

Example 14.16. Let $A = k$ be an algebraically closed field. In the following, we will study maps $\mathbb{P}^n \to \mathbb{P}^m$ given by global sections $s_0, \ldots, s_m$ of some $\mathcal{O}(d)$ with $d \geq 0$. We will write $[x_0, \ldots, x_n] \mapsto [s_0(x_0, \ldots, x_n), \ldots, s_m(x_0, \ldots, x_n)]$, as this is exactly what such a morphism does to closed points. [29]

(1) Assume $\operatorname{char}(k) \neq 2$. Consider the morphism

$$\begin{array}{rcl} \mathbb{P}^1 & \to & \mathbb{P}^2 \\ [x : y] & \mapsto & [x^2 y - y^3 : x^3 - xy^2 : y^3] \end{array}$$

The associated space of global sections of $\mathcal{O}(3)$ does not separate $[1 : 1]$ and $[1 : -1]$. Both $[1 : -1]$ and $[1 : 1]$ are mapped to $[0 : 0 : 1]$.

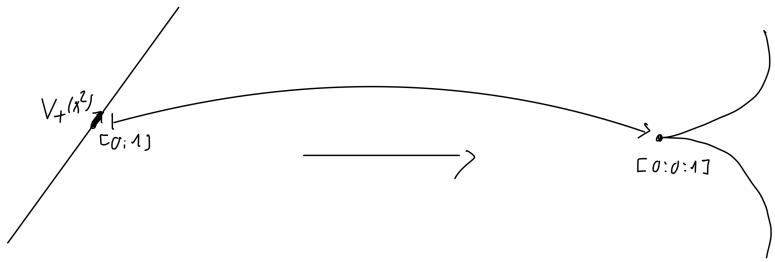


(2) Consider the morphism

$$\begin{array}{rcl} \mathbb{P}^1 & \to & \mathbb{P}^2 \\ [x : y] & \mapsto & [x^3 : x^2 y : y^3] \end{array}$$

The associated space of global sections of $\mathcal{O}(3)$ does not separate tangent vectors at $[0 : 1]$. The subscheme $V_+(x^2) \subseteq \mathbb{P}^1$ is "pinched" to its image $[0 : 0 : 1]$.

[29] Later, we will see as a consequence of $\operatorname{Pic}(\mathbb{P}^n) \cong \mathbb{Z}$ that every morphism between projective spaces is of this form.


$V_+(x^2)$
$[0:1]$
$[0:0:1]$

15. Lecture 19: (Quasi-)projective morphisms and (very) ample invertible sheaves

In the last lecture, we learned about the importance of globally generated invertible sheaves for the study of morphisms to projective space. An invertible sheaf $\mathcal{L}$ on a scheme X that arises as the inverse image of $\mathcal{O}(1)$ along an immersion to projective space is called *very ample*. In Theorem 15.3 below, we will see that if X is proper over a Noetherian affine scheme, then a very ample $\mathcal{L}$ is *ample*, i.e., for every coherent sheaf $\mathcal{F}$ on X, tensoring $\mathcal{F}$ with a sufficiently high power of $\mathcal{L}$ makes it globally generated. The goal of this lecture is to show that, conversely, every sufficiently high power of an ample invertible sheaf is very ample. In particular, in order to show that a proper scheme X over a Noetherian ring is projective, it suffices to produce an ample invertible sheaf on X.

Definition 15.1. Let $f : X \to Y$ be a morphism of schemes.

(1) f is called *projective* if there exists $n \in \mathbb{N}$ and a closed immersion $\iota : X \to \mathbb{P}^n_Y$ over Y.
(2) f is called *quasi-projective* if there exists $n \in \mathbb{N}$ and an immersion $\iota : X \to \mathbb{P}^n_Y$ over Y.
(3) f is called *locally projective* if there exists an open cover $Y = \cup V_i$ such that each f_{V_i} is projective.
(4) An invertible sheaf $\mathcal{L}$ on X is called *(f-relatively) very ample* if there exists an immersion $\iota : X \to \mathbb{P}^n_Y$ over Y such that $\iota^*\mathcal{O}_{\mathbb{P}^n_Y}(1) \cong \mathcal{L}$.

In particular, note that f is quasi-projective if and only if there exists an f-relatively very ample invertible sheaf on X. Moreover, f is projective if and only if it is proper and quasi-projective.

Definition 15.2. Let X be a Noetherian scheme. An invertible sheaf $\mathcal{L}$ on X is called *ample* if for every $\mathcal{F} \in \mathrm{Coh}(X, \mathcal{O}_X)$, there exists $n_0 \geq 0$ such that for all $n \geq n_0$, the sheaf $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is globally generated.

Theorem 15.3 (Serre)**.** *Let $f : X \to \operatorname{Spec} A$ be a proper morphism and assume that A is Noetherian. Let $\mathcal{L}$ be a (f-relatively) very ample sheaf on X. Then, $\mathcal{L}$ is ample.*

Proof. Choose an immersion $\iota : X \to \mathbb{P}^n_A$ such that $\iota^*\mathcal{O}_{\mathbb{P}^n_A}(1) \cong \mathcal{L}$. Since f is proper, ι is a closed immersion. Let $\mathcal{F} \in \mathrm{Coh}(X, \mathcal{O}_X)$. Then, $\iota_*\mathcal{F}$ is coherent [30] and there exists a natural isomorphism $\Gamma(\mathbb{P}^n_A, \iota_*\mathcal{F} \otimes \mathcal{O}_{\mathbb{P}^n_A}(m)) \cong \Gamma(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m})$ by the projection formula (Exercise 39), so $\mathcal{F} \otimes \mathcal{L}^{\otimes m}$ is globally generated if and only if $\iota_*\mathcal{F} \otimes \mathcal{O}_{\mathbb{P}^n_A}(m)$ is and we can assume without loss of generality that $X = \mathbb{P}^n_A$ and $\mathcal{L} = \mathcal{O}_{\mathbb{P}^n_A}(1)$.

Set $\mathcal{F}_i := \mathcal{F}|_{D_+(x_i)}$. Then, $\mathcal{F}_i = \widetilde{M_i}$ for some finitely generated $B_i := A[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}]$-module M_i. Choose generators $s_{ij} \in M_i$. By the Qcqs-lemma (Exercise 47), we can choose $m_0 \geq 0$ such that each $x_i^{m_0} s_{ij} = t_{ij}|_{D_+(x_i)}$ for some $t_{ij} \in \Gamma(\mathbb{P}^n_A, \mathcal{F}(m_0))$. Then, for every $m \geq m_0$, the global sections $x_i^{m-m_0} t_{ij}$ generate $\mathcal{F}(m)$. □

Remark 15.4. In fact, the proof of Theorem 15.3 shows that $\mathcal{F}(m)$ is generated by *finitely many* global sections for $m \geq m_0$. This can be used to show that $f_*\mathcal{F}$ is coherent if f is projective. We will come back to this later.

Lemma 15.5. *Let $\mathcal{L}$ be an invertible sheaf on a Noetherian scheme X. Then, the following are equivalent:*

[30] Locally, ι_* corresponds to restriction of scalars along a surjection $A \to B$, so coherence is preserved by ι_*.

(1) $\mathcal{L}$ *is ample.*
(2) $\mathcal{L}^{\otimes n}$ *is ample for all* $n > 0$.
(3) $\mathcal{L}^{\otimes n}$ *is ample for some* $n > 0$.

Proof. The only non-trivial claim is that $(3) \Rightarrow (1)$. Let $\mathcal{F} \in \mathrm{Coh}(X, \mathcal{O}_X)$. Set $\mathcal{F}' := \bigoplus_{i=0}^{n-1}(\mathcal{F} \otimes \mathcal{L}^{\otimes i})$. Then, $\mathcal{F}'$ is coherent, so there exists $m_0 \geq 0$ such that for all $m \geq m_0$ we have that $\mathcal{F}' \otimes \mathcal{L}^{\otimes mn} = \bigoplus_{i=0}^{n-1}(\mathcal{F} \otimes \mathcal{L}^{\otimes(mn+i)})$ is globally generated. Hence, $\mathcal{F} \otimes \mathcal{L}^{\otimes(mn+i)}$ is globally generated for all $m \geq m_0, i = 0, \ldots, (n-1)$, so $\mathcal{F} \otimes \mathcal{L}^{\otimes k}$ is globally generated for all $k \geq k_0 := m_0 n$. □

Theorem 15.6. *Let* $f : X \to \operatorname{Spec} A$ *be a morphism of finite type with* A *Noetherian. Let* $\mathcal{L}$ *be an invertible sheaf on* X*. Then,* $\mathcal{L}$ *is ample if and only if there exists* $n > 0$ *such that* $\mathcal{L}^{\otimes n}$ *is* f*-relatively very ample.*

Proof. First, assume that $\mathcal{L}^{\otimes n}$ is very ample. Then, there exists an immersion $\iota : X \to \mathbb{P}^n_A$ such that $\mathcal{L}^{\otimes n} \cong \iota^* \mathcal{O}_{\mathbb{P}^n}(1)$. Let $\overline{X}$ be the closure of X in $\mathbb{P}^n_A$ [31] Let $\mathcal{F} \in \mathrm{Coh}(X, \mathcal{O}_X)$. By Exercise 60, there exists $\mathcal{F}' \in \mathrm{Coh}(\overline{X}, \mathcal{O}_{\overline{X}})$ such that $\mathcal{F}'|_X \cong \mathcal{F}$. By Theorem 15.3, $\mathcal{O}_{\mathbb{P}^n}(1)|_{\overline{X}}$ is ample, so there exists $m_0 \geq 0$ such that $\mathcal{F}' \otimes \mathcal{O}_{\mathbb{P}^n}(m)|_{\overline{X}}$ is globally generated for all $m \geq m_0$, hence $\mathcal{F} \otimes \mathcal{L}^{\otimes mn}$ is globally generated for all $m \geq m_0$. Therefore, $\mathcal{L}^{\otimes n}$ is ample. By Lemma 15.5, this implies that $\mathcal{L}$ is ample.

Conversely, assume that $\mathcal{L}$ is ample. Recall that for $s \in \Gamma(X, \mathcal{L})$, we have the open subset $X_s := \{x \in X \mid s_x \notin \mathfrak{m}_x \mathcal{L}_x\}$.

<u>Claim: There exists $n > 0$ and $s_0, \ldots, s_m \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that X_{s_i} is affine and $X = \cup X_{s_i}$.</u>

Let $x \in X$ and choose an open affine neighborhood $x \in U \subseteq X$ such that $\mathcal{L}|_U \cong \mathcal{O}_U$. Let $Z := X \setminus U$ and let $\mathcal{I}_Z$ be the ideal sheaf of Z with its reduced induced scheme structure. Since X is Noetherian, $\mathcal{I}_Z$ is a coherent sheaf, so there exists $n > 0$ such that $\mathcal{I}_Z \otimes \mathcal{L}^{\otimes n}$ is globally generated. In particular, we find $s \in \Gamma(X, \mathcal{I}_Z \otimes \mathcal{L}^{\otimes n}) \subseteq \Gamma(X, \mathcal{L}^{\otimes n})$ such that $s_x \notin \mathfrak{m}_x \mathcal{L}_x^{\otimes n}$. Since s comes from a section of $\mathcal{I}_Z \otimes \mathcal{L}^{\otimes n}$, we have $s_z \in \mathfrak{m}_z \mathcal{L}_z^{\otimes n}$ for all $z \in Z$. So, $x \in X_s \subseteq U$. But $X_s \cap U$ is the principal open in U associated to the element $s|_U$, so X_s is affine.

Since X is quasi-compact, we can repeat the the above construction for any point in $x \in X$ to find finitely many global sections $s_i \in \Gamma(X, \mathcal{L}^{\otimes n_i})$ such that X_{s_i} is affine and $X = \cup X_{s_i}$. Replacing s_i by a tensor power does not change X_{s_i}, so we can assume that all the n_i are equal to some n. This proves the claim.

To simplify notation, we replace $\mathcal{L}$ by $\mathcal{L}^{\otimes n}$ to assume that $n = 1$ in the above claim. Let $B_i = \Gamma(X_{s_i}, \mathcal{O}_{X_{s_i}})$. Since f is of finite type, B_i is a finitely generated A-algebra. Choose finitely many generators b_{ij} of B_i as A-algebra. By the Qcqs lemma (Exercise 47) there exists $n > 0$ and $c_{ij} \in \Gamma(X, \mathcal{L}^{\otimes n})$ such that $c_{ij}|_{X_{s_i}} = s_i^n b_{ij}$. Let $\iota : X \to \mathbb{P}^N_A$ be the map determined by the s_i^n and c_{ij} (Theorem 14.3). Since the s_i^n generate $\mathcal{L}^{\otimes n}$, this is indeed a morphism.

[31] Defined by the quasi-coherent ideal sheaf $\ker(\mathcal{O}_{\mathbb{P}^n_A} \to \iota_* \mathcal{O}_X)$.

Write x_i, x_{ij} for the coordinates on $\mathbb{P}^N_A$ normalized so that $\iota^* x_i = s_i^n$ and $\iota^* x_{ij} = b_{ij}$. On $X_{s_i} = \iota^{-1}(D_+(x_i))$, the morphism ι is given by

$$\begin{aligned} A[\ldots, \frac{x_j}{x_i}, \ldots, \frac{x_{jk}}{x_i}, \ldots] &\to B_i \\ \frac{x_j}{x_i} &\mapsto \frac{s_j^n}{s_i^n} \\ \frac{x_{jk}}{x_i} &\mapsto \frac{c_{jk}}{s_i^n}. \end{aligned}$$

Since the $b_{ij} = \frac{c_{ij}}{s_i^n}$ generate B_i, this morphism is surjective. Hence, ι factors through a closed immersion $X \to \cup D_+(x_i) \subseteq \mathbb{P}^n_A$, so ι is an immersion. Since we used global sections of $\mathcal{L}^{\otimes n}$ to define ι, we have $\iota^* \mathcal{O}_{\mathbb{P}^n_A} \cong \mathcal{L}^{\otimes n}$. $\square$

16. Lectures 20 and 21: Picard groups and divisors

Even though we now understand why it is interesting to study globally generated and ample invertible sheaves on a scheme X, we do not yet know how to find or even classify invertible sheaves on any given scheme X. For example, we do not yet know how many non-isomorphic invertible sheaves there are on affine or projective space over a field.

In the next two lectures, we will see that there is a close connection between invertible sheaves on X and closed subschemes of codimension 1 (so-called *effective Weil divisors*) in X that can be used to classify invertible sheaves on X. First, we note that the isomorphism classes of invertible sheaves on X form a group.

Definition 16.1. Let X be a scheme. The *Picard group of* X is the group $\mathrm{Pic}(X)$ of isomorphism classes of invertible sheaves on X.

The group law on $\mathrm{Pic}(X)$ is given by the tensor product with neutral element given by the structure sheaf. If $\mathcal{L}$ is an invertible sheaf on X, then $\mathcal{L}^{-1} := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ is also invertible and the morphism

$$\begin{array}{rcl} \mathcal{L} \otimes \mathcal{L}^{-1} & \to & \mathcal{O}_X \\ s \otimes t^* & \mapsto & t^*(s) \end{array}$$

is an isomorphism: Indeed, if $\mathrm{Spec}\, A = U \subseteq X$ is an open affine where $\mathcal{L}|_U \cong \mathcal{O}_U$, then the above morphism can be identified with the isomorphism of A-modules $A \otimes_A \mathrm{Hom}_A(A, A) \to A$.

Definition 16.2. Let X be a topological space and A a ring.

(1) The *dimension* of X is

$$\dim(X) := \sup\{n \in \mathbb{N} \mid \exists \emptyset \subsetneq X_0 \subsetneq \ldots \subsetneq X_n \text{ with every } X_i \text{ closed and irreducible}\}$$

(2) For $Z \subseteq X$ closed and irreducible, the *codimension of Z in X* is

$$\mathrm{codim}(Z, X) := \sup\{n \in \mathbb{N} \mid \exists Z = Z_0 \subsetneq Z_1 \subsetneq \ldots \subsetneq Z_n \subseteq X \text{ with every } Z_i \text{ closed and irreducible}\}$$

(3) For $\mathfrak{p} \in \mathrm{Spec}\, A$, the *height* of $\mathfrak{p}$ is

$$\mathrm{ht}(\mathfrak{p}) := \sup\{n \in \mathbb{N} \mid \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \ldots \subsetneq \mathfrak{p}_n = \mathfrak{p} \text{ with every } \mathfrak{p}_i \text{ prime}\}$$

In particular, note that $\mathrm{ht}(\mathfrak{p}) = \mathrm{codim}(V(\mathfrak{p}), \mathrm{Spec}\, A)$ and $\dim \mathrm{Spec}\, A = \sup\{\mathrm{ht}(\mathfrak{p}) \mid \mathfrak{p} \in \mathrm{Spec}\, A\} =: \dim(A)$. We will need the following input from commutative algebra.

Theorem 16.3. *Let A be an integral domain of finite type over a field k. Then, for every $\mathfrak{p} \in \mathrm{Spec}\, A$, we have*

$$\mathrm{ht}(\mathfrak{p}) + \dim(A/\mathfrak{p}) = \dim(A) = \mathrm{tr.deg}(\mathrm{Frac}(A)/k),$$

where $\mathrm{tr.deg}(\mathrm{Frac}(A)/k)$ is the transcendence degree of $\mathrm{Frac}(A)$ over k.

Theorem 16.4 (Krull's principal ideal theorem)**.** *Let A be a Notherian ring and $a \in A$ a non-zero-divisor. Then, all minimal primes in $V(a)$ have height 1.*

Proposition 16.5 (Algebraic Hartog's lemma)**.** *Let A be a Noetherian normal integral domain. Then,*

$$A = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_{\mathfrak{p}},$$

where the intersection is taken inside $\mathrm{Frac}(A)$.

This has the following useful geometric consequence.

Corollary 16.6. *Let $\mathcal{F}$ be a locally free sheaf of finite rank on an integral Noetherian scheme X.*

(1) For all $\emptyset \neq V \subseteq U$ open, the restriction map $\rho : \mathcal{F}(U) \to \mathcal{F}(V)$ is injective. [32]

(2) If X is normal and every irreducible component Z of $U \backslash V$ has codimension $\operatorname{codim}(Z, U) \geq 2$, then ρ is bijective.

Proof. Since we can check injectivity (and surjectivity, once injectivity is given) of ρ on an open affine cover, we may assume that $U = \operatorname{Spec} A$ is affine and $\mathcal{F} = \mathcal{O}_X$.

For (1), let $\eta \in X$ be the generic point. Then, the composition

$$A = \mathcal{O}_X(U) \xrightarrow{\rho} \mathcal{O}_X(V) \to \mathcal{O}_{X,\eta} \cong \operatorname{Frac}(A)$$

is injective, hence so is ρ.

For (2), let $s \in \mathcal{O}_X(V)$. We consider A, $\mathcal{O}_X(V)$, and all localizations of A as subrings of $\operatorname{Frac}(A)$ via the injective restriction maps on $\mathcal{O}_X$. Then, $s \in A_{\mathfrak{p}}$ for all height 1 primes $\mathfrak{p}$ of A that are contained in V. But since Z has codimension at least 2 in U, every height 1 prime of A is contained in V. Hence, $s \in \bigcap_{\operatorname{ht}(\mathfrak{p})=1} A_{\mathfrak{p}} = A$, so s has a preimage under ρ. □

The final algebraic ingredient we need is the notion of regularity of local rings.

Definition 16.7. Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}_A$ and residue field k. Then, A is called *regular* if $\dim_k \mathfrak{m}_A/\mathfrak{m}_A^2 = \dim(A)$.

Proposition 16.8. *Let A be a Noetherian local ring of dimension 1. Then, the following are equivalent:*

(1) A is regular.

(2) A is normal.

(3) A is a discrete valuation ring.

Now, we are ready to define Weil divisors.

Definition 16.9 (Weil divisors)**.** Let X be a Noetherian scheme.

(1) A *prime divisor* on X is an integral closed subscheme $Z \subseteq X$ of codimension 1.

(2) A *Weil divisor* on X is a finite formal sum $\sum n_Z Z$ of prime divisors on X.

(3) The group of Weil divisors on X is denoted by $\operatorname{Div}(X)$.

(4) A Weil divisor $D = \sum n_Z Z$ is called *effective* if $n_Z \geq 0$ for all prime divisors Z. If D, D' are two Weil divisors with $D - D'$ effective, we write $D \geq D'$.

(5) The *support* of a Weil divisor $D = \sum n_Z Z$ is $\operatorname{supp}(D) := \cup_{n_Z \neq 0} Z$.

(6) If $U \subseteq X$ is open, the *restriction* of a Weil divisor $D = \sum n_Z Z$ to U is defined as $D|_U := \sum n_Z (Z \cap U)$ where we set $Z \cap U = 0$ if $Z \cap U = \emptyset$.

Next, we want to associate a Weil divisor to every pair $(\mathcal{L}, s)$, where $\mathcal{L}$ is an invertible sheaf and s is a non-zero *rational (or meromorphic) section* of $\mathcal{L}$, i.e., a section of $\mathcal{L}$ over some open subset $U \subseteq X$. For this, we need to be able to evaluate s at codimension 1 points of X. To do this, we need some extra assumptions on X.

Definition 16.10. Let X be a Noetherian scheme. We call X *regular in codimension* 1 (or simply (R1)) if for every $x \in X$ such that $\dim(\mathcal{O}_{X,x}) = 1$, the local ring $\mathcal{O}_{X,x}$ is regular (hence a DVR).

[32] This holds even without assuming that X is Noetherian.

For example, by Proposition 16.8, normal Noetherian schemes are (R1).

Convention. For the rest of this section, unless mentioned otherwise, we assume that X is Noetherian, integral, and (R1).

Let s be a non-zero section over $U \subseteq X$ of an invertible sheaf $\mathcal{L}$ on X. Then, we can define the valuation $v_Z(s)$ of s along a prime divisor $Z \subseteq X$ as follows:

Let $\eta_Z \in Z$ be the generic point of Z. Choose $\eta_Z \in V \subseteq X$ open such that there exists an isomorphism $\varphi : \mathcal{L}|_V \cong \mathcal{O}_V$. Let s' be the image of s under

$$\mathcal{L}(U) \to \mathcal{L}(U \cap V) \overset{\varphi(U\cap V)}{\to} \mathcal{O}_V(U \cap V) \hookrightarrow k(X).$$

Set $v_Z(s) := v(s')$, where v is the valuation associated to the discrete valuation ring $\mathcal{O}_{X,\eta_Z}$. Any two φ in the construction above differ by an automorphism of $\mathcal{O}_V$, so the integer $v_Z(s)$ is well-defined.

Definition 16.11. Let s be a rational section of an invertible sheaf $\mathcal{L}$ on X. We say that s has a *zero along* Z if $v_Z(s) > 0$ and that s has a *pole along* Z if $v_Z(s) < 0$.

Note that $v_Z(s) = v_Z(s|_V)$ for all $V \subseteq U$ open and if there exists an isomorphism $\varphi : \mathcal{L} \to \mathcal{M}$ with $\varphi(s) = t$, then $v_Z(s) = v_Z(t)$. We let $\sim$ be the equivalence relation on the set of pairs $(\mathcal{L}, s)$ generated by such isomorphisms and restrictions.

Next, note that the set $\{(\mathcal{L}, s)\}/\sim$ is a group with respect to the tensor product, since every rational section admits a rational inverse. In particular, forgetting the rational section yields a group homomorphism

$$\{(\mathcal{L}, s)\}/\sim \to \operatorname{Pic}(X)$$

which is surjective, because every invertible sheaf on X admits a non-zero rational section, and whose kernel is $(\{(\mathcal{O}_X, s)\}/\sim) \cong k(X)^\times$.

Now, we would like to define a group homomorphism

$$\begin{aligned} \operatorname{div} : \{(\mathcal{L}, s)\}/\sim &\to \operatorname{Div}(X) \\ (\mathcal{L}, s) &\mapsto \sum v_Z(s) Z. \end{aligned}$$

For this to be well-defined, we need the sum on the right side to be finite. This is the content of the following lemma.

Lemma 16.12. *Let s be a rational section of an invertible sheaf $\mathcal{L}$ on X. Then, there are only finitely many prime divisors $Z \subseteq X$ such that $v_Z(s) \neq 0$.*

Proof. Let $V \subseteq X$ be a non-empty open affine subset. Then, $\dim(X \setminus V) < \dim(X)$ since X is integral and $X \setminus V$ has only finitely many irreducible components, since X is Noetherian. In particular, there are at most finitely many prime divisors on X supported on Z, so we can replace X by V to assume without loss of generality that $X = \operatorname{Spec} A$ is affine, $\mathcal{L} = \mathcal{O}_X$, and $s \in k(X)$.

Since $s \in k(X)$, we can write $s = \frac{a}{b}$ for some $a, b \in A$ and then $v_Z(s) = v_Z(a) - v_Z(b)$, so we may assume without loss of generality that $s \in A$. Let $Z = V(\mathfrak{p})$ for some $\mathfrak{p} \in \operatorname{Spec} A$ of height 1. Since $s \in A$, we have $v_Z(s) \geq 0$ and $v_Z(s) > 0$ holds if and only if $s \in \mathfrak{p}$. In other words, by Krull's principal ideal theorem, the prime divisors Z such that $v_Z(s) > 0$ are in 1-to-1 correspondence with the minimal primes of A/s and since A is Noetherian, there are only finitely many such minimal primes. $\square$

Being a group homomorphism, div descends to a morphism from $\mathrm{Pic}(X)$ to the quotient of $\mathrm{Div}(X)$ by the divisors associated to pairs $\{(\mathcal{O}_X, s)\}$. The Weil divisors that arise from the structure sheaf in this way are called principal.

Definition 16.13. Let X be an integral normal Noetherian scheme that is regular in codimension 1.

(1) A Weil divisor D on X is called *principal* if it is of the form $\mathrm{div}(f)$ for some non-zero rational function $f \in k(X)^\times$.
(2) Two Weil divisors D and D' are called *linearly equivalent* if $D - D'$ is principal.
(3) The *(Weil divisor) class group* of X is the quotient $\mathrm{Cl}(X)$ of $\mathrm{Div}(X)$ by the subgroup of principal divisors.

Then, the map div descends to a homomorphism $\mathrm{Pic}(X) \to \mathrm{Cl}(X)$. In good situations, this homomorphism will be an isomorphism and we can use it to calculate the Picard group of X. We need the following analogue of Corollary 16.6, which is proved in the same way.

Lemma 16.14. *Let X be a normal integral Noetherian scheme. Let s be a rational section of an invertible sheaf $\mathcal{L}$ on X. If $v_Z(s) \geq 0$ for all prime divisors $Z \subseteq X$, then s is the restriction of a unique global section of $\mathcal{L}$.*

Proposition 16.15. *Let X be a normal integral Noetherian scheme. Then,* div *is injective.*

Proof. Assume $\mathrm{div}(\mathcal{L}, s) = 0$. By Lemma 16.14, we can assume that s is a global section of $\mathcal{L}$.

We claim that multiplication by s defines an isomorphism $\mathcal{O}_X \cong \mathcal{L}$. If this holds, then $0 = (\mathcal{O}_X, 1) \cong (\mathcal{L}, s)$, so div is injective. To check that multiplication by s is an isomorphism, we may assume $X = \mathrm{Spec}\ A$ and $\mathcal{L} = \mathcal{O}_X$, so that $s \in A$. Since $v_Z(s) = 0$ for all $Z \subseteq X$ prime, we have $v_Z(\frac{1}{s}) = 0$ for all $Z \subseteq X$, hence $s^{-1} \in A$ by Lemma 16.14. But then s is a unit in A, so multiplication by s is an isomorphism. □

The missing piece is a description of the image of div.

Definition 16.16. Let X be a scheme which is integral, Noetherian, and (R1) and let D be a Weil divisor on X. The *sheaf associated to D* is defined as

$$\mathcal{O}_X(D) : U \mapsto \{f \in k(X)^\times \mid \mathrm{div}|_U f + D|_U \geq 0\} \cup \{0\},$$

where $\mathrm{div}|_U f$ denotes the divisor associated to f considered as a rational function on U.

In other words, sections of $\mathcal{O}_X(D)$ over U are rational functions on U with poles and zeroes prescribed by D. Note that $1 \in k(X)$ defines a section of the sheaf $\mathcal{O}_X(D)$ over the complement of the support of D.

Lemma 16.17. *Let X be a normal integral Noetherian scheme and let D be a Weil divisor on X. Then, $\mathcal{O}_X(D)$ is quasi-coherent.*

Proof. By Theorem 11.14, it suffices to check that for all $\mathrm{Spec}\ A = U \subseteq X$ open and $a \in A \setminus \{0\}$, the map

$$\rho : \mathcal{O}_X(D)(U)_a \to \mathcal{O}_X(D)(D(a))$$

is an isomorphism. The map ρ is injective, because it is induced by the identity map on $k(X)$.

For the surjectivity, let $s \in \mathcal{O}_X(D)(D(a))$ and write $D = \sum n_Z Z$. Then, $v_Z(s) + n_Z \geq 0$ for all prime divisors $Z \subseteq X$ with $Z \cap D(a) \neq \emptyset$ (i.e. such that the generic point of Z is contained in $D(a)$). If $Z \cap D(a) = \emptyset$, then $v_Z(a) > 0$, so $sa^n \in \mathcal{O}_X(D)(U)$ for sufficiently large n. Therefore, ρ is surjective. □

If D is in the image of div, then $\mathcal{O}_X(D)$ is not only quasi-coherent, but even invertible.

Proposition 16.18. *Let X be a normal integral Noetherian scheme. Let $(\mathcal{L}, s)$ be a pair consisting of an invertible sheaf $\mathcal{L}$ and a non-zero rational section s. Then, there exists an isomorphism $\mathcal{O}_X(\operatorname{div}(\mathcal{L}, s)) \to \mathcal{L}$ that sends 1 to s.*

Proof. Let $U = X \setminus \operatorname{Supp}(\operatorname{div}(\mathcal{L}, s))$. Let $V \subseteq X$ be any open and $t \in \mathcal{O}_X(\operatorname{div}(\mathcal{L}, s))(V) \subseteq k(X)$. Then, $t|_{V\cap U}$ has no poles, so $t|_{V\cap U} \in \mathcal{O}_X(V \cap U)$ by Proposition 16.5. The section $st \in \mathcal{L}(V \cap U)$ has no poles anywhere on V, since $v_Z(st) = v_Z(s) + v_Z(t) \geq 0$ by definition of $\mathcal{O}_X(\operatorname{div}(s))$. By Lemma 16.14, this implies that st extends uniquely to a section of $\mathcal{L}$ over V. Mapping t to st in this way, we obtain a morphism

$$\mathcal{O}_X(\operatorname{div}(\mathcal{L}, s)) \to \mathcal{L}$$

which sends 1 to s.

This map is an isomorphism. To show this, we may assume that $\mathcal{L} = \mathcal{O}_X$. Then, for any $t \in \mathcal{O}_X(V)$, the rational function $\frac{t}{s} \in \mathcal{O}_X(\operatorname{div}(\mathcal{L}, s))$ is a preimage of t. □

Definition 16.19. A Weil divisor D on an integral Noetherian (R1) scheme is called *locally principal* if every point $x \in X$ has an open neighborhood $x \in U \subseteq X$ such that $D|_U$ is a principal divisor.

Corollary 16.20. *Let D be a locally principal Weil divisor on a normal integral Noetherian scheme X. Then, $\mathcal{O}_X(D)$ is invertible.*

Proof. First, note that for every $U \subseteq X$ open, we have $\mathcal{O}_U(D|_U) = \mathcal{O}_X(D)|_U$. If $D|_U$ is principal, then $\mathcal{O}_U(D|_U) \cong \mathcal{O}_U$ by Proposition 16.18, so D locally principal implies that $\mathcal{O}_X(D)$ is invertible. □

Proposition 16.21. *Let X be a normal integral Noetherian scheme. Let D be a Weil divisor on X such that $\mathcal{O}_X(D)$ is invertible. Then, $D = \operatorname{div}(\mathcal{O}_X(D), 1)$. In particular, D is locally principal.*

Proof. We may assume that $X = \operatorname{Spec} A$ is affine and that there exists an isomorphism $\varphi : \mathcal{O}_X \to \mathcal{O}_X(D)$. Considering the global sections of both sheaves as subsets of $k(X)$, the isomorphism φ is given by multiplication with some $s \in k(X)^\times$ and s generates $\mathcal{O}_X(D)(X)$ as an A-submodule of the A-module $k(X)$. Then, by definition, $\operatorname{div}(\mathcal{O}_X(D), 1) = \operatorname{div}(\mathcal{O}_X, s^{-1}) = -\operatorname{div}(\mathcal{O}_X, s)$, so we have to show that $D + \operatorname{div}(\mathcal{O}_X, s) = 0$. By definition of $\mathcal{O}_X(D)$, we have $D + \operatorname{div}(\mathcal{O}_X, s) \geq 0$.

Assume that there exists a prime divisor Z in the support of $D + \operatorname{div}(\mathcal{O}_X, s)$. There exists a rational function $f \in k(X)^\times$ with $v_Z(f) = 1$ [33] and after possibly shrinking X, we may assume that $v_W(f) = 0$ for all other prime divisors W of X. Then, $D + \operatorname{div}(\mathcal{O}_X, \frac{s}{f}) \geq 0$, so $\frac{s}{f} \in \mathcal{O}_X(D)(X)$, contradicting the fact that s generates $\mathcal{O}_X(D)(X)$. □

Summarizing, if X is integral normal and Noetherian, we have the following diagram

$$\begin{tikzcd}
\{(\mathcal{L}, s)\}/\sim \arrow[r, "\operatorname{div}"] \arrow[d] & \{D \in \operatorname{Div}(X) \mid D \text{ locally principal}\} \arrow[r] \arrow[d] & \operatorname{Div}(X) \arrow[d] \\
\operatorname{Pic}(X) \arrow[r] & \{D \in \operatorname{Cl}(X) \mid D \text{ locally principal}\} \arrow[r] & \operatorname{Cl}(X)
\end{tikzcd}$$

[33] Let η_Z be the generic point of Z. Then, $\mathcal{O}_{X,\eta_Z}$ is a discrete valuation ring. Then, we choose f as a generator of $\mathfrak{m}_{\eta_Z}$ and consider it as an element of $k(X)$.

where the two horizontal arrows on the left are isomorphisms (with inverse $D \mapsto (\mathcal{O}_X(D), 1)$).

In particular, if X is a normal integral Noetherian scheme such that every prime divisor $Z \subseteq X$ is locally principal, then div induces an isomorphism $\operatorname{Pic}(X) \cong \operatorname{Cl}(X)$. So, our next goal is to find a property of schemes that guarantess that all Weil divisors are locally principal.

Definition 16.22. A ring A is called *factorial* (or *unique factorization domain*) if it is an integral domain in which every non-zero element can be written as a product of irreducible elements and units.

An equivalent characterization of factorial rings that is closer to what we want is the following.

Proposition 16.23. *A Noetherian integral domain is factorial if and only if every height* 1 *prime ideal is principal.*

By a proof similar to the one that the integers are integrally closed in the rational numbers, one shows the following result.

Proposition 16.24. *Factorial rings are normal.*

As for examples of factorial rings, it is clear that discrete valuation rings are factorial, since every element in a DVR A is of the form $u\pi^n$, with π a generator of the maximal ideal and u a unit. By Proposition 16.8, this means that regular local rings of dimension 1 are factorial. In fact, this is true for regular local rings of arbitrary dimension, by the following theorem of Auslander and Buchsbaum.

Theorem 16.25 (Auslander-Buchsbaum)**.** *Regular Noetherian local rings are factorial.*

As explained in Example 6.13, "factorial" is not an affine-local property of rings. Therefore, a little care is needed when working with factoriality of schemes. To ensure that we can check factoriality on open affine covers despite this subtlety, we require that all stalks are factorial.

Definition 16.26. A scheme X is called *factorial* if $\mathcal{O}_{X,x}$ is factorial for every $x \in X$.

Proposition 16.27. *Let X be a factorial integral Noetherian scheme. Then, every Weil divisor D on X is locally principal. In particular,* div *is an isomorphism and* $\operatorname{Pic}(X) \cong \operatorname{Cl}(X)$.

Proof. Without loss of generality, $X = \operatorname{Spec} A$ and $D \subseteq X$ is a prime divisor cut out by a prime ideal $\mathfrak{p} \subseteq A$. On the complement $U = X \setminus D$, we have $D|_U = 0 = \operatorname{div}(\mathcal{O}_X, 1)|_U$, hence $D|_U$ is principal. Let $x \in D \subseteq X = \operatorname{Spec} A$. Then, $\operatorname{ht}(\mathfrak{p}A_x) = 1$ and A_x is factorial by assumption, so by Proposition 16.23, we can write $\mathfrak{p}A_x = (f_x)$ for some $f_x \in A_x$. Now, considering f_x as an element of $k(X)$, we have $v_D(f_x) = 1$. Thus, $\operatorname{div}(\mathcal{O}_X, f_x)|_V = D|_V$ for some open neighborhood $x \in V \subseteq X$. □

Corollary 16.28. *If A is a unique factorization domain, then* $\operatorname{Pic}(\operatorname{Spec} A) = 0$.

Proof. Since A is factorial, it is normal (Proposition 16.24). Hence, $\operatorname{Pic}(\operatorname{Spec} A) \cong \operatorname{Cl}(\operatorname{Spec} A)$. On the other hand, every height 1 prime ideal in A is principal. In other words, every prime divisor in $\operatorname{Spec} A$ is of the form $\operatorname{div}(\mathcal{O}_X, a)$ for a suitable $a \in A$. Hence, $\operatorname{Cl}(\operatorname{Spec} A) = 0$. □

Example 16.29. (1) If A is a Dedekind domain, then the ideal class group $\operatorname{Cl}(A)$ as defined in algebraic number theory is isomorphic to the Picard group of $\operatorname{Spec} A$. In particular, $\operatorname{Cl}(A) \cong \operatorname{Pic}(\operatorname{Spec} A) \cong \operatorname{Cl}(\operatorname{Spec} A)$.

(2) If k is a field, then $\mathrm{Pic}(\mathbb{A}^n_k) = \mathrm{Cl}(\mathbb{A}^n_k) = 0$, since $k[x_1, \ldots, x_n]$ is factorial.

Next, we want to calculate $\mathrm{Pic}(\mathbb{P}^n_k)$. This calculation can be reduced to the above example using the following.

Proposition 16.30. *Let X be a normal integral Noetherian scheme and $Z \subseteq X$ an integral closed subscheme. Let $U = X \setminus Z$. Then, the following hold:*

(1) The restriction map $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$ is surjective.
(2) If $\mathrm{codim}(Z, X) \geq 2$, then $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$ is an isomorphism.
(3) If $\mathrm{codim}(Z, X) = 1$ then there is an exact sequence

$$\mathbb{Z} \to \mathrm{Cl}(X) \to \mathrm{Cl}(U) \to 0$$

where the left map sends 1 to Z and the right map is the restriction map.

Proof. Restriction of divisors induces

(a) an isomorphism $\mathrm{Div}(X) \xrightarrow{\sim} \mathrm{Div}(U)$ if $\mathrm{codim}(Z, X) \geq 2$ (where the inverse image of a prime divisor is given by its closure).
(b) a short exact sequence

$$0 \to \mathbb{Z} \to \mathrm{Div}(X) \xrightarrow{\sim} \mathrm{Div}(U) \to 0$$

if $\mathrm{codim}(Z, X) = 1$.

Now, if $f \in k(X)^\times$ with $\mathrm{div}(f) = \sum n_i Z_i$, then $f|_U \in k(U)^\times = k(X)^\times$ and $\mathrm{div}(f|_U) = \sum n_i (Z_i \cap U)$. So, the restriction map $\mathrm{Div}(X) \xrightarrow{\sim} \mathrm{Div}(U)$ sends principal divisors to principal divisors. Then, to obtain claims (1),(2), and (3), we simply mod out by the subgroups of principal divisors in (a) and (b). □

Corollary 16.31. *Let k be a field. Then,* $\mathrm{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z} \cdot \mathcal{O}_{\mathbb{P}^n_k}(1)$.

Proof. Choosing $Z = V_+(x_0)$ in Proposition 16.30, we obtain that $\mathrm{Cl}(\mathbb{P}^n_k)$ is generated by the image of Z. Under the isomorphism $\mathrm{Cl}(\mathbb{P}^n_k) \cong \mathrm{Pic}(\mathbb{P}^n_k)$, Z corresponds to $\mathcal{O}_{\mathbb{P}^n_k}(1)$. Since $\mathcal{O}_{\mathbb{P}^n_k}(m)$ is non-trivial for any $n \neq 0$, the class of mZ in $\mathrm{Cl}(\mathbb{P}^n_k)$ is non-trivial for any $m \neq 0$. This proves the claim. □

In particular, by Theorem 14.3, we completely understand morphisms between projective spaces over a field at this point (see Exercise 61). For example, we have the following:

Corollary 16.32. *If k is a field, then*

$$\mathrm{Aut}_{(Sch/k)}(\mathbb{P}^n_k) \cong \mathrm{PGL}_{n+1}(k),$$

where the left-hand side denotes the group of automorphisms of $\mathbb{P}^n_k$ over $\mathrm{Spec}\, k$ and the right-hand side is the projective general linear group of dimension $n+1$ over k.

Proof. Let $\mathbb{P}^n_k = \mathrm{Proj}\, k[x_0, \ldots, x_n]$. The group $\mathrm{GL}_{n+1}(k)$ acts linearly on the coordinates x_i, hence there is a homomorphism $\mathrm{GL}_{n+1}(k) \to \mathrm{Aut}_{(Sch/k)}(\mathbb{P}^n_k)$. The elements of $\mathrm{GL}_{n+1}(k)$ that act trivially on $\mathbb{P}^n_k$ are precisely those that fix every $\frac{x_j}{x_i}$, hence they are precisely the scalar matrices. So, we get an injective homomorphism $\mathrm{PGL}_{n+1}(k) \hookrightarrow \mathrm{Aut}_{(Sch/k)}(\mathbb{P}^n_k)$.

Conversely, let $f : \mathbb{P}^n_k \to \mathbb{P}^n_k$ be an automorphism over k. Then, by Corollary 16.31, we have $f^*\mathcal{O}_{\mathbb{P}^n_k}(1) \cong \mathcal{O}(m)$ for some $m \in \mathbb{Z}$. But, by functoriality of the inverse image, the map $f^* : \mathrm{Pic}(\mathbb{P}^n_k) \to \mathrm{Pic}(\mathbb{P}^n_k)$ is an automorphism, so $m = \pm 1$. Since $\mathcal{O}_{\mathbb{P}^n_k}(-1)$ has no non-zero global

sections (Proposition 13.2), we must have $m = 1$. So, by Theorem 14.3, the map f is given by sections $s_0, \dots, s_n \in \Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$. Since $f(\mathbb{P}^n_k) = \mathbb{P}^n_k$, the s_i are k-linearly independent. On the other hand, $\dim_k \Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) = n+1$ by Proposition 13.2, so the s_i are a basis of $\Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1))$. Thus, there exists a matrix $A \in \mathrm{GL}_{n+1}(k)$ that changes the basis s_i to the standard basis x_i. In other words, f is in the image of $\mathrm{PGL}_{n+1}(k) \hookrightarrow \mathrm{Aut}_{(Sch/k)}(\mathbb{P}^n_k)$. $\square$

Remark 16.33. In contrast to Corollary 16.32, the precise structure of $\mathrm{Aut}_{(Sch/k)}(\mathbb{A}^n_k)$ is still unknown for $n \geq 3$.

17. Lecture 22: Cartier divisors and linear systems

In the last lectures, we studied the connection between equivalence classes of pairs $(\mathcal{L}, s)$, where s is a rational section of $\mathcal{L}$, and Weil divisors, which are formal sums of closed subschemes of codimension 1. If s is a global section of $\mathcal{L}$, there is a more immediate way of associating a closed subscheme of codimension 1 to the pair $(\mathcal{L}, s)$. This leads to the notion of *effective Cartier divisors*, which we study in this lecture.

Definition 17.1. Let X be a scheme.

(1) An *effective Cartier divisor* on X is a closed subscheme $D \subseteq X$ such that the associated ideal sheaf $\mathcal{I}_D$ is invertible.
(2) The *invertible sheaf associated to a Cartier divisor* D is $\mathcal{O}_X(D) := \mathcal{I}_D^{-1}$. We define $\mathcal{O}_X(-D) = \mathcal{I}_D$.

Associated to every effective Cartier divisor D, we have a short exact sequence

$$0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0,$$

where we abuse notation and denote by $\mathcal{O}_D$ the direct image of $\mathcal{O}_D$ under the closed immersion $\iota : D \to X$. Tensoring the above sequence with $\mathcal{O}_X(D)$, we obtain an *injective* map

$$\mathcal{O}_X \to \mathcal{O}_X(D),$$

since tensoring with a locally free sheaf is exact. We let $1_D \in \mathcal{O}_X(D)(X)$ be the associated global section.

Definition 17.2. Let X be a scheme, $\mathcal{L}$ an invertible sheaf on X, and $s \in \Gamma(X, \mathcal{L})$ a global section of $\mathcal{L}$.

(1) The global section s is called *regular* if the associated morphism $\mathcal{O}_X \to \mathcal{L}$ is injective.
(2) The *zero scheme* $Z(s) \subseteq X$ associated to s is the closed subscheme of X with ideal sheaf the image of the map $\mathcal{L}^{-1} \to \mathcal{O}_X$ induced by s.

Note that, by definition, we have $Z(1_D) = D$.

Lemma 17.3. *Let X be a scheme and let $Z \subseteq X$ be a closed subscheme. Then, the following are equivalent:*

(1) Z is an effective Cartier divisor.
(2) $\mathcal{I}_Z$ is locally generated by a non-zero-divisor.
(3) $Z = Z(s)$ for a regular section s of an invertible sheaf on X.

Proof. An ideal sheaf is invertible if and only if it is locally generated by a non-zero-divisor, so $(1) \Leftrightarrow (2)$. As noted above, $(1) \Rightarrow (3)$ is clear.

Finally, for $(3) \Rightarrow (1)$, we use that if $\mathcal{O}_X \xrightarrow{\cdot s} \mathcal{L}$ is injective, then so is $\mathcal{L}^{-1} \xrightarrow{\cdot s} \mathcal{O}_X$, hence $\mathcal{I}_{Z(s)} \cong \mathcal{L}^{-1}$ is invertible. □

Example 17.4. If X is integral, then a global section s of an invertible sheaf $\mathcal{L}$ on X is regular if and only if it is non-zero.

Indeed, let $\operatorname{Spec} A = U \subseteq X$ be a non-empty affine open such that $\mathcal{L}|_U \cong \mathcal{O}_U$. Then, $s|_U$, considered as an element of A, is regular if and only if $s|_U$ is a non-zero-divisor. Since X is integral, this happens if and only if $s|_U$ is non-zero. By Corollary 16.6, this holds if and only if $s \neq 0$.

If $D_1, D_2 \subseteq X$ are two effective Cartier divisors, then the morphism $\mathcal{I}_{D_1} \otimes_{\mathcal{O}_X} \mathcal{I}_{D_2} \to \mathcal{I}_{D_1} \to \mathcal{O}_X$ is injective, so

$$\mathcal{I}_{D_1} \otimes_{\mathcal{O}_X} \mathcal{I}_{D_2} \cong \mathcal{I}_{D_1} \cdot \mathcal{I}_{D_2},$$

where the right-hand side denotes the product of ideal sheaves. In particular, the closed subscheme $D_1 + D_2$ with ideal sheaf $\mathcal{I}_{D_1} \cdot \mathcal{I}_{D_2}$ is an effective Cartier divisor. [34]

This turns the set of effective Cartier divisors on X into a semi-group such that the map

$$\begin{array}{rcl} \{\text{Effective Cartier divisors on } X\} & \to & \{(\mathcal{L}, s)\}/\sim \\ \mathcal{I}_D & \mapsto & (\mathcal{O}_X(D), 1_D) \end{array}$$

is an injective homomorphism of semi-groups whose image consists of equivalence classes represented by pairs $(\mathcal{L}, s)$, where s is a regular global section of $\mathcal{L}$.

One can check the following:

Lemma 17.5. *Let X be an integral Noetherian (R1) scheme. Let $D \subseteq X$ be an integral effective Cartier divisors. Then,* $\operatorname{div}(\mathcal{O}_X(D), 1_D) = D$ *as Weil divisors.*

In particular, by Proposition 16.18, for an integral effective Cartier divisor $D \subseteq X$ on an integral normal Noetherian (R1) scheme X, the sheaf $\mathcal{O}_X(D)$ defined in this lecture is the same as the one of Definition 16.16 if we consider D as a prime divisor on X.

Remark 17.6. In many situations, *every* invertible sheaf $\mathcal{L}$ on a scheme X can be written as $\mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(-D')$, where D, D' are effective Cartier divisors on X. For example, we know that this holds if X is integral Noetherian and factorial by Lectures 20 and 21. Also, it holds if X is quasi-projective over a Noetherian ring (basically by Exercise 59). However, this does not work for general schemes.

Definition 17.7. Two effective Cartier divisors $D, D' \subseteq X$ are called *linearly equivalent*, denoted $D \sim D'$, if there is an isomorphism of invertible sheaves $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$.

For an integral scheme X, we can consider the subsheaf $\mathcal{O}_X^\times \subseteq \mathcal{O}_X$ of units. For every quasi-coherent sheaf $\mathcal{F}$ on X and any open $U \subseteq X$, the group $\mathcal{O}_X^\times(U)$ acts on $\mathcal{F}(U)$. The following follows from explicit computation.

Lemma 17.8. *Let X be an integral scheme. Then, $(\mathcal{L}, s) \cong (\mathcal{L}, t)$ for regular global sections s and t if and only if there exists $\lambda \in \Gamma(X, \mathcal{O}_X^\times)$ such that $t = \lambda s$.*

Corollary 17.9. *Let X be an integral scheme. Then, for any effective Cartier divisor D on X, there is a bijection*

$$\{\textit{effective Cartier divisors } D' \subseteq X \textit{ with } D' \sim D\} \leftrightarrow \frac{\Gamma(X, \mathcal{O}_X(D)) \setminus \{0\}}{\Gamma(X, \mathcal{O}_X^\times)}$$

Definition 17.10. Let X be an integral scheme over a field k.

(1) If $D \subseteq X$ is an effective Cartier divisor, then the *complete linear system $|D|$ associated to D* is the set of effective Cartier divisors linearly equivalent to D.

[34] Set-theoretically, this is just $D_1 \cup D_2$. As closed subschemes, this operation is more complicated. For example, if $X = \mathbb{A}^1_k$ and $D_1 = D_2 = V(x)$, then $D_1 + D_2 = V(x^2)$.

(2) A *linear system* V is the set of effective Cartier divisors corresponding to a finite-dimensional k-subspace W of $\Gamma(X, \mathcal{O}_X(D))$, where D is an effective Cartier divisor. The dimension of V is defined as $\dim W - 1$.
(3) The *base locus* $\mathrm{Bs}(V)$ of a linear system V is the scheme-theoretic intersection [35] of all effective Cartier divisors in V.
(4) A linear system is called *base-point free* if its base locus is empty.

Remark 17.11. In particular, using the correspondence of Theorem 14.3, a linear system of dimension n (more precisely, the subspace $W \subseteq \Gamma(X, \mathcal{O}_X(D))$ that induces V) determines a morphism $f : X \supseteq U \to \mathbb{P}^n_k$ (up to automorphisms of $\mathbb{P}^n_k$) from some open $U \subseteq X$. More precisely, $U = X \setminus \mathrm{Bs}(V)$, so if the linear system is base point free, then $U = X$. We call V *very ample* if f is an immersion.

Next, we want to calculate $\Gamma(X, \mathcal{O}_X^\times)$ in geometrically relevant cases.

Definition 17.12. Let $\mathcal{P}$ be a property of schemes (or morphisms). Let $f : X \to \operatorname{Spec} k$ be a scheme over a field k. We say that X (or f) is *geometrically* $\mathcal{P}$, if X_K (or f_K) is $\mathcal{P}$ for any field extension $k \subseteq K$.

Now, more precisely, we want to show the following.

Proposition 17.13. *Let X be a proper scheme over a field k. Assume that X is geometrically connected and geometrically reduced. Then, $\Gamma(X, \mathcal{O}_X) \cong k$. In particular, $\Gamma(X, \mathcal{O}_X^\times) = k^\times$.*

To prove this, we will reduce to the case where k is algebraically closed using the following useful statement.

Lemma 17.14. *Let X be a quasi-compact and quasi-separated scheme over a field k. Let $\mathcal{F}$ be a quasi-coherent sheaf on X and let A be a k-algebra. Then, the natural map $\mathcal{F}(X) \otimes_k A \to (f^*\mathcal{F})(X_A)$ induced by pullback of sections, where $f : X_A \to X$ is the projection, is an isomorphism.*

Proof. By Proposition 11.18, this holds if X is affine. If X is not affine, take a finite open affine cover $X = \cup_{i=1}^n X_i$ and consider the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}(X) \otimes_k A \arrow[r] \arrow[d, "g_1"] & \prod_{i=1}^n \mathcal{F}(X_i) \otimes_k A \arrow[r] \arrow[d, "g_2"] & \prod_{i=1}^n \mathcal{F}(X_i \cap X_j) \otimes_k A \arrow[d, "g_3"] \\
0 \arrow[r] & (f^*\mathcal{F})(X_A) \arrow[r] & \prod_{i=1}^n (f^*\mathcal{F})(X_{i,A}) \arrow[r] & \prod_{i=1}^n (f^*\mathcal{F})((X_i \cap X_j)_A)
\end{tikzcd}$$

The second row is exact because $f^*\mathcal{F}$ is a sheaf and the first row is exact because $\mathcal{F}$ is a sheaf and tensoring with A is exact. The morphism g_2 is an isomorphism by the affine case. Hence, g_1 is injective. Now, $X_i \cap X_j$ is quasi-compact and quasi-separated, so, by the same argument, g_3 is injective. Then, a diagram chase shows that g_1 is an isomorphism, as claimed. □

The final ingredient we need is the following:

Lemma 17.15. *Let $f : X \to Y$ be a surjective morphism of schemes over a scheme S. Assume that $\pi_X : X \to S$ is proper and $\pi_Y : Y \to S$ is separated and of finite type. Then, π_Y is proper.*

[35] i.e. the closed subscheme determined by the sum of all ideal sheaves of elements of V

Proof. Since π_Y is separated and of finite type, we only have to show that it is universally closed. So, let $T \to S$ be a morphism of schemes and let $Z \subseteq Y_T$ be a closed subset. By Exercise 30, $f_T : X_T \to Y_T$ is surjective. Moreover, $f_T^{-1}(Z) \subseteq X$ is closed, so $\pi_Y(Z) = \pi_X(f_T^{-1}(Z))$ is closed, as desired. □

Proof of Proposition 17.13. By Lemma 17.14, we have $\Gamma(X_{\bar{k}}, \mathcal{O}_{X_{\bar{k}}}) = \Gamma(X, \mathcal{O}_X) \otimes_k \bar{k}$, where $\bar{k}$ is an algebraic closure of k, so we may assume that k is algebraically closed.

Consider any morphism $f : X \to \mathbb{A}^1_k$. Since $X \to \operatorname{Spec} k$ is proper and $\mathbb{A}^1_k \to \operatorname{Spec} k$ is separated, $f(X)$ is closed. On the other hand, by Lemma 17.15, we have $f(X) \neq \mathbb{A}^1_k$, since $\mathbb{A}^1_k$ is not proper over k. Hence, $f(X)$ is a connected and reduced closed subscheme of $\mathbb{A}^1_k$ and not the whole space, hence a closed point. By Lemma 14.12, all closed points of $\mathbb{A}^1_k$ are k-rational.

Thus, f factors through a k-rational point $\operatorname{Spec} k \to X$, so $\Gamma(X, \mathcal{O}_X) = \operatorname{Hom}_{(Sch/k)}(X, \mathbb{A}^1_k) \cong \operatorname{Hom}_{(Sch/k)}(\operatorname{Spec} k, \mathbb{A}^1_k) = k$, as claimed. □

Corollary 17.16. *Let X be a geometrically integral scheme, proper over a field k. Then, for any effective Cartier divisor D on X, there is a bijection*

$$|D| \leftrightarrow \frac{\Gamma(X, \mathcal{O}_X(D)) \setminus \{0\}}{k^\times}.$$

In particular, if V is a (finite-dimensional) linear system on a geometrically integral proper scheme over a field k, then V can be naturally identified with the set of rational points of a projective space over k.

Remark 17.17. If X is geometrically integral and proper over k, and V is base-point free, then the elements of V are exactly the preimages of the k-rational hyperplanes in $\mathbb{P}^n_k$ under the morphism $f : X \to \mathbb{P}^n_k$ determined by V. Since the space of hyperplanes in $\mathbb{P}^n_k$ is isomorphic to $\mathbb{P}^n_k$, this explains why the set of Cartier divisors corresponding to V is the set of k-rational points of a projective space.

18. Lectures 23, 24 and 25: Differentials

Recall that we call a Noetherian local ring A regular, if the dimension of its Zariski cotangent space, considered as a vector space over the residue field, coincides with the Krull dimension of A. However, apart from this explicit definition, we do not have any tools to check whether a local ring is regular. For example, we do not yet know whether localizations of regular local rings are regular. The goal of these lectures is to introduce the cotangent sheaf $\Omega_{X/Y}$ of a morphism of schemes $f : X \to Y$. If Y is the spectrum of a perfect field and $x \in X$ is a closed point, it will turn out that the fiber of $\Omega_{X/Y}$ at x coincides with the Zariski cotangent space of x. In particular, this will allow us to study regularity of schemes using the quasi-coherent sheaf $\Omega_{X/Y}$.

Definition 18.1. Let A be a ring, B an A-algebra, and M a B-module. An *A-linear derivation* from B to M is a map $d : B \to M$ such that

(1) $d(b + b') = d(b) + d(b')$ for all $b, b' \in B$.
(2) $d(bb') = bd(b') + b'd(b)$ for all $b, b' \in B$.
(3) $d(a) = 0$ for all $a \in A$.

The set of all A-linear derivations from B to M is denoted by $\mathrm{Der}_A(B, M)$.

Definition 18.2. Let A be a ring and let B an A-algebra. The *module of relative differential forms* of B over A is a B-module $\Omega_{B/A}$ together with an A-linear derivation $d : B \to \Omega_{B/A}$ satisfying the following universal property:

For every B-module M and every A-linear derivation $d' : B \to M$, there exists a unique B-module homomorphism $f : \Omega_{B/A} \to M$ such that $d' = f \circ d$.

In other words, d gives an isomorphism of B-modules $\mathrm{Hom}_B(\Omega_{B/A}, M) \cong \mathrm{Der}_A(B, M)$.

The pair $(\Omega_{B/A}, d)$ exists: We can take for $\Omega_{B/A}$ the free B-module on the symbols db with $b \in B$ and divide out by the relations in Definition 18.1 and then define d as the map $b \mapsto db$.

Example 18.3. Let A be a ring and $B = A[x_1, \ldots, x_n]$. Then, $\Omega_{B/A}$ is the free B-module generated by the dx_i. You can either deduce this from the above explicit description or prove that the module $N :- \bigoplus_{i=1}^n Bdx_i$ together with the derivation $d : B \to N$ determined by $x_i \mapsto dx_i$ satisfies the universal property as follows:

Let M be a B-module and let $d' : B \to M$ be an A-linear derivation. A B-linear map $f : N \to M$ satisfies $f \circ d' = d$ if and only if $f(dx_i) = d'(x_i)$. Since N is free on the generators dx_i, we can always find a unique such f.

The formation of $\Omega_{B/A}$ commutes with base extension and localization in the sense of the following proposition, whose proof we omit.

Proposition 18.4. *Let A be a ring, C and B two A-algebras, let $S \subseteq B$ be a multiplicatively closed subset, and let $S' \subseteq A$ be a multiplicatively closed subset mapping to units in B. Then, there are natural isomorphisms:*

$$\Omega_{(B \otimes_A C)/A} \cong (\Omega_{B/A} \otimes_A C) \oplus (B \otimes_A \Omega_{C/A}) \tag{9}$$

$$\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A} \tag{10}$$

$$\Omega_{B/A} \cong \Omega_{B/S'^{-1}A} \tag{11}$$

Thus, if $f : X \to Y$ is a morphism of schemes, $\mathrm{Spec}\ A = V \subseteq Y$ is open affine and $U = f^{-1}(V) = \cup \mathrm{Spec}\ B_i$, then the $\widetilde{\Omega_{B_i/A}}$ glue to a quasi-coherent sheaf $\Omega_{U/A}$ on U by Proposition

18.4 (11). By Proposition 18.4 (10) and (11), the $\Omega_{U/A}$ glue to a quasi-coherent sheaf $\Omega_{X/Y}$ on X. Moreover, there is a map $d : \mathcal{O}_X \to \Omega_{X/Y}$ of sheaves of abelian groups which restricts to the universal derivation of $\Omega_{B_i/A}$ over $\operatorname{Spec} B_i$.

Definition 18.5. Let $f : X \to Y$ be a morphism of schemes. The quasi-coherent sheaf $\Omega_{X/Y}$ defined above is called the *sheaf of relative differentials (or cotangent sheaf) of X over Y*.

Proposition 18.6. *Let $f : X \to Y$ be a morphism of schemes.*

(1) Let $g : Z \to Y$ be a morphism of schemes. Then, there is a natural isomorphism

$$\Omega_{X \times_Y Z/Y} \cong \mathrm{pr}_1^* \Omega_{X/Y} \oplus \mathrm{pr}_2^* \Omega_{Z/Y},$$

where pr_1 and pr_2 are the two projections.

(2) Let $U \subseteq X$ an open subscheme. Then, there is a natural isomorphism

$$\Omega_{U/Y} \cong \Omega_{X/Y}|_U.$$

(3) Let $V \subseteq Y$ an open subscheme such that f factors through V. Then, there is a natural isomorphism

$$\Omega_{X/V} \cong \Omega_{X/Y}.$$

(4) (Cotangent sequence) Let $g : Y \to Z$ be a morphism of schemes. Then, there is a natural exact sequence of $\mathcal{O}_X$-modules

$$f^* \Omega_{Y/Z} \to \Omega_{X/Z} \to \Omega_{X/Y} \to 0$$

(5) (Conormal sequence) Let $Z \subseteq X$ be a closed subscheme with ideal sheaf $\mathcal{I}$. Then, there is a natural exact sequence of $\mathcal{O}_Z$-modules

$$\mathcal{I}/\mathcal{I}^2 \xrightarrow{\delta} \Omega_{X/Y} \otimes_{\mathcal{O}_X} \mathcal{O}_Z \to \Omega_{Z/Y} \to 0,$$

where $\delta(U)(s) = d(U)s \otimes 1$ for any $U \subseteq X$ open and $s \in \mathcal{I}(U)$.

(6) (Base change) Let $g : Y' \to Y$ be morphisms of schemes. Then, there is a natural isomorphism

$$\Omega_{X_{Y'}/Y'} \cong g'^* \Omega_{X/Y},$$

where $g' : X_{Y'} \to X$ is the projection.

Proof. For each claim, we may assume that all schemes involved are affine. Then, Claims (1), (2), and (3) follow from Proposition 18.4.

For Claim (4), let $X = \operatorname{Spec} C, Y = \operatorname{Spec} B, Z = \operatorname{Spec} A$. Then, the universal derivation $C \to \Omega_{C/B}$ induces a surjective C-linear map $\Omega_{C/A} \to \Omega_{C/B}$ sending dc to dc for $c \in C$. Its kernel is the C-submodule of $\Omega_{C/A}$ generated by symbols of the form db with $b \in B$. This is exactly the image of $\Omega_{B/A} \otimes_B C \to \Omega_{C/A}$.

We omit the proof of Claim (5). [36]

Finally, Claim (6) follows by combining Claims (1) and (4). □

Remark 18.7. In particular, note that it is easy to understand the restriction of $\Omega_{X/Y}$ to open subschemes (see (2)) compared to how we have to deal with closed subschemes (see (5)).

[36] See Matsumura, Commutative Algebra, Theorem 58

Corollary 18.8. *Let B be a localization of a finitely generated A-algebra. Then, $\Omega_{B/A}$ is a finitely generated B-module. In particular, if $f : X \to Y$ is a morphism of finite type and X is Noetherian, then $\Omega_{X/Y}$ is coherent.*

Proof. By Proposition 18.4, it suffices to prove the case where B is a finitely generated A-algebra. In this case, $\Omega_{B/A}$ is a quotient of $\Omega_{A[x_1,\dots,x_n]/A} \otimes_{A[x_1,\dots,x_n]} B$ by Proposition 18.6, hence it suffices to prove the case where B is a polynomial ring over A. This is Example 18.3. □

A useful tool for checking the injectivity of δ is the following proposition.

Proposition 18.9. *Let $f : X \to Y$ be a morphism of schemes and let $\iota : Z \hookrightarrow X$ be a closed subscheme with ideal sheaf $\mathcal{I}$. Assume that there exists a left-inverse $X \to Z$ to ι. Then, the conormal sequence*

$$0 \to \mathcal{I}/\mathcal{I}^2 \xrightarrow{\delta} \Omega_{X/Y} \otimes_{\mathcal{O}_X} \mathcal{O}_Z \to \Omega_{Z/Y} \to 0$$

is exact and locally split.

Proof. We may assume that $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, and $Z = \operatorname{Spec} C = \operatorname{Spec} B/I$ for some ideal $I \subseteq B$. Let $\alpha : B \to C$ be the quotient map and $\beta : C \to B$ a right-inverse, which exists by assumption. Consider the map

$$\begin{aligned} D : B &\to I/I^2 \\ b &\mapsto b - \beta(\alpha(b)). \end{aligned}$$

This D is additive and A-linear, and

$$bD(b') + b'D(b) - D(bb') = bb' - b\beta(\alpha(b')) - b'\beta(\alpha(b)) + \beta(\alpha(b))\beta(\alpha(b')) = D(b)D(b') = 0,$$

hence D is a derivation. Hence, by the universal property of $\Omega_{B/A}$, we obtain a B-linear map $\gamma : \Omega_{B/A} \to I/I^2$ such that $\gamma(db) = D(b)$. Now, for $s \in I$, we have $\gamma(sdb) = sD(b) = 0$, so γ induces a B-linear map $\overline{\gamma} : \Omega_{B/A} \otimes_B B/I \to I/I^2$. For $s \in I$ reducing to $\bar{s} \in I/I^2$ modulo I^2, we have

$$\overline{\gamma}(\delta(\bar{s})) = \overline{\gamma}(ds \otimes 1) = \gamma(ds) = D(s) = s - \beta(\alpha(s)) = s,$$

since $\alpha(s) = 0$. So, $\overline{\gamma}$ is a left-inverse to δ, hence the conormal sequence is left-exact and splits if X and Y are affine. □

Before we proceed, note that $\Omega_{X/Y,x} \cong \Omega_{\mathcal{O}_{X,x}/Y}$ by Proposition 18.6 (2) and Proposition 18.4 (10). The name cotangent sheaf suggests that there is a relation between the Zariski cotangent space at a point $x \in X$ and the fibers of $\Omega_{X/Y}$. By Proposition 18.6 (5) there exists, for every point $x \in X$ with maximal ideal $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$, a right-exact sequence of $k(x)$-vector spaces

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \to \Omega_{X/Y,x} \otimes_{\mathcal{O}_{X,x}} k(x) \to \Omega_{\operatorname{Spec} k(x)/Y} \to 0.$$

In good situations, this sequence is exact. Recall that a finitely generated field extension $K \subseteq L$ is called *separable* if there exists a transcendence basis $x_1, \dots, x_n$ of L over K such that L is finite and separable over $K(x_1, \dots, x_n)$. Furthermore, recall that every finite separable extension of fields can be generated by a single element.

Lemma 18.10. *In the above setting, if $\mathcal{O}_{X,x}$ is Noetherian, $Y = \operatorname{Spec} k$, where k is a field, and $k(x)$ is finitely generated and separable over k, then the canonical morphism*

$$\delta : \mathfrak{m}_x/\mathfrak{m}_x^2 \to \Omega_{X/Y,x} \otimes_{\mathcal{O}_{X,x}} k(x)$$

is split injective.

Proof. By Proposition 18.9, it suffices to show that the morphism $\mathcal{O}_{X,x} \to k(x)$ admits a right-inverse $\beta\colon k(x) \to \mathcal{O}_{X,x}$. If $k(x) = k(\bar{x}_1, \dots, \bar{x}_n)$ is purely transcendental over k, then for all $i = 1, \dots, n$ we let x_i be some lift of the generators $\bar{x}_i$ under the projection $\mathcal{O}_{X,x} \to k(x)$ and define $\beta(\bar{x}_i) := x_i$.

If instead $k(x) = k(\bar{x}_1, \dots, \bar{x}_n, \bar{y})$ we let $P(T) \in k(\bar{x}_1, \dots, \bar{x}_n)[T]$ be the minimal polynomial of $\bar{y}$. Note moreover that by the naturality of the conormal sequence, we may and do replace X by $\operatorname{Spec} \mathcal{O}_{X,x}/\mathfrak{m}_x^2$. By Proposition 18.9, it suffices to show that the morphism $\mathcal{O}_{X,x}/\mathfrak{m}_x^2 \to k(x)$ admits a right-inverse β. As before define $\beta(\bar{x}_i) := x_i$ and let $\beta(\bar{y}) := \bar{y} + b$, where $b \in \mathfrak{m}_x$ and hence $\bar{y} + b$ is a lift of $\bar{y}$ dependent on the choice of b.

The well-definedness of β depends on b; more precisely on making sure that $\beta(P(\bar{y})) = 0$. To this end, define $P^\beta(T) \in \mathcal{O}_{X,x}[T]$ to be the polynomial obtained by replacing $\bar{x}_i$ with x_i in the coefficients. Let $P'(T) := \partial P/\partial T$ and $P'^\beta(T) \in \mathcal{O}_{X,x}[T]$ be obtained by replacing the coefficients like before. Note that with the choice of b we have

$$\beta(P(\bar{y})) = P^\beta(\bar{y} + b) = P^\beta(\bar{y}) + P'^\beta(\bar{y})b \pmod{\mathfrak{m}_x^2}.$$

Indeed, the formula can be checked by a simple calculation on monomials in T. A few observations are in order: first $P^\beta(\bar{y}) = P(\bar{y}) \pmod{\mathfrak{m}_x}$. Since $P(\bar{y}) = 0$, we let $m := P^\beta(\bar{y}) \in \mathfrak{m}_x$. Second $P'^\beta(\bar{y})b = P'(\bar{y})b \pmod{\mathfrak{m}_x^2}$. Finally, since $\bar{y}$ is separable over $k(\bar{x_1}, \dots, \bar{x_n})$, we have $P'(\bar{y}) \neq 0$ in $k(x)$ and hence is invertible in $\mathcal{O}_{X,x}$. We let $u := P'(\bar{y})$. All in all, we obtain that $\beta(P(\bar{y})) = m + ub \pmod{\mathfrak{m}_x^2}$. Hence if we let $b = -m/u$, we obtain $\beta(P(\bar{y})) = 0$ in $\mathcal{O}_{X,x}/\mathfrak{m}_x^2$ and the inverse β is well-defined. □

Example 18.11. Let $k = \mathbb{F}_p(t)$, $X = \mathbb{A}^1_k = \operatorname{Spec} k[y]$, and let x be the closed point corresponding to the maximal ideal $(y^p - t)$. Then, $d(y^p - t) = py^{p-1}dy - dt = 0$, so $\delta = 0$. Therefore, the assumption that $k(x)$ is separable over k is essential.

Thus, to understand the fibers $\Omega_{X/k} \otimes_{\mathcal{O}_X} k(x)$ at points with $k(x)$ separable over k, it remains to understand $\Omega_{\operatorname{Spec} k(x)/k}$.

Lemma 18.12. *Let k be a field and K a finitely generated separable field extension of k. Then,* $\dim_K \Omega_{K/k} = \operatorname{tr.deg.} K/k$.

Proof. Choose a separating transcendence basis $x_1, \dots, x_n$ of K over k and let P be the minimal polynomial of a primitive element of K in the variable T over $k(x_1, \dots, x_n)$. Consider the conormal sequence

$$(P)/(P^2) \to \Omega_{k(x_1,\dots,x_n)[T]/k} \otimes K \to \Omega_{K/k} \to 0.$$

Since P is separable, $dP = P'dT$ with $P' \neq 0$, so $\Omega_{K/k} \cong \Omega_{k(x_1,\dots,x_n)/k} \otimes K$. Hence, it suffices to prove the claim if $K = k(x_1, \dots, x_n)$, where it follows from Example 18.3 and Proposition 18.4. □

Corollary 18.13. *Let X be a scheme of finite type over a field k and let $x \in X$ be a closed point such that $k(x)$ is separable over k. Then, there exists a natural isomorphism*

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x).$$

Proof. Since X is of finite type, $k(x)/k$ is in fact finite (see 14.12) and hence $\Omega_{k(x)/k} = 0$. □

Corollary 18.14 (Jacobian criterion)**.** *Let $A = k[x_1, \dots, x_n]/(f_1, \dots, f_m)$ with k a field. Then, the co-kernel of the Jacobian matrix at x*

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x) & \dots & \frac{\partial f_m}{\partial x_1}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_n}(x) & \dots & \frac{\partial f_m}{\partial x_n}(x) \end{pmatrix},$$

where $\frac{\partial f_i}{\partial x_j}$ is the formal deriative of f_i with respect to the variable x_j and $\frac{\partial f_i}{\partial x_j}(x)$ is the image of $\frac{\partial f_i}{\partial x_j}$ in $k(x)$, is isomorphic to $\Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x)$.

In particular, if $x \in X := \operatorname{Spec} A$ is a point such that $k(x)$ is separable over k, then the local ring $\mathcal{O}_{X,x}$ is regular if and only if the Jacobian matrix at x has co-rank $\dim \mathcal{O}_{X,x} + \operatorname{tr.deg.}(k(x)/k)$.

Proof. Let $I = (f_1, \dots, f_m)$. It suffices to observe that the image of the Jacobian matrix is exactly the image of the map induced by $\delta : I/I^2 \to \Omega_{k[x_1,\dots,x_n]/k} \otimes A$ on the fiber over x and then apply Lemma 18.12 and Lemma 18.10. □

This leaves us with the problem of deciding whether $k(x)$ is a separable extension of k. Fortunately, this is automatic for a large class of fields.

Definition 18.15. A field k is called *perfect* if every finite extension of k is separable.

Example 18.16. The following fields are perfect:

(1) Every field of characteristic 0.
(2) Finite fields.
(3) Algebraically closed fields.

The following fields are not perfect:

(4) $\mathbb{F}_p(t)$. Indeed, the field extension $\mathbb{F}_p(t) \subseteq \mathbb{F}_p(\sqrt[p]{t})$ is purely inseparable.

By the following algebraic result, the finiteness assumption can be dropped.

Theorem 18.17. *Let K be a perfect field. Then, every finitely generated field extension of K is separable.* [37]

To simplify the statements that follow, let us introduce some terminology.

Definition 18.18.
(1) A locally Noetherian scheme X is called *regular at $x \in X$* if $\mathcal{O}_{X,x}$ is regular. It is called *regular* if it is regular at every point.
(2) A scheme $X \to \operatorname{Spec} k$ of finite type over a field k is called *smooth over k at $x \in X$* if $\Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x)$ has dimension $\dim(\mathcal{O}_{X,x}) + \operatorname{tr.deg.}(k(x)/k)$. It is called *smooth over k* if it is smooth at every point.

From the Jacobian criterion, we deduce that regularity and smoothness coincide over perfect fields.

Corollary 18.19. *Let X be a scheme of finite type over a perfect field and let $x \in X$. Then, X is regular at x if and only of X is smooth over k at x.*

[37] See Matsumura, Commutative Ring Theory, Theorem 26.3 for a proof.

Example 18.20. Over imperfect fields, smoothness and regularity are not equivalent. For example, the field $\mathbb{F}_p(\sqrt[p]{t})$ is regular since it is a field, but it is not smooth over $\mathbb{F}_p(t)$. To see this, note that the extension $\mathbb{F}_p(t) \to \mathbb{F}_p(\sqrt[p]{t})$ is not separable and hence, albeit finite, may not have zero relative cotangent sheaf. [38] Note that $\mathbb{F}_p(\sqrt[p]{t}) \cong \mathbb{F}_p(t)[T]/(T^p - t)$. Therefore,

$$\Omega_{\mathbb{F}_p(\sqrt[p]{t})/\mathbb{F}_p(t)} \cong (\mathbb{F}_p(\sqrt[p]{t})dT)/(pT^{p-1}dT) \cong \mathbb{F}_p(\sqrt[p]{t})dT.$$

For a more geometric example, let k be an algebraically closed field of characteristic p and consider the closed subscheme X of $\mathbb{P}^2_k \times_k \mathbb{P}^2_k$ given by

$$x_0 y_0^p + x_1 y_1^p + x_2 y_2^p = 0.$$

Then, the generic fiber of the projection onto the x-coordinates is regular but not smooth (in fact, it is nowhere smooth). As an exercise, you can check that X itself is smooth over k!

Remark 18.21. Next term, we will prove the following statements:

(1) Smoothness "descends under faithfully flat morphisms". In the setting of schemes over fields, this means that smoothness and geometric smoothness coincide.
(2) Over any field, smoothness implies regularity. In particular, X is smooth if and only if it is geometrically regular.
(3) By Example 18.20, regularity is not stable under field extensions. Indeed, note that

$$\overline{\mathbb{F}_p(t)} \otimes_{\mathbb{F}_p(t)} \mathbb{F}_p(\sqrt[p]{t}) \cong \frac{\overline{\mathbb{F}_p(t)}[T]}{(T - \sqrt[p]{t})^p}$$

is not reduced and hence is not regular.

In particular, even though regularity is a useful concept (and, in certain situations, the only sensible notion of "non-singularity"), smoothness is more well-behaved.

Next, we want to prove that a connected scheme of finite type over a field which is smooth is automatically integral. Intuitively, this is because a point lying on two irreducible components is always singular. As a first step, we need the following result which says that, even though integrality is not a local property, it becomes one if we assume the total space to be connected and Noetherian:

Lemma 18.22. *Let X be a Noetherian scheme. Then, the following are equivalent:*

(1) X is integral.
(2) X is connected and for all closed points $x \in X$, the local ring $\mathcal{O}_{X,x}$ is an integral domain.

Proof. The implication $(1) \Rightarrow (2)$ is clear by Lemma 6.20.

For $(2) \Rightarrow (1)$, we know that X is reduced, so we only have to show that X is irreducible. By Exercise 18, we can write $X = \cup_{i=1}^n X_i$, where the X_i are closed irreducible components of X. Since X is connected, the X_i are not all disjoint, so assume $X_1 \cap X_2 \neq \emptyset$. Replace X by an open affine $\operatorname{Spec} A$ meeting $X_1 \cap X_2$. Then, $X_1 \cap X_2$ is closed in X, so it contains a closed point x. The irreducible components of X that pass through x are in 1-to-1 correspondence with minimal primes in $\mathcal{O}_{X,x}$, so $X_1 = X_2$ and thus X is irreducible. □

From Theorem 16.25 and the fact that regular local rings are integral, we deduce the following:

[38] In fact, one can show that a finite field extension is separable *if and only if* its module of relative differentials is 0.

Corollary 18.23. *Let X be a Noetherian scheme which is regular at all its closed points. Then, X is integral if and only if it is connected.*

Lemma 18.24. *Let X be a scheme of finite type over a perfect field k. Then, the following are equivalent:*

(1) X is smooth.
(2) X is smooth at every closed point.
(3) For every connected component $X_i \subseteq X$ the cotangent sheaf $\Omega_{X_i/k}$ is locally free of rank $\dim(X_i)$.

If any of these conditions holds, then every connected component of X is integral.

Proof. Clearly, $(1) \Rightarrow (2)$.

If (2) holds, then all closed points of X are regular by Corollary 18.19. Hence, by Corollary 18.23, all connected components of X are integral. In particular, for every connected component X_i of X and all $x \in X_i$, we have by Theorem 16.3, that

$$\dim(\mathcal{O}_{X,x}) + \mathrm{tr.deg}.k(x)/k = \mathrm{codim}(\overline{\{x\}}, X_i) + \dim(\overline{\{x\}}) = \dim(X_i).$$

By assumption, the function $\varphi(x) = \dim_{k(x)} \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x)$ is equal to $\dim(X_i)$ for every closed point $x \in X_i$ of every connected component X_i of X. By semi-continuity of φ (Exercise 42 (i)), this implies that $\varphi(x) \leq \dim(X_i)$ for all points $x \in X_i$. On the other hand, for the generic point $\eta \in X_i$, the cotangent sequence shows that $\varphi(\eta) \geq \mathrm{tr.deg}\, k(\eta)/k = \dim(X_i)$. Hence, again by semi-continuity, we have $\varphi(x) = \dim(X_i)$ for all points of X_i. By Exercise 42 (iii), this implies that $\Omega_{X_i/k}$ is locally free of rank $\dim(X_i)$ on X_i.Therefore, $(2) \Rightarrow (3)$.

Finally, if (3) holds, then, by Exercise 42, $\dim_{k(x)} \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x) = \dim(X_i)$ for all points $x \in X_i$. Write $X_i = \cup X_{ij}$, where the X_{ij} are irreducible components, and assume that $\dim(X_i) = \dim(X_{i1})$. Then, for every closed point $x \in X_{i1}$, we have

$$\dim(\mathcal{O}_{X,x}) = \dim(X_i) = \dim_{k(x)} \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x),$$

hence x is smooth as a point of X. Hence, $\mathcal{O}_{X,x}$ is regular by Corollary 18.14. This implies that $X_{i1} \cap X_{ij} = \emptyset$ for $j \neq 1$, because at any closed point $x \in X_{i1} \cap X_{ij}$, the ideals of X_{i1} and X_{ij} would yield distinct minimal primes in $\mathcal{O}_{X,x}$, contradicting the fact that $x \in X$ is regular. Hence, X_i is integral, so for every $x \in X_i$, we have

$$\dim(\mathcal{O}_{X,x}) + \mathrm{tr.deg}.k(x)/k = \dim(X_i) = \dim_{k(x)} \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} k(x),$$

so every X_i is smooth, hence so is X. This shows that $(3) \Rightarrow (1)$. □

Corollary 18.25. *Let X be an integral scheme of finite type over a perfect field k. Then, there exists an open dense subscheme $U \subseteq X$ such that U smooth over k.*

Proof. Let η be the generic point of X. Then, $\mathcal{O}_{X,\eta} = k(X)$ is regular, so $\Omega_{X/k,\eta}$ is free of rank $\dim(X)$. Hence, $\Omega_{X/k}$ is free of rank $\dim(U)$ in an open neighborhood U of η. So, by Lemma 18.24, U is smooth over k. □

For the next theorem about exactness of the conormal sequence for smooth closed subschemes of smooth schemes, we will use the following lemma from commutative algebra.

Lemma 18.26. *Let A and B be regular Noetherian local rings with maximal ideals $\mathfrak{m}$ and $\mathfrak{n}$, respectively. Let $\varphi : A \to B$ be a surjective local homomorphism with kernel I. Then, the following hold:*

(1) There exists a system of parameters[39] $(x_1, \dots, x_d) = \mathfrak{m}$ *such that* $(x_1, \dots, x_c) = I$.
(2) I/I^2 *is a free* A/I*-module of rank* $c = \dim(A) - \dim(B)$.

Proof. Write $B = A/I$, so that $\mathfrak{n} = \mathfrak{m}/I$ and let $d = \dim(A), d - c = \dim(B)$. Set $k = A/\mathfrak{m}$. Since $\mathfrak{n}^2 = (I + \mathfrak{m}^2)/I$ and thus $\mathfrak{n}/\mathfrak{n}^2 \cong \mathfrak{m}/(I + \mathfrak{m}^2)$, the short exact sequence of k-vector spaces

$$0 \to (I + \mathfrak{m}^2)/\mathfrak{m}^2 \to \mathfrak{m}/\mathfrak{m}^2 \to \mathfrak{n}/\mathfrak{n}^2 \to 0$$

together with Nakayama's lemma shows that we can choose $x_1, \dots, x_c \in I$ mapping to linearly independent elements in $\mathfrak{m}/\mathfrak{m}^2$ and complete to $x_1, \dots, x_d \in \mathfrak{m}$ inducing a basis of $\mathfrak{m}/\mathfrak{m}^2$. This is (1).

For (2), consider the surjective A-linear map

$$\begin{array}{rcl} \varphi : (A/I)^{\oplus c} & \to & I/I^2 \\ e_i & \mapsto & x_i. \end{array}$$

We want to show that φ is injective. Let $(a_1, \dots, a_n, 0, \dots, 0) \in \ker(\varphi)$ be a non-zero element with n minimal, so that $\sum_{i=1}^n a_i x_i \in I^2$. Thus, in $C := A/(x_1, \dots, x_{n-1})$, we have $a_n x_n \in (x_n^2)$. Thus, there exists $c \in C$ with $(a_n - c x_n) x_n = 0$. Since x_n is part of a system of parameters, it is not a zero divisor in C, hence $a_n - c x_n = 0$ and so $a_n \in (x_n)$. But then $(a_1, \dots, a_{n-1}, 0, \dots) \in \ker(\varphi)$, contradicting the minimality of n. □

Theorem 18.27. *Let X be an integral scheme of finite type over a perfect field k. Assume that X is smooth. Let $Z \subseteq X$ be a connected closed subscheme with sheaf of ideals $\mathcal{I}_Z$ and assume that all irreducible components of Z have the same dimension. Then, Z is smooth if and only if*

(1) $\Omega_{Z/k}$ *is locally free, and*
(2) the conormal sequence

$$0 \to \mathcal{I}_Z/\mathcal{I}_Z^2 \to \Omega_{X/k} \otimes \mathcal{O}_Z \to \Omega_{Z/k} \to 0$$

is exact.

Moreover, if these conditions hold, then $\mathcal{I}_Z/\mathcal{I}_Z^2$ *is locally free of rank* $\operatorname{codim}(Z, X)$.

Proof. First, assume that (1) and (2) hold. In particular, $\Omega_{Z/k}$ is locally free. Let r be the rank of $\Omega_{Z/k}$. By Lemma 18.24, it remains to show that $r = \dim(Z)$.

Since $\Omega_{Z/k}$ is locally free and the conormal sequence is exact, it splits locally on Z, so $\mathcal{I}_Z/\mathcal{I}_Z^2$ is locally free of rank $\dim(X) - r$. Hence, by Nakayama's lemma, $\mathcal{I}_Z$ can be generated locally by $\dim(X) - r$ elements. So, by Krull's principal ideal theorem 16.4, $\dim(Z) \geq r$. On the other hand, for every closed point $z \in Z$, we have $\dim_{k(z)} \mathfrak{m}_z/\mathfrak{m}_z^2 = r$ by Corollary 18.13, so $r = \dim(Z)$, as desired.

Conversely, assume that Z is smooth. By Lemma 18.24, (1) holds. So, it remains to show that the sequence

$$0 \to \mathcal{I}_Z/\mathcal{I}_Z^2 \xrightarrow{\delta} \Omega_{X/k} \otimes \mathcal{O}_Z \xrightarrow{\varphi} \Omega_{Z/k} \to 0$$

is exact on the left, or, in other words, that the surjective map $\psi : \mathcal{I}_Z/\mathcal{I}_Z^2 \to \operatorname{Ker}(\varphi)$ is an isomorphism. Replacing Z by an irreducible component, we may assume by Lemma 18.24 that Z is

[39] Recall that a system of parameters of a regular local ring A of dimension d is any set of d generators of the maximal ideal. Equivalently, by Nakayama's lemma, this is a set of elements of the maximal ideal whose residue classes generate the Zariski cotangent space. Any such system of parameters is a "regular sequence", that is, for all $n \geq 1$, x_n is not a zero divisor in $A/(x_1, \dots, x_{n-1})$.

integral. By Lemma 18.10, ψ is an isomorphism at the generic point of Z, since k is perfect. Thus, $\mathrm{Ker}(\psi) \subseteq \mathcal{I}_Z/\mathcal{I}_Z^2$ is supported on a proper closed subset of Z.

Recall from Exercise 41 that the support of a coherent sheaf on a Noetherian scheme is equal to the vanishing locus of its annihilator and we are saying that the annihlator of $\mathrm{Ker}(\psi)$ is non-zero. [40] If we can prove that $\mathcal{I}_Z/\mathcal{I}_Z^2$ is locally free on Z, then it is torsion-free and hence so its subsheaf $\mathrm{Ker}(\psi)$. But the support of a torsion-free sheaf on an integral scheme is either empty or the whole scheme, so this will conclude the proof.

To prove that $\mathcal{I}_Z/\mathcal{I}_Z^2$ is locally free of rank $\mathrm{codim}(Z, X)$, we can localize at a point $z \in Z$. Then, we are in the situation where X is the spectrum of a regular local ring A with maximal ideal $\mathfrak{m}$, Z is determined by an ideal $I \subseteq \mathfrak{m}$ such that A/I is also a regular local ring, and we are claiming that I/I^2 is a free A/I-module. This is Lemma 18.26 (2). □

Definition 18.28. Let X be an integral smooth scheme of finite type over a perfect field k. The *canonical sheaf* ω_X of X is given by $\det \Omega_X$.

Proposition 18.29 (Adjunction)**.** *Let D be a smooth integral effective Cartier divisor on a smooth integral scheme X of finite type over a perfect field k. Then, $\omega_D \cong (\omega_X \otimes \mathcal{O}_X(D))|_D$.*

Proof. By Theorem 18.27, we have a short exact sequence

$$0 \to \mathcal{O}_X(-D)/\mathcal{O}_X(-2D) \to \Omega_{X/k} \otimes \mathcal{O}_D \to \Omega_{D/k} \to 0. \tag{12}$$

Now, $\mathcal{O}_X(-D)/\mathcal{O}_X(-2D) \cong \mathcal{O}_D \otimes \mathcal{O}_X(-D)$. Thus, taking determinants (Exercise 75) in the above sequence and using that taking tensor products, exterior powers, and duals of locally free sheaves commutes with the inverse image functor (since the corresponding constructions on the level of modules commute with extension of scalars), we obtain the claim. □

Theorem 18.30 (Euler sequence)**.** *Let A be a ring, $Y = \mathrm{Spec}\, A$, and $X = \mathbb{P}^n_A$. Then, there is an exact sequence of sheaves on X*

$$0 \to \Omega_{X/Y} \to \mathcal{O}_X(-1)^{\oplus n+1} \to \mathcal{O}_X \to 0.$$

Proof. Let $B = A[x_0, \dots, x_n]$. Choose a basis $e_0, \dots, e_n$ of the free B-module $B(-1)^{\oplus n+1}$, consider the graded morphism of graded B-modules $\varphi : B(-1)^{n+1} \to B$ that sends e_i to x_i, and let M be its kernel. The induced map of sheaves $\mathcal{O}_X(-1)^{\oplus n+1} \to \mathcal{O}_X$ is surjective since the x_i are in the image, so it remains to give an isomorphism $\Omega_{X/Y} \cong \widetilde{M}$.

We will do this on the affine charts $U = D_+(x_i)$. First, let us describe the modules $\Omega_{X/Y}|_U$ and $\widetilde{M}|_U$:

The sheaf $\widetilde{M}|_U$ is associated to the $B_{(x_i)}$-module $M_{(x_i)}$, which is a free $B_{(x_i)}$-module with basis $\{\frac{1}{x_i}e_j - \frac{x_j}{x_i^2}e_i\}$.

On the other hand, the sheaf $\Omega_{X/Y}|_U$ is associated to the $B_{(x_i)}$-module $\Omega_{B_{(x_i)}/A}$ and $B_{(x_i)} = A[\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i}]$. By Example 18.3, $\Omega_{B_{(x_i)}/A}$ is the free $B_{(x_i)}$-module generated by the $d\left(\frac{x_j}{x_i}\right)$.

Therefore, we can define an isomorphism

$$\begin{aligned} \psi_i : \Omega_{B_{(x_i)}/A} &\to M_{(x_i)} \\ d\left(\frac{x_j}{x_i}\right) &\mapsto \frac{1}{x_i^2}(x_i e_j - x_j e_i). \end{aligned}$$

[40] This is why one also says that such a sheaf $\mathrm{Ker}(\psi)$ is torsion

It remains to check that the ψ_i coincide on $D_+(x_i) \cap D_+(x_j)$, so that they glue to the desired isomorphism $\Omega_{X/Y} \cong \widetilde{M}$. Note that, on $D_+(x_i) \cap D_+(x_j)$, we have

$$\frac{x_k}{x_i} = \frac{x_k}{x_j} \cdot \frac{x_j}{x_i},$$

for any k, so, applying the derivation d, we obtain

$$d\left(\frac{x_k}{x_i}\right) - \frac{x_k}{x_j} d\left(\frac{x_j}{x_i}\right) = \frac{x_j}{x_i} d\left(\frac{x_k}{x_j}\right).$$

Hence, for all i, j, k, we have

$$\begin{aligned}
\psi_i\left(d\left(\frac{x_k}{x_j}\right)\right) &= \frac{x_i}{x_j}\psi_i\left(\frac{x_j}{x_i} d\left(\frac{x_k}{x_j}\right)\right) \\
&= \frac{x_i}{x_j}\psi_i\left(d\left(\frac{x_k}{x_i}\right) - \frac{x_k}{x_j} d\left(\frac{x_j}{x_i}\right)\right) \\
&= \frac{x_i}{x_j}\left(\frac{1}{x_i^2}(x_i e_k - x_k e_i) - \frac{x_k}{x_j}\frac{1}{x_i^2}(x_i e_j - x_j e_i)\right) \\
&= \frac{1}{x_j^2}(x_j e_k - x_k e_j) = \psi_j\left(d\left(\frac{x_k}{x_j}\right)\right).
\end{aligned}$$

Hence, ψ_i and ψ_j agree on $D_+(x_i) \cap D_+(x_j)$, as desired. □

Corollary 18.31. *Let k be a field. Then, $\omega_{\mathbb{P}^n_k} \cong \mathcal{O}_{\mathbb{P}^n_k}(-n-1)$.*

Proof. Taking determinants in the Euler sequence yields, by Exercise 75, an isomorphism

$$\det(\mathcal{O}_{\mathbb{P}^n_k}(-1)^{\oplus n+1}) \otimes \det(\mathcal{O}_{\mathbb{P}^n_k}) \cong \omega_{\mathbb{P}^n_k}.$$

Exercise 75 applied to a direct sum of line bundles shows that in that case, the determinant is just the tensor products of the summands, so

$$\omega_{\mathbb{P}^n_k} \cong \mathcal{O}_{\mathbb{P}^n_k}(-n-1),$$

as claimed. □

Example 18.32. Let $X = V_+(f_d) \subseteq \mathbb{P}^n_k$ be a smooth hypersurface of degree d over a field k, that is, f_d is a homogeneous polynomial of degree d. Then, $\omega_X \cong \mathcal{O}_X(d-n-1)$. In particular, if $d < n+1$, then ω_X^{-1} is very ample, if $d = n+1$, then ω_X is trivial, and if $d > n+1$, then ω_X is very ample.

This is a special case of a much deeper trichotomy in birational geometry that separates (up to birational modifications, fibrations, introducing singularities, and many more complexities) smooth projective varieties over a field k into the Fano (ω_X^{-1} ample), Calabi–Yau (ω_X trivial), and canonically polarized (ω_X ample) cases.

As an example of varieties that are birational to ones with ample anti-canonical sheaf ω_X^{-1}, we say that X is *rational* if it is birational to $\mathbb{P}^{n-1}_k$. If X is rational, then there exists an open subset of X that can be parametrized in the sense that there is a bijection from an open subset of k^n with an open subset of $X(k)$:

(1) If $d = 1$, then X is a hyperplane, hence $X \cong \mathbb{P}^{n-1}_k$ and $\omega_X \cong \mathcal{O}_{\mathbb{P}^{n-1}_k}(-n) \cong (\mathcal{O}_{\mathbb{P}^n_k}(-n-1+1))|_X$.

(2) If $d = 2$, and $\mathrm{char}(k) \neq 2$, then by Exercise 67, we can assume $f_2 = \sum_{i=0}^{n} a_i x_i^2$ and if k is algebraically closed, we can further assume that $a_i = 1$. Now, for $n \geq 2$, any quadric hypersurface is birational to $\mathbb{P}^{n-1}$: For example, choosing instead the equation $x_0 x_1 + \sum_{i=2}^{n} x_i^2$ for X, the projection from the point $[1 : 0 : \ldots : 0]$ sends $[x_0 : \ldots : x_n] \in X$ to $[x_1 : \ldots : x_n]$. On an open set, this map has the inverse

$$\begin{array}{ccc} \mathbb{P}^{n-1} & \dashrightarrow & X \\ [x_1 : \ldots : x_n] & \mapsto & [-\dfrac{\sum_{i=2}^{n} x_i^2}{x_1} : x_1 : \ldots : x_n]. \end{array}$$

(3) For $d = 3$, an intriguing geometric question is to ask for which n some (or all) X are rational:

- If $n = 2$, then $\omega_X \cong \mathcal{O}_X$. If X were rational, then, since $\dim(X) = 1$, we would have $X \cong \mathbb{P}^1_k$, contradicting $\omega_X \cong \mathcal{O}_X$. Hence, these curves are never rational.
- If $n = 3$, then X is a cubic surface. Next term, we will see that, if k is algebraically closed, then X is rational and contains exactly 27 lines.
- If $n = 4$, then X is a cubic threefold. By a result of Clemens and Griffiths, smooth cubic threefolds are never rational.
- If $n = 5$, then X is cubic fourfold. The rationality of a smooth cubic fourfold is still wide open and an active area of research. A proof of the non-rationality of a very general cubic fourfold over the complex numbers has recently been announced in the preprint "Birational Invariants from Hodge Structures and Quantum Multiplication" by Katzarkov, Kontsevich, Pantev, and Yu. For a nice exposition to this topic see https://www.math.brown.edu/bhassett/papers/CIMEsurvey/CubicLectures6.pdf.

19. Lecture 26: Curves and function fields

For the rest of this term, we will apply the theory we developed so far to curves. In this lecture, we will see that the category of regular proper curves X over a field k is equivalent to the category of finitely generated field extensions of transcendence degree 1 over k. On the one hand, this means that all the geometric information about X is already encapsulated in $k(X)$. On the other hand, it means that we can study the field $k(X)$ by exploiting geometric properties of X. First, let us clarify what we mean by a curve.

Definition 19.1. Let k be a field.

(1) A *variety* over k is an integral scheme X over k such that the morphism $X \to \operatorname{Spec} k$ is separated and of finite type.
(2) A variety X over k is called *complete* if it is proper over k.
(3) A *curve* over k is a variety of dimension 1.

Proposition 16.8 yields the following criterion for regularity of curves.

Lemma 19.2. *A curve X over k is regular if and only if it is normal. If X is normal, then $\mathcal{O}_{X,x}$ is a discrete valuation ring for every closed point $x \in X$.*

As announced above, studying regular complete curves is equivalent to studying function fields of transcendence degree 1 over k. To show this, we need the following fact from commutative algebra (see e.g. Theorem 4.14 in "Commutative Algebra with a View Toward Algebraic Geometry" by D. Eisenbud).

Lemma 19.3. *Let A be an integral domain, finitely generated over a field k. Let K be the fraction field of A and $K \subseteq L$ a finite extension. Then, the integral closure B of A in L is a finitely generated A-module, a finitely generated k-algebra, normal, and has fraction field L.*

Theorem 19.4. *There is an equivalence of categories*

$$\begin{array}{rcl} \{\textit{Regular complete curves over } k\} & \longrightarrow & \left\{\begin{array}{l}\textit{Finitely generated field extensions of } k \\ \textit{of transcendence degree } 1 \textit{ over } k\end{array}\right\} \\ X & \longmapsto & k(X), \end{array}$$

where the morphisms on the left are dominant morphisms of schemes and the morphisms on the right are field extensions.

Proof. Curves are integral and dominant morphisms of curves induce extensions of function fields, so the functor is well-defined.

<u>Claim 1: $X \mapsto k(X)$ is essentially surjective</u>

Let K be a finitely generated field of transcendence degree 1 over k. Choose $x \in K$ transcendental over k and let A be the integral closure of $k[x]$ in K. Note that K is finite over $k[x]$, since K is finitely generated of transcendence degree 1 over k. Hence, by Lemma 19.3, $\operatorname{Spec} A$ is a curve over k with function field K. Since A is of finite type over k, we find a closed immersion $\iota : \operatorname{Spec} A \to \mathbb{A}^n_k$. Let X be the normalization of the closure of $\iota(\operatorname{Spec} A)$ in $\mathbb{P}^n_k$, with its reduced induced scheme structure. Then, X is normal, hence regular by Lemma 19.2, and finite over a projective scheme over k by Lemma 19.3, hence complete. By construction, $k(X) = K$.

<u>Claim 2: $X \mapsto k(X)$ is fully faithful</u>

Let $f, g : X \to Y$ be two dominant morphisms of curves. If the induced field extensions $k(Y) \to k(X)$ are the same, then the equalizer of f and g contains the generic point of X. By Exercise 33, the equalizer of f and g is closed, hence $f = g$. This shows that $X \mapsto k(X)$ is faithful.

Fullness follows from Exercise 36. □

In the proof of Theorem 19.4, we showed that every regular complete curve X admits a finite morphism to a projective (not necessarily regular) curve. By the following lemma, this implies that X is itself projective.

Lemma 19.5. *Let $f : X \to Y$ be a finite morphism of Noetherian schemes. Let $\mathcal{L}$ be an ample invertible sheaf on Y. Then, $f^*\mathcal{L}$ is ample.*

Proof. Let $\mathcal{F} \in \mathrm{Coh}(X, \mathcal{O}_X)$. Then, $f_*\mathcal{F}$ is coherent, so there exists $n_0 \geq 0$ such that $f_*\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ is globally generated for all $n \geq n_0$. By the projection formula (Exercise 39), we have $f_*\mathcal{F} \otimes \mathcal{L}^{\otimes n} \cong f_*(\mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n})$. Since f^* is right-exact, we obtain that $f^*(f_*(\mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n}))$ is globally generated for all $n \geq n_0$. As f is finite, it is affine, so the natural map $f^*(f_*(\mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n})) \to \mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n}$ is surjective. This corresponds to the easy algebraic fact that for every morphism of rings $A \to B$ and every B-module N, the map

$$\begin{array}{rcl} {}_AN \otimes_A B & \to & N \\ n \otimes b & \mapsto & bn \end{array}$$

is surjective.

Hence, $\mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n})$ is globally generated for all $n \geq n_0$, so $f^*\mathcal{L}$ is ample. □

Remark 19.6. More generally, one can show that the pullback of an ample sheaf along any affine morphism between Noetherian schemes is again ample.

Corollary 19.7. *Every regular complete curve X over a field is projective.*

20. Lecture 27: Degrees on curves and Riemann–Roch

In this lecture, we study the notions of degrees of divisors on and morphisms between regular complete curves. We will encounter a ramification formula similar to the formula for ramification of primes in rings of integers of number fields. At the end of the lecture, we will be able to phrase the Riemann–Roch theorem, which gives a way of calculating the size of the space of global sections of an invertible sheaf $\mathcal{L}$ on a smooth complete curve X if the degree of $\mathcal{L}$ and ω_X are known. The proof of this theorem will have to wait until after we developed the theory of sheaf cohomology next term.

First, we observe that dominant morphisms between regular complete curves are in fact finite.

Lemma 20.1. *Let $f : X \to Y$ be a dominant morphism between curves over a field k. Assume that X is regular and complete. Then, f is finite and Y is complete.*

Proof. Since $X \to \operatorname{Spec} k$ is proper and $Y \to \operatorname{Spec} k$ is separated, f is proper by Theorem 9.9, so the image of f is closed. Since f is dominant, this implies that f is surjective, hence Y is proper by Lemma 17.15.

Since taking integral closures commutes with localization, the integral closures of open affines $\operatorname{Spec} A \subseteq Y$ in $k(X)$ glue to a scheme X' together with a morphims $\pi : X' \to Y$. By Lemma 19.3, π is finite, X' is normal, and $k(X') = k(X)$. In particular, X' is complete. By Theorem 19.4, this implies that X and X' are isomorphic over Y, hence f is finite. □

Remark 20.2. The proof above shows that if $\operatorname{Spec} A \subseteq Y$ is an open affine, then we can describe $f^{-1}(\operatorname{Spec} A) \subseteq X$ as the integral closure of A in $k(X)$. Localizing further, we obtain that for every closed point $y \in Y$, the fiber product $f^{-1}(\operatorname{Spec} \mathcal{O}_{Y,y})$ is the spectrum of the integral closure of $\mathcal{O}_{Y,y}$ in $k(X)$.

Definition 20.3. Let $f : X \to Y$ be a dominant morphism of curves over a field k.

(1) The *degree* of f is $\deg(f) := [k(X) : k(Y)]$.
(2) f is called *separable* if $k(Y) \subseteq k(X)$ is a separable field extension and *inseparable* otherwise.

Recall that a *uniformizer* of a discrete valuation ring is a generator of the maximal ideal.

Proposition 20.4. *Let $f : X \to Y$ be a finite morphism of curves over a field k. Assume that Y is regular. Then, $f_*\mathcal{O}_X$ is locally free of rank $\deg(f)$.*

Proof. By Exercise 42, and since Y is integral, it suffices to show that for every point $y \in Y$, we have $\varphi(y) = \dim_{k(y)}(f_*\mathcal{O}_X)_y \otimes_{\mathcal{O}_{Y,y}} k(y) = \deg(f)$. Assume that y is a closed point. The ring $B = (f_*\mathcal{O}_X)_y$ is a finitely generated module over the principal ideal domain $A = \mathcal{O}_{Y,y}$. By the structure theorem for modules over principal ideal domains, we have $B \cong A^n \oplus \bigoplus A/(t^{n_i})$ for some $n_i \geq 0$, where t is a uniformizer of A. Since X is integral, B is integral, so $B \cong A^n$ as A-module, because every element of $A/(t^{n_i})$ is annihilated by some power of t. Hence, $\varphi(y) = n = \varphi(\eta) = \deg(f)$, where η is the generic point of η. This finishes the proof. □

The following results implies that, for a finite morphism f of regular complete curves, the degree of f coincides with the size of the fibers if we measure them appropriately.

Lemma 20.5. *Let $f : X \to Y$ be a finite morphism of regular curves over a field k. Let $y \in Y$ be a closed point and let $t \in \mathcal{O}_{Y,y}$ be a uniformizer. Then,*

$$\deg(f) = \sum_{f(x)=y} v_x(f_x^\sharp t)[k(x) : k(y)],$$

where v_x is the valuation of $\mathcal{O}_{X,x}$.

Proof. We may assume that $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, f is given by $f^\sharp : A \to B$ and $B \cong A^{\deg(f)}$ as A-module by Proposition 20.4. Moreover, we may assume that there is an element $t \in A$ such that $v_y(t) = 1$ and $v_{y'}(t) = 0$ for all $y' \neq y$ and so that t_y coincides with the chosen uniformizer of $\mathcal{O}_{Y,y}$. Hence, there exists an isomorphism of $k(y)$-vector spaces

$$k(y)^{\deg(f)} \cong A^{\deg(f)} \otimes_A k(y) \cong B \otimes_A k(y).$$

Note that $k(y) = A/(t)$ and $B \otimes_A k(y) = B/(f^\sharp t)$. By the Chinese remainder theorem, we can write

$$B/(f^\sharp t) \cong \prod_{f(x)=y} \mathcal{O}_{X,x}/((f^\sharp t)_x) = \prod_{f(x)=y} \mathcal{O}_{X,x}/(f_x^\sharp(t_y)).$$

Now, each $\mathcal{O}_{X,x}$ is a discrete valuation ring with maximal ideal $\mathfrak{m}_x$, so for every $n \geq 1$, we have $\dim_{k(y)} \mathfrak{m}_x^n/\mathfrak{m}_x^{n+1} = \dim_{k(y)} k(x)$, hence $\dim_{k(y)} \mathcal{O}_{X,x}/(f_x^\sharp(t_y)) = v_x(f_x^\sharp(t_y))[k(x) : k(y)]$. This finishes the proof. □

Definition 20.6. Let $f : X \to Y$ be a finite morphism of regular curves over a field k.

(1) The *ramification index* of f at a closed point $x \in X$ is defined as $e_f(x) := v_x(f_x^\sharp t)$, where t is a uniformizer of $\mathcal{O}_{Y,f(x)}$.
(2) A closed point $x \in X$ is called *ramification point of f* (and f is called *ramified at x*) if $e_f(x) > 1$.
(3) The *inertia degree* of f at a closed point $x \in X$ is defined as $f_f(x) := [k(x) : k(f(x))]$.

Remark 20.7. Note the analogy between Lemma 20.5 and the ramification formula for primes in rings of integers of numbers fields. The similarity stems from the fact that Lemma 20.5 holds more generally for finite morphisms between schemes which are locally spectra of Dedekind domains and we can recover the number field and the curve case as special cases. If you are interested in number theory, you may want to try to generalize the results of this lecture to the arithmetic setting.

Note that if $f : X \to Y$ is a finite morphism of regular complete curves over a field k and $(\mathcal{L}, s)$ is a pair consisting of an invertible sheaf $\mathcal{L}$ on Y and s is a rational section of $\mathcal{L}$, then we can define the pullback of $(\mathcal{L}, s)$ to be $(f^*\mathcal{L}, f^*s)$. Passing to Weil divisors and applying the proof of the above lemma, we obtain the following definition of pullbacks of Weil divisors along morphisms of regular complete curves.

Definition 20.8 (Pullbacks)**.** Let $f : X \to Y$ be a finite morphism of regular curves over a field k. Let $D = \sum_{p \in Y} a_p p$ be a Weil divisor on Y. Then, the *pullback* of D along f is defined as

$$f^*D := \sum_{p \in Y} \sum_{f(q)=p} a_p e_f(q) f_f(q) q$$

Definition 20.9 (Degree)**.** Let $D = \sum_{p \in X} a_p p$ be a Weil divisor on a regular curve X over a field k. The *degree* of D is defined as

$$\deg(D) := \sum a_p [k(p) : k].$$

Then, the following is immediate from Lemma 20.5.

Proposition 20.10. *Let $f : X \to Y$ be a finite morphism of regular curves over a field k. Let D be a Weil divisor on Y. Then*

$$\deg(f^* D) = \deg(f) \cdot \deg(D).$$

Proposition 20.11. *Let D be a principal divisor on a regular complete curve X over a field k. Then,* $\deg(D) = 0$.

Proof. Write $D = D_+ - D_-$ with D_+ and D_- effective and without common points. By Corollary 17.9, there exists an invertible sheaf $\mathcal{L}$ on X and global sections s_+ and s_- of $\mathcal{L}$ such that $\operatorname{div}(s_+) = D_+$ and $\operatorname{div}(s_-) = D_-$. Note that s_+ and s_- are global sections of the same sheaf, since D is a principal divisor. The sections generate $\mathcal{L}$ globally, since D_+ and D_- have no common points. Let $f : X \to \mathbb{P}^1_k$ be the morphism determined by s_+, s_- via Theorem 14.3. Note that our assumption that X is complete guarantees that f is finite, by Lemma 20.1. Since $\deg(V_+(x_0)) = \deg(V_+(x_1)) = 1$, we have

$$\deg(D_+) = \deg(f^* V_+(x_0)) = \deg(f) = \deg(f^* V_+(x_1)) = \deg(D_-),$$

so $\deg(D) = 0$. □

In particular, it makes sense to talk about the degree of an invertible sheaf $\mathcal{L}$ on a regular complete curve X: Write $\mathcal{L} = \mathcal{O}_X(D)$ for some Weil divisor D and set $\deg(\mathcal{L}) := \deg(D)$.

Lemma 20.12. *Let X be a regular complete curve over a field k and let $\mathcal{L}$ be an invertible sheaf on X. If $\deg(\mathcal{L}) < 0$, then $H^0(X, \mathcal{L}) = 0$.*

Proof. Assume $0 \neq s \in H^0(X, \mathcal{L})$. Then, the Weil divisor $\operatorname{div}(s)$ is effective, that is, it has only non-negative coefficients, since $s_x \in \mathcal{L}_x$ and hence $v_x(s_x) \geq 0$ for all $x \in X$ closed. Hence, $\deg(\mathcal{L}) = \deg(\operatorname{div}(s)) \geq 0$. □

Much more involved is the following famous theorem, which we will prove next term. It shows that there is a close connection between the degree and the size of the space of global sections of an invertible sheaf on a curve. In the statement, $h^0(X, \mathcal{F}) := \dim_k H^0(X, \mathcal{F})$.

Theorem 20.13 (Riemann–Roch)**.** *Let X be a smooth complete geometrically integral curve over a perfect field k and let $\mathcal{L}$ be an invertible sheaf on X. Then, $h^0(X, \mathcal{L})$ is finite and*

$$h^0(X, \mathcal{L}) - h^0(X, \omega_X \otimes_{\mathcal{O}_X} \mathcal{L}^{-1}) = \deg(\mathcal{L}) - h^0(X, \omega_X) + 1.$$

For the rest of this term, we will use the above theorem as a black box and instead enjoy the beautiful geometry that can be deduced from it. First of all, we see that the number $h^0(X, \omega_X)$ is an important invariant of X.

Definition 20.14. Let X be a smooth complete curve over a perfect field k. The number $g(X) := h^0(X, \omega_X)$ is called (geometric) *genus* of X.

Corollary 20.15. *Let X be a smooth complete geometrically integral curve over a perfect field k. Then,* $\deg(\omega_X) = 2g(X) - 2$.

Proof. Since X is geometrically integral and complete, we have $h^0(X, \mathcal{O}_X) = 1$ by Proposition 17.13. Then, the corollary is obtained by setting $\mathcal{L} = \omega_X$ in Theorem 20.13. □

Example 20.16 (Curves of genus 0)**.** Let X be a smooth complete geometrically integral curve of genus 0 over a perfect field k. Then, $\deg(\omega_X) = -2$. Let $\mathcal{L} = \omega_X^{-1}$. Then, $\deg(\mathcal{L}) = 2$. By Riemann–Roch, we have $h^0(X, \mathcal{L}) = 3$. Let D be an effective Cartier divisor given by a global section of $\mathcal{L}$. Then, $|D|$ can contain the following three types of effective divisors:

(1) $P + Q$ with $P \neq Q$ and $k(P) = k(Q) = k$.
(2) $2P$ with $k(P) = k$.
(3) P with $[k(P) : k] = 2$.

We claim that $|D|$ is base-point free. Indeed, if $|D|$ has a base-point P, then we can write $|D| = P + |D'|$ for some complete linear system D'. Clearly, $\deg(D') \leq 1$, so by Riemann–Roch, we have $h^0(X, \mathcal{O}_X(D')) \leq 2$, contradicting $h^0(X, \mathcal{L}) = 3$. Hence, $|D|$ defines a morphism $f : X \to \mathbb{P}^2_k$. We let Y be the image of f and denote the induced morphism $f : X \to Y$ by the same letter.

Let us describe Y if k is algebraically closed. Since we used the complete linear system to define f, Y is not contained in a line. In particular, $\dim(Y) \neq 0$. Hence, Y is a Cartier divisor in $\mathbb{P}^2_k$, so $\mathcal{O}_{\mathbb{P}^2_k}(Y) \cong \mathcal{O}_{\mathbb{P}^2_k}(d)$ for some $d \in \mathbb{Z}$. Again because Y is not contained in a line, we must have $d \geq 2$, so $Y = V_+(f_d)$ for some homogeneous polynomial f_d of degree d.

Since Y is integral, it is smooth on an open dense subset V. Take a line $\ell = V_+(g) \subseteq \mathbb{P}^2_k$ that avoids the non-smooth points of Y and let $\iota : Y \to \mathbb{P}^2_k$ be the immersion. Then, $Z(\iota^* g)$ is a divisor of degree d on Y. Hence, $Z(f^* g)$ is a divisor of degree $\deg(f) \cdot d$ on X. But $f^* g \in H^0(X, \mathcal{L})$, so $Z(f^* g)$ is linearly equivalent to D. We conclude that $\deg(f) \cdot d = 2$, hence $\deg(f) = 1$ and $d = 2$.

Now, assume additionally that $\mathrm{char}(k) \neq 2$. Then, we can write $Y = \sum_{i=0}^{r} x_i^2$ with $r \in \{0, 1, 2\}$. If $r = 0$, then Y is non-reduced and if $r = 1$, then Y is reducible, both of which are absurd. Hence, $r = 2$ and Y is smooth. Since $\deg(f) = 1$, this implies that f is an isomorphism.

To summarize, we have shown that every smooth complete curve of genus 0 over an algebraically closed field k of characteristic different from 2 is isomorphic to a smooth conic in $\mathbb{P}^2_k$, hence isomorphic to $\mathbb{P}^1_k$. [41] Divisors of type (1) in $|D|$ correspond to lines in $\mathbb{P}^2_k$ meeting X in two distinct points. Divisors of type (2) correspond to tangent lines in $\mathbb{P}^2_k$. Divisors of type (3) do not exist if k is algebraically closed (if k is not algebraically closed, they correspond to lines that meet X in a single point whose residue field is a degree 2 extension of k).

[41] In fact, the first part of this statement is true over arbitrary fields and we will see next term that it follows from what we did above. However, not every smooth conic $X$is isomorphic to $\mathbb{P}^1_k$. Clearly, a necessary condition for the existence of such an isomorphism is the existence of a k-rational point on X. That this is also sufficient is the content of Exercise 77.

21. Lecture 28: Morphisms and embeddings of curves

In this lecture, we study the behaviour of cotangent sheaves for finite separable morphisms between curves. In particular, we prove the Hurwitz formula which relates the genera of the curves with the degree of the ramification divisor of such a morphism. As a corollary, we prove Lüroth's theorem which, algebraically speaking, states that every finite index subfield of $k(x)$ is isomorphic to $k(x)$. Finally, we apply the local criterion of Proposition 14.15 to the case of curves, where, together with the Riemann–Roch theorem, it is particulary powerful. As a corollary, we obtain results about embeddings of curves into projective spaces.

Proposition 21.1. *Let $f : X \to Y$ be a finite morphism of smooth curves over a a perfect field k. Assume that f is separable. Then, the cotangent sequence*

$$0 \to f^*\Omega_{Y/k} \to \Omega_{X/k} \to \Omega_{X/Y} \to 0$$

is exact.

Proof. Since the cotangent sequence is exact on the right, we only have to show that $f^*\Omega_{Y/k} \to \Omega_{X/k}$ is injective. Since X and Y are smooth curves, both sheaves are invertible. Hence, by Corollary 16.6, it suffices to show that the sequence is exact at the generic point η of X. Since f is separable, $k(X)$ is finite and separable over $k(Y)$, hence $\Omega_{X/Y,\eta} = \Omega_{k(X)/k(Y)} = 0$ by Lemma 18.12. Hence, $f^*\Omega_{Y/k,\eta} \to \Omega_{X/k,\eta}$ is surjective, hence an isomorphism. □

In particular, $\mathrm{supp}(\Omega_{X/Y})$ is a finite union of closed points. Since $\Omega_{X/Y}$ is coherent and $\mathcal{O}_{X,x}$ is a discrete valuation ring for every $x \in X$ closed, we can choose a uniformizer $t \in \mathcal{O}_{X,x}$ and write $\Omega_{X/Y,x} \cong \mathcal{O}_{X,x}/(t^n)$ for some $n \geq 0$. The number n is the length of $\Omega_{X/Y,x}$ (i.e. the maximal length of a strictly increasing chain of submodules) as a module over $\mathcal{O}_{X,x}$.

Definition 21.2. Let $f : X \to Y$ be a finite separable morphism of smooth curves over a a perfect field k. The *ramification divisor* R_f of f is

$$R_f = \sum_{x \in X} \mathrm{length}_{\mathcal{O}_{X,x}}(\Omega_{X/Y,x})x.$$

Theorem 21.3 (Hurwitz formula)**.** *Let $f : X \to Y$ be a finite separable morphism of smooth complete geometrically integral curves over a perfect field k. Then,*

$$2g(X) - 2 = \deg(f)(2g(Y) - 2) + \deg(R_f).$$

Proof. By definition of R_f, we have $\Omega_{X/Y} \cong \mathcal{O}_{R_f}$, if we consider R_f as an effective Cartier divisor on X. Tensoring the exact sequence of Proposition 21.1 with $\Omega_{X/k}^{-1}$, we obtain

$$0 \to f^*\Omega_{Y/k} \otimes_{\mathcal{O}_X} \Omega_{X/k}^{-1} \to \mathcal{O}_X \to \mathcal{O}_{R_f} \to 0.$$

Hence, $f^*\Omega_{Y/k} \otimes_{\mathcal{O}_X} \Omega_{X/k}^{-1} \cong \mathcal{O}_X(-R_f)$. Taking degrees on both sides, we obtain

$$-\deg(R_f) = \deg(f^*\Omega_{Y/k}) - \deg(\Omega_{X/k}) = \deg(f)(2g(Y) - 2) - (2g(X) - 2),$$

using Corollary 20.15 and Proposition 20.10. Reorganizing the equation, we obtain the Hurwitz formula. □

Corollary 21.4. *Let $f : X \to Y$ be a finite separable morphism of smooth complete geometrically integral curves over a perfect field k. Then, $g(X) \geq g(Y)$.*

Proof. Without loss of generality, we may assume $g(Y) \geq 1$. Then, the Hurwitz formula yields

$$g(X) = (\deg(f)g(Y) - \deg(f)) + 1 + \frac{1}{2}\deg(R_f).$$

If $g(Y) \geq 2$, then the term in the brackets is at least as big as $g(Y)$ and the remaining summands are non-negative, so $g(X) \geq g(Y)$. If $g(Y) = 1$, the right-hand side of the equation is at least 1, so again $g(X) \geq g(Y)$. □

Corollary 21.5 (Lüroth's theorem)**.** *Let $f : X \to Y$ be a finite morphism of smooth complete geometrically integral curves over a perfect field k. Assume $X \cong \mathbb{P}^1_k$. Then, $Y \cong \mathbb{P}^1_k$.*

Proof. Taking separable closures, we find $k(Y) \subseteq L \subseteq k(X)$ with $k(X)$ purely inseparable over L and L separable over $k(Y)$. Let Z be the regular complete curve with $L = k(Z)$. Note that Z is smooth since k is perfect and Z is geometrically integral, since L is, being a subfield of the geometrically integral field $k(X)$.

Since $X \cong \mathbb{P}^1_k$, we have $X(k) \neq \emptyset$ and hence $Y(k), Z(k) \neq \emptyset$. By Corollary 21.4 and Exercise 77, it thus suffices to show that $g(Z) = 0$.

Since $X \cong \mathbb{P}^1_k$, we find an element $t \in k(X)$ such that $k(X) = k(t)$. Since $L \subseteq k(t)$ is purely inseparable, we have

$$(k(t))^{p^n} = k(t^{p^n}) \subseteq L \subseteq k(t),$$

where $p^n = [k(t) : L]$ and where the first equality holds because k is perfect. But now $[k(t) : k(t^{p^n})] = p^n$, so $[L : k(t^{p^n})] = 1$, hence $L = k(t^{p^n})$. Finally, note that $k(x^{p^n}) \cong k(x)$ as field extensions of k, hence $Z \cong \mathbb{P}^1_k$. □

Proposition 21.6. *Let D be an effective divisor on a smooth complete geometrically integral curve X over an algebraically closed field k. Then,*

(1) the complete linear system $|D|$ is base-point free if and only if for every closed point $x \in X$,

$$h^0(X, \mathcal{O}_X(D - x)) = h^0(X, \mathcal{O}_X(D)) - 1.$$

(2) the complete linear system $|D|$ is very ample if and only if for any two closed points $x_1, x_2 \in X$ (not necessarily distinct),

$$h^0(X, \mathcal{O}_X(D - x_1 - x_2)) = h^0(X, \mathcal{O}_X(D)) - 2$$

Proof. For (1), choose a closed point $x \in X$ and consider the exact sequence

$$0 \to \mathcal{O}_X(D - x) \to \mathcal{O}_X(D) \to \mathcal{O}_x(D) \to 0.$$

Since $x \cong \operatorname{Spec} k$, we have $\mathcal{O}_x(D) \cong \mathcal{O}_x$, so $h^0(x, \mathcal{O}_x(D)) = 1$. Now, x is not a base-point of $|D|$ if and only if the map $H^0(X, \mathcal{O}_X(D)) \to H^0(x, \mathcal{O}_x(D))$ is non-zero. Since the kernel of this map is $H^0(X, \mathcal{O}_X(D - x))$ and the target is 1-dimensional, this happens if and only $h^0(X, \mathcal{O}_X(D - x)) = h^0(X, \mathcal{O}_X(D)) - 1$.

For (2), we may assume that $|D|$ has no base points. Indeed, we have $h^0(X, \mathcal{O}_X(D - x_1 - x_2)) \in \{h^0(X, \mathcal{O}_X(D - x_1)), h^0(X, \mathcal{O}_X(D - x_1)) - 1\}$ by the exact sequence

$$0 \to \mathcal{O}_X(D - x_1 - x_2) \to \mathcal{O}_X(D - x_1) \to \mathcal{O}_{x_2} \to 0,$$

so if the condition in (2) holds, then the condition in (1) holds as well and $|D|$ is base-point free. Moreover, if $|D|$ is very ample, then it is clearly base-point free.

Now, global sections of $\mathcal{O}_X(D)$ separate x_1 from x_2 if and only if x_2 is not a base-point of the linear system $|D - x_1|$, hence, by (1), if and only if

$$h^0(X, \mathcal{O}_X(D - x_1 - x_2)) = h^0(X, \mathcal{O}_X(D - x_1)) - 1 = h^0(X, \mathcal{O}_X(D)) - 2.$$

Next, for the separation of tangent vectors, note that for every $x \in X$, we have $\dim_k \mathfrak{m}_x/\mathfrak{m}_x^2 = 1$. So, $|D|$ separates tangent vectors at x if and only if there exists a global section of $\mathcal{O}_X(D)$ that has a zero of order 1 at x. In other words, this happens if and only if there exists a divisor D' in $|D|$ such that $D' - x$ is effective but $D' - 2x$ is not. Rephrasing again, this happens if and only if x is not a base point of the linear system $|D' - x|$. Hence, by (1), this happens if and only if

$$h^0(X, \mathcal{O}_X(D - 2x)) = h^0(X, \mathcal{O}_X(D - x)) - 1.$$

Since $|D|$ is base-point free, we have $h^0(X, \mathcal{O}_X(D - x)) = h^0(X, \mathcal{O}_X(D)) - 1$, and the proof is finished. □

Corollary 21.7. *Let D be a divisor on a smooth complete geometrically integral curve X over an algebraically closed field k.*

(1) If $\deg(D) \geq 2g(X)$, then $|D|$ is base-point free.
(2) If $\deg(D) \geq 2g(X) + 1$, then $|D|$ is very ample.
(3) D is ample if and only if $\deg(D) > 0$.

Proof. Follows from Proposition 21.6 and Riemann–Roch, using that invertible sheaves of negative degree do not admit non-zero global sections. □

Corollary 21.8. *Let X be a smooth complete geometrically integral curve over an algebraically closed field k. Then, there is a closed immersion $X \to \mathbb{P}^{g(X)+1}$.*

Remark 21.9. In fact, X embeds into $\mathbb{P}^3_k$. In principle, if we had one or two more lectures, we could prove this with what we know by now. The idea is as follows:

Given X in $\mathbb{P}^n_k$, we can consider the *secant variety* $\mathrm{Sec}(X)$, which is the union of all secant lines to X, that is, lines in $\mathbb{P}^n_k$ through two (not necessarily distinct) closed points on X. Then, $\dim(\mathrm{Sec}(X)) \leq 3$, since X has dimension 1, so the choice of a secant line depends on two parameters and every secant line contains a 1-dimensional family of points. So, if $n \geq 4$, we find a point $p \in \mathbb{P}^4_k \setminus \mathrm{Sec}(X)$. The map $X \to \mathbb{P}^{n-1}_k$ given by the projection from p separates points, because p does not lie on any line through distinct points of X, and it separates tangent vectors, since p does not lie on any tangent line to X. Hence, $X \to \mathbb{P}^{n-1}_k$ is a closed immersion. Repeating this process, we arrive at $n = 3$.

Now, given $X \subseteq \mathbb{P}^3_k$, we can try to project further to $\mathbb{P}^2_k$. Since $\mathrm{Sec}(X)$ may have dimension 3, such a projection will, in general, not be a closed immersion. However, one can find a point $p \in \mathbb{P}^3_k$ which lies on only finitely many secant lines and such that these finitely many secant lines join only two distinct points of X. The resulting projected curve $X' \subseteq \mathbb{P}^2_k$ will have normalization X, since $X \to X'$ has degree 1, and its singular points will be *nodes*, that is, they look locally (in a suitable sense) like the intersection of two lines in $\mathbb{P}^2_k$.

Remark 21.10. More generally, the same type of argument shows that every smooth projective variety of dimension d embeds into $\mathbb{P}^{2d+1}$.

22. Lecture 29: Elliptic curves

In this lecture, we will study elliptic curves. These curves have applications in numerous other areas of mathematics.

Definition 22.1. An *elliptic curve* over a perfect field k is a smooth complete geometrically integral curve E of genus 1 over k together with a rational point $O \in E(k)$.

Recall that the Picard group $\mathrm{Pic}(E)$ is a group with respect to tensor products of invertible sheaves. Let $\mathrm{Pic}^0(E) \subseteq \mathrm{Pic}(E)$ be the subgroup of invertible sheaves of degree 0. Consider the map of sets

$$\begin{array}{rcl} E(k) & \to & \mathrm{Pic}^0(E) \\ P & \mapsto & \mathcal{O}_E(P-O). \end{array}$$

Proposition 22.2. *The above map is a bijection.*

Proof. Let $\mathcal{L} \in \mathrm{Pic}^0(E)$. Then, $\deg(\mathcal{L}(O)) = 1$. By Riemann–Roch, we have

$$h^0(E, \mathcal{L}(O)) = 1,$$

so there exists a unique effective divisor D of degree 1 with $\mathcal{O}_X(D) \cong \mathcal{L}(O)$. But an effective divisor of degree 1 is nothing but a k-rational point P, and we have $\mathcal{O}_X(P-O) \cong \mathcal{L}$. □

Corollary 22.3. *Let (E, O) be an elliptic curve. Then, $E(k)$ admits a unique group structure compatible with the bijection $E(k) \to \mathrm{Pic}^0(E)$.*

Now, we can try to describe this group structure geometrically.

Proposition 22.4. *Let (E, O) be an elliptic curve over an algebraically closed field. Then, there exists a closed immersion $\iota : E \hookrightarrow \mathbb{P}^2_k$ that identifies E with a cubic curve and such that there exists a line in $\mathbb{P}^2_k$ that meets E only in O.*

Proof. Consider the linear system $|3O|$. Since $\deg(3O) = 3$, Proposition 21.6 shows that $|3O|$ is very ample. By Riemann–Roch, we have $h^0(E, \mathcal{O}_E(3O)) = 3$, hence $|3O|$ yields a closed immersion $\iota : E \to \mathbb{P}^2_k$. Since $|3O|$ contains $3O$, there is a global section s of $\mathcal{O}_E(3O)$ with associated zero locus $3O$. We can choose the immersion to $\mathbb{P}^2_k$ so that $\iota^* x_0 = s$. Then, $3O = Z(\iota^* x_0) = V_+(x_0) \cap \iota(E)$ and $V_+(x_0)$ is the line in the statement. □

Let $P, Q \in E$ be two closed points. By Riemann–Roch, there exists a unique point R' such that $P + Q + R' \sim 3O$. Geometrically, using that the elements of $|3O|$ are exactly the intersections of E with lines in $\mathbb{P}^2_k$, R' is the third intersection point of the line through P and Q with E. Similarly, by Riemann–Roch, there exists a unique point R such that $R' + R + O \sim 3O$. Geometrically, R is the third intersection point of the line through R and O with E. Now, we have

$$R - O \sim O - R' \sim P + Q - 2O,$$

so $\mathcal{O}_E(R-O) \cong \mathcal{O}_E(P+Q-2O)$, hence R is the sum of P and Q with respect to the group law on (E, O). Similarly, one shows that the inverse of $P \in E$ with respect to the group law on (E, O) is the third intersection point of the line through P and O with E.

Let us be even more explicit. By Proposition 22.4 and using x, y, z as coordinates on $\mathbb{P}^2_k$, we can describe E as the vanishing locus of a cubic equation

$$F = ax^3 + bx^2y + cx^2z + dxy^2 + exyz + fxz^2 + gy^3 + hy^2z + iyz^2 + jz^3.$$

Choose coordinates such that $O = [0 : 1 : 0]$ and such that the tangent line to O in $\mathbb{P}^2_k$ is given by $z = 0$. In terms of F, $[0 : 1 : 0] \in E$ is equivalent to $g = 0$, $T_O(E) = V_+(z)$ is equivalent to $d = 0$ and $h \neq 0$. We know that there is a line in $\mathbb{P}^2_k$ that meets E only in O. This line has to be the tangent to E (for all other lines through O, the restriction of the equation of the line to E is non-zero in $\mathfrak{m}_O/\mathfrak{m}_O^2$, hence not in $\mathfrak{m}_O^2$, so the associated divisor contains O only once). Hence,

$$T_O(E) \cap E = V_+(z) \cap V_+(F) = V_+(z, ax^3 + bx^2y + dxy^2 + gy^3),$$

so $b = 0$ and $a \neq 0$. Rescaling x and y, we may assume that $a = h = 1$. Now, our F looks as follows:

$$F = x^3 + cx^2z + exyz + fxz^2 + y^2z + iyz^2 + jz^3.$$

If $\mathrm{char}(k) \neq 2$, we can complete the square in y to assume $e = i = 0$. If, additionally, $\mathrm{char}(k) \neq 3$, we can also simplify the cubic in x to assume $c = 0$. So, in these characteristics, we arrive at the *Weierstraß equation*

$$y^2z = x^3 + Axz^2 + Bz^3$$

with $A, B \in k$. Conversely, by the Jacobian criterion, such an equation defines a smooth curve if and only if $4A^3 + 27B^2 \neq 0$.

One can check that two smooth cubic curves E and E' given by Weierstraß equations with coefficients A, B and A', B' are isomorphic (over $k = \bar{k}$) if and only if

$$1728\frac{4A^3}{4A^3 + 27B^2} = 1728\frac{4A'^3}{4A'^3 + 27B'^2}.$$

The scalar $j(E) := 1728\frac{4A^3}{4A^3+27B^2} \in k$ is called *j-invariant* of E.

Remark 22.5. If we believe the statement about the j-invariant detecting whether two smooth cubic curves are isomorphic, then the above discussion shows that the set of isomorphism classes of elliptic curves over $k = \bar{k}$ is in bijection with the k-rational points of $\mathbb{A}^1_k$. Hence, we can think of $\mathbb{A}^1_k$ as a parameter space, or (coarse) *moduli space* for elliptic curves.

Remark 22.6. Assume (E, O) is given by a Weierstraß equation. Given $P = [a : b : c] \in E$, we have $-P = [a : -b : c]$, since the line through P and $[0 : 1 : 0]$ is given by the equation $cx - az = 0$. Similarly, one can describe the addition using explicit formulas.

Remark 22.7. Note that, in particular, $\mathrm{Pic}^0(E)$ is uncountable if k is, so the Picard group of a scheme is usually not a finitely generated abelian group.

Remark 22.8. All of the above can be carried out over arbitrary fields. Over finite fields k, the group $E(k)$ is a finite abelian group. While it is easy to calculate multiples of points on an elliptic curve, division is (computationally) hard if $E(k)$ is large. This is what makes elliptic curves relevant in cryptography.

Remark 22.9. If $k = \mathbb{C}$, then $E(\mathbb{C})$ considered as a complex manifold with the analytic topology, is isomorphic to the quotient of $\mathbb{C}$ by a full lattice $\Lambda = \mathbb{Z} + \tau\mathbb{Z} \subseteq \mathbb{C}$ where $\tau \in \mathbb{C} \setminus \mathbb{R}$, that is, $E(\mathbb{C})$ is a complex torus.

Conversely, every complex torus is an elliptic curve over $\mathbb{C}$ in our sense. Given $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$, we consider

$$\wp(z) := \frac{1}{z^2} + (\sum_{0\neq\lambda\in\Lambda} \frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2})$$

$$g_2 := 60 \sum_{0\neq\lambda\in\Lambda} \frac{1}{\lambda^4}$$

$$g_3 := 120 \sum_{0\neq\lambda\in\Lambda} \frac{1}{\lambda^6}$$

where $\wp(z)$ is a meromorphic function called *Weierstrass $\wp$-function* and g_i is the i-th Eisenstein series. The function $\wp$ satisfies the differential equation

$$\wp'(z) = 4\wp(z)^3 - g_2\wp(z) - g_3$$

and it can be shown that the map

$$\begin{array}{ccc} \mathbb{C} & \to & \mathbb{P}^2_{\mathbb{C}}(\mathbb{C}) \\ z & \mapsto & \begin{cases} [\wp(z) : \wp'(z) : 1] \text{ if } z \notin \Lambda \\ [0:1:0] \text{ else} \end{cases} \end{array}$$

induces a biholomorphic map between the torus $\mathbb{C}/\Lambda$ and $E(\mathbb{C})$, where E is the elliptic curve given by the Weierstrass equation

$$y^2 z = 4x^3 + g_2 x + g_3.$$

In higher dimensions, a general complex torus is not algebraic, i.e., does not embed into complex projective space.

23. Lecture 30: Theta characteristics

Throughout this section, we work over an algebraically closed field k of characteristic different from 2. All our curves will be smooth, complete, and geometrically integral.

Recall from Corollary 20.15 that the canonical sheaf ω_X of a curve X has degree $2g(X) - 2$. In particular, it always even, so ω_X may be divisible by 2 in $\mathrm{Pic}(X)$.

Definition 23.1. Let X be a curve. A *theta characteristic on* X is an invertible sheaf θ with $\theta^{\otimes 2} \cong \omega_X$. It is called *even* if $h^0(X, \theta)$ is even and *odd* otherwise. It is called *effective* if $h^0(X, \theta) \neq 0$.

Example 23.2.
- If $g(X) = 0$, then $X \cong \mathbb{P}^1_k$, $\mathrm{Pic}(X) \cong \mathbb{Z}$ and $\omega_X \cong \mathcal{O}_X(-2)$, so $\mathcal{O}_X(-1)$ is the unique theta characteristic on X.
- If $g(X) = 1$, then $\omega_X \cong \mathcal{O}_X$. Thus, theta characteristics are in bijection with the 2-torsion subgroup $E(k)[2] \subseteq E(k)$ and using the group law of the previous section, we see that $E(k)[2] \cong (\mathbb{Z}/2\mathbb{Z})^2$. Since every theta characteristic on X has degree 0, it is effective if and only if it is trivial. So, there are 3 even and 1 odd theta characteristic on X.
- If $g(X) = 2$, then
$$h^0(X, \omega_X) = 2$$
by Riemann–Roch and ω_X is globally generated by Proposition 21.6. The associated complete linear system thus induces a map $\pi\colon X \to \mathbb{P}^1_k$ of degree 2 realizing X as a hyperelliptic curve.

  Since $\mathrm{char}(k) \neq 2$, the map π is separable, so by the Hurwitz formula it has exactly 6 ramification points $p_1, \dots, p_6 \in X$. As $2p_i$ is in the linear system corresponding to ω_X, the sheaf $\mathcal{O}_X(p_i)$ is an effective (and odd, since $|p_i|$ is 0-dimensional by Riemann–Roch) theta characteristic on X. Conversely, if θ is an effective theta characteristic, then $\theta \cong \mathcal{O}_X(p)$ for a point $p \in X$ with $\mathcal{O}_X(2p) \cong \omega_X$, so p is a ramification point of π.

  Thus, X has precisely 6 effective theta characteristics, they are all odd, and they correspond to the ramification points of the canonical map to $\mathbb{P}^1_k$.

The natural questions are:

(1) Does every curve have a theta characteristic?
(2) If yes, how many are there?
(3) How many of them are even/odd?

This is where we need some input from Abelian varieties, or more precisely Jacobians of curves.

Definition 23.3. A *group scheme* over k is a k-scheme G together with k-morphisms

$$\begin{aligned} m\colon G \times G &\to G \\ i\colon G &\to G \\ e\colon \mathrm{Spec}\, k &\to G \end{aligned}$$

such that for every k-scheme S, the tuple $(G(S), m(S), i(S), e(S))$ is a group. G is called *Abelian variety* if it is smooth and proper over k.

If $g \in G(k)$ is a k-rational point, then the translation morphism

$$G = \mathrm{Spec}\, k \times G \xrightarrow{g \times \mathrm{id}} G \times G \xrightarrow{m} G$$

is an isomorphism. In particular, all connected components of G are isomorphic to the *identity component* G^0, which is the component containing the point e. We will use the following results as blackboxes:

Theorem 23.4. *Let X be a smooth and proper variety over k. Then, there is a group scheme $\mathrm{Pic}_{X/k}$ locally of finite type over k, called* Picard scheme of X over k *such that there is an isomorphism of groups*

$$\mathrm{Pic}_{X/k}(k) \cong \mathrm{Pic}(X).$$

Theorem 23.5. *Let X be a curve. Then, the following hold:*

(1) $\mathrm{Jac}(X) := \mathrm{Pic}^0_{X/k}$ *is an Abelian variety, called* Jacobian of X
(2) For every $d \in \mathbb{Z}$, there is a unique connected component $\mathrm{Pic}^d_{X/k}$ of $\mathrm{Pic}_{X/k}$ such that $\mathrm{Pic}^d_{X/k}(k)$ is the subset of $\mathrm{Pic}(X)$ consisting of invertible sheaves of degree d.
(3) There are no other connected components of $\mathrm{Pic}_{X/k}$.

Theorem 23.6. *Let A be an Abelian variety over k and let n be an integer coprime to $\mathrm{char}(k)$. Then,*

$$A(k)[n] \cong (\mathbb{Z}/n\mathbb{Z})^{2\dim(A)}.$$

Back to theta characteristics, we can use these results to prove the following statement:

Corollary 23.7. *Let X be a curve and $\mathcal{L} \in \mathrm{Pic}(X)$ an invertible sheaf of degree d and let n be a positive integer prime to $\mathrm{char}(k)$ and with $n \mid d$. Then, there are exactly $n^{2g(X)}$ invertible sheaves $\mathcal{M} \in \mathrm{Pic}(X)$ with $\mathcal{M}^{\otimes n} \cong \mathcal{L}$.*

Proof. Multiplication by n on $\mathrm{Pic}(X)$ can be realized as the map on k-points induced by the composition

$$[n]\colon \mathrm{Pic}_{X/k} \xrightarrow{\Delta^n} \prod_{i=1}^{n} \mathrm{Pic}_{X/k} \xrightarrow{m^n} \mathrm{Pic}_{X/k}.$$

Now, $[n]^{-1}(\mathrm{Pic}^d_{X/k}) \subseteq \mathrm{Pic}^{d/n}_{X/k}$, so we have an induced map $\mathrm{Pic}^{d/n}_{X/k} \to \mathrm{Pic}^d_{X/k}$.

This morphism is proper, as a morphism between proper k-schemes. If the fiber of $[n]$ over $\mathcal{L}$ is non-empty, then it consists of exactly $n^{2g(X)}$ points. Indeed, if $\mathcal{M}^{\otimes n} \cong \mathcal{L}$ and $\mathcal{N}^{\otimes n} \cong \mathcal{O}_X$, then $(\mathcal{M} \otimes \mathcal{N})^{\otimes n} \cong \mathcal{L}$ and any two preimages of $\mathcal{L}$ differ by such an $\mathcal{N}$. In particular, the fibers of $[n]$ over k-points are finite. Zariski's main theorem, which we will prove next term, implies that a proper morphism of finite type k-schemes with finite fibers over k-points is in fact finite, so the map $\mathrm{Pic}^{d/n}_{X/k} \to \mathrm{Pic}^d_{X/k}$ is finite between proper varieties of the same dimension and hence it is surjective. □

In particular, every curve X of genus g admits exactly 2^{2g} theta characteristics. Let $\mathrm{TChar}(X)$ be the set of theta characteristics on X and $\mathrm{Jac}(X)[2]$ the set of 2-torsion points on the Jacobian of X.

On $\mathrm{Jac}(X)[2]$, one can define the *Weil pairing*

$$\langle\ ,\ \rangle\colon \mathrm{Jac}(X)[2] \times \mathrm{Jac}(X)[2] \quad \to \quad \mathbb{F}_2$$

as follows: Given $\mathcal{L}, \mathcal{M} \in \mathrm{Jac}(X)[2]$, choose divisors D, E with $\mathcal{L} \cong \mathcal{O}(D)$ and $\mathcal{M} \cong \mathcal{O}(E)$ such that D and E have only points of multiplicity 1 and points in common. Then, choose rational

functions $f, g \in k(X)$ with $\operatorname{div}(f) = 2D$ and $\operatorname{div}(g) = 2E$. Then, by what is called *Weil's reciprocity law*, one has

$$\frac{f(E)}{g(D)} := \frac{\prod_{x\in E}(f(x))}{\prod_{x\in D}(g(x))} = \pm 1.$$

Now, one sets

$$\langle \mathcal{L}, \mathcal{M} \rangle := \begin{cases} 1 \text{ if } \frac{f(E)}{g(D)} = -1 \\ 0 \text{ else.} \end{cases}$$

It turns out that the Weil pairing is a well-defined non-degenerate symplectic form on $2^{2g(X)}$-dimensional $\mathbb{F}_2$-vector space $\operatorname{Jac}(X)[2]$.

On the other hand, for any $\theta \in \operatorname{TChar}(X)$, one can define a quadratic form

$$\begin{aligned} q_\theta : \operatorname{Jac}(X)[2] &\to \mathbb{F}_2 \\ \mathcal{L} &\mapsto h^0(X, \theta \otimes \mathcal{L}) + h^0(X, \theta) \bmod 2 \end{aligned}$$

Recall that one can associate to any quadratic form q on a vector space V a symmetric bilinear form b, called polarization of q, as follows

$$b(v, w) := q(v + w) - q(v) - q(w)$$

and if V is of characteristic 2, then $b(v, v) = q(2v) - 2q(v) = 0$, so the form b is symplectic.

Theorem 23.8 (Riemann–Mumford relations)**.** *The q_θ are distinct quadratic forms with polarization equal to the Weil pairing.*

Given a quadratic form q on a vector space V of dimension $2g$ over $\mathbb{F}_2$ whose associated polarization b is non-degenerate, one can find a symplectic basis $e_1, \dots, e_g, f_1, \dots, f_g$, i.e., such that $b(e_i, e_j) = b(f_i, f_j) = 0$ and $b(e_i, f_j) = \delta_{ij}$. The *Arf-invariant* of q is then defined as

$$\operatorname{Arf}(q) = \sum_{i=1}^{g} q(e_i)q(f_i) \in \mathbb{F}_2.$$

Combinatorics show that there are exactly 2^{2g} quadratic forms with the same associated non-degenerate polarization and that $2^{2g-1}(2^g+1)$ of them have Arf-invariant 0 while the other $2^{2g-1}(2^g-1)$ ones have Arf-invariant 1.

The connection to theta characteristics is that θ is even if and only if the Arf-invariant of q_θ is 0. We thus conclude:

Corollary 23.9. *On a curve X, there are exactly $2^{g-1}(2^g + 1)$ even and $2^{g-1}(2^g - 1)$ odd theta characteristics.*

Instead of giving details for the above, let us see some applications:

Example 23.10. Let X be a curve of genus 3. Then, X has $2^6 = 64$ theta characteristics, with 36 of them even and 28 of them odd. Note that all of them have degree 2.

Let D be a divisor on X with $\omega_X \cong \mathcal{O}_X(D)$. Then, D has degree 4 and $h^0(X, \mathcal{O}_X(D)) = 4$. Thus, $|D|$ yields a rational map $f\colon X \dashrightarrow \mathbb{P}^2_k$. We claim that $|D|$ is base-point free. Indeed, if were not, then the image of f would be either a cubic or a conic, and (the normalizations of) both of these curves cannot have genus 3 by the genus-degree formula. Considering the intersection with a general line, we thus have

$$4 = \deg(f) \cdot d.$$

Hence, either $\deg(f) = d = 2$, so that X is a hyperelliptic curve, or $\deg(f) = 1$ and $d = 4$, so that X is isomorphic to a plane quartic curve. Recall that, in the latter case, X is not a hyperelliptic curve in any way by Exercise 81.

Now, we claim that X is hyperelliptic if and only if it admits a theta characteristic θ with $h^0(X, \theta) \geq 2$: By the above, if X is hyperelliptic, then we can choose this hyperelliptic map π so that the composition $X \xrightarrow{\pi} \mathbb{P}^1 \xrightarrow{v_2} \mathbb{P}^2$ with the second Veronese embedding is induced by the canonical system. Then,

$$(\pi^*\mathcal{O}(1))^{\otimes 2} = \pi^*(\mathcal{O}(2)) \cong \pi^* v_2^* \mathcal{O}(1) \cong \omega_X,$$

so $\theta := \pi^*\mathcal{O}(1)$ is a theta characteristic with $h^0(X, \theta) \geq 2$. Conversely, pick a theta characteristic θ with $h^0(X, \theta) \geq 2$. Let D be a divisor associated to θ. Since $\deg(D) = 2$ and X is not isomorphic to $\mathbb{P}^1$, any 1-dimensional linear subsystem of $|D|$ is base-point free. Thus, such a system gives a morphism $\pi\colon X \to \mathbb{P}^1$ of degree 2, so X is a hyperelliptic curve.

From now on, assume $X \subseteq \mathbb{P}^2_k$, embedded via the complete linear system $|K_X|$ associated to the canonical sheaf ω_X. Let θ be an odd theta characteristic. By the above, θ has a unique (up to scaling) non-zero global section and we let $D = P + Q$ be the associated divisor. Since $2D \sim K_X$ and the embedding $X \subseteq \mathbb{P}^2_k$ is given by the complete linear system, we deduce that $2D$ is the intersection of X with a line ℓ. In other words, the equation of the line ℓ restricted to X has valuation 2 at P and Q (resp. 4 if $P = Q$), so ℓ is the tangent line to both P and Q. We say that ℓ is a *bitangent* to X. Conversely, any bitangent ℓ' to X intersects X in a divisor of the form $2P' + 2Q'$ and $P' + Q'$ is an odd theta characteristic of X. We conclude:

Every smooth plane quartic curve admits precisely 28 bitangents.

Remark 23.11. Analogously, one can show that any smooth sextic curve on $\mathbb{P}(1, 1, 2) \subseteq \mathbb{P}^3$ admits exactly 120 tritangent planes.

Remark 23.12. Let $X \subseteq \mathbb{P}^3_k$ be a smooth cubic surface and let $\pi\colon X \dashrightarrow \mathbb{P}^2_k$ be the projection from a general point $x \in X$. Then, π lifts to a morphism $\psi : Y \to \mathbb{P}^2_k$ from the blow-up Y of X at x. The map ψ is a double cover of $\mathbb{P}^2_k$ branched over a plane quartic curve and one can show that π identifies lines on X (and the exceptional divisor of $Y \to X$) with the 28 bitangent lines to X. This is one approach of showing that every smooth cubic surface contains exactly 27 lines.

Remark 23.13. The numbers $27, 56 = 2 \cdot 28$, and $240 = 2 \cdot 120$ are the numbers of roots in the three exceptional finite root lattices E_6, E_7, E_8. One can show that smooth cubic surfaces, double covers of $\mathbb{P}^2_k$ branched over a smooth quartic, and double covers of $\mathbb{P}(1, 1, 2)$ branched over a smooth sextic, are blowups of $\mathbb{P}^2_k$ in $6, 7$, and 8 general points, respectively, and the root lattices appear naturally when one studies the intersection theory of curves on these blowups.

24. Appendix A: Auxiliary results from commutative algebra

Here, we collect some non-trivial auxiliary results from commutative algebra.

24.1. Properties of Artinian rings.

Definition 24.1. A ring A is called *Artinian* if it satisfies the descending chain condition for ideals.

The following properties of an Artinian ring A should be known from commutative algebra:

(1) Every prime ideal in A is maximal.
(2) A has finitely many maximal ideals.
(3) A is a product of finitely many Artinian local rings.
(4) A is Noetherian of dimension 0 (and this is in fact equivalent to being Artinian).

These standard results imply the following:

Proposition 24.2. *Let K be a field and let A be a finitely generated K-algebra. Then, the following are equivalent:*

(1) A is Artinian.
(2) $\dim_K A < \infty$.
(3) $\operatorname{Spec} A$ is finite.

Proof. $(1) \Rightarrow (3)$ follows from the fact that A has finitely many prime ideals, all of which are maximal.

For $(3) \Rightarrow (1)$, we use that A is finitely generated over a field, hence Jacobson, so its nilradical $N = \sqrt{0}$ is the intersection of the maximal ideals in A. Since $\operatorname{Spec} A$ is finite, there are only finitely many primes in A, so the chinese remainder theorem yields $A/N = \prod_{i=1}^n A/\mathfrak{m}_i$, where the $\mathfrak{m}_i$ are the maximal ideals in A. In particular, we see that $\operatorname{Spec} A = \{\mathfrak{m}_1, \dots, \mathfrak{m}_n\}$, so all prime ideals in A are maximal, i.e. $\dim(A) = 0$. Hence, A is Artinian.

For $(1) \Rightarrow (2)$, we use that A is a product of Artinian local rings to reduce to the case where A is local with maximal ideal $\mathfrak{m}$. By Hilbert's Nullstellensatz, $L = A/\mathfrak{m}$ is finite-dimensional over K. Since A is Noetherian and 0-dimensional, $\mathfrak{m}$ is nilpotent, so we have an ascending chain

$$0 = \mathfrak{m}^n \subseteq \mathfrak{m}^{n-1} \subseteq \ldots \subseteq A.$$

Every quotient $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a vector space over L and it has to be finite-dimensional over L, for otherwise we could extend the above sequence to an infinite chain of ideals in A, which is impossible, since A is Noetherian. Since L is finite over K, this implies that each $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a finite-dimensional K-vector space. So, as K-vector spaces, we have $A \cong \bigoplus_{i=0}^n \mathfrak{m}^i/\mathfrak{m}^{i+1}$ and thus $\dim_K A < \infty$.

Finally, $(2) \Rightarrow (1)$ is immediate, since finite-dimensionality implies the descending chain condition for ideals. $\square$

REFERENCES

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